HIGHER-REGULATOR AND DELIGNE STRUCTURES ON PICARD MODULAR SURFACES RELATED BY A NONCRITICAL GU(2,1) L-VALUE

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Allowed standard prerequisites (editorial).

- Motivic/absolute Hodge cohomology, Shimura varieties, GU(2,1) representations and Deligne periods. Original complete reference data and author-local constructions/formulations retained.

1. Introduction

1.1. **Motivation.** The analytic class number formula is a deep relationship between an analytic invariant and an arithmetic invariant of a number field F. It relates the leading Taylor coefficient at 0 of the Dedekind zeta-function $\zeta_F(s)$ of F to the Dirichlet regulator

$$r\colon O_F^\times \to \mathbb{R}^{r_1+r_2},$$

where $O_F^\times$ is the unit group and r_1 (resp., r_2) is the number of real (resp., complex) places of the number field F. In the 1980s, Beilinson made deep conjectures about special values of motivic L-functions generalizing the classical analytic class formula [Bei84a]. In order to do so, Beilinson replaced the units $O_K^\times$ by the motivic cohomology $\mathrm{H}^i_M(X,\mathbb{Q}(n))$ where X is a smooth $\mathbb{Q}$-scheme and replaced the Dirichlet regulator by the higher regulator map

$$r_H\colon \mathrm{H}^i_M(X,\mathbb{Q}(n)) \to \mathrm{H}^i_H(X,\mathbb{R}(n))$$

from motivic cohomology to absolute Hodge cohomology. For an introduction to Beilinson's conjectures, the interested reader may consult [Nek94].

In this paper, we establish a connection between motivic classes in motivic cohomology with non-trivial coefficients of Picard modular surfaces and the motivic L-functions of some cuspidal automorphic representations of the unitary similitude group $\mathrm{GU}(2,1)$, through which we show that the motivic classes are non-trivial. It answers a question raised in [LSZ22], in which the authors constructed an Euler system for $\mathrm{GU}(2,1)$ from the motivic classes. Along the way, we also prove a vanishing on the boundary result for the motivic classes. We hope this work could shed some light on proving more general relationships between special values of motivic L-functions associated to automorphic representations and mixed motives.

1.2. **Statement of the main results.** Now we introduce the main results. These can be divided into two parts. In the first part, we construct motivic classes $\mathcal{E}is_M(\phi_f)$ in the motivic cohomology with non-trivial coefficients of Picard modular surfaces $\mathrm{H}^3_M(S,V(2))$ and then we prove the classes are in a "nice" subspace of the cohomology. Along the way, we also prove the regulators of the classes $\mathcal{E}is_H(\phi_f)$ are in some "nice" subspace of the absolute Hodge cohomology $\mathrm{H}^3_M(S,V(2))$. In the second part, we prove a relation between the image of the motivic class and a non-critical value of the motivic L-function associated to some "nice" irreducible cuspidal automorphic representation π. The relation resembles Beilinson's conjectures on special values of L-functions.

Now, we state the results more precisely. Let $G=\mathrm{GU}(2,1)$ and let H be a subgroup of G that can be embedded into G and is an extension of the group GL_2. You can find the precise

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definition of H in Definition 3.2. The relation between the groups GL_2, H and G can be summarized in following diagram:

$$\begin{array}{ccc} H & \hookrightarrow & G \\ \downarrow & & \\ \mathrm{GL}_2 & & \end{array} .$$

These maps of algebraic groups induce morphisms of the corresponding Shimura varieties[1] :

$$\begin{array}{ccc} M = \mathrm{Sh}_H & \xrightarrow{\iota} & S = \mathrm{Sh}_G \\ \downarrow p & & \\ \mathrm{Sh}_{\mathrm{GL}_2} & & \end{array} .$$

Let V be an algebraic representation of G (see Definition 3.7), V_2 be the standard representation of GL_2 and W be an algebraic representation of H (see Definition 3.10). They correspond to motivic sheaves V on Sh_G, V_2 on $\mathrm{Sh}_{\mathrm{GL}_2}$ and W on Sh_H (see Lemma 4.12 and Proposition 4.4). Then we have the corresponding morphisms of motivic cohomologies:

$$\begin{array}{ccc} \mathrm{H}^1_M(M, W(1)) & \xrightarrow{\iota^*} & H^3_M(S, V(2)) \\ {\scriptstyle p^*}\uparrow & & \\ \mathrm{H}^1_M(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & & \end{array} .$$

Beilinson constructed the *Eisenstein symbol* [Bei86a, §3]. For each nonnegative integer n, the Eisenstein Symbol is a $\mathbb{Q}$-linear map

$$Eis^n_M \colon \mathcal{B}_n \to \mathrm{H}^1_M(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)),$$

where the source $\mathcal{B}_n$ is the space of locally constant $\mathbb{Q}$-valued functions on $\mathrm{GL}_2(\mathbb{A}_f)$ satisfying some equivariant properties (see Section 7.6).

If we compose the Eisenstein symbol with the pullback $p^* \colon \mathrm{H}^1_M(M, \mathrm{Sym}^n V_2(1)) \to \mathrm{H}^1_M(M, W(1))$ and then with the Gysin map $\iota_* \colon \mathrm{H}^1_M(M, W(1)) \to \mathrm{H}^3_M(S, V(2))$, we get the map $\mathcal{E}is^n_M \colon \mathcal{B}_n \to \mathrm{H}^3_M(S, V(2))$. Hence, for $\phi_f \in \mathcal{B}_n$, we have a motivic class $\mathcal{E}is^n_M(\phi_f) \in \mathrm{H}^3_M(S, V(2))$. In other words, we have the following diagram:

$$\begin{array}{ccccccc} \mathcal{B}_n & \xrightarrow{Eis^n_M} & \mathrm{H}^1_M(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p^*} & \mathrm{H}^1_M(M, W(1)) & \xrightarrow{\iota_*} & \mathrm{H}^3_M(S, V(2)) \\ \phi_f & \longmapsto & Eis^n_M(\phi_f) & & & \longmapsto & \mathcal{E}is^n_M(\phi_f) \end{array} .$$

This construction has been given in [LSZ22, Definition 9.2.3].

Applying the Beilinson regulator r_H, we have the following diagram of absolute Hodge cohomology:

$$\begin{array}{ccccccc} \mathcal{B}_{n,\mathbb{R}} & \xrightarrow{Eis^n_H} & \mathrm{H}^1_H(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p^*} & \mathrm{H}^1_H(M, W(1)) & \xrightarrow{\iota_*} & \mathrm{H}^3_H(S, V(2)) \\ \phi_f & \longmapsto & Eis^n_H(\phi_f) & & & \longmapsto & \mathcal{E}is^n_H(\phi_f) \end{array} .$$

1.2.1. *Geometric results.* Our first geometric theorem states that the Hodge class $\mathcal{E}is^n_H(\phi_f)$ is in a "nice" subspace of $\mathrm{H}^3_H(S, V(2))$, whose proof follows from the method used in [Lem15]:

Theorem 1.1 (Theorem 5.15)**.** *For $V = V^{a,b}\{r, s\}$ as in Definition 3.7, under certain conditions (see Theorem 5.15), the map $\mathcal{E}is^n_H \colon \mathcal{B}_{n,\mathbb{R}} \to \mathrm{H}^3_H(S, V(2))$ factors through the inclusion*

$$\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbf{1}, \mathrm{H}^2_{B,!}(S, V(2))_{\mathbb{R}}) \hookrightarrow \mathrm{H}^3_H(S, V(2))_{\mathbb{R}},$$

[1] In the introduction, we ignore the level structures. Rigorously speaking, S (resp., M) is defined to be the Weil restriction to $\mathbb{Q}$ of Sh_G (resp., Sh_H) (see Convention 3.25 for more details). But in the introduction, we ignore this technical detail.

where $\mathrm{MHS}^+_{\mathbb{R}}$ *is the abelian category of mixed* $\mathbb{R}$*-Hodge structures,* $\mathbf{1}$ *denotes the trivial Hodge structure that is the unit of* $\mathrm{MHS}^+_{\mathbb{R}}$ *and* $\mathrm{H}^2_{B,!}(S,V(2))$ *is the interior Betti cohomology with coefficients in* V *defined by*

$$\mathrm{H}^2_{B,!}(S,V(2))_{\mathbb{R}} := \mathrm{Im}(\mathrm{H}^2_{B,c}(S,V(2))_{\mathbb{R}} \to \mathrm{H}^2_B(S,V(2))_{\mathbb{R}}).$$

Remark 1.2. The interior Betti cohomology $\mathrm{H}^2_{B,!}(S,V(2))_{\mathbb{C}}$ is the cohomology group containing the "cohomological" cuspidal automorphic representations π of $G(\mathbb{A})$ (see Proposition 3.27 and its proof). Hence, Theorem 1.1 tells us that the regulator of the motivic class $\mathcal{E}is^n_H(\phi_f)$ is in a "nice" subspace that is related to automorphic representations. It gives one hope that the class $\mathcal{E}is^n_H(\phi_f)$ can be related to the motivic L-function of some automorphic representation.

Our second geometric theorem is a "motivic lifting" of the first theorem that says that the motivic class $\mathcal{E}is^n_M(\phi_f)$ is in a "nice" subspace of $\mathrm{H}^3_M(S,V(2))$:

Theorem 1.3 (Theorem 6.20)**.** *Under the same conditions of the previous theorem, the map* $\mathcal{E}is^n_M\colon \mathcal{B}_n \to \mathrm{H}^3_M(S,V(2))$ *factors through the inclusion*

$$\mathrm{H}^3_M(\mathrm{Gr}_0M_{gm}(V(2)),\mathbb{Q}(0)) \hookrightarrow \mathrm{H}^3_M(S,V(2)),$$

where $\mathrm{H}^3_M(\mathrm{Gr}_0M_{gm}(V(2)),\mathbb{Q}(0))$ *(see Definition 6.27) is the motivic cohomology of the interior motive* $\mathrm{Gr}_0M_{gm}(S,V(2))$ *(see Definition 6.17).*

Remark 1.4.

- The interior Betti cohomology is the realization of the interior motive $\mathrm{Gr}_0\mathrm{M}_{\mathrm{gm}}(S,V(2))$ through the Betti realization functor.
- Under the Beilinson regulator r_H, the image of $\mathrm{H}^3_M(\mathrm{Gr}_0M_{gm}(V(2)),\mathbb{Q}(0))$ lies in
$$\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbf{1},\mathrm{H}^2_{B,!}(S,V(2))_{\mathbb{R}}).$$
Hence, Theorem 1.3 can be viewed as a "motivic lifting" of Theorem 1.1.
- Theorem 1.3 tells us that the motivic class $\mathcal{E}is^n_M(\phi_f)$ is in a "nice" subspace that is related to the Grothendick motive associated to some automorphic representation [Wil15, Theorem 5.6]. It gives one hope to relate the motivic class $\mathcal{E}is^n_M(\phi_f)$ to the motivic L-function of some automorphic representation.
- In [Kin98], the author raised the question of proving the motivic version of vanishing on the boundary for Eisenstein class on Shimura varieties. In the case of Hilbert modular surfaces, a result of motivic vanishing on the boundary has been proved in [Wil12, Corollary 3.14] using a different formalism from us.

1.2.2. *Results on special values of L-functions.* Let $\pi = \pi_f \otimes \pi_\infty$ be some "cohomological" irreducible cuspidal automorphic representation π of $G(\mathbb{A})$, where we denote by $E(\pi_f)$ its rational field (see Section 10). Let $M(\pi_f,V(2))$ be the Grothendieck motive associated to π [Wil15, Theorem 5.6] after a Tate twist and $M_B(\pi_f,V(2))$ be its Betti realization. Then the absolute Hodge cohomology of $M_B(\pi_f,V(2))$ is a rank-1 $E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}$-module. In it, we define two rational structures $\mathcal{K}(\pi_f,V(2))$ and $\mathcal{D}(\pi_f,V(2))$, where the rational structure $\mathcal{K}(\pi_f,V(2))$ (maybe trivial) is defined using the motivic class $\mathcal{E}is_M(\phi_f)$ and $\mathcal{D}(\pi_f,V(2))$ (non-trivial) is defined using the comparison between the Betti and de Rham realizations (see Section 7.1). Our theorem says that the difference between the two rational structures can be measured using a non-critical value of the motivic L-function $L(M_{\text{ét}}(\pi_f,V(2)),s)$ (see Definition 10.10) of π:

Theorem 1.5 (Theorem 10.16)**.** *Under the conditions in the statement of Theorem 10.16, the relation between* $\mathcal{K}(\pi_f,V(2))$ *and* $\mathcal{D}(\pi_f,V(2))$ *is as follows:*

$$\mathcal{K}(\pi_f,V(2)) = C\cdot L(M_{\acute{e}t}(\pi_f,V(2)),0)\mathcal{D}(\pi_f,V(2)),$$

where $C \in (E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C})^\times$ *and* C *can be expressed explicitly in terms of a comparison of Whittaker periods (see Defintion 9.9) and Deligne periods (see Definition 9.2).*

Remark 1.6.

- That the motivic L-function $L(M(\pi_f,V(2)),s)$ is being evaluated at $s=0$ is exactly what Beilinson conjectured. [Nek94, Conjecture (6.1)]

- The C in Theorem 1.5 should be in $(E(\pi_f)\otimes_{\mathbb{Q}}\overline{\mathbb{Q}})^\times$ according to Beilinson's conjectures.
- Theorem 1.5 is a weight $w \leq -3$ counterpart of the main theorem proved in [PS18], where the authors constructed a motivic class ξ in the motivic cohomology with trivial coefficients $\mathrm{H}^3_M(S,\mathbb{Q}(2))_{\overline{\mathbb{Q}}}$ and proved a relation between ξ and a non-critical L-value of some cuspidal automorphic representation π[2].
- A direct corollary of Theorem 1.5 is that the map $\mathcal{E}is_M$ is non-trivial (see Corollary 10.18). This answers a question raised in [LSZ22], where the authors needed the non-triviality of the motivic classes $\mathcal{E}is(\phi_f)$ for $\phi_f \in \mathcal{B}_n$ as an input to construct an Euler system for $\mathrm{GU}(2,1)$.
- Theorem 1.5 is deduced from an automorphic version of the theorem (see Theorem 10.4) and a comparison between automorphic L-functions and motivic L-functions after many normalizations (see Proposition 10.14).

1.3. **An overview of the proof.**

1.3.1. *An outline of the proof of the geometric results.* We first give an outline of the proof of vanishing on the boundary of absolute Hodge cohomology (Theorem 1.1, Theorem 5.15).

For each of the two Shimura varieties $X = \{M, S\}$, we denote by X^* its Baily–Borel compactification (see Section 5.1 for more details) and let $\partial X = X^* - X$ be the cusps of the compactification. Hence, we have the following commutative diagram:

$$\begin{array}{ccccc} M & \xrightarrow{j'} & M^* & \xleftarrow{i'} & \partial M \\ \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ S & \xrightarrow{j} & S^* & \xleftarrow{i} & \partial S, \end{array}$$

where j and j' are open immersions, i and i' are closed immersions and ι, p and q are closed immersions[3].

By properties of the derived category of real algebraic mixed Hodge modules (see [HW98, Definition A.2.4]), we have the following exact sequence (see Proposition 5.17):

$$0 \to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbf{1}, \mathrm{H}^2_{B,!}(S,V(2))_{\mathbb{R}}) \to \mathrm{H}^3_H(S,V(2)) \to \mathrm{H}^1_H(\partial S, i^*j_*V(2)). \tag{1.1}$$

By the functoriality, we can get the following commutative diagram (see Proposition 5.18):

$$\begin{array}{ccc} & & \mathcal{B}_{n,\mathbb{R}} \\ & & \downarrow \\ \mathrm{H}^1_H(M,W(1)) & \longrightarrow & \mathrm{H}^0_H(\partial M, i'^*j'_*W(1)) \\ \downarrow & & \downarrow{\scriptstyle \theta^H} \\ \mathrm{H}^3_H(S,V(2)) & \longrightarrow & \mathrm{H}^1_H(\partial S, i^*j_*V(2)). \end{array} \tag{1.2}$$

By exact sequence (1.1) and commutative diagram (1.2), in order to prove that the map $\mathcal{E}is^n_H\colon \mathcal{B}_{n,\mathbb{R}} \to \mathrm{H}^3_H(S,V(2))$ factors through the inclusion

$$\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbf{1}, \mathrm{H}^2_{B,!}(S,V(2))_{\mathbb{R}}) \hookrightarrow \mathrm{H}^3_H(S,V(2)),$$

it suffices to prove that the map θ^H is zero, which is carried out in Proposition 5.26. The key input of the proof of that fact is the explicit computation of degeneration of Hodge structures over M and S (see Lemma 5.7, Lemma 5.10), which is based on a theorem of Burgos-Wildeshaus

[2]There is a gap in the proof of [PS18], which seems could be fixed by the tool of tempered currents developed recently in [BGCLRJ24]

[3]It holds after a careful choice of level structures.

(see Theorem 5.2) and Kostant's theorem (see Theorem 5.5). Finally, we remark that the degeneration of Hodge structures over S has been computed in [Anc17].

We now outline the proof of vanishing on the boundary of motivic cohomology (Theorem 1.3, Theorem 6.20). The proof is parallel to the proof in the absolute Hodge cohomology case. However, we need to translate everything into the language of motives.

First, we have a motivic version of the exact sequence (1.1) (see Proposition 6.28):

$$0 \to \mathrm{H}^3_M(\mathrm{Gr}_0 M_{gm}(V(2)), \mathbb{Q}(0)) \to \mathrm{H}^3_M(S, V(2)) \to \mathrm{H}^3_M(\partial S, i^* j_* V(2)). \tag{1.3}$$

Second, by functoriality of the triangulated category of mixed motives, we have the motivic version of the commutative diagram (1.1) (see Proposition 6.30):

$$\begin{array}{ccc} & & \mathcal{B}_n \\ & & \downarrow \\ \mathrm{H}^1_M(M, W(1)) & \longrightarrow & \mathrm{H}^1_M(\partial M, i'^* j'_* W(1)) \\ \downarrow & & \downarrow{\scriptstyle \theta^M} \\ \mathrm{H}^3_M(S, V(2)) & \longrightarrow & \mathrm{H}^3_M(\partial S, i^* j_* V(2)). \end{array} \tag{1.4}$$

Finally, similar to the absolute Hodge cohomology case, in order to prove that the map $\mathcal{E}is^n_M \colon \mathcal{B}_n \to \mathrm{H}^3_M(S, V(2))$ factors through the inclusion

$$\mathrm{H}^3_M(\mathrm{Gr}_0 M_{gm}(V(2)), \mathbb{Q}(0)) \hookrightarrow \mathrm{H}^3_M(S, V(2)),$$

it suffices to prove that θ^M is zero, whose proof is based on the fact that θ^H is zero and weight conservativity of Hodge realization functor for Voevodsky motives of Abelian type [Wil15, Theorem 1.12.] (see Proposition 6.32).

1.3.2. *An outline of the proof of the L-value results.* In order to prove the relation

$$\mathcal{K}(\pi_f, V(2)) = C' \cdot \mathcal{D}(\pi_f, V(2)),$$

in the rank 1 $E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{R}$-module $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}})$ for some $C' \in (E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{R})^\times$, it suffices to find an $E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{R}$-linear form $\psi \colon M_B(\pi_f, V(2))^-_{\mathbb{R}}(-1) \to E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{C}$ which is trivial on $F^0 M_{dR}(\pi_f, V(2))_{\mathbb{R}}$ such that

$$\psi(\tilde{v}_K) = C' \cdot \psi(\tilde{v}_D),$$

where v_K (resp., v_D) is a $E(\pi_f)$-generator of $\mathcal{K}(\pi_f, V(2))$ (resp., $\mathcal{D}(\pi_f, V(2))$) and $\tilde{v}_K$ (resp., $\tilde{v}_D$) is a lifting of v_K (resp., v_D) through the exact sequence (7.1) (see Lemma 7.8).

By the Poincaré duality pairing

$$\langle \cdot, \cdot \rangle_B \colon \mathrm{H}^2_{B,!}(S, V(2)) \otimes \mathrm{H}^2_{B,!}(S, D) \to \mathbb{Q}(0),$$

where D is the contragredient representation of V and the construction of a differential form $\Omega \in M(\tilde{\pi}_f |\mu|^{-2}, D)^+_{\mathbb{C}}$, which is carried out in Section 7.2, we construct a linear form Ψ and we get

$$\mathcal{K}(\pi_f, V(2)) = \frac{\langle \Omega, \tilde{v}_K \rangle_B}{\langle \Omega, \tilde{v}_D \rangle_B} \mathcal{D}(\pi_f, V(2)).$$

Hence, we are left to compute $\langle \Omega, \tilde{v}_K \rangle_B$ and $\langle \Omega, \tilde{v}_D \rangle_B$.

The way to compute the pairing $\langle \Omega, \tilde{v}_K \rangle_B$ is as follows. We first map the class v_K to Deligne–Beilinson cohomology through the natural map $r_{H \to D}$ from absolute Hodge cohomology to Deligne–Beilinson cohomology (see definition of Deligne–Beilinson cohomology in Proposition 7.30), in which the class will be represented by a pair of tempered currents (see Remark 7.51)

$$(\iota_* T_{p^* Eis^n_H(\phi_f)}, \iota_* T_{p^* Eis^n_B(\phi_f)})$$

and this can be used to compute the pairing $\langle \Omega, \tilde{v}_K \rangle_B$ (see Lemma 7.52 for details). Finally, we express the pairing in terms of an integral of differential forms over Shimura varieties Sh_H (see Proposition 7.55), which is expressed explicitly in Propositin 7.59.

Remark 1.7.
- The tool of tempered currents defined in [BGCLRJ24] is a key ingredient of this step. Hence, we recall the key definitions in Section 7.4.
- In order to apply the tool of Deligne–Beilinson cohomology, we need to apply "Liebermann's trick" (see Definition 7.34) because we are working with non-trivial coefficients.

By a careful choice of test vectors (see Proposition 8.6), we can express the above explicit integral as a zeta integral $I(\varphi, \Phi, \nu, s)$ (see Definition 8.5) constructed by Gelbart and Piatetski-Shapiro, which gives an integral repersentation of the twisted standard automorphic L-function of the contragredient $\tilde{\pi}$ of π. We then prove the algebraicity of the nonarchimedean local Zeta integral at rational points in Propositon 8.21 and compute the archimedean Zeta integral explicitly in Proposition 8.34. Combining all of these with an identity (see Lemma 10.3) between automorphic L-functions of an automorphic representation and it congragredient, we can express the pairing $\langle \Omega, \tilde{v}_K \rangle_B$ in terms of an automorphic L-value of $L(s, \pi, \mathrm{std})$ (see Definition 8.13) at non-critical points and Whittaker periods (see Definition 9.9).

By the definition of Deligne periods and Deligne $E(\pi_f)$-structure (see Definition 7.4), we express the pairing $\langle \Omega, \tilde{v}_D \rangle_B$ in terms of Deligne periods (see Proposition 9.7).

Combining the above computation of $\langle \Omega, \tilde{v}_K \rangle_B$ and $\langle \Omega, \tilde{v}_D \rangle_B$, we prove a relation

$$\mathcal{K}(\pi_f, V(2)) = C' \cdot \mathcal{D}(\pi_f, V(2)),$$

where C' is expressed in terms of a non-critical value of $L(s, \pi, \mathrm{std})$ at $s = 0$, Whittaker periods and Deligne periods (see Theorem 10.4). Finally, we relate the automorphic L-function

$$L(s, \pi, \mathrm{std})$$

to the motivic L-function $L(M_{\text{ét}}(\pi_f, V(2)), s)$ (see Proposition 10.14). The fact that $C' \in (E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{C})^\times$ follows from $L(M_{\text{ét}}(\pi_f, V(2)), 0) \neq 0$ (see the proof of Corollary 10.18).

1.4. **Structure of the paper.** In Section 3.1-Section 3.2, we define certain algebraic groups, Lie groups and their representations. In Section 3.3-Section 3.4, we define the corresponding Shimura varieties and recall the needed properties of cohomology of Picard modular surfaces. In Section 3.5-Section 3.6, we compute the L-packet of discrete series at the archimedean place and compute the Hodge decomposition of corresponding motives associated to automorphic representations. In Section 4.1-Section 4.3, we recall the definition of motivic cohomology, absolute Hodge cohomology and Beilinson regulators. In Section 4.4, we recall the construction of the motivic classes in [LSZ22] and their image under the Beilinson regulator, which we call the Hodge classes. In Section 5.1-Section 5.4, we prove that the Hodge classes belong to a "nice" subspace of absolute Hodge cohomology. In Section 6.1-Section 6.4, we prove that the motivic classes belong to a "nice" subspace of motivic cohomology. In Section 7.1-Section 7.3, we explain how to associate a differential form to the contragredient automorphic representation and use it and Poincaré duality to construct a linear form. In Section 7.4, we recall the definition of Deligne–Beilinson cohomology using tempered currents. In Section 7.5-Section 7.7, we give an explicit expression of the classes in Deligne–Beilinson cohomology and an explicit expression of its pairing with the above differential form. In Section 8.1-Section 8.2, we express the above integral in terms of a zeta integral. In Section 8.3-Section 8.5, we unfold the zeta integral and recall an algebraicity result of nonarchimedean zeta integrals and compute the archimedean zeta integral explicitly. In Section 9.1, we compute the pairing of the Deligne rational structure with the diffferential form. In Section 9.2, we recall the definition of the Whittaker period. In Section 10.1-Section 10.2, we prove the main results of the paper.

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2. Notations and Conventions

General notations.

- Let $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ be the set of rational numbers, real numbers and complex numbers.
- Let $\mathbb{Q}_+^\times$ be the set of non-zero positive rational numbers.
- We use $\overline{\mathbb{Q}}$ to denote the set of algebraic numbers and use $\overline{\mathbb{Q}}^\times$ to denote the set of non-zero algebraic numbers.
- $=_{\overline{\mathbb{Q}}^\times}$: for $A, B \in \mathbb{C}$, $A =_{\overline{\mathbb{Q}}^\times} B$ means that there exists a $c \in \overline{\mathbb{Q}}^\times$ such that $A = c \cdot B$.
- For a number field F, we use $\mathbb{A}_F$ to denote the ring of adeles and use $\mathbb{A}_{F,f}$ to denote the ring of finite adeles. We use $\mathbb{A}$ to denote the ring of adeles of $\mathbb{Q}$ and use $\mathbb{A}_f$ to denote the ring of finite adeles of $\mathbb{Q}$.
- We use the notation $\hat{\mathbb{Z}}$ to denote the ring $\varprojlim \mathbb{Z}/N\mathbb{Z}$ and use $\hat{\mathbb{Z}}^\times$ to denote the abelian group $\varprojlim (\mathbb{Z}/N\mathbb{Z})^\times$, where N runs over positive integers.
- We use i to denote the complex number $\sqrt{-1}$.

Algebraic geometry.

- $\mathcal{O}_X$: When X is a scheme (resp., a complex manifold), we denote by $\mathcal{O}_X$ the structure sheaf (resp., structure sheaf of holomorphic functions) of X.
- Let $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathrm{G_m}$ be the Deligne torus. Following Deligne and Pink, we use the following convention for the equivalence of categories between algebraic representations of $\mathbb{S}$ in finite-dimensional $\mathbb{R}$-vector spaces and (semisimple) $\mathbb{R}$-mixed Hodge structures. Let (ρ, V) be such a representation of $\mathbb{S}$. The summand $V^{p,q} \subset V$ of Hodge type (p,q) is the summand of V such that $\rho(z_1, z_2)$ acts by $z_1^{-p} z_2^{-q}$ for $(z_1, z_2) \in \mathbb{S}(\mathbb{C})$. In particular, any algebraic representation V of $\mathbb{S}$ with central character c corresponds to a pure Hodge structure of weight $-c$.
- For a $\mathbb{C}$-scheme X, we use the notation X^{an} to denote the complex analytic variety associated to X through GAGA.
- For a field extension E of F and a scheme over the field E, we use the notation $\mathrm{Res}_{E/F}(X)$ to denote the Weil restriction of X to F. To be more precise, for any F-scheme S, the S-points of the scheme $\mathrm{Res}_{E/F}(X)$ are the $S \times_{\mathrm{Spec}(F)} \mathrm{Spec}(E)$-points of X.
- When we say Betti cohomology of an algebraic variety over $\mathbb{Q}$, we always mean Betti cohomology of the $\mathbb{C}$-points of the algebraic variety. In order to simplify notation, we avoid any notation for taking $\mathbb{C}$-points.

Group Theory.

- For a positive integer n, we use S_n to denote the permuation group of n elements.
- For an affine group scheme G over a ring k, sometimes we write $g \in G$ to replace

$$g \in G(R),$$

  for a k-algebra R.
- For a reductive group G over $\mathbb{Q}$, we denote by A_G the connected component of the identity of the real points of the maximal $\mathbb{Q}$-split center of G.
- For a reductive group G over $\mathbb{Q}$, let K be a maximal compact group of $G(\mathbb{R})$ and denote $K_G = A_G K$.
- For a Lie group G, we denote by G^+ the connected component of the identity in G.

Representation theory.

- For a representation (π, V) of a group G, sometimes we use π to denote the space V when there is no confusion.
- A representation of a p-adic group is always a smooth admissible $\mathbb{C}$-representation.
- For a representation V of a group (algebraic group, real Lie group, p-adic group or adelic group), we use $V^\vee$ to denote the contragredient representation of V.
- We use the notation Ind to the denote unnormalized induction functor.
- For a representation π of a group G over a field E, which is an extension of a field F, we use the notation $\mathrm{Res}_{E/F}\pi$ to denote the restriction of the representation to the field F.

Automorphic Forms.

- For a reductive group G over $\mathbb{Q}$ and a quasi-character ξ (not necessarily unitary) of A_G, let $\mathrm{L}^2(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$ be the space of functions $\phi\colon G(\mathbb{Q})\backslash G(\mathbb{A}) \to \mathbb{C}$ such that
$$\phi(zx) = \xi(z)\phi(x), \ \ z \in A_G, x \in G(\mathbb{A}),$$
and which are square integrable modulo A_G.
- Let $\mathrm{C}^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$, respectively $\mathrm{C}_c^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$ be the space of functions $\phi\colon G(\mathbb{Q})\backslash G(\mathbb{A}) \to \mathbb{C}$ such that
  - $\phi(zx) = \xi(z)\phi(x), \ \ z \in A_G, x \in G(\mathbb{A})$,
  - the restriction of ϕ to $G(\mathbb{R})$ is smooth, respectively, smooth and compactly supported modulo A_G,
  - the restriction of ϕ to $G(\mathbb{A}_f)$ is locally constant and compactly supported.
- Let $\mathrm{C}^\infty_{(2)}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$ be the space
$$\mathrm{L}^2(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi) \cap \mathrm{C}^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi).$$
We have inclusions of $((\mathfrak{g}_\mathbb{C}, K_G) \times G(\mathbb{A}_f))$-modules
$$\mathrm{C}_c^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi) \subset \mathrm{C}^\infty_{(2)}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi) \subset \mathrm{C}^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi).$$
- Let $\mathrm{C}^\infty_{\mathrm{cusp}}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi) = \mathrm{L}^2_{\mathrm{cusp}}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$ be the space of cusp forms in
$$\mathrm{C}^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi).$$
Smooth truncation to a large compact set modulo A_G induces a map of $((\mathfrak{g}_\mathbb{C}, K_G) \times G(\mathbb{A}_f))$-modules
$$\mathrm{C}^\infty_{\mathrm{cusp}}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi) \to \mathrm{C}_c^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi).$$
- An irreducible cuspidal automorphic representation $\pi = \pi_f \otimes \pi_\infty$ is an irreducible $((\mathfrak{g}_\mathbb{C}, K_G) \times G(\mathbb{A}_f))$-submodule of $\mathrm{C}^\infty_{\mathrm{cusp}}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$.

3. Preliminaries

3.1. **Algebraic groups, algebraic representations and branching laws.** In this subsection, we define the reductive group $\mathrm{GU}(2,1)$ and a subgroup H of G, together with their representations.

3.1.1. *Algebraic groups.*

Notation 3.1. Let E be an imaginary quadratic field of discriminant $-D$, and let $x \mapsto \bar{x}$ be the non-trivial Galois automorphism of E over $\mathbb{Q}$. Let $\mathcal{O}$ be the ring of integers of E. We fix an identification of $E \otimes_\mathbb{Q} \mathbb{R}$ with $\mathbb{C}$ such that the imaginary part of $\delta = \sqrt{-D}$ is positive.

Definition 3.2. (1) Let $J \in \mathrm{GL}_3(E)$ be the Hermitian matrix
$$J = \begin{pmatrix} 0 & 0 & \delta^{-1} \\ 0 & 1 & 0 \\ -\delta^{-1} & 0 & 0 \end{pmatrix}, \quad \text{where } \delta = \sqrt{-D},$$

and let $G = \mathrm{GU}(J)$ be the group scheme over $\mathbb{Z}$ such that for all $\mathbb{Z}$-algebras R, we have

$$G(R) = \{(g, \mu) \in \mathrm{GL}_3(\mathcal{O} \otimes R) \times R^\times | \, {}^t\bar{g} J g = \mu J\}.$$

This is a unitary similitude group and we denote by $\mu\colon G \to \mathrm{G_m}$ the similitude character. The center Z_G of the group G can be identified with $\mathrm{Res}_{\mathcal{O}/\mathbb{Z}}(\mathrm{G_m})$ via

$$z \in \mathrm{Res}_{\mathcal{O}/\mathbb{Z}}(\mathrm{G_m})(R) \mapsto \begin{pmatrix} z & 0 & 0 \\ 0 & z & 0 \\ 0 & 0 & z \end{pmatrix} \in Z_G(R). \tag{3.1}$$

The group G is quasi-split over $\mathbb{Q}$ and is split when base change from $\mathbb{Q}$ to E:

$$G_E \cong \mathrm{GL}_{3,E} \times \mathrm{G_{m},}_E .$$

We define $G_0 = \ker(\mu)$ as the unitary group. Hence, we have

$$G(R) = Z_G(R) G_0(R),$$

for all $\mathbb{Z}$-algebras R. Note that G is reductive over $\mathbb{Z}_l$ for $l \nmid D$ (even for $l = 2$).

(2) Let H be the group scheme over $\mathbb{Z}$ such that for a $\mathbb{Z}$-algebras R, we have

$$H(R) = \{(g, z) \in \mathrm{GL}_2(R) \times (\mathcal{O} \otimes R)^\times | \det(g) = z\bar{z}\}.$$

The center Z_H of the group H is

$$Z_H(R) = \{(\begin{pmatrix} t & 0 \\ 0 & t \end{pmatrix}, z) \in \mathrm{GL}_2(R) \times (\mathcal{O} \otimes R)^\times | t^2 = z\bar{z}\}. \tag{3.2}$$

The group H is quasi-split and is split when base change from $\mathbb{Q}$ to E:

$$H_E \cong \mathrm{GL}_{2,E} \times \mathrm{G_{m},}_E .$$

(3) We have the following two maps:

$$\iota\colon H \hookrightarrow G, \quad (\begin{pmatrix} a & b \\ c & d \end{pmatrix}, z) \mapsto (\begin{pmatrix} a & 0 & b \\ 0 & z & 0 \\ c & 0 & d \end{pmatrix}, z\bar{z})$$

and

$$p\colon H \twoheadrightarrow \mathrm{GL}_2, \quad (\begin{pmatrix} a & b \\ c & d \end{pmatrix}, z) \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The relation between the groups GL_2, H and G can be summarized in the diagram:

$$\begin{array}{ccc} H & \hookrightarrow & G \\ \downarrow & & \\ \mathrm{GL}_2 & & \end{array} .$$

(4) Via the map ι, we define the group scheme Z as

$$Z(R) := Z_G(R) \cap H(R) = \{(\begin{pmatrix} t & 0 \\ 0 & t \end{pmatrix}, t) \in \mathrm{GL}_2(R) \times R^\times\}, \tag{3.3}$$

which is the common center of G and H.

The definition of H is natural for the following reasons.
Let $J_2 \in \mathrm{GL}_2(E)$ be the Hermitian matrix

$$J_2 = \begin{pmatrix} 0 & \delta^{-1} \\ -\delta^{-1} & 0 \end{pmatrix}, \quad \text{where } \delta = \sqrt{-D},$$

and let $\mathrm{GU}(J_2)$ be the group scheme over $\mathbb{Z}$ such that for all $\mathbb{Z}$-algebras R, we have

$$G(R) = \{(g, \mu) \in \mathrm{GL}_2(\mathcal{O} \otimes R) \times R^\times | \, {}^t\bar{g} J_2 g = \mu J_2\}.$$

Let $\mathrm{GU}(J_2)'$ be the subgroup scheme of $\mathrm{GU}(J_2)$ defined by

$$\mathrm{GU}(J_2)' = \{g \in \mathrm{GU}(J_2) | \det(g) = \mu(g)\}.$$

Fact 3.3. *We have*

$$\mathrm{GU}(J_2)' = \mathrm{GL}_2,$$

through which the similitude character μ is identified with the determinant character det*. Hence, we use the notation μ to denote the determinant character for H and* GL_2*.*

Proof. For $g \in \mathrm{GU}(J_2)'$, we have ${}^t\bar{g}J_2 g = J_2$, and if we left multiply both sides by $\begin{pmatrix}\delta & 0\\ 0 & \delta\end{pmatrix}$, we have

$${}^t\bar{g}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix} g = \mu\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}.$$

Hence, for $g = \begin{pmatrix}a & b\\ c & d\end{pmatrix}$, we have

$$\begin{aligned}\begin{pmatrix}-\bar{c} & \bar{a}\\ -\bar{d} & \bar{b}\end{pmatrix} = \begin{pmatrix}\bar{a} & \bar{c}\\ \bar{b} & \bar{d}\end{pmatrix}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix} &= \mu\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}d & -b\\ -c & a\end{pmatrix}\frac{1}{ad-bc}\\ &= \begin{pmatrix}-c & a\\ -d & b\end{pmatrix}\frac{\mu}{ad-bc} = \begin{pmatrix}-c & a\\ -d & b\end{pmatrix}.\end{aligned}$$

So we have $a = \bar{a}, b = \bar{b}, c = \bar{c}$ and $d = \bar{d}$, which means $g \in \mathrm{GL}_2$.

For $g \in \mathrm{GL}_2$, the above process can be reversed. □

Remark 3.4. The natural subgroup $\mathrm{GU}(J_2)' \boxtimes \mathrm{Res}_{\mathcal{O}/\mathbb{Z}}(\mathrm{G_m})$ of G, which is defined by

$$\mathrm{GU}(J_2)' \boxtimes \mathrm{Res}_{\mathcal{O}/\mathbb{Z}}(\mathrm{G_m})(R) = \{(g, z) \in \mathrm{GU}(J_2)'(R) \times (\mathcal{O}\otimes R)^\times | \mu(g) = z\bar{z}\}$$

for all $\mathbb{Z}$-algebras R, can be identified with H.

Convention 3.5. When we consider H as an algebraic group over $\mathbb{Q}$, we are free to use the notation

$$H = \mathrm{GU}(J_2)' \boxtimes E^\times = \mathrm{GL}_2 \boxtimes E^\times.$$

3.1.2. *Algebraic representations.*

Convention 3.6. In this subsubsection, the algebraic groups G, H and GL_2 are all defined over $\mathbb{Q}$.

We first define algebraic representations of G.

Definition 3.7. (1) Let $\chi_i, i = 1, \dots, 4$, be the four characters of the diagonal torus T/E of G such that

$$\chi_1 : \begin{pmatrix}x & 0 & 0\\ 0 & z & 0\\ 0 & 0 & \frac{z\bar{z}}{\bar{x}}\end{pmatrix} \mapsto x,\ \chi_2 : \begin{pmatrix}x & 0 & 0\\ 0 & z & 0\\ 0 & 0 & \frac{z\bar{z}}{\bar{x}}\end{pmatrix} \mapsto \bar{x},$$

$$\chi_3 : \begin{pmatrix}x & 0 & 0\\ 0 & z & 0\\ 0 & 0 & \frac{z\bar{z}}{\bar{x}}\end{pmatrix} \mapsto \frac{\det}{\mu},\ \chi_4 : \begin{pmatrix}x & 0 & 0\\ 0 & z & 0\\ 0 & 0 & \frac{z\bar{z}}{\bar{x}}\end{pmatrix} \mapsto \frac{\overline{\det}}{\mu}.$$

Here, $x, \bar{x}, z, \bar{z}$ are 4 independent variables.

(2) In this paper, we also use the notation $(\mu_1, \mu_2, \mu_3; d)$ to denote the character of the diagonal torus T such that

$$t = \begin{pmatrix}a & 0 & 0\\ 0 & b & 0\\ 0 & 0 & c\end{pmatrix} \mapsto a^{\mu_1} b^{\mu_2} c^{\mu_3} \mu(t)^d.$$

We denote by $\lambda(\mu_1, \mu_2, \mu_3; d)$ the irreducible algebraic representation of G with highest weight $(\mu_1, \mu_2, \mu_3; d)$. When the action of μ is clear, sometimes we write (μ_1, μ_2, μ_3) for $\lambda(\mu_1, \mu_2, \mu_3; d)$.

(3) For $a_1, a_2 \geq 0$, let V^{a_1, a_2} denote the representation of G of highest weight $a_1\chi_1 + a_2\chi_2$.

(4) For each representation V of G, we write $V\{a_3, a_4\}$ for its twists by $\chi_3^{a_3}\chi_4^{a_4}$.

Each irreducible representation of G is $V^{a_1,a_2}\{a_3,a_4\}$ for some $a_1,a_2,a_3,a_4\in\mathbb{Z}$ with $a_1\geq 0, a_2\geq 0$.

Proposition 3.8. *(1) Each irreducible representation of G is $V^{a_1,a_2}\{a_3,a_4\}$ for some $a_1,a_2,a_3,a_4\in\mathbb{Z}$ with $a_1\geq 0, a_2\geq 0$.*
(2) The highest weight of $V^{a_1,a_2}\{a_3,a_4\}$ is $(a_1+a_3-a_4, a_3-a_4, a_3-(a_2+a_4); 2a_4+a_2-a_3)$.
(3) The contragredient representation D^{a_1,a_2} of V^{a_1,a_2} is $V^{a_2,a_1}\{-(a_1+a_2),-(a_1+a_2)\}$.

Proof. (1) The special unitary group $SU(J)$ is the subgroup of G such that $\det=\mu=1$, which is a semisimple algebraic group. It is enough to consider the algebraic representation of $\mathrm{SU}(J)\cong \mathrm{SL}_{3,E}$. For $\mathrm{SL}_{3,E}$, the fundamental weights are $\omega_1=(1,0)$ and $\omega_2=(1,1)$. For any $t=\begin{pmatrix} x & 0 & 0\\ 0 & z & 0\\ 0 & 0 & \frac{1}{xz}\end{pmatrix}\in T_E$, we have $\chi_1(t)=x=\omega_1(t)$ and $\chi_2(t)=\bar{x}=xz=\omega_2(t)$. Hence, by highest weight theory, every irreducible algebraic representation of $\mathrm{SU}(J)_E$ is determined by

$$a_1\omega_1+a_2\omega_2=a_1\chi_1+a_2\chi_2,$$

for $a_1\geq 0$ and $a_2\geq 0$. Hence, every irreducible algebraic representation of G is of the form $V^{a_1,a_2}\{a_3,a_4\}$ for some $a_1,a_2,a_3,a_4\in\mathbb{Z}$ with $a_1\geq 0, a_2\geq 0$.

(2) The highest weight of $V^{a_1,a_2}\{a_3,a_4\}$ is

$$\begin{aligned} t=\left(\begin{pmatrix} x & 0 & 0\\ 0 & z & 0\\ 0 & 0 & \frac{z\bar{z}}{\bar{x}}\end{pmatrix}, z\bar{z}\right)\in T_E &\mapsto x^{a_1}(\bar{x})^{a_2}(\frac{xz}{\bar{x}})^{b_1}(\frac{\bar{x}\bar{z}}{x})^{b_2}\\ &=x^{a_1+b_1-b_2}\bar{x}^{a_2+b_2-b_1}z^{b_1}\bar{z}^{b_2}\\ &=x^{a_1+b_1-b_2}z^{b_1-b_2}(\frac{z\bar{z}}{\bar{x}})^{b_1-(a_2+b_2)}(z\bar{z})^{2b_2+a_2-b_1}. \end{aligned}$$

(3) First, we have

$$\begin{aligned} a_1\omega_1+a_2\omega_2&=a_1(\frac{2}{3},-\frac{1}{3},-\frac{1}{3})+a_2(\frac{1}{3},\frac{1}{3},-\frac{2}{3})\\ &=(\frac{2a_1+a_2}{3},\frac{-a_1+a_2}{3},\frac{-a_1-2a_2}{3}). \end{aligned}$$

Hence, when restrict to $\mathrm{SU}(J)$, the highest weight of $D^{a_1,a_2}=(V^{a_1,a_2})^\vee$ is

$$(\frac{a_1+2a_2}{3},\frac{a_1-a_2}{3},\frac{-2a_1-a_2}{3})=a_2\omega_1+a_1\omega_2.$$

Second, for the t in (2), we have

$$\begin{aligned} \chi_1(t)=x&=x^{\frac{2}{3}}z^{-\frac{1}{3}}(\frac{z\bar{z}}{\bar{x}})^{-\frac{1}{3}}\cdot(x\cdot z\cdot(\frac{z\bar{z}}{\bar{x}}))^{\frac{1}{3}}\\ &=\omega_1(t)\chi_3(t)^{\frac{2}{3}}\chi_4(t)^{\frac{1}{3}}. \end{aligned}$$

Similarly, we have

$$\chi_2(t)=\omega_2(t)\chi_3(t)^{\frac{1}{3}}\chi_4(t)^{\frac{2}{3}}.$$

So the highest weight of D^{a_1,a_2} is $\omega_1^{a_2}\omega_2^{a_1}\chi_3^{\frac{-2a_1-a_2}{3}}\chi_4^{\frac{-a_1-2a_2}{3}}$ which is

$$\chi_1^{a_2}\chi_2^{a_1}\chi_3^{-(a_1+a_2)}\chi_4^{-(a_1+a_2)}.$$

□

Remark 3.9. For later use, we record the highest weight and central character in the following table.

(3.4)

Representation	Highest weight	central character
$V=V^{a,b}\{r,s\}$	$(a+r-s, r-s, r-s-b; 2s-r+b)$	$a+b+r+s$
$D=V^\vee$	$(b+s-r, s-r, s-r-a; -2s+r-b)$	$-a-b-r-s$

Now we define irreducible representations of H.

Definition 3.10. (1) For each representation W of H, we write $W\{a_3, a_4\}$ for its twists by $\chi_3^{a_3}\chi_4^{a_4}$ through $\iota\colon H \hookrightarrow G$.

(2) In this paper, we also use the notation $(\mu_1, \mu_2; d)$ to denote the character of the diagonal torus

$$t = (\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}; z) \in H \mapsto a^{\mu_1} b^{\mu_2} z^d.$$

We denote by $\lambda'(\mu_1, \mu_2; d)$ the irreducible algebraic representation of H with highest weight $(\mu_1, \mu_2; d)$.

Convention 3.11. When the action of $\mathrm{Res}_{E/\mathbb{Q}}\mathrm{G_m}$ in H is clear, we sometimes write (μ_1, μ_2) instead of $(\mu_1, \mu_2; d)$.

Remark 3.12. Along the map ι, the restriction of χ_3 and χ_4 is

$$\chi_3\colon (\begin{pmatrix} a & b \\ c & d \end{pmatrix}, z) \in H \mapsto z$$

$$\chi_4\colon (\begin{pmatrix} a & b \\ c & d \end{pmatrix}, z) \in H \mapsto \bar{z}.$$

So the representation $W\{a_3, a_4\}$ is the representation W with a twist by the character $z \mapsto z^{a_3}\bar{z}^{a_4}$.

Hence, we can get the following proposition:

Proposition 3.13. *(1) For $b_1 \geq 0$, let W^{b_1} be the representation $\mathrm{Sym}^{b_1} V_2$ of H, where V_2 denotes the pullback of the standard representation of GL_2 through the map $p\colon : H \twoheadrightarrow \mathrm{GL}_2$.*

Then every irreducible representation of H has the form $W^{b_1}\{b_2, b_3\}$ for some $b_1, b_2, b_3 \in \mathbb{Z}$ with $b_1 \geq 0$.

(2) The contragredient of W^{b_1} is $W^{b_1}\{-b_1, -b_1\}$.

3.1.3. *Branching law.* The following is a corollary of the classical branching law for $\mathrm{GL}_2 \subset \mathrm{GL}_3$ [GW09, Lemma 8.3.1].

Proposition 3.14. *For an irreducible representation $V^{a_1,a_2}\{b_1, b_2\}$ of G satisfying:*

$$0 \leq -b_1 \leq a_1,$$
$$0 \leq -b_2 \leq a_2,$$

we have an embedding in the category of representations of H:

$$W^n \hookrightarrow \iota^* V^{a_1,a_2}\{b_1, b_2\},$$

where $n = a_1 + a_2 + b_1 + b_2$ and ι^* denotes the restriction of the representation of G to H through ι.

Proof. The highest weight of W^n is $(a_1 + a_2 + b_1 + b_2, 0)$ and the highest weight of $V^{a_1,a_2}\{b_1, b_2\}$ is $(a_1 + b_1 - b_2, b_1 - b_2, b_1 - (a_2 + b_2); 2b_2 + a_2 - b_1)$.

First, the common center of G and H is isomorphic to $\mathrm{G_m}$ via

$$z \in \mathrm{G_m} \mapsto (\begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}, z) \in H.$$

Its action on W^n is via

$$z \mapsto z^n$$

and its action on $\iota^* V^{a_1,a_2}\{b_1, b_2\}$ is

$$z \mapsto z^{a_1+a_2+b_1+b_2}.$$

Hence, their central actions are compatible.

Second, the highest weight of $W^n\{-(2b_2+a_2-b_1),-(2b_2+a_2-b_1)\}$ is $(a_1+2b_1-b_2,b_1-2b_2-a_2)$. Up to conjugation, by the classical branching $\mathrm{GL}_2\subset\mathrm{GL}_3$ [GW09, Lemma 8.3.1], in order to have the branching

$$W^n\{-(2b_2+a_2-b_1),-(2b_2+a_2-b_1)\}\hookrightarrow(\iota^*V^{a_1,a_2}\{b_1,b_2\})\{-(2b_2+a_2-b_1),-(2b_2+a_2-b_1)\}$$

we must have

$$a_1+b_1-b_2\geq a_1+2b_1-b_2\geq b_1-b_2\geq b_1-2b_2-a_2\geq b_1-(a_2+b_2),$$

which is equivalent to

$$\begin{aligned}0\leq -b_1\leq a_1,\\ 0\leq -b_2\leq a_2.\end{aligned}$$

□

Remark 3.15. In this paper, we only consider the case $W^n\hookrightarrow\iota^*V^{a,b}\{r,s\}$ when $n=a+b+r+s$ and will write W instead of W^n and V instead of $V^{a,b}\{r,s\}$ to simplify notations. We will also denote by D the contragredient of V.

Notation 3.16.

- Throughout the paper, we always let $n=a+b+r+s$.
- Throughout the paper, we always use V to denote the algebraic representation $V=V^{a,b}\{r,s\}$ of G, D to denote the contragredient of V and $W=W^n$ to denote the algebraic representation of H except in the situation where we will explain the meaning.
- For a representation V of G, H or GL_2 and an integer d, we use the notation $V(d)$ denote the representation V with a twist of the character μ^d.

3.2. **Lie groups and representations.**

3.2.1. *Lie groups and Lie algebras.*

Notation 3.17. To describe the maximal compact subgroup of $G(\mathbb{R})$ easily, when considering representations of Lie groups, we use the Hermitian form

$$J'=\begin{pmatrix}1&0&0\\0&1&0\\0&0&-1\end{pmatrix}$$

to define the group $G=\mathrm{GU}(J')$ and use the Hermitian form

$$J_2'=\begin{pmatrix}1&0\\0&-1\end{pmatrix}$$

to define the group $\mathrm{GU}(J_2')'$, and then define the group $H=\mathrm{GU}(J_2')'\boxtimes(\mathrm{Res}_{E/\mathbb{Q}}(\mathrm{G_m}))$. The definition of G (resp., H) is the same as in Definition 3.2 except that we use J' (resp., J_2') instead of J (resp., J_2).

Proposition 3.18. *(1) If we let*

$$C=\begin{pmatrix}\frac{D^{\frac14}}{\sqrt{2}}&0&\frac{D^{\frac14}}{\sqrt{2}}\\0&1&0\\-\frac{D^{\frac14}}{\sqrt{-2}}&0&\frac{D^{\frac14}}{\sqrt{-2}}\end{pmatrix},$$

then we have the following isomorphism of real Lie groups:

$$\begin{aligned}\mathrm{GU}(J)(\mathbb{R})&\xrightarrow{\cong}\mathrm{GU}(J')(\mathbb{R})\\ g&\longmapsto C^{-1}gC\end{aligned}.$$

(2) If we let

$$C_2 = \begin{pmatrix} \frac{D^{\frac{1}{4}}}{\sqrt{2}} & \frac{D^{\frac{1}{4}}}{\sqrt{2}} \\ -\frac{D^{\frac{1}{4}}}{\sqrt{-2}} & \frac{D^{\frac{1}{4}}}{\sqrt{-2}} \end{pmatrix},$$

then we have the following isomorphism

$$\begin{array}{ccc} \mathrm{GU}(J_2)'(\mathbb{R}) & \xrightarrow{\cong} & \mathrm{GU}(J_2')'(\mathbb{R}) \\ g & \longmapsto & C_2^{-1} g C_2 \end{array}.$$

Proof. The proof follows from direct computation. □

Convention 3.19. In this paper, we write $G(\mathbb{R})$ (resp., $H(\mathbb{R})$ and $\mathrm{GL}_2(\mathbb{R})$) for $\mathrm{GU}(J')(\mathbb{R})$ (resp., $(\mathrm{GU}(J_2')' \boxtimes E^\times)(\mathbb{R})$ and $\mathrm{GU}(J_2')'(\mathbb{R})$).

Let K_G be the real Lie group $A_G(\mathrm{U}(2) \times \mathrm{U}(1))(\mathbb{R})$, K_H be the real Lie group $A_H(\mathrm{U}(1) \times \mathrm{U}(1))(\mathbb{R})$ and K_{GL_2} be the real Lie group $A_{\mathrm{GL}_2}\mathrm{U}(1)(\mathbb{R})$. Let $\mathfrak{g}$ be the Lie algebra of $G(\mathbb{R})$, $\mathfrak{h}$ be the Lie algebra of $H(\mathbb{R})$ and $\mathfrak{gl}_2$ be the Lie algebra of $\mathrm{GL}_2(\mathbb{R})$. Then by the Iwasawa decomposition, we have the following decomposition of complexified Lie algebra:

$$\begin{aligned} \mathfrak{g}_{\mathbb{C}} &= \mathfrak{k}_{G,\mathbb{C}} \oplus \mathfrak{p}_{\mathbb{C}}, \\ \mathfrak{h}_{\mathbb{C}} &= \mathfrak{k}_{H,\mathbb{C}} \oplus \mathfrak{p}_{H,\mathbb{C}}, \\ \mathfrak{gl}_{2,\mathbb{C}} &= \mathfrak{k}_{H,\mathbb{C}} \oplus \mathfrak{p}_{H,\mathbb{C}}, \end{aligned}$$

where $\mathfrak{k}_{\mathbb{C}}$, $\mathfrak{k}_{H,\mathbb{C}}$ and $\mathfrak{k}_{\mathrm{GL}_2,\mathbb{C}}$ are the complexified Lie algebras of K_G, K_H and K_{GL_2}, which is

$$\begin{aligned} \mathfrak{k}_{\mathbb{C}} &= \{ \begin{pmatrix} * & * & 0 \\ * & * & 0 \\ 0 & 0 & * \end{pmatrix} \} \subset \mathfrak{g}_{\mathbb{C}}, \\ \mathfrak{k}_{H,\mathbb{C}} &= \{ \begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix} \} \oplus \mathfrak{gl}_{1,\mathbb{C}} \subset \mathfrak{h}_{\mathbb{C}}, \\ \mathfrak{k}_{\mathrm{GL}_2,\mathbb{C}} &= \{ \begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix} \} \subset \mathfrak{gl}_{2,\mathbb{C}}; \end{aligned}$$

and $\mathfrak{p}_{\mathbb{C}}$, $\mathfrak{p}_{H,\mathbb{C}}$ and $\mathfrak{p}_{\mathrm{GL}_2,\mathbb{C}}$ are the sub Lie algebras of $\mathfrak{g}_{\mathbb{C}}$, $\mathfrak{h}_{\mathbb{C}}$ and $\mathfrak{gl}_{2,\mathbb{C}}$:

$$\begin{aligned} \mathfrak{p}_{\mathbb{C}} &= \{ \begin{pmatrix} 0 & 0 & * \\ 0 & 0 & * \\ * & * & 0 \end{pmatrix} \} \subset \mathfrak{g}_{\mathbb{C}}, \\ \mathfrak{p}_{H,\mathbb{C}} &= \{ \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \} \subset \mathfrak{h}_{\mathbb{C}}, \\ \mathfrak{p}_{\mathrm{GL}_2,\mathbb{C}} &= \{ \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \} \subset \mathfrak{gl}_{2,\mathbb{C}}. \end{aligned}$$

We also have a natural embedding $\iota_*\colon \mathfrak{h}_{\mathbb{C}} \hookrightarrow \mathfrak{g}_{\mathbb{C}}$ induced by the map $\iota\colon H \hookrightarrow G$ that maps $\mathfrak{p}_{H,\mathbb{C}}$ to $\mathfrak{p}_{\mathbb{C}}$. Explicitly, the map is given by

$$\begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix} \in \mathfrak{p}_{H,\mathbb{C}} \mapsto \begin{pmatrix} 0 & 0 & A \\ 0 & 0 & 0 \\ B & 0 & 0 \end{pmatrix} \in \mathfrak{p}_{\mathbb{C}}.$$

3.2.2. *Representation Theory of compact Lie groups.* We recall some basic facts of representation theory of the compact Lie group $(\mathrm{U}(2) \times \mathrm{U}(1))(\mathbb{R})$.

The compact torus T_c is $(\mathrm{U}(1) \times \mathrm{U}(1) \times \mathrm{U}(1))(\mathbb{R})$. Hence, the dominant weights of T_c are parametrized by the triples $(\lambda_1, \lambda_2, \lambda_3)$ such that $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{Z}$ and $\lambda_1 \geq \lambda_2 \geq \lambda_3$. By highest weight theory of representations of compact Lie groups, for each dominant weight $(\lambda_1, \lambda_2, \lambda_3)$, there is an unique irreducible representation $\tau_{(\lambda_1,\lambda_2,\lambda_3)}$ of $(\mathrm{U}(2) \times \mathrm{U}(1))(\mathbb{R})$ whose highest weight is $(\lambda_1, \lambda_2, \lambda_3)$.

By the Weyl dimension formula, we have

$$\dim \tau_{(\lambda_1,\lambda_2,\lambda_3)} = \lambda_1 - \lambda_2 + 1.$$

We denote $d = \lambda_1 - \lambda_2 + 1$. Actually, there exists a basis $(\nu_s)_{0\leq s\leq d}$ of $\tau_{(\lambda_1,\lambda_2,\lambda_3)}$ such that

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \nu_s = (s+\lambda_2)\nu_s, \tag{3.5}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \nu_s = (-s+\lambda_1)\nu_s, \tag{3.6}$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \nu_s = (s+1)\nu_{s+1}, \tag{3.7}$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \nu_s = (d-s+1)\nu_{s-1}. \tag{3.8}$$

Convention 3.20.
- We will use K_G (resp., K_H and K_{GL_2}) to denote $(\mathrm{U}(2)\times\mathrm{U}(1))(\mathbb{R})$ (resp., $(\mathrm{U}(1)\times\mathrm{U}(1))(\mathbb{R})$ and $\mathrm{U}(1)(\mathbb{R})$) when the action A_G is clear.
- We will use the coordinate system $(\lambda_1,\lambda_2,\lambda_3)$ in subsection 3.5 and 3.6 to denote the Blattner parameter and Harish-Chandra parameter of a discrete series representation of a real reductive Lie group.

3.3. **Shimura varieties.** In this subsection, we define the Shimura varieties associated to G, H and GL_2.

Notation 3.21. Throughout the paper, we let $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}}\,\mathrm{G_m}$ be the Deligne torus.

Definition 3.22. (1) Let X_G be the $G(\mathbb{R})$-conjugacy class of the morphism $h\colon \mathbb{S}\to G_{\mathbb{R}}$ given by

$$z = x+iy \mapsto \begin{pmatrix} x & 0 & y \\ 0 & z & 0 \\ -y & 0 & x \end{pmatrix}.$$

The pair (G, X_G) is a (pure) Shimura datum associated to G.

The Hodge cocharacter $\mu_{\mathbb{C}}$ of $\mathrm{G_{m,\mathbb{C}}}$ associated to h is given on $\mathbb{C}$-points by

$$z \mapsto \begin{pmatrix} z & 0 & 0 \\ 0 & z & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Since it is not complex conjugation invariant, we can see that the reflex field of (G, X_G) is E with a the fixed complex embedding of E into $\mathbb{C}$ [LR92, Lemma 4.2., P15]. For any neat compact open subgroup L of $G(\mathbb{A}_f)$, there is a smooth *quasi-projective* E-scheme $\mathrm{Sh}_G(L)$ such that as complex analytic varieties, we have

$$\mathrm{Sh}_G(L)^{an}_{\mathbb{C}} = G(\mathbb{Q})\backslash(X_G\times G(\mathbb{A}_f)/L),$$

where $\mathrm{Sh}_G(L)^{an}_{\mathbb{C}}$ is the analytification of the base change of $\mathrm{Sh}_G(L)$ to $\mathbb{C}$.

(2) Let X_H be the $H(\mathbb{R})$-conjugacy class of the morphism $h\colon \mathbb{S}\to H_{\mathbb{R}}$ given by

$$z = x+iy \mapsto (\begin{pmatrix} x & y \\ -y & x \end{pmatrix}, z).$$

The pair (H, X_H) is a (pure) Shimura datum associated to H.

The Hodge cocharacter $\mu_{\mathbb{C}}$ of $\mathrm{G_{m,\mathbb{C}}}$ associated to h is given on $\mathbb{C}$-points by

$$z \mapsto (\begin{pmatrix} z & 0 \\ 0 & 1 \end{pmatrix}, z).$$

The reflex field of (H, X_H) is E with a fixed complex embedding of E into $\mathbb{C}$. For any neat compact open subgroup K of $H(\mathbb{A}_f)$, there is a smooth *quasi-projective* E-scheme $\mathrm{Sh}_H(K)$ such that as complex analytic varieties, we have

$$\mathrm{Sh}_H(K)^{an}_{\mathbb{C}} = H(\mathbb{Q})\backslash(X_H\times H(\mathbb{A}_f)/K),$$

where $\mathrm{Sh}_H(K)^{an}_{\mathbb{C}}$ is the analytification of the base change of $\mathrm{Sh}_H(K)$ to $\mathbb{C}$.

(3) Let X_{GL_2} be the $\mathrm{GL}_2(\mathbb{R})$-conjugacy class of the morphism $h\colon \mathbb{S} \to \mathrm{GL}_{2,\mathbb{R}}$ that is given by

$$z \mapsto \begin{pmatrix} x & y \\ -y & x \end{pmatrix},$$

The pair $(\mathrm{GL}_2, X_{\mathrm{GL}_2})$ is a (pure) Shimura datum associated to GL_2.

The Hodge cocharacter $\mu_{\mathbb{C}}$ of $\mathrm{G}_{\mathrm{m},\mathbb{C}}$ associated to h is given on $\mathbb{C}$-points by

$$z \mapsto \begin{pmatrix} z & 0 \\ 0 & 1 \end{pmatrix}.$$

The reflex field of $(\mathrm{GL}_2, X_{\mathrm{GL}_2})$ is $\mathbb{Q}$. For any neat compact open subgroup K' of $\mathrm{GL}_2(\mathbb{A}_f)$, there is a smooth *quasi-projective* $\mathbb{Q}$-scheme $\mathrm{Sh}_{\mathrm{GL}_2}(K')$ such that as complex analytic varieties, we have

$$\mathrm{Sh}_{\mathrm{GL}_2}(K')^{an}_{\mathbb{C}} = \mathrm{GL}_2(\mathbb{Q})\backslash(X_{\mathrm{GL}_2} \times \mathrm{GL}_2(\mathbb{A}_f)/K'),$$

where $\mathrm{Sh}_{\mathrm{GL}_2}(K')^{an}_{\mathbb{C}}$ means the analytification of the base change of $\mathrm{Sh}_{\mathrm{GL}_2}(K')$ to $\mathbb{C}$.

Remark 3.23.

- Our choice of Shimura datum is standard, bu is a little different from [LSZ22, 8.1.1] (see the difference in [LSZ22, Remark 8.1.1]).
- The maps $\iota\colon H \hookrightarrow G$ and $p\colon H \twoheadrightarrow \mathrm{GL}_2$ of algebraic groups induce the following E-morphisms of Shimura varieties for suitable neat compact open subgroup L, K and K' of $G(\mathbb{A}_f)$, $H(\mathbb{A}_f)$ and $\mathrm{GL}_2(\mathbb{A}_f)$:

$$\begin{array}{ccc} \mathrm{Sh}_H(K) & \xhookrightarrow{\ \iota\ } & \mathrm{Sh}_G(L) \\ \downarrow{\scriptstyle p} & & \\ \mathrm{Sh}_{\mathrm{GL}_2}(K')_E & & \end{array}, \tag{3.9}$$

where ι is a closed immersion and $\mathrm{Sh}_{\mathrm{GL}_2}(K')_E$ is the base change of $\mathrm{Sh}_{\mathrm{GL}_2}(K')$ to E.

Fact 3.24.

- *The decomposition*

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{k}_{G,\mathbb{C}} \oplus \mathfrak{p}_{G,\mathbb{C}}$$

can be further decomposed as

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{k}_{G,\mathbb{C}} \oplus \mathfrak{p}^+_{G,\mathbb{C}} \oplus \mathfrak{p}^-_{G,\mathbb{C}},$$

where $\mathfrak{p}^+_{G,\mathbb{C}}$ *is*

$$\{\begin{pmatrix} 0 & 0 & * \\ 0 & 0 & * \\ 0 & 0 & 0 \end{pmatrix}\} \subset \mathfrak{g}_{\mathbb{C}},$$

on which $(z, \bar{z}) \in \mathbb{S}(\mathbb{C})$ *acts by* $\frac{z}{\bar{z}}$ *and* $\mathfrak{p}^-_{\mathbb{C}}$ *is*

$$\{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ * & * & 0 \end{pmatrix}\} \subset \mathfrak{g}_{\mathbb{C}},$$

on which $(z, \bar{z}) \in \mathbb{S}(\mathbb{C})$ *acts by* $\frac{\bar{z}}{z}$.

- *Similarly, we have the decomposition for* H*:*

$$\mathfrak{h}_{\mathbb{C}} = \mathfrak{k}_{H,\mathbb{C}} \oplus \mathfrak{p}^+_{H,\mathbb{C}} \oplus \mathfrak{p}^-_{H,\mathbb{C}},$$

where $\mathfrak{p}^+_{H,\mathbb{C}}$ *is*

$$\{\begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}\} \subset \mathfrak{h}_{\mathbb{C}}$$

and $\mathfrak{p}^-_{H,\mathbb{C}}$ *is*

$$\{\begin{pmatrix} 0 & 0 \\ * & 0 \end{pmatrix}\} \subset \mathfrak{h}_{\mathbb{C}}.$$

- *We have a decomposition for* GL_2*:*

$$\mathfrak{gl}_{2,\mathbb{C}} = \mathfrak{k}_{\mathrm{GL}_2,\mathbb{C}} \oplus \mathfrak{p}^+_{\mathrm{GL}_2,\mathbb{C}} \oplus \mathfrak{p}^-_{\mathrm{GL}_2,\mathbb{C}},$$

where $\mathfrak{p}^+_{\mathrm{GL}_2,\mathbb{C}}$ *is*

$$\{\begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}\} \subset \mathfrak{gl}_{2,\mathbb{C}}$$

and $\mathfrak{p}^-_{\mathrm{GL}_2,\mathbb{C}}$ *is*

$$\{\begin{pmatrix} 0 & 0 \\ * & 0 \end{pmatrix}\} \subset \mathfrak{gl}_{2,\mathbb{C}}\,.$$

Convention 3.25. (1) We use $M(K)$ (resp., $S(L)$) to denote $\mathrm{Res}_{E/\mathbb{Q}}(\mathrm{Sh}_H(K))$ (resp., $\mathrm{Res}_{E/\mathbb{Q}}(\mathrm{Sh}_G(L))$), where $\mathrm{Res}_{E/\mathbb{Q}}$ is the Weil restriction. Hence, we have

$$S(L)^{an}_{\mathbb{C}} = \mathrm{Sh}_G(L)^{an}_{\mathbb{C}} \bigsqcup \overline{\mathrm{Sh}_G(L)^{an}_{\mathbb{C}}},$$
$$M(K)^{an}_{\mathbb{C}} = \mathrm{Sh}_H(K)^{an}_{\mathbb{C}} \bigsqcup \overline{\mathrm{Sh}_H(K)^{an}_{\mathbb{C}}},$$

where $\overline{\mathrm{Sh}_G(L)^{an}_{\mathbb{C}}}$ (resp., $\overline{\mathrm{Sh}_H(K)^{an}_{\mathbb{C}}}$) is the complex conjugate of $\mathrm{Sh}_G(L)^{an}_{\mathbb{C}}$ (resp., $\mathrm{Sh}_H(K)^{an}_{\mathbb{C}}$).

(2) Since we compute complex regulators in this paper, an explicit choice of a level structure of a Shimura variety is not necessary. Hence, we will sometimes ignore the level structure in the notation related to Shimura varieties and write S, M and $\mathrm{Sh}_{\mathrm{GL}_2}$ instead of $S(L)$, $M(K)$ and $\mathrm{Sh}_{\mathrm{GL}_2}(K')$ when there is no confusion. If reader prefers, this can be explained as a Shimura variety with suitable neat level structure.

(3) Applying the Weil restriction functor $\mathrm{Res}_{E/\mathbb{Q}}$ to the diagram (3.9), we get the following $\mathbb{Q}$-morphisms of Shimura varieties:

$$\begin{array}{ccc} M & \xhookrightarrow{\ \iota\ } & S \\ \downarrow{\scriptstyle p} & & \\ \mathrm{Sh}_{\mathrm{GL}_2} & & \end{array}, \tag{3.10}$$

where ι is a closed immersion.

3.4. **Cohomology of Picard modular surfaces.**

Convention 3.26. (1) For any $V \in \mathrm{Rep}_E(G)$ (resp., $W \in \mathrm{Rep}_E \in \mathrm{Rep}_E(H)$), we denote by $\overline{V}$ (resp., $\overline{W}$) the twist of V (resp., W) by the non-trivial element of $\mathrm{Gal}(E/\mathbb{Q})$.

(2) For any morphism $(f\colon V \to V') \in \mathrm{Rep}_E(G)$ (resp., $(g\colon W \to W') \in \mathrm{Rep}_E(H)$), we denote by $\overline{f}\colon \overline{V} \to \overline{V'}$ (resp., $\overline{g}\colon \overline{W} \to \overline{W'}$) the twist of f (resp., g) by the non-trivial element of $\mathrm{Gal}(E/\mathbb{Q})$.

(3) We denote by $\mathrm{Rep}_{\mathbb{Q}}(G)$ (resp., $\mathrm{Rep}_{\mathbb{Q}}(H)$) the cateogry whose objects are $(V, \overline{V})$ (resp., $(W, \overline{W})$), where $V \in \mathrm{Rep}_E(G)$ (resp., $W \in \mathrm{Rep}_E(H)$); and whose morphisms are $(f, \overline{f})$ (resp., $(g, \overline{g})$), where $(f : V \to V') \in \mathrm{Rep}_E(G)$ (resp., $(g : W \to W') \in \mathrm{Rep}_E(H)$).

(4) We denote by $V \in \mathrm{Rep}_{\mathbb{Q}}(G)$ (resp., $W \in \mathrm{Rep}_{\mathbb{Q}}(H)$) the pair $(V, \overline{V})$ (resp., $(W, \overline{W})$). When talking about weights of $V \in \mathrm{Rep}_{\mathbb{Q}}(G)$ (resp., $W \in \mathrm{Rep}_{\mathbb{Q}}(H)$), we offen mean the weights of $V \in \mathrm{Rep}_E(G)$ (resp., $W \in \mathrm{Rep}_E(H)$).

(5) For any $V \in \mathrm{Rep}_{\mathbb{Q}}(G)$ (resp., $W \in \mathrm{Rep}_{\mathbb{Q}}(H)$), there is a canonical local system over $S(L)$ (resp., $M(K)$), which is also denoted by V (resp., W).

(6) We denote by $\mathrm{Rep}_{\mathbb{C}}(G)$ (resp., $W \in \mathrm{Rep}_{\mathbb{C}}(H)$ the category whose objects are $V \otimes_{\mathbb{Q}} \mathbb{C}$ (resp., $W \otimes_{\mathbb{Q}} \mathbb{C}$) such that $V \in \mathrm{Rep}_{\mathbb{Q}}(G)$ (resp., $W \in \mathrm{Rep}_{\mathbb{Q}}(H)$).

For any $V \in \mathrm{Rep}_{\mathbb{C}}(G)$, let ξ be the restriction of the inverse of the central character of V to A_G and let L be an compact open subgroup of $G(\mathbb{A}_f)$. Let $\mathcal{A}^*_c(S(L), V(2))$ be the de Rham complex of C^∞-differential forms with compact support on $S(L)^{an}_{\mathbb{C}}$ with values in the local system $V(2)$. Similarly, let $\mathcal{A}^*(S(L), V(2))$ be the usual de Rham complex and let $\mathcal{A}^*_{(2)}(S(L), V(2))$ be the

complex of square integrable differential forms. If $\circ$ is the symbol c, (2) or the empty symbol, we define

$$\mathcal{A}^*_\circ(S, V(2)) = \varinjlim_L \mathcal{A}^*_\circ(S(L), V(2)).$$

Let $\mathfrak{g}$ be the Lie algebra of $G(\mathbb{R})$ and $K_G = A_G(\mathrm{U}(2) \times \mathrm{U}(1))(\mathbb{R})$ be the subgroup of $G(\mathbb{R})$ which is maximal modulo the center. For any $(\mathfrak{g}_\mathbb{C}, K_G)$-module M, let $\mathcal{C}^*(\mathfrak{g}_\mathbb{C}, K_G; M)$ be the $(\mathfrak{g}_\mathbb{C}, K_G)$-complex of M. Then we have $G(\mathbb{A}_f)$-equivariant isomorphisms of complexes: [BW00, VII §2]

$$\mathcal{A}^*_c(S_\mathbb{Q}, V(2)) \cong \mathcal{C}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2) \otimes_\mathbb{C} C^\infty_c(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)),$$
$$\mathcal{A}^*(S_\mathbb{Q}, V(2)) \cong \mathcal{C}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2) \otimes_\mathbb{C} C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)),$$

that are compatible with the inclusions $\mathcal{A}^*_c(S_\mathbb{Q}, V(2)) \subset \mathcal{A}^*(S_\mathbb{Q}, V(2))$ and

$$\mathcal{C}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2) \otimes_\mathbb{C} C^\infty_c(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)) \subset \mathcal{C}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2) \otimes_\mathbb{C} C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)).$$

If we take cohomology on both sides, we obtain $G(\mathbb{A}_f)$-equivariant isomorphisms

$$\mathrm{H}^*_{dR,c}(S, V(2)) \cong \mathrm{H}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2) \otimes_\mathbb{C} C^\infty_c(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)),$$
$$\mathrm{H}^*_{dR}(S, V(2)) \cong \mathrm{H}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2) \otimes_\mathbb{C} C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)),$$

which are compatible with the maps from cohomology with compact support to cohomology without support.

Let $\mathrm{H}^*_{(2)}(S, V(2))$ be the cohomology of the complex $\mathcal{A}^*_{(2)}(S, V(2))$ which is the L^2 cohomology of S with coefficients in $V(2)$. According to [Bor83], we have a $G(\mathbb{A}_f)$-equivariant isomorphism

$$\mathrm{H}^*_{(2)}(S, V(2)) \cong \mathrm{H}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2) \otimes_\mathbb{C} C^\infty_{(2)}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)),$$

where $C^\infty_{(2)}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$ denotes the subspace of functions in $C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$ which are square integrable modulo the center of G. We also use the notation $\mathrm{H}^*_{\mathrm{cusp}}(S, V(2))$ to denote

$$\mathrm{H}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2) \otimes_\mathbb{C} C^\infty_{\mathrm{cusp}}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)),$$

where $C^\infty_{cusp}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$ denotes the subspace of functions in $C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$ which are cuspidal.

By definition, we have the following maps:

$$C^\infty_{\mathrm{cusp}}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi) \to C^\infty_c(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi) \to C^\infty_{(2)}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi) \to C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi),$$

where the first map is defined by smooth truncation to a large set which is compact modulo the center, and the second and the third arrows are the natural inclusions. These maps induce maps of the corresponding cohomology groups:

$$\mathrm{H}^*_{\mathrm{cusp}}(S, V(2)) \to \mathrm{H}^*_{dR,c}(S, V(2)) \to \mathrm{H}^*_{(2)}(S, V(2)) \to \mathrm{H}^*_{dR}(S, V(2)).$$

Let

$$\mathrm{H}^*_{dR,!}(S, V(2)) = \mathrm{Im}(\mathrm{H}^*_{dR,c}(S, V(2)) \to \mathrm{H}^*_{dR}(S, V(2)).$$

be the interior de Rham cohomology. In general, by [Bor74, Theorem 7.5], the map

$$\mathrm{H}^*_{\mathrm{cusp}}(S, V(2)) \to \mathrm{H}^*_{dR,!}(S, V(2))$$

is injective.

Proposition 3.27. *If the algebraic representation V is regular, we have*

$$\mathrm{H}^*_{cusp}(S, V(2)) = \mathrm{H}^*_{(2)}(S, V(2)) = \mathrm{H}^*_{dR,!}(S, V(2)) = \mathrm{H}^2_{dR,!}(S, V(2)).$$

Proof. For the equality besides the last one, by the fact that the map

$$\mathrm{H}^*_{\mathrm{cusp}}(S, V(2)) \to \mathrm{H}^*_{dR,!}(S, V(2))$$

is injective, it suffices to prove

$$\mathrm{H}^*_{\mathrm{cusp}}(S, V(2)) = \mathrm{H}^*_{(2)}(S, V(2)).$$

By [Bor80, Theorem 4] (which applies for $\operatorname{rank} G = \operatorname{rank} K_G$), we have

$$\mathrm{H}^*_{dR,!}(S, V(2)) = \bigoplus_{\pi=\pi_\infty\otimes\pi_f} m(\pi)\mathrm{H}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2)\otimes_\mathbb{C}\pi_\infty)\otimes\pi_f,$$

where π runs over the discrete spectrum of $L^2(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$, and ξ is the restriction of the central character of $V^\vee$ to A_G. Let $\pi = \pi_f \otimes \pi_\infty$ be such an automorphic representation. By [VZ84, Theorem 5.6][4], the assumption $\mathrm{H}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2)\otimes_\mathbb{C}\pi_\infty) \neq 0$ implies $\pi_\infty = A_\mathfrak{q}(\lambda)$, which is a cohomological induction from a θ-stable parabolic subalgebra $\mathfrak{q}$. Since the highest weight of V is regular, the only $\mathfrak{q}$ that satisfies the condition in [VZ84, Theorem 5.6] is the Borel. In our case, the torus of the Borel is compact, so $A_q(\lambda)$ must be a discrete series. Discrete series are tempered, so by [BW00, III, Corollary 5.2], π_∞ only contributes to middle degree, that is

$$\mathrm{H}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2)\otimes_\mathbb{C}\pi_\infty) = \mathrm{H}^2(\mathfrak{g}_\mathbb{C}, K_G; V(2)\otimes_\mathbb{C}\pi_\infty).$$

Finally, for π appearing in the discrete spectrum, by Wallach's theorem [Wal84, Theorem 4.3] π must be cuspidal. □

Remark 3.28. The condition that V is regular is very important for Proposition 3.27 to hold.

3.5. Discrete series L-packets.

Convention 3.29. In this subsection, V is any object in $\mathrm{Rep}_\mathbb{C}(G)$.

The $((\mathfrak{g}_\mathbb{C}, K_G)\times G(\mathbb{A}_f))$-module $C^\infty_{\mathrm{cusp}}(G(\mathbb{Q})\backslash G(\mathbb{A}), \xi)$ can be decomposed into a direct sum

$$C^\infty_{\mathrm{cusp}}(G(\mathbb{Q}\backslash G(\mathbb{A}), \xi) = \bigoplus_\pi m(\pi)\pi,$$

where the sum runs over the irreducible cuspidal representations π of G and $m(\pi)$ is the multiplicity of π. This induces a decomposition

$$\mathrm{H}^2_{B,!}(S, V(2)) \cong \mathrm{H}^2_{dR,!}(S, V(2)) = \bigoplus_{\pi=\pi_\infty\otimes\pi_f} m(\pi)\mathrm{H}^2(\mathfrak{g}_\mathbb{C}, K_G; V(2)\otimes_\mathbb{C}\pi_\infty)\otimes\pi_f. \tag{3.11}$$

Recall that we have the following theorem about the multiplicity $m(\pi)$:

Theorem 3.30. [Mok15], [CHL11, §1.1], [LSZ22, Theorem 2.6.3] *The multiplicity $m(\pi)$ of the cohomological π (i.e., the π with $\mathrm{H}^*(\mathfrak{g}_\mathbb{C}, K_G; V(2)\otimes_\mathbb{C}\pi_\infty) \neq 0$, for some cohomological degree $*$) is* 1.

Definition 3.31. The *discrete series L-packet* $P(V(2))$ associated to $V(2)$ is the set of isomorphism classes of discrete series π_∞ of $G(\mathbb{R})^+$ whose Harish-Chandra parameter and central character are opposed to the ones of $V(2)$.

Proposition 3.32. *The π such that*

$$\mathrm{H}^2(\mathfrak{g}_\mathbb{C}, K_G; V(2)\otimes_\mathbb{C}\pi_\infty) \neq 0$$

are those π such that $\pi_\infty|_{G(\mathbb{R})^+} \in P(V)$ [5].

Proof. See the proof of Proposition 3.27. □

According to [BW00, Chapter II, §3, Proposition 3.1], for any representation π_∞ of $G(\mathbb{R})^+$, the $(\mathfrak{g}_\mathbb{C}, K_G)$-complex $\pi_\infty \otimes_\mathbb{C} V(2)$ has zero differential map. Hence, we have

$$\mathrm{H}^2(\mathfrak{g}_\mathbb{C}, K_G; V(2)\otimes_\mathbb{C}\pi_\infty) = \mathrm{Hom}_{K_G}(\wedge^2\mathfrak{g}_\mathbb{C}/\mathfrak{k}_{G,\mathbb{C}}, V(2)\otimes_\mathbb{C}\pi_\infty).$$

We have the decomposition:

$$\wedge^2\mathfrak{g}_\mathbb{C}/\mathfrak{k}_{G,\mathbb{C}} = \wedge^2\mathfrak{p}^+_{G,\mathbb{C}} \oplus \wedge^2\mathfrak{p}^-_{G,\mathbb{C}} \oplus \mathfrak{p}^+_{G,\mathbb{C}}\otimes\mathfrak{p}^-_{G,\mathbb{C}}.$$

[4] Although π_∞ is not unitary as a representation of $G(\mathbb{R})^+$, if we restrict to the semisimple part we could make it unitary. Hence, we can use the theorem in [VZ84], because the authors assume the real Lie group is semisimple.

[5] The archimedean part π_∞ of a smooth automorphic representation π is a $(\mathfrak{g}_\mathbb{C}, K_G)$-module. By the exponential map, π_∞ is equivalent to an representation of $G(\mathbb{R})^+$.

And we have the following identities as representations of K_G[6] :

$$\wedge^2 \mathfrak{p}^+_{G,\mathbb{C}} = \tau_{(1,1,-2)},$$
$$\wedge^2 \mathfrak{p}^-_{G,\mathbb{C}} = \tau_{(-1,-1,2)},$$
$$\mathfrak{p}^+_{G,\mathbb{C}} \otimes \mathfrak{p}^-_{G,\mathbb{C}} = \tau_{(1,-1,0)} \oplus \tau_{(0,0,0)}.$$

We also have the following proposition:

Proposition 3.33. *The dimension of the $\mathbb{C}$-vector space*

$$\mathrm{H}^2(\mathfrak{g}_{\mathbb{C}}, K_G; V(2) \otimes_{\mathbb{C}} \pi_\infty) = \mathrm{Hom}_{K_G}(\wedge^2 \mathfrak{g}_{\mathbb{C}} / \mathfrak{k}_{G,\mathbb{C}}, V(2) \otimes_{\mathbb{C}} \pi_\infty)$$

is less than or equal to 1.

Proof. This is a direct consequence of [BW00, II, Proposition 3.1. and Theorem 5.3.]. □

Finally, we can compute explicitly the elements of $P(V(2))$.

Proposition 3.34. *If the highest weight of $V(2)$ is $(\lambda_1, \lambda_2, \lambda_3; d)$ such that $\lambda_1 \geq \lambda_2 \geq \lambda_3$, we have*

$$P(V(2)) = \{\pi_1, \pi_2, \pi_3, \bar{\pi}_1, \bar{\pi}_2, \bar{\pi}_3\},$$

where for $i = 1, 2, 3$, $\bar{\pi}_i$ is the complex conjugate of π_i and the Harish-Chandra parameters and Blattner parameters[7] *of π_1, π_2 and π_3 are listed in the following table.*

(3.12)

	Harish-Chandra parameters	*Blattner parameters*
π_1	$(1-\lambda_1, -\lambda_2, -1-\lambda_1)$	$(1-\lambda_3, 1-\lambda_2, -2-\lambda_1)$
π_2	$(1-\lambda_3, -1-\lambda_1, -\lambda_2)$	$(1-\lambda_3, -1-\lambda_1, -\lambda_2)$
π_3	$(-\lambda_2, -1-\lambda_1, 1-\lambda_3)$	$(-1-\lambda_2, -1-\lambda_1, 2-\lambda_3)$

Proof. For a dominant root system with positive roots

$$\Delta^+ = \{(1,-1,0), (0,1,-1), (1,0,-1)\},$$

we have $\rho = (1, 0, -1)$. Hence, the infinitesimal character of $V(2)$ is $(\lambda_1 + 1, \lambda_2, \lambda_3 - 1)$. By [BW00, II, Proposition 3.1], the infinitesimal character of the representation π_∞ of $G(\mathbb{R})^+$ with

$$\mathrm{H}^*(\mathfrak{g}_{\mathbb{C}}, K_G; V(2) \otimes_{\mathbb{C}} \pi_\infty) \neq 0$$

must be the negative of the infinitesimal character of $V(2)$. Hence, the representations in $P(V(2))$ must have infinitesimal character $(1-\lambda_3, -\lambda_2, -1-\lambda_1)$. Then by [Kna86, Theorem 9.20], we can write down all the discrete series with fixed infinitesimal character. □

Remark 3.35. The representations π_1, π_2 and π_3 contribute to $\mathrm{H}^2_{dR,!}(\mathrm{Sh}_G(L)^{an}_{\mathbb{C}}, V(2))$ while the representations $\bar{\pi}_1$, $\bar{\pi}_2$ and $\bar{\pi}_3$ contribute to $\mathrm{H}^2_{dR,!}(\overline{\mathrm{Sh}_G(L)^{an}_{\mathbb{C}}}, V(2))$

3.6. **Hodge decomposition.** In this subsection, we compute the Hodge decomposition of the Betti realization of the pure motive associated to an automorphic representation π.

Convention 3.36. In this subsection, V is any object in $\mathrm{Rep}_{\mathbb{Q}}(G)$. We use $V_{\mathbb{C}}$ to denote $V \otimes_{\mathbb{Q}} \mathbb{C}$.

Recall that we have the following theorem.

Theorem 3.37 ([BHR94],Theorem 3.2.2 and (2.3.3))**.** *For any cuspidal automorphic representation $\pi = \pi_f \otimes \pi_\infty$ such that $\pi_\infty \in P(V_{\mathbb{C}}(2))$, π_f is defined over a number field $E(\pi_f)$.*

Convention 3.38. In this subsection, we use π_f to denote its $E(\pi_f)$-model.

[6]When considering $(\mathfrak{g}_{\mathbb{C}}, K_G)$-cohomology, the action of A_G is determined by the action of K because the centeral character is fixed, hence, we ignore the action of A_G in the notation of representations of K_G.

[7]We ignore the action of similitude factor in the notation of Harish-chandra and Blattner parameters because it is determind by the fixed central character.

Definition 3.39. We define

$$M_{dR}(\pi_f, V(2)) := \mathrm{Hom}_{\mathbb{Q}[G(\mathbb{A}_f)]}(\mathrm{Res}_{E(\pi_f)/\mathbb{Q}}\,\pi_f, \mathrm{H}^2_{dR,!}(S, V(2)))$$

and similarly

$$M_B(\pi_f, V(2)) := \mathrm{Hom}_{\mathbb{Q}[G(\mathbb{A}_f)]}(\mathrm{Res}_{E(\pi_f)/\mathbb{Q}}\,\pi_f, \mathrm{H}^2_{B,!}(S, V(2))),$$

where $\mathrm{H}^2_{dR,!}(S, V(2))$ and $\mathrm{H}^2_{B,!}(S, V(2))$ are the interior de Rham and Betti cohomology with coefficients in V. These are $\mathbb{Q}$-vector spaces with a $\mathbb{Q}$-linear action of $E(\pi_f)$.

Remark 3.40. Since $M_B(\pi_f, V(2))$ is a $\mathbb{Q}$-vector space, define its base change to $\mathbb{R}$ and $\mathbb{C}$:

$$M_B(\pi_f, V(2))_{\mathbb{R}} := M_B(\pi_f, V(2)) \otimes_{\mathbb{Q}} \mathbb{R},$$
$$M_B(\pi_f, V(2))_{\mathbb{C}} := M_B(\pi_f, V(2)) \otimes_{\mathbb{Q}} \mathbb{C}.$$

After base change to $\mathbb{C}$, we have the comparison isomorphism:

$$I_\infty \colon M_B(\pi_f, V(2))_{\mathbb{C}} \xrightarrow{\cong} M_{dR}(\pi_f, V(2))_{\mathbb{C}}$$

We also have

$$\mathrm{rank}_{E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C}}\, M_B(\pi_f, V(2))_{\mathbb{C}} = \mathrm{rank}_{E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C}}\, M_{dR}(\pi_f, V(2))_{\mathbb{C}} = 6.$$

Finally, we state a proposition about the Hodge decomposition of $M_B(\pi_f, V)_{\mathbb{C}}$.

Proposition 3.41. *If the highest weight of $V(2)$ is $(\lambda_1, \lambda_2, \lambda_3; d)$ such that $\lambda_1 \geq \lambda_2 \geq \lambda_3$, we have*

$$\begin{aligned} &M_B(\pi_f, V(2))_{\mathbb{C}} \\ \cong &M_B^{2-\lambda_2-\lambda_3-d,-\lambda_1-d} \oplus M_B^{1-\lambda_1-\lambda_3-d,1-\lambda_2-d} \oplus M_B^{-\lambda_1-\lambda_2-d,2-\lambda_3-d} \\ \oplus &M_B^{-\lambda_1-d,2-\lambda_2-\lambda_3-d} \oplus M_B^{1-\lambda_2-d,1-\lambda_1-\lambda_3-d} \oplus M_B^{2-\lambda_3-d,-\lambda_1-\lambda_2-d}, \end{aligned}$$

where $M_B^{s,t}$ is a 1-dimensional $\mathbb{C}$-subspace with Hodge type (s,t). So $M_B(\pi_f, V(2))$ is a pure $\mathbb{Q}$-Hodge structure with weight $2-(\lambda_1+\lambda_2+\lambda_3+2d)$. Moreover, we have

$$\begin{aligned} M_B^{2-\lambda_2-\lambda_3-d,-\lambda_1-d} &= \bigoplus_{\sigma: E(\pi_F)\hookrightarrow\mathbb{C}} \mathrm{H}^2(\mathfrak{g}_{\mathbb{C}}, K_G; V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_1), \\ M_B^{1-\lambda_1-\lambda_3-d,1-\lambda_2-d} &= \bigoplus_{\sigma: E(\pi_F)\hookrightarrow\mathbb{C}} \mathrm{H}^2(\mathfrak{g}_{\mathbb{C}}, K_G; V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_2), \\ M_B^{-\lambda_1-\lambda_2-d,2-\lambda_3-d} &= \bigoplus_{\sigma: E(\pi_F)\hookrightarrow\mathbb{C}} \mathrm{H}^2(\mathfrak{g}_{\mathbb{C}}, K_G; V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_3), \end{aligned}$$

where

$$P(V_{\mathbb{C}}(2)) = \{\pi_1, \pi_2, \pi_3, \bar{\pi}_1, \bar{\pi}_2, \bar{\pi}_3\}$$

is defined in Proposition 3.34.

Proof. For $i = 1, 2, 3$, if the Hodge type of π_i is (s,t), then the Hodge type of $\bar{\pi}_i$ is (t,s). So we only need to compute the Hodge type of π_1, π_2, π_3. Since complexification of h in the definition of the Shimura datum of $\mathrm{Sh}_G(L)$ is

$$h_{\mathbb{C}} \colon (z, \bar{z}) \in \mathbb{S}(\mathbb{C}) \mapsto \begin{pmatrix} z & 0 & 0 \\ 0 & z & 0 \\ 0 & 0 & \bar{z} \end{pmatrix},$$

the contribution from the local system V only depends on K_G-types. Then we compute the Hodge type of π_1, π_2, π_3 case by case.

(1) For π_1: The positive roots in [Kna86, Theorem 9.20] for π_1 are

$$\Delta^+ = \{(1,-1,0),(0,1,-1),(1,0,-1)\}.$$

Hence the lowest weight of $V_{\mathbb{C}}(2)$ is $(\lambda_3,\lambda_2,\lambda_1)$, the minimal K_G-type of $V_{\mathbb{C}}(2)$ is $\tau_{(\lambda_2,\lambda_3,\lambda_1)}$. Since the minimal K_G-type of π_1 is $\tau_{(1-\lambda_3,1-\lambda_2,-2-\lambda_1)}$, and the minimal K_G-type of $V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_1$ is $\tau_{(1,1,-2)}$. Hence, only the K_G-type $\tau_{(1,1,-2)}$ contributes to

$$\mathrm{H}^2(\mathfrak{g}_{\mathbb{C}},K_G;V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_1)=\mathrm{Hom}_{K_G}(\wedge^2\mathfrak{g}_{\mathbb{C}}/\mathfrak{k}_{G,\mathbb{C}},V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_1).$$

Since the contributed K_G-type of $V_{\mathbb{C}}(2)$ is $\tau_{(\lambda_2,\lambda_3,\lambda_1)}$ and

$$\tau_{(1,1,-2)}\subset\wedge^2\mathfrak{p}^+_{G,\mathbb{C}}\subset\wedge^2\mathfrak{g}_{\mathbb{C}}/\mathfrak{k}_{G,\mathbb{C}},$$

the contributed Hodge type of π_1 is

$$(2-\lambda_2-\lambda_3-d,-\lambda_1-d).$$

(2) For π_2: The positive roots in [Kna86, Theorem 9.20] for π_2 are

$$\Delta^+ = \{(1,-1,0),(0,-1,1),(1,0,-1)\}.$$

Hence the lowest weight of $V_{\mathbb{C}}(2)$ is $(\lambda_3,\lambda_1,\lambda_2)$, so the minimal K_G-type of $V_{\mathbb{C}}(2)$ is $\tau_{(\lambda_1,\lambda_3,\lambda_2)}$. Since the minimal K_G-type of π_2 is $\tau_{(1-\lambda_3,-1-\lambda_1,-\lambda_2)}$, the minimal K_G-type of $V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_2$ is $\tau_{(1,-1,0)}$. Hence, only the K_G-type $\tau_{(1,-1,0)}$ contributes to

$$\mathrm{H}^2(\mathfrak{g}_{\mathbb{C}},K_G;V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_2)=\mathrm{Hom}_{K_G}(\wedge^2\mathfrak{g}_{\mathbb{C}}/\mathfrak{k}_{G,\mathbb{C}},V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_2).$$

Since the contributed K_G-type of $V_{\mathbb{C}}(2)$ is $\tau_{(\lambda_1,\lambda_3,\lambda_2)}$ and

$$\tau_{(1,-1,0)}\subset\mathfrak{p}^+_{G,\mathbb{C}}\otimes\mathfrak{p}^-_{G,\mathbb{C}}\subset\wedge^2\mathfrak{g}_{\mathbb{C}}/\mathfrak{k}_{G,\mathbb{C}},$$

the contributed Hodge type of π_1 is

$$(1-\lambda_1-\lambda_3-d,1-\lambda_2-d).$$

(3) For π_3: The positive roots in [Kna86, Theorem 9.20] for π_3 are

$$\Delta^+ = \{(-1,0,1),(0,-1,1),(1,-1,0)\}.$$

Hence the lowest weight of $V_{\mathbb{C}}(2)$ is $(\lambda_2,\lambda_1,\lambda_3)$, so the minimal K_G-type of $V_{\mathbb{C}}(2)$ is $\tau_{(\lambda_1,\lambda_2,\lambda_3)}$. Since the minimal K_G-type of π_3 is $\tau_{(-1-\lambda_2,-1-\lambda_1,2-\lambda_3)}$, the minimal K_G-type of $V_{\mathbb{C}}(2)\otimes\pi_3$ is $\tau_{(-1,-1,0)}$. Hence, only the K_G-type $\tau_{(-1,-1,0)}$ contributes to

$$\mathrm{H}^2(\mathfrak{g}_{\mathbb{C}},K_G;V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_3)=\mathrm{Hom}_{K_G}(\wedge^2\mathfrak{g}_{\mathbb{C}}/\mathfrak{k}_{G,\mathbb{C}},V_{\mathbb{C}}(2)\otimes_{\mathbb{C}}\pi_3).$$

Since the contributed K_G-type of $V_{\mathbb{C}}(2)$ is $\tau_{(\lambda_1,\lambda_2,\lambda_3)}$ and

$$\tau_{(-1,-1,0)}\subset\wedge^2\mathfrak{p}^-_{G,\mathbb{C}}\subset\wedge^2\mathfrak{g}_{\mathbb{C}}/\mathfrak{k}_{G,\mathbb{C}},$$

the contributed Hodge type of π_3 is

$$(-\lambda_1-\lambda_2-d,2-\lambda_3-d).$$

□

Corollary 3.42. *If $V=V^{a,b}\{r,s\}$, we have*

$$\begin{aligned}&M_B(\pi_f,V(2))_{\mathbb{C}}\\ \cong&M_B^{-r,-2-a-b-s}\oplus M_B^{-1-a-r,-1-b-s}\oplus M_B^{-2-a-b-r,-s}\\ \oplus&M_B^{-2-a-b-s,-r}\oplus M_B^{-1-b-s,-1-a-r}\oplus M_B^{-s,-2-a-b-r}.\end{aligned}$$

Part 1. The vanishing on the boundary

4. Construction of the motivic classes

4.1. **Relative motives and motivic cohomology.** In this subsection, we first recall some basic facts about motivic categories. We then use motivic categories to give the definition of motivic cohomology and collect the needed facts about it. Finally, we give the Gysin map of motivic cohomology corresponding to the map of the Shimura varieties that we are interested in.

4.1.1. *Motivic categories.* Let Λ be a commutative ring with a unit and let $\mathrm{DM}_{B,c}(S,\Lambda)$ be the *triangulated category of constructible Beilinson motives over a base excellent scheme S with Λ-coefficients* defined in [CD19, Def.15.1.1], which is a symmetric monoidal category and we denote by 1_S the unit of the category, which is the Tate motive [CD19, Introduction B, page xxi]. We write $\mathrm{DM}_{B,c}(S)$ for $\mathrm{DM}_{B,c}(S,\mathbb{Q})$.

Fact 4.1. *We have the following properties:*

- [CD19, Theorem 15.2.4] *The triangulated category $\mathrm{DM}_{B,c}(S)$ has the formalism of Grothendieck's six-functors*
$$(f^*, f_*, f_!, f^!, \underline{\mathrm{Hom}}, \otimes)$$
and dualizing functor $\mathbb{D}$.
- [CD19, Remark 11.1.14] *Over a field F, we have the following equivalence of triangulated categories:*

$$\mathrm{DM}_{B,c}(\mathrm{Spec}(F)) \cong \mathrm{DM}_{gm}(F), \tag{4.1}$$

where the right hand side is the triangulated category of geometric motives [Voe00, Definition 2.1.1, Page 5]. *We use the following isomorphism: for any smooth $f\colon X \to \mathrm{Spec}(F)$, the motive $f_*1_X \in \mathrm{DM}_{B,c}(\mathrm{Spec}(F))$ is mapped to the dual of the motive $M_{gm}(X) \in \mathrm{DM}_{gm}(F)$ defined in* [Voe00, Definition 2.1.1].

Proposition 4.2 ([PL19, Proposition 1.7])**.** *For a regular immersion $f\colon Y \to X$ with $\dim X - \dim Y = d$ and any $M \in \mathrm{DM}_{B,c}(S)$, we have the following absolute purity isomorphism:*
$$f^*(M) \cong f^!M(d)[2d].$$

Remark 4.3. It follows from [Sta25, Tag 0E9J] that if Y and X are regular schemes, then f is a regular immersion.

Let Λ be a field. Corti and Hanamura [CH00, Definition 2.9] defined the category of relative Chow motives $\mathrm{CHM}(S,\Lambda)$ with Λ-coefficients over a base scheme S generalizing the category of Chow motives over a field. We write $\mathrm{CHM}(S)$ for $\mathrm{CHM}(S,\mathbb{Q})$. The relationship between $\mathrm{CHM}(S)$ and $\mathrm{DM}_{B,c}(S)$ proved by Jin is as follows:

Proposition 4.4 ([Fan16, Theorem 3.17])**.** *Let $\mathrm{Chow}(S)$ be the heart of the Chow weight structure on $\mathrm{DM}_{B,c}(S)$ constructed in* [H11, Théorème 3.3.]. *Then there is a natural functor*
$$F\colon \mathrm{CHM}(S) \to \mathrm{Chow}(S),$$
which is an equivalence of the two categories $\mathrm{CHM}(S)$ and $\mathrm{Chow}(S)$.

Convention 4.5. This proposition tells us that for any object $V \in \mathrm{CHM}(S)$, $F(V)$ is an object of the category $\mathrm{DM}_{B,c}(S)$. In order to simplify notation, we write V for $F(V)$.

Finally, we recall the definition of motives of Abelian type which will be used in the proof of vanishing on the boundary for motivic cohomology.

Definition 4.6 ([Wil15, Definition 1.1])**.** Let F be a perfect field and $\bar{F}$ be its algebraic closure.

(1) The category of *Chow motives of Abelian type* over $\bar{F}$ is defined as a strict full additive tensor sub-category $\mathrm{CHM}^{Ab}(\bar{F})$ of $\mathrm{CHM}(\bar{F})$ generated by the following:

(a) Shifts of Tate motives $\mathbb{Z}(m)[2m]$, for $m \in \mathbb{Z}$,
(b) Chow motives of Abelian varieties over $\bar{F}$.

(2) Define the category of *Chow motives of Abelian type over* $\bar{F}$ as the (strict) full (additive tensor) sub-category $\mathrm{CHM}^{Ab}(F)$ of $\mathrm{CHM}(F)$ whose base change to $\bar{F}$ lies in $\mathrm{CHM}^{Ab}(\bar{F})$.
(3) The category $\mathrm{DM}_{gm}^{Ab}(F)$ is defined as the (strict) full triangulated subcategory of $\mathrm{DM}_{gm}(F)$ generated by $\mathrm{CHM}^{Ab}(F)$.

Notation 4.7. We denote by $\mathrm{DM}_{B,c}^{Ab}(\mathrm{Spec}(\mathbb{Q}))$ the sub-category of $\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))$ that is isomorphic to $\mathrm{DM}_{gm}^{Ab}(F)$ via the isomorphim in equation (4.1).

4.1.2. *The localization exact triangle.* We recall the localization exact triangle of Beilinson motives that will be used in the proof of vanishing on the boundary for motivic cohomology.

Let

$$U \xrightarrow{j} X \xleftarrow{i} Y$$

be a diagram in $\mathrm{Sch}(\mathbb{Q})$ where j is an open immersion and i is the complementary reduced closed immersion. We assume that U has the same dimension as X and that Y has codimension c in X.

Proposition 4.8 ([CD19, Proposition 2.3.3])**.** *Let N be an object of $\mathrm{DM}_{B,c}(S)$. We have the following exact triangle*

$$j_!j^*N \longrightarrow N \longrightarrow i_*i^*N \longrightarrow j_!j^*N[1]$$

in $\mathrm{DM}_{B,c}(S)$.

4.1.3. *Motivic cohomology.*

Definition 4.9 (Motivic Cohomology)**.** If W is an object of $\mathrm{DM}_{B,c}(S)$ and 1_S is the unit object of $\mathrm{DM}_{B,c}(S)$, then the *motivic cohomology* is defined as the Λ-module

$$\mathrm{H}^*_{\mathrm{M}}(S, W) = \mathrm{Hom}_{\mathrm{DM}_{B,c}(S)}(1_S, W[*]).$$

Remark 4.10.
- It follows from [CD19, Corrollary 14.2.14] that this definition is compatible with the K-theoretical one: for any regular scheme X, we have a canonical isomorphism
$$\mathrm{H}^q_{\mathrm{M}}(X, \mathbb{Q}(p)) \cong \mathrm{K}_{2p-q}(X)^{(p)}_{\mathbb{Q}},$$
where the right-hand side is the p-th graded piece of the γ-filtration of an algebraic K-group.
- Recall that in [Wil09, Page 21], for any $M \in \mathrm{DM}_{gm}(\mathbb{Q})$, the motivic cohomology $\mathrm{H}^i_M(M, \mathbb{Q}(j))$ is defined as

$$\mathrm{H}^i_M(M, \mathbb{Q}(j)) := \mathrm{Hom}_{\mathrm{DM}_{gm}(\mathbb{Q})}(M, \mathbb{Q}(i)[j]). \tag{4.2}$$

We can see that this definition is compatible with our definition through the isomorphism in equation (4.1).

Notation 4.11. Let $\mathcal{G}$ temporarily denote any of the three groups $\{\mathrm{GL}_2, H, G\}$, so we have the associated Shimura variety $\mathrm{Sh}_{\mathcal{G}}$.

Lemma 4.12 ([Anc15, Theorem 8.6])**.** *Let F be a number field. There is an additive functor*

$$\mu_M \colon \mathrm{Rep}_F(\mathcal{G}) \to \mathrm{CHM}(\mathrm{Sh}_{\mathcal{G}}, F)$$

from the category of representations of $\mathcal{G}$ over F to the category of relative Chow motives over $\mathrm{CHM}(\mathrm{Sh}_{\mathcal{G}}, F)$ *with the following properties:*

(1) The functor μ_M preserves tensor product and duals;
(2) For the similitude character $\mu \colon \mathcal{G} \to \mathbf{G}_m$, $\mu_M(\mu)$ is the Lefschetz motive $\mathbb{Q}(1)$[8];

[8]The Lefschetz motive $\mathbb{Q}(1)$ has weight -2, and its image under the Hodge realization functor is $\mathbb{R}(1) := 2\pi i$.

(3) If V denotes the defining representation of $\mathcal{G}$, then $\mu_M(V) = \mathrm{h}^1(A)(1)$, where A is the universal PEL *abelian variety over* $\mathrm{Sh}_{\mathcal{G}}$*;*

Remark 4.13. We give four remarks about Lemma 4.12.

(1) The construction by Ancona in Lemma 4.12 works much more generally for arbitrary PEL Shimura varieties, but in this paper, we shall only need it for the above three groups.
(2) Our normalization is compatible with Ancona's original normalization, but dual to the normalization of [LSZ22, Theorem 8.3.1] where they send the multiplier representation to $\mathbb{Q}(-1)$ and the defining representation to $\mathrm{h}^1(A)$.
(3) The Ancona functor μ_M is faithful. [Tor20, REMARK 5.10].
(4) For any $V \in \mathrm{Rep}_F(\mathcal{G})$ and any integer d, we have $\mu_M(V(d)) = \mu_M(V)(d)$. Hence, we can use the twist (d) before or after applying ancona functor.

By applying Torzewski's general theorem [Tor20, Theorem 9.7] in the setting of G and H, we get the following result:

Proposition 4.14 ("Branching" for motivic sheaves)**.** *For the map of Shimura varieties $\iota : M \to S$, we have the following commutative diagram of functors*

$$\begin{array}{ccc} \mathrm{Rep}_F(G) & \xrightarrow{\mu_M} & \mathrm{CHM}(S,F) \\ \downarrow{\scriptstyle \iota^*} & & \downarrow{\scriptstyle \iota^*} \\ \mathrm{Rep}_F(H) & \xrightarrow{\mu_M} & \mathrm{CHM}(M,F), \end{array}$$

where the left-hand ι^* denotes the restriction of representations, and the right-hand ι^* denotes the pullback of relative Chow motives.

Convention 4.15.
- For any $\mathcal{G}$, we write $\mathrm{Rep}(\mathcal{G})$ for $\mathrm{Rep}_{\mathbb{Q}}(\mathcal{G})$. For any $V \in \mathrm{Rep}(\mathcal{G})$, we also write V for $\mu_M(V)$.
- Recall that we define the elements of $\mathrm{Rep}(G)$ and $\mathrm{Rep}(H)$ as the Weil restriction of elements of $\mathrm{Rep}_E(G)$ and $\mathrm{Rep}_E(H)$.
- Recall that we always let $M = \mathrm{Sh}_H$ and $S = \mathrm{Sh}_G$.

From the above discussion, for a representation of $\mathcal{G}$ denoted as V, we get a relative Chow motive $V \in \mathrm{CHM}(\mathrm{Sh}_{\mathcal{G}})$ over Shimura variety $\mathrm{Sh}_{\mathcal{G}}$, which can also be viewed as an object of $\mathrm{DM}_{B,c}(\mathrm{Sh}_{\mathcal{G}})$. So for V, we can define the motivic cohomology $\mathrm{H}^*_{\mathrm{M}}(\mathrm{Sh}_{\mathcal{G}}, V(*))$ [9].

Proposition 4.16 (Gysin morphism for motivic cohomology)**.** *For the map of Shimura varieties $\iota\colon M \to S$ defined earlier and for the representations $W \in \mathrm{Rep}(H)$ and $V \in \mathrm{Rep}(G)$ satisfying $W \hookrightarrow \iota^* V$ in $\mathrm{Rep}(H)$, we have the Gysin morphism*

$$\iota_*\colon \mathrm{H}^1_{\mathrm{M}}(M, W(1)) \to \mathrm{H}^3_{\mathrm{M}}(S, V(2)),$$

which is induced by

$$(\iota_* W(1)[1] \to V(2)[3]) \in \mathrm{DM}_{B,c}(S). \tag{4.3}$$

Proof. From the above discussions, we have a map $W \hookrightarrow \iota^* V$ in $\mathrm{DM}_{B,c}(M)$, where ι^* is the pullback functor on the category of Beilinson motives. By the absolute purity isomorphism [PL19, Proposition 1.7], the right hand side is $\iota^* V \cong \iota^! V(1)[2]$. By adjunction, we get the morphism $\iota_! W \to V(1)[2]$ in $\mathrm{DM}_{B,c}(S)$. Since ι is proper, we have $\iota_* = \iota_!$. So we get the morphism

$$\iota_* W(1)[1] \to V(2)[3] \in \mathrm{DM}_{B,c}(S).$$

Then by applying the functor $N \longmapsto \mathrm{Hom}_{\mathrm{DM}_{B,c}(S)}(1_S, N)$ to the morphism, we get

$$\mathrm{Hom}_{\mathrm{DM}_{B,c}(S)}(1_S, \iota_* W(1)[1]) \to \mathrm{Hom}_{\mathrm{DM}_{B,c}(S)}(1_S, V(2)[3]).$$

[9] In this paper, in order to apply Beilinson regulator, we define all motivic cohomology groups to be $\mathbb{Q}$-vector spaces, which is the same as Weil restriction to $\mathbb{Q}$ of the motivic cohomology groups defined in [LSZ22].

The right-hand side is $\mathrm{H}^3_{\mathrm{M}}(S, V(2))$, and by adjunction and $\iota^* 1_S = 1_M$, the left-hand side is

$$\mathrm{Hom}_{\mathrm{DM}_{B,c}(M)}(1_M, W(1)[1]),$$

which is $\mathrm{H}^1_{\mathrm{M}}(M, W(1))$. □

4.2. **Mixed Hodge modules and absolute Hodge cohomology.** In this subsection, we first recall some basic facts about mixed Hodge modules, which can be viewed as a relative version of mixed Hodge structures, "archimedean" analogues of mixed l–adic perverse sheaves, or the Hodge realization of the mixed motivic sheaves which have not been defined. We then use mixed Hodge modules to define absolute Hodge cohomology, which can be seen as the Hodge realization of the motivic cohomology defined in Definition 4.9, and collect the needed facts about it. Finally, we give the Gysin map of absolute Hodge cohomology corresponding to the map of the Shimura varieties that we are interested in. For this part, we mainly follow [Lem15, §2.2].

4.2.1. *Mixed Hodge modules.* Let $A \subset \mathbb{R}$ be a subfield and $\mathrm{Sch}(\mathbb{Q})$ be the category of quasi-projective $\mathbb{Q}$-schemes. For $X \in \mathrm{Sch}(\mathbb{Q})$, we have the abelian category $\mathrm{MHM}_A(X/\mathbb{R})$ of real algebraic mixed A-Hodge modules [HW98, Definition A.2.4]. Let $\mathrm{D}^b_c(X^{\mathrm{an}}_{\mathbb{C}}, A)$ be the bounded derived category of sheaves for the analytic topology of A-vector spaces with constructible cohomology objects and let $\mathrm{Perv}_A(X^{\mathrm{an}}_{\mathbb{C}}) \subset \mathrm{D}^b_c(X^{\mathrm{an}}_{\mathbb{C}}, A)$ be the abelian category of perverse sheaves on $X^{\mathrm{an}}_{\mathbb{C}}$.

Fact 4.17. *We have the following properties:*

(1) [Bei87] *The natural functor* $\mathrm{D}^b(\mathrm{Perv}_A(X^{\mathrm{an}}_{\mathbb{C}})) \to \mathrm{D}^b_c(X^{\mathrm{an}}_{\mathbb{C}}, A)$ *is an equivalence of categories.*

(2) [Sai90b, Theorem 0.1] *There is a faithful and exact functor*

$$rat\colon \mathrm{MHM}_A(X/\mathbb{R}) \to \mathrm{Perv}_A(X^{\mathrm{an}}_{\mathbb{C}}).$$

We also use the same notation to denote its derived functor

$$rat\colon \mathrm{D}^b(\mathrm{MHM}_A(X/\mathbb{R})) \to \mathrm{D}^b_c(X^{\mathrm{an}}_{\mathbb{C}}, A).$$

Also, for $M \in \mathrm{D}^b(\mathrm{MHM}_A(X/\mathbb{R}))$, *we have* $rat(\mathcal{H}^i M) = {}^p\mathcal{H}^i rat(M)$, *where* $\mathcal{H}^i M$ *is the standard cohomology functor and* ${}^p\mathcal{H}^i$ *is the perverse cohomology functor.*

(3) [Sai90a], [Sai90b], [HW98, Theorem A.2.5] *We have six-functors formalism with*

$$(f^*, f_*, f_!, f^!, \underline{\mathrm{Hom}}, \otimes)$$

and the dualizing functor $\mathbb{D}$ *on the derived category* $\mathrm{D}^b(\mathrm{MHM}_A(X/\mathbb{R}))$, *which is a symmetric monoidal category. Furthermore, these functors commute with the functor rat.*

We now collect some facts about the relationship between variations of Hodge structures and mixed Hodge modules.

Assume that X is smooth and pure of dimension d. Then for any local system V of A-vector spaces on $X^{\mathrm{an}}_{\mathbb{C}}$, by the definition of perverse sheaves the complex $V[d]$ concentrated in degree $-d$ is an object of $\mathrm{Perv}_A(X^{\mathrm{an}}_{\mathbb{C}})$.

Let $\mathrm{MHM}_A(X/\mathbb{R})^s$ be the full subcategory of $\mathrm{MHM}_A(X/\mathbb{R})$ whose objects are those $M \in \mathrm{MHM}_A(X/\mathbb{R})$ such that $rat(M) = V[d]$ for some local system V. Denote by $\mathrm{Var}_A(X/\mathbb{R})$ the category of real admissible polarizable variations of mixed A-Hodge structures over X [HW98, Definition A.2.4 b].

Fact 4.18 ([HW98, Definition A.2.4.b])**.** *There is an equivalence of categories*

$$\mathrm{Var}_A(X/\mathbb{R}) \cong \mathrm{MHM}_A(X/\mathbb{R})^s.$$

From this, we have the following equivalence of abelian categories:

$$\mathrm{MHM}_A(\mathrm{Spec}(\mathbb{Q})/\mathbb{R}) \cong \mathrm{MHS}^+_A,$$

where the right hand side denotes the abelian category of mixed polarizable A-Hodge structures [HW98, Lemma A.2.2].

Convention 4.19. (1) We use the convention that a variation of mixed Hodge structure is viewed as a mixed Hodge module but not a shift of mixed Hodge modules, which is the same as the one used in [BW04a]. When X is smooth and pure of dimension d, the embedding

$$\iota\colon \mathrm{Var}_A(X/\mathbb{R}) \to \mathrm{D}^{\mathrm{b}}(\mathrm{MHM}_A(X/\mathbb{R}))$$

is characterized by the fact: for any object $V \in \mathrm{Var}_A(X/\mathbb{R})$, the complex $rat(\iota(V))[-d]$ is a single non-trivial constituent in degree zero, which means that it is a local system. When the context is clear, we always write V for $\iota(V)$.

(2) In order to simplify terminology, when $A = \mathbb{R}$ we will write "variation of Hodge structure" instead of "real admissible polarizable variation of mixed $\mathbb{R}$-Hodge structure" and write "mixed Hodge modules" for "real algebraic mixed $\mathbb{R}$-Hodge modules".

Proposition 4.20 ([Blo06, Proposition 2.3.12, Proposition 2.3.14, Lemme 2.3.7])**.** *Under the embedding*

$$\iota\colon \mathrm{Var}_{\mathbb{R}}(X/\mathbb{R}) \to \mathrm{D}^{\mathrm{b}}(\mathrm{MHM}_{\mathbb{R}}(X)/\mathbb{R}),$$

we have the following three consequences.

(1) For any $V, W \in \mathrm{Var}_{\mathbb{R}}(X/\mathbb{R})$, we have

$$\iota(V \otimes W)[d] = \iota(V) \otimes \iota(W),$$

where the left tensor product is the tensor product in $\mathrm{Var}_{\mathbb{R}}(X/\mathbb{R})$ and the right tensor product is the tensor product in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(X)/\mathbb{R})$.

(2) Let $f\colon X \to Y$ be a morphism between smooth algebraic varieties with $\dim X - \dim Y = d$. For any $V \in \mathrm{Var}_{\mathbb{R}}(X/\mathbb{R})$, we have the identity:

$$\iota((f^s)^* V) = f^* \iota(V)[d],$$

where $(f^s)^*$ is pullback in $\mathrm{Var}_{\mathbb{R}}(X/\mathbb{R})$ and f^* is the pullback in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(X)/\mathbb{R})$. In particular, if $s\colon X \to \mathrm{Spec}(\mathbb{Q})$ is a smooth scheme which is pure of dimension d, $A_X(n)$ (resp., $A(n)$) is the Tate variation on X (resp., $\mathrm{Spec}(\mathbb{Q})$) viewed as an object of $\mathrm{MHM}_A(X/\mathbb{R})$ (resp., MHS_A^+), we have the equality $A_X(n) = s^* A(n)[d]$ in $\mathrm{MHM}_A(X/\mathbb{R})$.

(3) For any $V \in \mathrm{Var}_{\mathbb{R}}(X/\mathbb{R})$, the following identity holds:

$$\mathbb{D}(\iota(V)) = \iota(V^\vee)(d),$$

where $(\cdot)^\vee$ denotes the dualizing functor in $\mathrm{Var}_{\mathbb{R}}(X/\mathbb{R})$ and $\mathbb{D}$ stands for the dualizing functor in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(X)/\mathbb{R})$.

Remark 4.21. From the proposition, we can see that the unit 1_X in the symmetric monoidal category $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(X)/\mathbb{R})$ is $A(n)_X[-d]$.

Convention 4.22. When the scheme X is clear from the context, we write $A(n)$ for $A(n)_X$.

Theorem 4.23 ([Sai88, Theorem 2])**.** *Let $X \in \mathrm{Sch}(\mathbb{Q})$ be smooth and pure of dimension d. A variation of Hodge structure of weight w is a mixed Hodge module of weight $w + d$ via the above identification.*

Finally, we state the absolute purity theorem for mixed Hodge modules, which will be used in the construction of the Gysin map for absolute Hodge cohomology.

Proposition 4.24 ([Blo06, Proposition 2.3.16])**.** *Let $f\colon X \to Y$ be a morphism of pure relative dimension d between smooth algebraic varieties. Then for any $V \in \mathrm{Var}_{\mathbb{R}}(Y/\mathbb{R})$, we have the identity:*

$$f^! V = f^* V(d)[2d].$$

4.2.2. *The localization exact triangle.* We recall the localization exact triangle of mixed Hodge modules that will be used in the proof of vanishing on the boundary for absolute Hodge cohomology.

Let

$$U \xrightarrow{j} X \xleftarrow{i} Y$$

be a diagram in $\mathrm{Sch}(\mathbb{Q})$ where j is an open immersion and i is the complementary reduced closed immersion. We assume that U has the same dimension as X and that Y has codimension c in X.

Proposition 4.25 ([Sai90b, (4.4.1)])**.** *Let N be an object of $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(X/\mathbb{R}))$, we have the following exact triangle*

$$j_!j^*N \longrightarrow N \longrightarrow i_*i^*N \longrightarrow j_!j^*N[1]$$

in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(X/\mathbb{R}))$.

Definition 4.26 (Singular (Betti) cohomology)**.** (1) For $X \in \mathrm{Sch}(\mathbb{Q})$ with the structure map $s\colon X \to \mathrm{Spec}(\mathbb{Q})$, by [HW98, Corollary A.1.7.c], $\mathcal{H}^i s_*s^*A(0)$ is the i-th singular cohomology of the underlying topological space of $X^{\mathrm{an}}_{\mathbb{C}}$ with coefficients in A. It also has the mixed Hodge structure constructed by Deligne with the involution induced by the complex conjugation on $X(\mathbb{C})$. From this, we get that if X is smooth of pure dimension d, and if $A(0)$ denotes the trivial Tate variation of Hodge structures on X, then the *i-th singular cohomology* is $\mathcal{H}^{i-d}s_*A(0)$.

(2) For any X and any $M \in \mathrm{MHM}_A(X/\mathbb{R})$, we define the *$i$-th singular cohomology* of X with M as coefficients, $\mathrm{H}^i(X, M)$, as the group $\mathcal{H}^{i-d}s_*M$; similarly, we define the compactly supported cohomology $\mathrm{H}^i_c(X, M)$ as $\mathcal{H}^{i-d}s_!M$.

Corollary 4.27. *Let M be an object of $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(U/\mathbb{R}))$ and assume X is proper. We have the following long exact sequence of mixed Hodge structures*

$$\cdots \to \mathrm{H}^i_c(U, M) \to \mathrm{H}^i(U, M) \to \mathrm{H}^{i-c}(Y, i^*j_*M) \to \mathrm{H}^{i+1}_c(U, M) \to \cdots . \tag{4.4}$$

Proof. Let $N = j_*M$. Then apply the above triangle to get the exact triangle:

$$j_!M \longrightarrow j_*M \longrightarrow i_*i^*j_*M \longrightarrow j_!M[1] .$$

Let $s\colon X \to \mathrm{Spec}(\mathbb{Q})$ be the structure morphism of X and apply the functor s_* to the exact triangle and then take cohomology $\mathcal{H}^{i-d}$, where $d = \dim X$. □

4.2.3. *Absolute Hodge cohomology.* We now give the definition of absolute Hodge cohomology and construct its Gysin morphism in the case that we are interested in.

Definition 4.28 (Absolute Hodge cohomology)**.** Let X be the scheme as above. The *i-th real absolute Hodge cohomology* $\mathrm{H}^i_H(X, M)$[10] of X with coefficients in M is

$$\begin{aligned}\mathrm{H}^i_H(X, M) :=& \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_A(X/\mathbb{R}))}(A(0)_X, M[i])\\ =& \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_A(X/\mathbb{R}))}(s^*A(0)[d], M[i])\\ \cong& \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHS}^+_A)}(A(0), s_*M[i-d]),\end{aligned}$$

where the second isomorphism comes from adjunction.

Remark 4.29. (1) This definition is the "archimedean" analogue of the definition of motivic cohomology given in Definition 4.9.

[10]The standard notation for real absolute Hodge cohomology is $\mathrm{H}^i_H(X/\mathbb{R}, M)$, which is good to distinguish with complex absolute Hodge cohomology. However, in our paper, we only work with real absolute Hodge cohomology, so we use $\mathrm{H}^i_H(X, M)$ to denote real absolute Hodge cohomology.

(2) If the abelian category $\mathrm{MHS}^+_{\mathbb{R}}$ has cohomological dimension 1 [Bei86b, Corollary 1.10], then by the Leray spectral sequence, we have the short exact sequence

$$0 \to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^{i-1}(X, M)) \to \mathrm{H}^i_H(X, M) \to \mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^i(X, M)) \to 0 \tag{4.5}$$

for all i and all $M \in \mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(X/\mathbb{R}))$.

Now we come to the setting of Shimura varieties.

Notation 4.30.
- Recall that there is a $\mathbb{R}$-linear tensor functor associating an algebraic representation of the group underlying a given Shimura variety to the corresponding variation of Hodge structure on the Shimura variety, and we denote the functor as μ_H (see [BW04a, 2]).
- Similar to the last section, we let $\mathcal{G}$ temporarily denote any of the three groups $\{\mathrm{GL}_2, H, G\}$, and we have the associated Shimura variety $\mathrm{Sh}_{\mathcal{G}}$. So if $V \in \mathrm{Rep}_{\mathbb{Q}}(\mathcal{G})$[11], we have the variation of Hodge stucture $\mu_H(V) \in \mathrm{Var}_{\mathbb{R}}(\mathrm{Sh}_{\mathcal{G}})$.
- Compose with the embedding $\mathrm{Var}_{\mathbb{R}}(\mathrm{Sh}_{\mathcal{G}}/\mathbb{R}) \to \mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\mathrm{Sh}_{\mathcal{G}})/\mathbb{R})$, we can view $\mu_H(V)$ as an object of $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\mathrm{Sh}_{\mathcal{G}}/\mathbb{R}))$ which has a single non-trivial constituent in degree 0. We will write V instead of $\mu_H(V)$ when there is no confusion. In other words, we can view $V \in \mathrm{Rep}_{\mathbb{Q}}(\mathcal{G})$ as an object of $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\mathrm{Sh}_{\mathcal{G}}/\mathbb{R}))$, and we can then define the absolute Hodge cohomology $\mathrm{H}^i_H(\mathrm{Sh}_{\mathcal{G}}, V)$.
- Recall that we always let $M = \mathrm{Sh}_H$ and $S = \mathrm{Sh}_G$.

Remark 4.31. Unlike the motivic setting, the "branching" for variation of Hodge structrues is direct from the definition: for the map of Shimura varieties $\iota\colon M \to S$, we have the following commutative diagram of functors

$$\begin{array}{ccc} \mathrm{Rep}(G) & \xrightarrow{\mu_H} & \mathrm{Var}_{\mathbb{R}}(S) \\ \downarrow{\scriptstyle \iota^*} & & \downarrow{\scriptstyle \iota^*} \\ \mathrm{Rep}(H) & \xrightarrow{\mu_H} & \mathrm{Var}_{\mathbb{R}}(M), \end{array}$$

where the left-hand ι^* denotes the restriction of representations, and the right-hand ι^* denotes the pullback of variation of Hodge structures.

Just as with motivic cohomolgy, we have the following Gysin morphism.

Proposition 4.32 (Gysin morphism for absolute Hodge cohomology)**.** *For the map of Shimura varieties $\iota\colon M \to S$ defined as before and for the representations $W \in \mathrm{Rep}_{\mathbb{Q}}(H)$ and $V \in \mathrm{Rep}_{\mathbb{Q}}(G)$ satisfying $W \hookrightarrow \iota^* V$ in $\mathrm{Rep}_{\mathbb{Q}}(H)$, we have the Gysin morphism*

$$\iota_*\colon \mathrm{H}^1_H(M, W(1)) \to \mathrm{H}^3_H(S, V(2)),$$

which is induced by

$$(\iota_* W(1) \to V(2)[1]) \in \mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R})). \tag{4.6}$$

Proof. From the above discussions, we have a map $W \hookrightarrow (\iota^s)^* V$ in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(M)/\mathbb{R})$, where ι^* is the pullback functor for $\mathrm{Var}_{\mathbb{R}}(\mathrm{Sh}_{\mathcal{G}})$. It follows from Proposition 4.20 that $(\iota^s)^* V = \iota^* V[-1]$, where ι^* is the pullback functor for $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R}))$. By the absolute purity isomorphism (see Proposition 4.24), the right-hand side is $\iota^* V \cong \iota^! V(1)[2]$. By adjunction, we get the morphism $\iota_! W \to V(1)[1]$ in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R}))$. Since ι is proper, we have $\iota_* = \iota_!$, and we get the morphism

$$(\iota_* W(1) \to V(2)[1]) \in \mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R})).$$

Finally, applying the functor $N \mapsto \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}(S/\mathbb{R}))}(\mathbb{R}(0)_S, N[2])$ to the map, we get the Gysin map $\mathrm{H}^1_H(M, W(1)) \to \mathrm{H}^3_H(S, V(2))$. □

[11] For H and G, the meaning of $V \in \mathrm{Rep}_{\mathbb{Q}}(G)$ (reps. $V \in \mathrm{Rep}_{\mathbb{Q}}(H)$) is that there exists some $V' \in \mathrm{Rep}_E(G)$ (resp., $V' \in \mathrm{Rep}_E(H)$) such that $V = \mathrm{Res}_{E/\mathbb{Q}} V'$.

4.3. **Beilinson regulator and functoriality.** In this subsection, we give the definition of Beilinson regulator using the Hodge realization functor for smooth quasi-projective varieties[12] over $\mathbb{Q}$ and state its functorial properties.

Definition 4.33 (Hodge realization)**.** For a smooth quasi-projective variety S over $\mathbb{Q}$, let $S_{\mathbb{C}}$ and $S_{\mathbb{R}}$ be the base change of S to $\mathbb{C}$ and $\mathbb{R}$. The *Hodge realization functor* is defined as the composition of the following functors:

$$R_H\colon \mathrm{DM}_{B,c}(S) \xrightarrow{f_1^*} \mathrm{DM}_{B,c}(S_{\mathbb{R}}) \xrightarrow{f_2^*} \mathrm{DM}_{B,c}(S_{\mathbb{C}}) \xrightarrow{R_H^{Bou}} \mathrm{D}(\mathrm{MHM}(S_{\mathbb{C}})),$$

where $f_1\colon S_{\mathbb{R}} \to S$ and $f_2\colon S_{\mathbb{C}} \to S_{\mathbb{R}}$ are the canonical map and the functor R_H^{Bou} is the Hodge realization functor from the triangulated category of constructible Beilinson motives $\mathrm{DM}_{B,c}(S_{\mathbb{C}})$ over $S_{\mathbb{C}}$ to the (unbounded) derived category $\mathrm{D}(\mathrm{MHM}(S_{\mathbb{C}}))$ of algebraic mixed Hodge modules over $S_{\mathbb{C}}$ defined in [Bou22, Definition 169, Theorem 43].

It follows from the étale descent property of constructible Beilinson motives (see [CD19, Theorem 14.3.4]), the definition of the cateogory $\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R})$ (see [HW98, Definition A.2.4]), and the functoriality of R_H (see [Bou22, Theorem 43 (ii0)]) that the image of R_H lies in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R}))$. Hence, we get the covariant Hodge realization functor

$$R_H\colon \mathrm{DM}_{B,c}(S) \to \mathrm{D}(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R})).$$

Remark 4.34.

- As the functor R_H^{Bou} (see [Bou22, Page 1]) is an extension of the functor

$$\begin{aligned} \mathrm{Variety}/S_{\mathbb{C}} &\longrightarrow \mathrm{D}(\mathrm{MHM}(S_{\mathbb{C}})) \\ (f\colon X \mapsto S) &\longmapsto f_*\mathbb{R}_X, \end{aligned}$$

  where $\mathrm{Variety}/S_{\mathbb{C}}$ is the category of $\mathbb{C}$-varieties over $S_{\mathbb{C}}$, we see that when we restrict R_H^{Bou} to the sub-category $\mathrm{CHM}(S_{\mathbb{C}})$, the image lies in $\mathrm{D}^b(\mathrm{MHM}(S_{\mathbb{C}}))$.
- We normalize the realization functor such that for any V in the category of relative Chow motives $\mathrm{CHM}(S)$ over S with $\dim S = d$, we have

$$R_H(V) = V[-d] \in \mathrm{D}^{\mathrm{b}}(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R})).$$

Definition 4.35 (Beilinson regulator)**.** For a smooth quasi-projective variety S over $\mathbb{Q}$, $\{i,j\} \in \mathbb{Z}$ and $W \in \mathrm{DM}_{B,c}(S)$, the *Beilinson regulator* r_H is defined as follows:

$$\begin{aligned} r_H\colon \mathrm{H}^i_{\mathrm{M}}(S, W(j)) &= \mathrm{Hom}_{\mathrm{DM}_{B,c}(S)}(1_S, W(j)[i]) \\ \to \mathrm{H}^i_H(S, W(j)) &= \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R}))}(R_H(1_S), R_H(W)(j)[i]), \end{aligned}$$

where R_H is the Hodge realization functor defined above.

Remark 4.36.

- When $\dim S = d$, we have

$$R_H(1_S) = \mathbb{R}(0)_S[-d],$$

  which is the unit of $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R}))$.

The following proposition tells us that our definition is compatible with Beilinson's original definition (see [Bei84b, §2]).

Proposition 4.37. *When $W = \mathbb{Q}$, if we choose a canonical isomorphism*

$$\mathrm{H}^i_{\mathrm{M}}(S, \mathbb{Q}(j)) \cong \mathrm{K}_{2j-i}(S)^{(j)}_{\mathbb{Q}},$$

then the regulator map

$$r_H\colon \mathrm{H}^i_{\mathrm{M}}(S, \mathbb{Q}(j)) \to \mathrm{H}^i_H(S, \mathbb{R}(j))$$

is the same as the original definition of Beilinson in [Bei84b, §2].

[12] This level of generality is enough for our applications.

Proof. It follows from the functoriality of R_H that the Beilinson regulator can be defined as

$$\begin{aligned} r_H\colon \mathrm{H}^i_{\mathrm{M}}(S,\mathbb{Q}(j)) &= \mathrm{Hom}_{\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))}(1_{\mathrm{Spec}(\mathbb{Q})}, a_*1_S(j)[i]) \\ &\to \mathrm{H}^i_H(S,\mathbb{R}(j)) = \mathrm{Hom}_{\mathrm{D^b}(\mathrm{MHS}^+_{\mathbb{R}})}(R_H(1_{\mathrm{Spec}(\mathbb{Q})}), R_H(a_*1_S)(j)[i]), \end{aligned}$$

where $a\colon S \to \mathrm{Spec}(\mathbb{Q})$ is the structure morphism of S. It can be seen from the definition of Huber's contravariant Hodge realization functor $R^{Hu}_H\colon \mathrm{DM}_{gm}(\mathbb{Q}) \to \mathrm{D^b}(\mathrm{MHS}^+_{\mathbb{R}})$ (see [Hub00, Theorem 2.3.3] and [Hub04, Theorem B.2.2]) that in this case r_H is the same as

$$\begin{aligned} r^{Hu}_H\colon \mathrm{H}^i_{\mathrm{M}}(S,\mathbb{Q}(j)) &\cong \mathrm{Hom}_{\mathrm{DM}_{gm}(\mathbb{Q})}((a_*1_S)^*, 1_{\mathrm{Spec}(\mathbb{Q})}(j)[i]) \\ &\to \mathrm{H}^i_H(S,\mathbb{R}(j)) = \mathrm{Hom}_{\mathrm{D^b}(\mathrm{MHS}^+_{\mathbb{R}})}(R^{Hu}_H(1_{\mathrm{Spec}(\mathbb{Q})}), R^{Hu}_H(a_*1_S)(j)[i]), \end{aligned}$$

where the notation $(\cdot)^*$ means taking dual. Finally, we have the commutative diagram

$$\begin{array}{ccccc} & & \mathrm{K}_{2j-i}(S) & & \\ & \swarrow & \downarrow{\scriptstyle p} & \searrow{\scriptstyle r^B_H} & \\ \mathrm{H}^i_{\mathrm{M}}(S,\mathbb{Q}(j)) & \xrightarrow{id} & \mathrm{H}^i_{\mathrm{M}}(S,\mathbb{Q}(j)) & \xrightarrow{r^{Hu}_H} & \mathrm{H}^i_H(S,\mathbb{R}(j)), \end{array}$$

where r^B_H is Beilinson's original definition of his regulator in [Bei84b, §2], and the map p is defined in [Hub00, Corollary 4.2.2]. The commutativity of the right triangle follows from [Hub95, Definition 18.2.6, Proposition 18.2.8] and [Hub04, Corollary 4.2.3] and the commutativity of the left triangle comes from the canonical choice of $\mathrm{H}^i_{\mathrm{M}}(S,\mathbb{Q}(j)) \cong \mathrm{K}_{2j-i}(S)^{(j)}_{\mathbb{Q}}$. Thus, the proposition is proved. □

By the definition of the Beilinson regulator and functoriality of the Hodge realization functor, we have the following functorial properties of the Beilinson regulator.

Proposition 4.38. *(1) For the morphism of Shimura varieties $p\colon M \to \mathrm{Sh}_{\mathrm{GL}_2}$ induced by the projection $H \to \mathrm{GL}_2$, we have the commutative diagram.*

$$\begin{array}{ccc} \mathrm{H}^1_M(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p^*} & \mathrm{H}^1_M(M, W(1)) \\ \downarrow{\scriptstyle r_H} & & \downarrow{\scriptstyle r_H} \\ \mathrm{H}^1_H(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p^*} & \mathrm{H}^1_H(M, W(1)) \end{array},$$

where p^* is the pullback of the corresponding cohomomlogy theory.

(2) For the closed immersion of Shimura varieties $\iota\colon M \to S$ and V, W as in Proposition 4.16 and Proposition 4.32, we have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{H}^i_M(M, W(1)) & \xrightarrow{\iota_*} & \mathrm{H}^3_M(S, V(2)) \\ \downarrow{\scriptstyle r_H} & & \downarrow{\scriptstyle r_H} \\ \mathrm{H}^1_H(M, W(1)) & \xrightarrow{\iota_*} & \mathrm{H}^3_H(S, V(2)) \end{array}.$$

4.4. **Motivic classes and their Hodge realizations.** In this subsection, we give the constructions of the motivic classes that we are interested in and compute their Hodge realizations.

4.4.1. *Eisenstein symbols.* We first recall the construction of Eisenstein symbols.

Recall that the *Eisenstein symbol* [Bei86a, §3] is a $\mathbb{Q}$-linear map

$$Eis^n_M\colon \mathcal{B}_n \longrightarrow \mathrm{H}^{n+1}_M(E^n, \mathbb{Q}(n+1))$$

where $\mathrm{H}^{n+1}_M(E^n, \mathbb{Q}(n+1))$ is the direct limit over compact open subgroups $K \subset \mathrm{GL}_2(\mathbb{A}_f)$ of the motivic cohomology $\mathrm{H}^{n+1}_M(E^n_K, \mathbb{Q}(n+1))$ of the n-th fiber product of the universal elliptic curve E_K/M_K over the modular curve of level K.

Definition 4.39. The notation $\mathcal{B}_n$ stands for the space of locally constant $\mathbb{Q}$-valued functions ϕ_f on $\mathrm{GL}_2(\mathbb{A}_f)$ statisfying the following conditions:

(1) for all $a, d \in \mathbb{Q}$ such that $ad > 0$ and for all $b \in \mathbb{A}_f$, we have

$$\phi_f(\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} g) = a^{-1} d^{n+1} \phi_f(g), \tag{4.7}$$

(2)

$$\phi_f(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} g) = \phi_f(g),$$

(3) for all $k \in \hat{\mathbb{Z}}^\times$, we have

$$\phi_f(\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix} g) = \phi_f(g).$$

Remark 4.40.

- The source of the Eisenstein symbol can identified with a space of $\mathbb{Q}$-valued functions $\mathcal{F}^n$ on $\mathrm{GL}_2(\mathbb{A}_f)$ in [Bei86a, 2.1.2, P7] by the following map

$$\Psi_f \in \mathcal{F}^n \mapsto (\phi_f \colon g \mapsto \phi_f(g) = \Psi_f({}^t g)) \in \mathcal{B}^n.$$

- At present, many of the results relating special value of L-functions to regulators rely on Eisenstein symbols ([Bei86a], [Den89], [Den90], [Kin98], [Lem15], [Lem17], [Kat04], ...).

Notation 4.41. By the definition of the motivic sheaf $\mathrm{Sym}\, V_2(1)$, the motivic cohomology $\mathrm{H}^1_M(M, \mathrm{Sym}^n V_2(1))$ is a direct summand of $\mathrm{H}^{n+1}_M(E^n, \mathbb{Q}(n+1))$. Moreover, the Eisenstein symbol map factors through the inclusion $\mathrm{H}^1_M(M, \mathrm{Sym}^n V_2(1)) \subset \mathrm{H}^{n+1}_M(E^n, \mathbb{Q}(n+1))$ and we will also use the notation Eis^n_M to denote the map

$$Eis^n_M \colon \mathcal{B}_n \longrightarrow \mathrm{H}^1_M(M, \mathrm{Sym}^n V_2(1)).$$

Let ν be a finite-order Hecke character, so its archimedean component ν_∞ is a unitary character.

Definition 4.42. Let $I_n(\nu)$ be the space of locally constant $\bar{\mathbb{Q}}$-valued functions f such that for all $a, d \in \mathbb{Q}_+^\times$, $\alpha, \delta \in \hat{\mathbb{Z}}^\times$, $b \in \mathbb{A}_f$ and $g \in \mathrm{GL}_2(\mathbb{A}_f)$, the function f satisfies that

$$f(\begin{pmatrix} a\alpha & b \\ & d\delta \end{pmatrix} g) = a^{-1} d^{n+1} \nu(\alpha) f(g), \tag{4.8}$$

Thus, the space $I_n(\nu)$ is endowed with the action of $\mathrm{GL}_2(\mathbb{A}_f)$ by right translation.

Lemma 4.43. *There is a $\mathrm{GL}_2(\mathbb{A}_f)$-equivariant decomposition*

$$\mathcal{B}_{n,\bar{\mathbb{Q}}} = \bigoplus_{\nu, \mathrm{sgn}(\nu) = (-1)^n} I_n(\nu),$$

where the direct sum is indexed by all finite-order Hecke characters of sign $(-1)^n$.

Proof. The proof follows from the proof of [Lem17, Lemma 4.3]. Let T_2 be the diagonal maximal torus of GL_2. We are interested in the action of $T_2(\mathbb{A}_f)$ on $\mathcal{B}_{n,\bar{\mathbb{Q}}}$ by left translation. It follows from the decomposition $\mathbb{A}_f^\times = \mathbb{Q}_+^\times \hat{\mathbb{Z}}^\times$ and equation (4.7) that we only need to consider the action of $T_2(\hat{\mathbb{Z}}^\times)$. Since each function in $\mathcal{B}_{n,\bar{\mathbb{Q}}}$ is locally constant, it is fixed by an compact open subgroup of $\mathrm{GL}_2(\hat{\mathbb{Z}}^\times)$. Hence, $\mathcal{B}_{n,\bar{\mathbb{Q}}}$ can be decomposed into a direct sum of finite-dimensional $\bar{\mathbb{Q}}$-vector spaces that are stable under the action of $T_2(\hat{\mathbb{Z}}^\times)$. For each of the finite-dimensional vector spaces V, V is a direct sum of finite order Hecke characters

$$\chi \colon \begin{pmatrix} \alpha & 0 \\ 0 & \delta \end{pmatrix} \mapsto \nu(\alpha)\chi(\delta)$$

because the action is continuous. By equation (4.8), we can see that χ is trivial. Finally, if we plug $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ into equation (4.7), then we get that ν has to have sign $(-1)^n$. □

4.4.2. *Motivic classes and their image under the Beilinson regulator.* Using the Eisenstein symbol, we will give the construction of motivic classes, which has been done in [LSZ22, Definition 9.2.3] and compute their Hodge realizations.

Construction 4.44. We compose the Eisenstein symbol with the pullback

$$p^*\colon \mathrm{H}^1_M(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) \to \mathrm{H}^1_M(M, W(1))$$

and the Gysin map

$$\iota_*\colon \mathrm{H}^1_M(M, W(1)) \to \mathrm{H}^3_M(S, V(2))$$

to get the map

$$\mathcal{E}is^n_M\colon \mathcal{B}_n \to \mathrm{H}^3_M(S, V(2)).$$

Hence, for any $\phi_f \in \mathcal{B}_n$, we a have motivic class $\mathcal{E}is^n_M(\phi_f) \in \mathrm{H}^3_M(S, V(2))$. In summary, we have the following diagram[13]:

$$\begin{array}{ccccccc} \mathcal{B}_n & \xrightarrow{Eis^n_M} & \mathrm{H}^1_M(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p^*} & \mathrm{H}^1_M(M, W(1)) & \xrightarrow{\iota_*} & \mathrm{H}^3_M(S, V(2)) \\ \phi_f & \longmapsto & Eis^n_M(\phi_f) & & & \longmapsto & \mathcal{E}is^n_M(\phi_f) \end{array}.$$

By Proposition 4.38, we have the following commutative diagram:

$$\begin{array}{ccccccc} \mathcal{B}_n & \xrightarrow{Eis^n_M} & \mathrm{H}^1_M(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p^*_M} & \mathrm{H}^1_M(M, W(1)) & \xrightarrow{\iota_{M,*}} & \mathrm{H}^3_M(S, V(2)) \\ \downarrow{\scriptstyle r_H} & & \downarrow{\scriptstyle r_H} & & \downarrow{\scriptstyle r_H} & & \downarrow{\scriptstyle r_H} \\ \mathcal{B}_{n,\mathbb{R}} & \xrightarrow{Eis^n_H} & \mathrm{H}^1_H(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p^*_H} & \mathrm{H}^1_H(M, W(1)) & \xrightarrow{\iota_{H,*}} & \mathrm{H}^3_H(S, V(2)) \end{array},$$

where we use the subscript M and H to distinguish maps of motivic cohomology and absolute Hodge cohomology. Hence, the image of the motivic classes $\mathcal{E}is^n_M(\phi_f)$ under the Beilinson regulator r_H is

$$\begin{aligned} \mathcal{E}is^n_H(\phi_f) &= r_H(\iota_{M,*} \circ p^*_M \circ Eis^n_M(\phi_f)) \\ &= \iota_{H,*} \circ p^*_H \circ Eis^n_H(r_H(\phi_f)). \end{aligned}$$

Convention 4.45. In the rest of the paper, when the context is clear, we will ignore the subscript M and H, and only write ι_*, p^* and Eis^n.

5. The vanishing on the boundary of absolute Hodge cohomolgy

5.1. **The Baily–Borel compactification of Picard modular surfaces.** In this subsection, we recall the basics of the The Baily–Borel compactification of Picard modular surfaces.

The boundary of the Baily-Borel compactification of a Shimura variety is stratified by Shimura varieties associated to standard (containing the Borel) admissible parabolic subgroups [Pin90, 4.5 Definition] of the underlying reductive group G.

Let us recall the structure of standard admissible parabolic subgroups in the cases that we are interested in.

- The reductive group is H (Definition 3.2): Recall that H is $\mathrm{GL}_2 \boxtimes E^\times$[14], where the notation $\boxtimes$ denotes a pair with the same norm. If we denote by $B_2 = T_2 N'$ the standard Borel of GL_2 together with its Levi decomposition, then the standard admissible parabolic subgroups Q' of H are isomorphic to $B_2 \boxtimes E^\times$, whose Levi decomposition is denoted as $Q' = M'N'$. The canonical normal subgroup P'_1 of Q' defined in [Pin90, 4.7] is

$$P'_1 = \{\begin{pmatrix} * & * \\ 0 & 1 \end{pmatrix}\} \boxtimes E^\times \subseteq H,$$

[13] Our construction is the Weil restriction to $\mathbb{Q}$ of the construction over E in [LSZ22, Definition 9.2.3.] and it is more convenient for Beilinson's conjectures.

[14] In the following sections, we use $E^\times$ to denote $\mathrm{Res}_{E/\mathbb{Q}}\,\mathrm{G_m}$.

whose Levi is

$$M_1' = \{\begin{pmatrix} * & 0 \\ 0 & 1 \end{pmatrix}\} \boxtimes E^\times \subseteq H.$$

- The reductive group is G (Definition 3.2): The standard admissible parabolic subgroups Q of G is isomorphic to the standard Borel $B = \begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix}$ of G, whose Levi decomposition is denoted by $Q = MN$ (see [Anc17, Proposition 3.3]). The canonical normal subgroup P_1 of Q defined in [Pin90, 4.7] is

$$P_1 = \{\begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & 1 \end{pmatrix}\} \subseteq G,$$

whose Levi is

$$M_1 = \{\begin{pmatrix} * & 0 & 0 \\ 0 & * & 0 \\ 0 & 0 & 1 \end{pmatrix}\} \subseteq G$$

(see [Anc17, Lemma 3.8]).

For each of the two Shimura varieties $X = \{M, S\}$, we denote by X^* its Baily–Borel compactification and let $\partial X = X^* - X$ be the cusps of the compactification. Then we have the commutative diagram.

$$\begin{array}{ccccc} M & \xrightarrow{j'} & M^* & \xleftarrow{i'} & \partial M \\ \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ S & \xrightarrow{j} & S^* & \xleftarrow{i} & \partial S \end{array}, \tag{5.1}$$

where j and j' are open immersions, i and i' are closed immersions and p and q are closed immersions induced by the closed immersion ι [15]. We also have

$$\dim \partial S = 0, \dim \partial M = 0.$$

5.2. **Degeneration of Hodge structures.** In this subsection, we compute the degeneration of Hodge structures over Shimura varieties M and S, which will be used in the proof of vanishing on the boundary.

Notation 5.1. In the remaining part of this section, we use μ (instead of μ_H) to denote the tensor functor that associates an algebraic representation of a reductive group to the corresponding variation of Hodge structure on the corresponding Shimura variety.

5.2.1. *The theorem of Burgos-Wildehaus and Kostant.* We first recall the theorem of Burgos-Wildehaus in the setting of M and S.

Theorem 5.2 ([BW04b], Theorem 2.6, 2.9)**.** *Let E (resp., E') be an algebraic representation of G (resp., H). In the derived category $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S))$ (resp., $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(M))$), we have*

$$i^* j_* \mu(E) \cong \bigoplus_n \mathcal{H}^n i^* j_* \mu(E)[-n]$$

$$(resp., i'^{,*} j'_* \mu(E') \cong \bigoplus_n \mathcal{H}^n i'^{,*} j'_* \mu(E')[-n]).$$

Furthermore, we have

$$\mathcal{H}^n i^* j_* \mu(E) \cong \mu(\mathrm{H}^{n+2}(N, E))$$

$$(resp., \mathcal{H}^n i'^{,*} j'_* \mu(E') \cong \mu(\mathrm{H}^{n+1}(N', E')).$$

[15] For this to hold, we need to choose level structures carefully.

Remark 5.3. (1) The group G_1 (resp., G_1') acts on the group cohomology of the unipotent group $\mathrm{H}^q(N,E)$ (resp., $\mathrm{H}^q(N',E')$) via its action both on N (resp., N') and E (resp., E'), so the second isomorphism can be viewed an isomorphism of representations of G_1 (resp., G_1').

(2) The theorem of Burgos-Wildehaus and Kostant works for general Shimura varieties.

Now, let us introduce the necessary notations for Kostant's theorem.

Notation 5.4. (1) Recall that for any unipotent group N and any representation E of N, we have $\mathrm{H}^q(N,E)=\mathrm{H}^q(\mathfrak{n},E)$ where $\mathfrak{n}$ is the Lie algebra of N and the right hand side is Lie algebra cohomology.

(2) Let Q be a parabolic subgroup of a reductive group G with Levi decomposition $Q=MN$, and $\mathfrak{q}=\mathfrak{n}\oplus\mathfrak{m}$ where $\mathfrak{q}=\mathrm{Lie}(Q)$, $\mathfrak{n}=\mathrm{Lie}(N)$ and $\mathfrak{m}=\mathrm{Lie}(M)$. Let $\mathfrak{h}$ be the Cartan subalgebra of $\mathfrak{g}=\mathrm{Lie}(G)$ corresponding to a fixed Borel and $\mathrm{R}^+(\mathfrak{g},\mathfrak{h})$ be the set of positive roots. We also write $\mathrm{R}(\mathfrak{n},\mathfrak{h})$ as the subset of $\mathrm{R}^+(\mathfrak{g},\mathfrak{h})$ appearing in $\mathfrak{n}$. Denote by ρ the half-sum of positive roots, and by $\mathrm{W}(\mathfrak{g},\mathfrak{h})$ the Weyl group. For any $w\in W(\mathfrak{g},\mathfrak{h})$, we write
- $\mathrm{R}^+(w)=\{\alpha\in\mathrm{R}^+(\mathfrak{g},\mathfrak{h})|w^{-1}\alpha\notin\mathrm{R}^+(\mathfrak{g},\mathfrak{h})\}$,
- $\mathrm{l}(w)=|\mathrm{R}^+(w)|$,
- $\mathrm{W}'=\{w\in\mathrm{W}(\mathfrak{g},\mathfrak{h})|\mathrm{R}^+(w)\subset\mathrm{R}(\mathfrak{n},\mathfrak{h})\}$.

Theorem 5.5 ([Vog81], Theorem 3.2.3)**.** *Let E_λ be an irreducible representation of $\mathfrak{g}$ of highest weight λ. Then*

$$\mathrm{H}^q(\mathfrak{n},E_\lambda)=\bigoplus_{\{w\in\mathrm{W}'|\mathrm{l}(w)=q\}}F_{w(\lambda+\rho)-\rho}$$

as representations of G_1, where F_μ is an irreducible representation of $\mathfrak{m}$ of highest weight μ.

5.2.2. *Degeneration of Hodge structures on M and S.* Now we give the explicit description of degeneration of Hodge structures on the Shimura varieties M and S.

We first consider the case of M.

Notation 5.6. Let us denote by $\lambda'(d_1,d_2)$ the algebraic representation of

$$G_1'=\{(x;z)\in\mathrm{G_m}\boxtimes E^\times|x=z\bar{z}\}$$

with (highest) weight $(x;z)\mapsto z^{d_1}\bar{z}^{d_2}$.

Lemma 5.7. *Consider the diagram*

$$M\xrightarrow{j'}M^*\xleftarrow{i'}\partial M,\ .$$

As variations of mixed Hodge structures on ∂M, we have

$$i'^*j'_*\mu(W)\cong\bigoplus_{n=-1}^{0}\mathcal{H}^ni'^*j'_*\mu(W)[-n], \tag{5.2}$$

where

$$\begin{aligned}&\mathcal{H}^{-1}i'^*j'_*\mu(W)=\mu(\lambda'(a+b+r+s,a+b+r+s)),\\&\mathcal{H}^0i'^*j'_*\mu(W)=\mu(\lambda'(-1,-1)),\\&\mathcal{H}^ni'^*j'_*\mu(W)=0,\ \textit{for } n<-1 \textit{ or } n>0.\end{aligned}$$

The Hodge type of the corresponding variation of mixed Hodge structures are listed below:

VMHS	*Hodge type*
$\mu(\lambda'(a+b+r+s,a+b+r+s))$	$(-(a+b+r+s),-(a+b+r+s)))$
$\mu(\lambda'(-1,-1))$	$(1,1)$,

(5.3)

where "VMHS" stands for variations of mixed Hodge structures.

Proof. It follows from Theorem 5.2 and Theorem 5.5 that

$$\mathcal{H}^n i'^* j'_* \mu(W) = \begin{cases} \mu(\mathrm{H}^{n+1}(N', W)) & \text{when } n \geq -1, \\ 0 & \text{when } n < -1. \end{cases}$$

Moreover, in our case when $\mathrm{W}' = \mathrm{W} = S_2$ and $\rho = (\frac{1}{2}, -\frac{1}{2}; 0)$, we have the following table of elements $w \in \mathrm{W}'$:

w	$\mathrm{l}(w)$	$w(\lambda + \rho) - \rho$
1	0	$(\lambda_1, \lambda_2; d) \mapsto (\lambda_1, \lambda_2; d)$
(12)	1	$(\lambda_1, \lambda_2; d) \mapsto (\lambda_2 - 1, \lambda_1 + 1; d)$

From this, we can see that

$$\mathcal{H}^n i'^* j'_* \mu(W) = 0, \text{ for } n > 0.$$

For $W = \mathrm{Sym}^n V_2$ where $n = a + b + r + s$, we get as algebraic representations of G'_1,

$$\begin{aligned} \mathrm{H}^0(N', W) &= \lambda'(a + b + r + s, a + b + r + s), \\ \mathrm{H}^1(N', W) &= \lambda'(-1, -1). \end{aligned}$$

□

Remark 5.8. This computation is essentially the same as [Lem15, Lemma 4.4].

Now, let us consider the case of S.

Notation 5.9. Let us denote by $\lambda(\mu_1, \mu_2; d)$ the algebraic representation of

$$G_1 = \{ \begin{pmatrix} x & & \\ & z & \\ & & 1 \end{pmatrix} \} \subseteq G$$

with (highest) weight $x^{\mu_1} z^{\mu_2} (z\bar{z})^d$.

Lemma 5.10. *Consider the diagram*

$$S \xrightarrow{j} S^* \xleftarrow{i} \partial S.$$

As variations of mixed Hodge structures on ∂S, we have

$$i^* j_* \mu(V) \cong \bigoplus_{n=-2}^{1} \mathcal{H}^n i^* j_* \mu(V)[-n], \tag{5.4}$$

where

$$\begin{aligned} \mathcal{H}^{-2} i^* j_* \mu(V) &= \mu(\lambda(a + r - s, r - s; 2s - r + b)), \\ \mathcal{H}^{-1} i^* j_* \mu(V) &= \mu(\lambda(r - s - 1, a + r - s; 2s - r + b)) \\ &\quad \oplus \mu(\lambda(a + r - s, r - s - b - 1; 2s - r + b)), \\ \mathcal{H}^0 i^* j_* \mu(V) &= \mu(\lambda(r - s - b - 2, a + r - s + 1; 2s - r + b)) \\ &\quad \oplus \mu(\lambda(r - s - 1, r - s - b - 1; 2s - r + b)), \\ \mathcal{H}^1 i^* j_* \mu(V) &= \mu(\lambda(r - s - b - 2, r - s; 2s - r + b), \\ \mathcal{H}^n i^* j_* \mu(V) &= 0, \text{ for } n < -2 \text{ or } n > 1. \end{aligned}$$

The Hodge type of the corresponding variation of mixed Hodge structures are listed as follows, where "VMHS" stands for variations of mixed Hodge structures.

(5.5)

VMHS	*Hodge type*
$\mu(\lambda(a+r-s, r-s; 2s-r+b))$	$(-(a+b+r), -(a+b+s))$
	$(-(a+b+s), -(a+b+r))$
$\mu(\lambda(r-s-1, a+r-s; 2s-r+b))$	$(-(a+b+r), -(b+s-1))$
	$(-(b+s-1), -(a+b+r))$
$\mu(\lambda(a+r-s, r-s-b-1; 2s-r+b))$	$(-(a+r-1), -(a+b+s))$
	$(-(a+b+s), -(a+r-1))$
$\mu(\lambda(r-s-b-2, a+r-s+1; 2s-r+b))$	$(-(a+r-1), -(s-2))$
	$(-(s-2), -(a+r-1))$
$\mu(\lambda(r-s-1, r-s-b-1; 2s-r+b))$	$(-(r-2), -(s+b-1))$
	$(-(s+b-1), -(r-2))$
$\mu(\lambda(r-s-b-2, r-s; 2s-r+b)$	$(-(r-2), -(s-2))$
	$(-(s-2), -(r-2))$

Proof. It follows from Theorem 5.2 that

$$\mathcal{H}^n i^* j_* \mu(V) = \mu(\mathrm{H}^{n+2}(N, V)).$$

It follows from the fact $\mathrm{R}(\mathfrak{n}, \mathfrak{h}) = \mathrm{R}^+(\mathfrak{g}, \mathfrak{h})$ that $\mathrm{W}' = \mathrm{W}$. By $\rho = (1, 0, -1; 0)$, we have the following table about elements $w \in \mathrm{W}'$:

w	$\mathrm{l}(w)$	$w(\lambda+\rho)-\rho$
1	0	$(\lambda_1, \lambda_2, \lambda_3; d) \mapsto (\lambda_1, \lambda_2, \lambda_3; d)$
(12)	1	$(\lambda_1, \lambda_2, \lambda_3; d) \mapsto (\lambda_2 - 1, \lambda_1 + 1, \lambda_3; d)$
(13)	3	$(\lambda_1, \lambda_2, \lambda_3; d) \mapsto (\lambda_3 - 2, \lambda_2, \lambda_1 + 2; d)$
(23)	1	$(\lambda_1, \lambda_2, \lambda_3; d) \mapsto (\lambda_1, \lambda_3 - 1, \lambda_2 + 1; d)$
(132)	2	$(\lambda_1, \lambda_2, \lambda_3; d) \mapsto (\lambda_3 - 2, \lambda_1 + 1, \lambda_2 + 1; d)$
(123)	2	$(\lambda_1, \lambda_2, \lambda_3; d) \mapsto (\lambda_2 - 1, \lambda_3 - 1, \lambda_1 + 2; d)$

So if $n < -2$ or $n > 1$, we have

$$\mathcal{H}^n i^* j_* \mu(V) = 0.$$

Therefore, for $V = V^{a,b}\{r, s\} = V(a+r-s, r-s, r-s-b; 2s-r+b)$ and $-2 \leq n \leq 1$, we get the results listed in the proposition. □

Remark 5.11. The computation is carried out in [Anc17, §4]. Our parametrization is different from [Anc17], so we reproduce it here.

5.3. **A vanshing theorem for Betti cohomology.** We recall the following vanishing theorem.

Theorem 5.12 ([Sap05b], Theorem 5)**.** *Let G be a reductive algebraic group defined over $\mathbb{Q}$ and let D be the associated symmetric space, which can be written as $D = G(\mathbb{R})/K_\infty A_G$, where K_∞ is a maximal compact of $G(\mathbb{R})$ and A_G is the identity component of $\mathbb{R}$-points of a maximal $\mathbb{Q}$-split torus in the center of G. Let $\Gamma \subset G(\mathbb{Q})$ be a neat arithmetic subgroup, $X = \Gamma\backslash D$ be the associated locally symmetric space and E be the local system on X associated to an irreducible algebraic representation of G.*

If D is a Hermitian or equal-rank symmetric space and the highest weight of E is regular, then $\mathrm{H}^i(X, E) = 0$ for all $i < \frac{1}{2}\dim X$, where $\dim X$ is the real dimension of the locally symmetric space X.

Remark 5.13. This result is announced without a detailed proof in [Sap05b] and proved in the preprint [Sap05a].

5.4. **Vanishing on the boundary for absolute Hodge cohomology.** In this subsection, we state and prove the theorem of vanishing on the boundary for absolute Hodge cohomology.

Convention 5.14.
- In this subsection, Betti cohomology, compactly supported cohomology and interior cohomology all use $\mathbb{R}$-coefficients. We will not stress this in the notation used for cohomology.
- For Betti and compactly supported cohomology, we do not write the subscript B but for interior cohomology, we write the subscript B.
- In this subsection, we ignore the notation μ and write V (resp., W) instead of $\mu(V)$ (resp. $\mu(W)$).

5.4.1. *The theorem.* We first give the statement of the theorem.

Theorem 5.15. *Under the conditions*

(1) $0 \leq -r \leq a$ *and* $0 \leq -s \leq b$,
(2) $a > 0$ *and* $b > 0$,
(3) $r \neq 0$ *or* $s \neq 0$,

the map $\mathcal{E}is_H^n \colon \mathcal{B}_{n,\mathbb{R}} \to \mathrm{H}^3_H(S, V(2))$ *factors through the inclusion*

$$\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbf{1}, \mathrm{H}^2_{B,!}(S, V(2))) \hookrightarrow \mathrm{H}^3_H(S, V(2)),$$

where $\mathrm{MHS}^+_{\mathbb{R}}$ *is the abelian category of mixed* $\mathbb{R}$*-Hodge structures and* $\mathbf{1}$ *denotes the trivial Hodge structure that is the unit of* $\mathrm{MHS}^+_{\mathbb{R}}$.

5.4.2. *The proof the theorem.* We give a proof of Theorem 5.15 here and the proof is divided into three steps.

Step I. In the first step, the goal is to prove the exactness of the sequence in Proposition 5.17.

Lemma 5.16. *We have*

$$\begin{aligned}
&\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial S, i^*j_*V(2))) = 0,\\
&\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^1(\partial S, i^*j_*V(2))) = 0,\\
&\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^3(S, V(2))) = 0.
\end{aligned}$$

Proof. First, by equation (5.4) and $\dim \partial S = 0$, we have

$$\begin{aligned}
\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial S, i^*j_*V(2))) &= \mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathcal{H}^0 i^*j_*V(2)),\\
\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^1(\partial S, i^*j_*V(2))) &= \mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathcal{H}^1 i^*j_*V(2)).
\end{aligned}$$

By Table (5.5) in Lemma 5.10, we can see that there is no part with Hodge type $(0,0)$ contained in $\mathcal{H}^0 i^*j_*V(2)$ and $\mathcal{H}^1 i^*j_*V(2)$. Hence, we get

$$\begin{aligned}
&\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial S, i^*j_*V(2))) = 0,\\
&\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^1(\partial S, i^*j_*V(2))) = 0.
\end{aligned}$$

Second, by the exact sequence in equation (4.4) and Theorem 5.12, we have

$$\mathrm{H}^3(S, V(2)) \cong \mathrm{H}^1(\partial S, i^*j_*V(2))).$$

Hence, we conclude that

$$\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^3(S, V(2))) = 0.$$

□

Proposition 5.17. *We have the following exact sequence*

$$0 \longrightarrow \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbf{1}, \mathrm{H}^2_{B,!}(S, V(2))) \longrightarrow \mathrm{H}^3_H(S, V(2)) \xrightarrow{\partial_S^H} \mathrm{H}^1_H(\partial S, i^*j_*V(2)),$$

where i^* *and* j_* *denote the pullback and pushforward of the mixed Hodge module.*

Proof. Apply the choices $U = S$, $c = 2$, $Y = \partial S$ and $M = V(2)$ in the exact sequence in equation (4.4), to get the exact sequence

$$0 \to \mathrm{H}^2_c(S, V(2)) \to \mathrm{H}^2(S, V(2)) \to \mathrm{H}^0(\partial S, i^* j_* V(2)) \to \mathrm{H}^3_c(S, V(2)). \tag{5.6}$$

By Poincaré duality, we have

$$\mathrm{H}^3_c(S, V(2)) \cong \mathrm{H}^1(S, V^\vee(-2)) = 0,$$

where the last equality holds since $V^\vee$ is regular. So we have the short exact sequence

$$0 \to \mathrm{H}^2_{B,!}(S, V(2)) \to \mathrm{H}^2(S, V(2)) \to \mathrm{H}^0(\partial S, i^* j_* V(2)) \to 0.$$

Apply the functor $\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), -)$ to this short exact sequence to get the exact sequence

$$\begin{aligned}\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial S, i^* j_* V(2))) &\to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^2_{B,!}(S, V(2))) \\ &\to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^2(S, V(2))) \to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial S, i^* j_* V(2)).\end{aligned}$$

Since $\mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial S, i^* j_* V(2))) = 0$ by Lemma 5.16, we have the following exact sequence

$$\begin{aligned}0 &\to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^2_{B,!}(S, V(2))) \\ &\to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^2(S, V(2))) \to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial S, i^* j_* V(2)).\end{aligned}$$

Also by Lemma 5.16 and the exact sequence (4.5), we get

$$\mathrm{H}^3_H(S, V(2)) \cong \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^2(S, V(2))) \tag{5.7}$$

and

$$\mathrm{H}^1_H(\partial S, i^* j_* V(2)) \cong \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial S, i^* j_* V(2)), \tag{5.8}$$

so we get the desired exact sequence

$$0 \longrightarrow \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^2_{B,!}(S, V(2))) \longrightarrow \mathrm{H}^3_H(S, V(2)) \longrightarrow \mathrm{H}^1_H(\partial S, i^* j_* V(2)).$$

□

Step II. In the second step, the goal is to prove the diagram in Proposition 5.18 commutes.

Proposition 5.18. *The following diagram is commutative,*

$$\begin{array}{ccc} \mathcal{B}_{n,\mathbb{R}} & & \\ \downarrow{\scriptstyle p^* \circ Eis^n_H} & & \\ \mathrm{H}^1_H(M, W(1)) & \xrightarrow{\partial^H_M} & \mathrm{H}^0_H(\partial M, i'^* j'_* W(1))) \\ \downarrow{\scriptstyle \iota^H_*} & & \downarrow{\scriptstyle \theta^H} \\ \mathrm{H}^3_H(S, V(2)) & \xrightarrow{\partial^H_S} & \mathrm{H}^1_H(\partial S, i^* j_* V(2)) \end{array},$$

where ι^H_* is the Gysin morphism defined in Proposition 4.32, ∂^H_M and ∂^H_S are the corresponding boundary maps and the map θ^H is induced from the map

$$(\partial \iota^H_* : q_* i'^* j'_* W(1) \to i^* j_* V(2)[1]) \in \mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial S/\mathbb{R})), \tag{5.9}$$

which is degenerated from ι^H_*.

Proof. It follows from Proposition 4.32 that the Gysin morphism $\iota_*\colon \mathrm{H}^1_H(M, W(1)) \to \mathrm{H}^3_H(S, V(2))$ is induced by

$$(\iota_* W(1) \to V(2)[1]) \in \mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R})).$$

Apply the functor j_* to the morphism, and by the fact that $j_*\iota_* W(1) = p_* j'_* W(1)$, we get the map

$$(p_* j'_* W(1) \to j_* V(2)[1]) \in \mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S^*/\mathbb{R})).$$

Then we apply the functor $i_* i^*$ to the morphism and have a morphism

$$(i_* i^* p_* j'_* W(1) \to i_* i^* j_* V(2)[1]) \in \mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S^*/\mathbb{R})).$$

From the equation $i^* p_* = q_* i'^*$ which follows from the proper base change theorem of mixed Hodge modules for the proper map p, we get the following map

$$(i_* q_* i'^* j'_* W(1) \to i_* i^* j_* V(2)[1]) \in \mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S^*/\mathbb{R})).$$

By functoriality of the natural transformation $\mathrm{id} \to i_* i^*$ from adjunction, we can see the following commutative diagram in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S^*/\mathbb{R}))$:

$$\begin{array}{ccc} p_* j'_* W(1) & \longrightarrow & i_* q_* i'^* j'_* W(1) \\ \downarrow & & \downarrow \\ j_* V(2)[1] & \longrightarrow & i_* i^* j_* V(2)[1] \end{array}.$$

Finally, apply the functor $\mathrm{Hom}_{\mathrm{D^b}(\mathrm{MHM}_{\mathbb{R}}(S^*/\mathbb{R}))}(\mathbb{R}(0)_{S^*}, -[2])$ to the above commutative diagram to get the commutative diagram

$$\begin{array}{ccc} \mathrm{H}^1_H(M, W(1)) & \longrightarrow & \mathrm{H}^0_H(\partial M, i'^* j'_* W(1)) \\ \downarrow & & \downarrow \\ \mathrm{H}^3_H(S, V(2)) & \longrightarrow & \mathrm{H}^1_H(\partial S, i^* j_* V(2)) \end{array}.$$

□

Remark 5.19. From the construction, the boundary map θ^H_S in Proposition 5.18 is the same as the θ^H_S in Proposition 5.17. Hence, we use the same notation to for them.

Step III. In the third step, the goal is to prove the map θ^H_S in Proposition 5.18 is zero, which is stated in Proposition 5.26.

We first recall some notation.

Notation 5.20.

- We have the following commutative diagram:

$$\begin{array}{ccccc} M & \xrightarrow{j'} & M^* & \xleftarrow{i'} & \partial M \\ \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ S & \xrightarrow{j} & S^* & \xleftarrow{i} & \partial S \end{array}.$$

- Recall that the group homomorphism underlying the morphism between the boundary components $q\colon \partial M \to \partial S$ is the map

$$\begin{aligned} \iota\colon M_1' = \mathrm{G_m} \boxtimes E^\times &\longrightarrow M_1 \\ (x; z) &\longmapsto \begin{pmatrix} x & & \\ & z & \\ & & 1 \end{pmatrix}. \end{aligned} \tag{5.10}$$

- We denote by $\lambda(\mu_1,\mu_2;d)$ the algebraic representation of M_1 with weight

$$\begin{pmatrix} x & & \\ & z & \\ & & 1 \end{pmatrix} \mapsto x^{\mu_1} z^{\mu_2} (z\bar{z})^d, \tag{5.11}$$

  and we use $\lambda'(d_1,d_2)$ to represent the algebraic representation of M_1' with weight $(x;z)\mapsto z^{d_1}\bar{z}^{d_2}$.
- We let $n=a+b+r+s$.

Lemma 5.21. *The source* $\mathrm{H}^0_H(\partial M, i'^* j'_* W(1))$ *of the map* θ_S^H *is isomorphic to*

$$\mathrm{H}^0_H(\partial M, \mathcal{H}^0 i'^* j'_* W(1)).$$

Proof. It follows from Lemma 5.7 that

$$i^* j_* \mu(W) \cong \bigoplus_{n=-1}^{0} \mathcal{H}^n i^* j_* \mu(W)[-n].$$

Hence, we have

$$\mathrm{H}^0_H(\partial M, i'^* j'_* W(1)) \cong \mathrm{H}^0_H(\partial M, \mathcal{H}^0 i'^* j'_* W(1)) \oplus \mathrm{H}^1_H(\partial M, \mathcal{H}^{-1} i'^* j'_* W(1)).$$

We can see

$$\mathrm{H}^1_H(\partial M, \mathcal{H}^{-1} i'^* j'_* W(1)) = 0$$

from the fact that $\mathcal{H}^{-1} i'^* j'_* W(1)$ is regular and $\dim \partial M = 0$. □

Remark 5.22. From Lemma , we can see that it suffices to prove the map

$$\theta_H \colon \mathrm{H}^0_H(\partial M, \mathcal{H}^0 i'^* j'_* W(1)) \to \mathrm{H}^1_H(\partial S, i^* j_* V(2))$$

is zero.

Lemma 5.23. *We have the following pullback formulas in* $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))$*:*

$$\begin{aligned}
&q^*\mathcal{H}^{-2} i^* j_* V(2) = \mu(\lambda'(a+b+r+2, a+b+s+2))[1],\\
&q^*\mathcal{H}^{-1} i^* j_* V(2) = \mu(\lambda'(a+b+r+2, b+s+1))[1] \oplus \mu(\lambda'(a+r+1, a+b+s+2))[1],\\
&q^*\mathcal{H}^{0} i^* j_* V(2) = \mu(\lambda'(a+r+1, s))[1] \oplus \mu(\lambda'(r, b+s+1))[1],\\
&q^*\mathcal{H}^{1} i^* j_* V(2) = \mu(\lambda'(r, s))[1].
\end{aligned}$$

Proof. First, it follows from [Blo06, Proposition 2.3.12] that for $-2 \le l \le 1$, we have

$$q^*\mathcal{H}^l i^* j_* V(2) = (q^s)^* \mathcal{H}^l i^* j_* V(2)[1],$$

where $(q^s)^*$ is the pullback functor in the category of variation of Hodge structures. The desired formulas are now direct consequences of Lemma 5.10 and equation (5.10). □

Lemma 5.24. *For* $-2 \le l \le 0$*, we have the following vanishing:*

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))}(\mathcal{H}^0 i'^* j'_* W(1), q^*\mathcal{H}^l i^* j_* V(2)[1-l]) = 0.$$

Proof. From Lemma 5.7 and Lemma 5.23, there exist $X, Y \in \mathrm{Var}_{\mathbb{R}}(M)$ such that

$$\begin{aligned}
&\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))}(\mathcal{H}^0 i'^* j'_* W(1), q^*\mathcal{H}^l i^* j_* V(2)[1-l])\\
=&\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))}(X, Y[2-l]).
\end{aligned}$$

From the geometry of ∂M and the definition of the derived category,

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))}(X, Y[2-l]) \cong \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHS}^+_{\mathbb{R}})}(X, Y[2-l]) \cong \mathrm{Ext}^{2-l}_{\mathrm{MHS}^+_{\mathbb{R}}}(X, Y).$$

Then the lemma follows from the fact that the cohomological dimension of the category $\mathrm{MHS}^+_{\mathbb{R}}$ is 1. □

Lemma 5.25. *We have the following vanishing result:*

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))}(\mathcal{H}^0 i'^* j'_* W(1), q^*\mathcal{H}^{-1} i^* j_* V(2)) = 0.$$

Proof. It follows from Lemma 5.7 and Lemma 5.23 that

$$\begin{aligned}
&\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))}(\mathcal{H}^0 i'^* j'_* W(1), q^* \mathcal{H}^1 i^* j_* V(2)) \\
=&\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))}(\mu(\lambda'(0,0)), \mu(\lambda'(r,s))[1]) \\
=&\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHS}^+_{\mathbb{R}})}(\mathbb{R}(0), \mu(\lambda'(r,s))[1]) \\
=&\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mu(\lambda'(r,s))).
\end{aligned}$$

The weight of $\mu(\lambda'(r,s))$ is greater than zero, so by [HW98, Theorem A.2.10],

$$\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mu(\lambda'(r,s))) = 0.$$

□

Proposition 5.26. *The map θ^H in Proposition 5.18 is zero.*

Proof. From Proposition 5.18 and Lemma 5.22, in order to prove $\theta^H_S = 0$ it suffices to prove

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial S/\mathbb{R}))}(q_* \mathcal{H}^0 i'^* j'_* W(1), i^* j_* V(2)[1]) = 0.$$

Since $q\colon \partial M \to \partial S$ is a morphism from a 0-dimensional scheme to a 0-dimensional scheme, it is both an open immersion and a closed immersion. Therefore, we have

$$q_! = q_*, q^! = q^*,$$

from which we get the following isomorphisms:

$$\begin{aligned}
&\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial S/\mathbb{R}))}(q_* \mathcal{H}^0 i'^* j'_* W(1), i^* j_* V(2)[1]) \\
\cong&\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial S/\mathbb{R}))}(q_! \mathcal{H}^0 i'^* j'_* W(1), i^* j_* V(2)[1]) \\
\cong&\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))}(\mathcal{H}^0 i'^* j'_* W(1), q^! i^* j_* V(2)[1]) \\
\cong&\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(\partial M/\mathbb{R}))}(\mathcal{H}^0 i'^* j'_* W(1), q^* i^* j_* V(2)[1]),
\end{aligned}$$

where the second isomorphism comes from the adjunction between $q_!$ and $q^!$. It follows from Lemma 5.10 that

$$i^* j_* V(2) = \bigoplus_{l=-2}^{1} \mathcal{H}^l i^* j_* V(2)[-l].$$

Finally, by Lemma 5.24 and Lemma 5.25, we complete the proof of the proposition. □

The following corollary follows directly from Proposition 5.26 and Lemma 5.22.

Corollary 5.27. *For any*

$$f \in \mathrm{H}^0_H(\partial M, i'^* j'_* W(1)) = \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S^*/\mathbb{R}))}(\mathbb{R}(0)_{S^*}, i_* q_* i'^* j'_* W(1)[2]),$$

the element

$$i_*(\partial \iota^H_*) \circ f \in \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S^*/\mathbb{R}))}(\mathbb{R}(0)_{S^*}, i_* i^* j_* V(2)[3]) = \mathrm{H}^1_H(\partial S, i^* j_* V(2))$$

is 0.

Remark 5.28. Although we have proved that the map

$$\theta^H_S\colon \mathrm{H}^0_H(\partial M, \mathcal{H}^0 i'^* j'_* W(1)) \to \mathrm{H}^1_H(\partial S, i^* j_* V(2))$$

is zero, the source and the target may not be zero for the following reasons.

- It can be seen that

  $$\mathrm{H}^0_H(\partial M, \mathcal{H}^0 i'^* j'_* W(1)) = \mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial M, \mathcal{H}^0 i'^* j'_* W(1))).$$

  Then by Lemma 5.7, we get

  $$\mathrm{H}^0_H(\partial M, \mathcal{H}^0 i'^* j'_* W(1)) = \mathrm{Hom}_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathbb{R}(0)) = \mathbb{R}.$$

  Hence, we have shown that $\mathrm{H}^0_H(\partial M, i'^* j'_* W(1)) \neq 0$.

- It follows from Lemma 5.10 that

$$\mathrm{H}^1_H(\partial S, i^*j_*V(2)) = \bigoplus_{l=-2}^{1} \mathrm{H}_H^{-l+1}(\partial S, \mathcal{H}^i i^*j_*V(2)).$$

We can see that

$$\begin{aligned}\mathrm{H}^1_H(\partial S, \mathcal{H}^0 i^*j_*V(2)) &= \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHS}^+_{\mathbb{R}})}(\mathbb{R}(0), \mathrm{H}^0(\partial S, \mathcal{H}^0 i^*j_*V(2))[1]) \\ &= \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^0(\partial S, \mathcal{H}^0 i^*j_*V(2))) \\ &= \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), H_1) \oplus \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), H_2),\end{aligned}$$

where $H_1 \in \mathrm{MHS}^+_{\mathbb{R}}$ has Hodge type $(-(a+r+1), -s)$ and $(-s, -(a+r+1))$, and $H_2 \in \mathrm{MHS}^+_{\mathbb{R}}$ has Hodge type $(-r, -(b+s+1))$ and $(-(b+s+1), -r)$. It follows from [HW98, Theorem A.2.10.] that if $a+r+s > 0$, then $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), H_1) \neq 0$, and if $b+r+s > 0$, then $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), H_2) \neq 0$. Hence, we conclude that when V is sufficiently regular, $\mathrm{H}^1_H(\partial S, \mathcal{H}^0 i^*j_*V(2)) \neq 0$.

Finally, by combining Proposition 5.17, Proposition 5.18 and Proposition 5.26, we complete the proof of Theorem 5.15.

6. The vanishing on the boundary of motivic cohomology

6.1. Weight structures.

6.1.1. *Abstract weight structure.* First we collect some definitions and facts about weight structrue on an abstract triangulated category. We mainly follow [Wil09, §1].

Definition 6.1 (Weight structure)**.** Let $\mathcal{C}$ be a triangulated category. A *weight structure* on $\mathcal{C}$ is a pair $w = (\mathcal{C}_{w\leq 0}, \mathcal{C}_{w\geq 0})$ of full-subcategories of $\mathcal{C}$ such that if we put

$$\mathcal{C}_{w\leq n} := \mathcal{C}_{w\leq 0}[n], \ \ \mathcal{C}_{w\geq n} := \mathcal{C}_{w\geq 0}[n]$$

for $n \in \mathbb{Z}$, the following conditions are satisfied.

(1) The categories $\mathcal{C}_{w\leq 0}$ and $\mathcal{C}_{w\geq 0}$ are Karoubi-closed: for any object M of $\mathcal{C}_{w\leq 0}$ (resp., $\mathcal{C}_{w\geq 0}$), any direct sum of M formed in $\mathcal{C}$ is an object of $\mathcal{C}_{w\leq 0}$ (resp., $\mathcal{C}_{w\geq 0}$).

(2) (Semi-invariance with respect to shifts) We have the following inclusions:

$$\mathcal{C}_{w\leq 0} \subset \mathcal{C}_{w\leq 1}, \ \ \mathcal{C}_{w\geq 0} \supset \mathcal{C}_{w\geq 1}$$

of full sub-categories of $\mathcal{C}$.

(3) (Orthogonality) For any pair of objects $M \in \mathcal{C}_{w\leq 0}$ and $N \in \mathcal{C}_{w\geq 1}$, we have

$$\mathrm{Hom}_{\mathcal{C}}(M, N) = 0.$$

(4) (Weight filtration) For any object $M \in \mathcal{C}$, there exists an exact triangle

$$A \to M \to B \to A[1]$$

in $\mathcal{C}$ such that $A \in \mathcal{C}_{w\leq 0}$ and $B \in \mathcal{C}_{w\geq 1}$. We shall refer to any such exact triangle as a weight filtration of M.

Remark 6.2.

- By (2), we have

$$\mathcal{C}_{w\leq n} \subset \mathcal{C}_{w\leq 0}$$

for any negative integer n, and

$$\mathcal{C}_{w\geq n} \subset \mathcal{C}_{w\geq 0}$$

for any postive integer n. There are analogues of the other conditins for all the categories $\mathcal{C}_{w\leq n}$ and $\mathcal{C}_{w\geq n}$. In particular, they are all Karoubi-closed, and for any object $M \in \mathcal{C}$, there is an exact triangle

$$A \to M \to B \to A[1]$$

in $\mathcal{C}$ such that $A \in \mathcal{C}_{w\leq n}$ and $B \in \mathcal{C}_{w\geq n+1}$.

- In (4), "the" weight filtration is not assumed to be unique.
- Recall that on a triangulated category $\mathcal{C}$, there is a notion of *t-structure* [BBD82, Déf. 1.3.1]. It consists of a pair $t = (\mathcal{C}^{t\leq 0}, \mathcal{C}^{t\geq 0})$ of full sub-categories satisfying formal analogues of conditions (2)-(4), but putting

$$\mathcal{C}^{t\leq n} := \mathcal{C}^{t\leq 0}[-n], \;\; \mathcal{C}^{t\geq n} := \mathcal{C}^{t\geq 0}[-n],$$

for any $n \in \mathbb{Z}$. Note that in the context of t-structures, the analogues of (4) is unique up to isomorphism, and the analogue of (1) is implied by the other conditions.

Definition 6.3 (Heart)**.** [Bon10, Definition 1.2.1] Let $w = (\mathcal{C}_{w\leq 0}, \mathcal{C}_{w\geq 0})$ be a weight structure on $\mathcal{C}$. The *heart* of w is the full additive subcategory $\mathcal{C}_{w=0}$ of $\mathcal{C}$ whose objects lie both in $\mathcal{C}_{w\leq 0}$ and in $\mathcal{C}_{w\geq 0}$.

Definition 6.4 (Weight exact)**.** [Bon10, Definition 4.4.4] Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor between two triangulated categories with weight structures $(\mathcal{C}_{w\leq 0}, \mathcal{C}_{w\geq 0})$ and $(\mathcal{D}_{w\leq 0}, \mathcal{D}_{w\geq 0})$. The functor F is called weight exact if F maps $\mathcal{C}_{w\leq 0}$ to $\mathcal{D}_{w\leq 0}$ and maps $\mathcal{C}_{w\geq 0}$ to $\mathcal{D}_{w\geq 0}$.

Proposition 6.5. *Let $w = (\mathcal{C}_{w\leq 0}, \mathcal{C}_{w\geq 0})$ be a weight structure on $\mathcal{C}$. There is an exact triangle*

$$L \to M \to N \to L[1]$$

in $\mathcal{C}$.

(1) If both L and N belong to $\mathcal{C}_{w\leq 0}$, then so does M.
(2) If both L and N belong to $\mathcal{C}_{w\geq 0}$, then so does M.

Proof. See [Bon10, Proposition 1.3.3.3]. □

From now on, we consider a fixed weight structure w on a triangulated category $\mathcal{C}$.

Definition 6.6. Let $M \in \mathcal{C}$ and $m \leq n$ be two integers. A *weight filtration of M avoiding weights $m, m+1, \ldots, n-1, n$* is an exact triangle

$$M_{w\leq m-1} \to M \to M_{\geq n+1} \to M_{w\leq m-1}[1]$$

in $\mathcal{C}$, with $M_{w\leq m-1} \in \mathcal{C}_{w\leq m-1}$ and $M_{\geq n+1} \in \mathcal{C}_{\geq n+1}$.

Proposition 6.7. *Assume that $m \leq n$, and that $M, N \in \mathcal{C}$ admit weight filtrations*

$$M_{w\leq m-1} \xrightarrow{x_-} M \xrightarrow{x_+} M_{\geq n+1} \longrightarrow M_{w\leq m-1}[1]$$

and

$$N_{w\leq m-1} \xrightarrow{y_-} N \xrightarrow{y_+} N_{\geq n+1} \longrightarrow N_{w\leq m-1}[1]$$

avoiding weights $m, \ldots, n$. Then any morphism $M \to N$ in $\mathcal{C}$ extends uniquely to a morphism of exact triangles:

$$\begin{array}{ccccccc} M_{w\leq m-1} & \xrightarrow{x_-} & M & \xrightarrow{x_+} & M_{\geq n+1} & \longrightarrow & M_{w\leq m-1}[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ N_{w\leq m-1} & \xrightarrow{y_-} & N & \xrightarrow{y_+} & N_{\geq n+1} & \longrightarrow & N_{w\leq m-1}[1] \end{array}.$$

Corollary 6.8. *Assume that $m \leq n$. If $M \in \mathcal{C}$ admits a weight filtration avoiding weights $m, \ldots, n$, then it is unique up to a unique isomorphism.*

Definition 6.9. Assume that $m \leq n$. We say that $M \in \mathcal{C}$ *does not have weights $m, \cdots n$* or M *is without weights $m \ldots n$*, if it admits a weight filtration avoiding $m, \ldots, n$.

6.1.2. *Motivic weight structure.* We recall Hébert's construction of motivic weight structure by applying the abstract machine to the triangulated category of constructible Beilinson motives.

Theorem 6.10 ([H11, Théorème 3.3])**.** *Let $S \in \mathrm{Sch}(\mathbb{Q})$ be an excellent scheme. Set*

$$\mathrm{DM}_{B,c}(S) \supset \mathcal{H}_S := \{f_* 1_X(n)[2n] | n \in \mathbb{Z}, f\colon X \to S \text{ is proper with regular source } X\}.$$

There exists a unique weight structure $w = (\mathrm{DM}_{B,c}(S)_{w\leq 0}, \mathrm{DM}_{B,c}(S)_{w\geq 0})$ on $\mathrm{DM}_{B,c}(S)$ such that

$$\mathcal{H}_S \subset \mathrm{DM}_{B,c}(S)_{w=0}.$$

Remark 6.11.

- If $S = \mathrm{Spec}(\mathbb{Q})$, the weight structure
$$w = (\mathrm{DM}_{gm}(\mathbb{Q})_{w\leq 0}, \mathrm{DM}_{gm}(\mathbb{Q})_{w\geq 0})$$
is constructed by Bondarko in [Bon10, Proposition 6.5.3].
- The existence of this weight structure was also proved independently by Bondarko in [Bon14].
- It can be seen from the definition that $\mathrm{CHM}(S) \subset \mathcal{H}_S$.

6.2. **Intersection motives, interior motives and boundary motives.** In this subsection, we recall the definition of intersection motives, interior motives and boundary motives of Picard modular surfaces and some of their properties.

Notation 6.12. We use the same notation as in section 5: for each of the two Shimura varieties $X = \{M, S\}$, we denote by X^* its Baily–Borel compactification and let $\partial X = X^* - X$ be the cusps of the compactification. Then we have the following commutative diagram:

$$\begin{array}{ccccc} M & \xrightarrow{j'} & M^* & \xleftarrow{i'} & \partial M \\ \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ S & \xrightarrow{j} & S^* & \xleftarrow{i} & \partial S \end{array} , \tag{6.1}$$

where j and j' are open immersions, i and i' are closed immersions and p and q are closed immersions induced by the closed immersion ι. We also have

$$\dim \partial S = 0, \dim \partial M = 0.$$

6.2.1. *Boundary motives.* Let $g\colon \partial S \to \mathrm{Spec}(\mathbb{Q})$ be the structure morphism of ∂S.

Definition 6.13 (Boundary motives)**.** For $V(2) \in \mathrm{Rep}_{\mathbb{Q}}(G)$, the boundary motive $\partial M_{gm}(V(2))$ is defined as

$$\partial M_{gm}(V(2)) := (g_* i^* j_* V(2))^* \in \mathrm{DM}_{gm}(\mathbb{Q}),$$

where the notation $(\cdot)^*$ denotes taking dual.

Remark 6.14. It can be seen from the proof of [Wil20, Corollary 3.10] that the definition of the boundary motive is the same as the one given in [Wil06, Definition 2.1].

Notation 6.15. Denote by $\mathrm{DM}_{B,c}(S)_{w=0,\partial w\neq 0,1}$ the full sub-category of $\mathrm{DM}_{B,c}(S)_{w=0}$ of objects N such that $i^* j_* N$ is without weights 0 and 1.

Proposition 6.16. *For $V = V^{a,b}\{r, s\}$ and $a > 0$ and $b > 0$, we have*

$$V(2) \in \mathrm{DM}_{B,c}(S)_{w=0,\partial w\neq 0,1}.$$

Proof. First, the $k = \min(k_1 - k_2, k_2)$ in [Wil09, Page 21] is $\min(a, b)$ in our convention, and the condition $k_1 > k_2 > 0$ in [Wil09] is equivalent to our condition $a > 0$ and $b > 0$. Second, it follows from [Wil15, Theorem 3.8] that the boundary motive $\partial M_{gm}(V(2))$ is without weights -1 and 0. Finally, it can be seen from the fact that $g_* i^* j_* V(2) = g_! i^* j_* V(2) = \partial M_{gm}(V(2))^*$ and the fact that $g_!$ is weight exact [H11, Théorème 3.3(i)] that $i^* j_* V(2)$ is without weights 0 and 1. □

6.2.2. *Intersection motives and interior motives.* Let $a\colon S^* \to \mathrm{Spec}(\mathbb{Q})$ be the structure morphism of S^*.

Recall that in [Wil20, Definition 2.4], the author defines the motivic intermediate extension functor

$$j_{!*}\colon \mathrm{DM}_{B,c}(S)_{w=0,\partial w\neq 0,1} \to \mathrm{DM}_{B,c}(S^*)_{w=0}.$$

Definition 6.17. (1) (Intersection motives) For $V = V^{a,b}\{r,s\}$ and $a>0$ and $b>0$, the *intersection motive* of S relative to S^* is defined as

$$a_*j_{!*}V(2) \in \mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))_{w=0}.$$

(2) (Interior motives) For $V = V^{a,b}\{r,s\}$ and $a>0$ and $b>0$, the *interior motive* $\mathrm{Gr}_0M_{gm}(V(2))$ of S is defined as

$$\mathrm{Gr}_0M_{gm}(V(2)) := (a_*j_{!*}V(2))^* \in \mathrm{DM}_{gm}(\mathbb{Q})_{w=0},$$

where the notation $(\cdot)^*$ denotes taking dual.

Remark 6.18.
- It follows from [Wil20, Corollary 3.10(b)] that the definition of interior motives is the same as the one defined in [Wil15, Corollary 3.9]
- By [Wil20, Remark 3.13(a)], the interior cohomology $\mathrm{H}^2_{B,!}(S,V(2))$ is the Hodge realization of the interior motive $\mathrm{Gr}_0M_{gm}(V(2))$.

Now we recall a few properties of intersection motives.

Proposition 6.19. *Let $k=\min(a,b)$, which is greater than 0.*

(1) There are canonical exact triangles in $\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))$

$$a_*j_{!*}V(2) \xrightarrow{\iota_1} a_*j_*V(2) \longrightarrow C_{\geq(k+1)} \longrightarrow a_*j_{!*}V(2)[1] \tag{6.2}$$

and

$$a_*j_!V(2) \xrightarrow{\iota_2} a_*j_{!*}V(2) \longrightarrow C_{\leq -k} \longrightarrow a_*j_!V(2)[1]\ , \tag{6.3}$$

where $C_{\geq(k+1)} \in \mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))_{w\geq k+1}$ and $C_{\leq -k} \in \mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))_{w\leq -k}$.

(2) We have the following commutative diagram in $\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))$*:*

$$\begin{array}{ccc} a_*j_!V(2) & \xrightarrow{\quad m \quad} & a_*j_*V(2) \\ & {\scriptstyle \iota_2}\searrow \quad \nearrow{\scriptstyle \iota_1} & \\ & a_*j_{!*}V(2) & \end{array}\ , \tag{6.4}$$

where m is the map induced by the first arrow in the exact triangle of Proposition 4.8, and ι_1 and ι_2 are the corresponding maps in (1).

Proof. The two exact triangles follow from [Wil15, Corollary 3.9] and Definition 6.17, and the commutative diagram of (2) follows from [Wil09, Theorem 2.4(c)] and Definition 6.17. □

6.3. **Vanishing on the boundary for motivic cohomology.** In this subsection, we state the theorem of vanishing on the boundary for motivic cohomology.

Theorem 6.20. *Under the conditions*

(1) $0 \leq -r \leq a$ and $0 \leq -s \leq b$,
(2) $a>0$ and $b>0$,
(3) $r\neq 0$ or $s\neq 0$,

the map $\mathcal{E}is^n_M\colon \mathcal{B}_n \to \mathrm{H}^3_M(S,V(2))$ factors through the inclusion

$$\mathrm{H}^3_M(\mathrm{Gr}_0M_{gm}(V(2)),\mathbb{Q}(0)) \hookrightarrow \mathrm{H}^3_M(S,V(2)).$$

Remark 6.21. Using the Beilinson regulator r_H, $\mathrm{H}^3_M(\mathrm{Gr}_0M_{gm}(V(2)),\mathbb{Q}(0))$ has image in

$$\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbf{1},\mathrm{H}^2_{B,!}(S,V(2))).$$

Hence, Theorem 6.20 can be viewed as a "motivic lifting" of Theorem 5.15.

6.4. **The proof of motivic vanishing.** In this subsection, we prove the theorem of vanishing on the boundary for motivic cohomology.

6.4.1. *Conversativity and weight convervativity.* Before going into the steps of the proof, we first recall the conservativity and weight conservativity theorem of the Hodge realization functor over motives of Abelian type.

Theorem 6.22 (Conservativity, [Wil15, Theorem 1.12.]). *For $M \in \mathrm{DM}_{B,c}^{Ab}(\mathrm{Spec}(\mathbb{Q}))$, if*

$$H^n R_H(M) = 0$$

for all $n \in \mathbb{Z}$, then $M = 0$.

A direct corollary of the conservativity theorem is the following.

Corollary 6.23. *For all $M, N \in \mathrm{DM}_{B,c}^{Ab}(\mathrm{Spec}(\mathbb{Q}))$, if*

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHS}_{\mathbb{R}}^+)}(R_H(M), R_H(N)) = 0,$$

then

$$\mathrm{Hom}_{\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))}(M, N) = 0.$$

Theorem 6.24 (Weight conservativity, [Wil15, Theorem 1.13]). *Let $M \in \mathrm{DM}_{B,c}^{Ab}(\mathrm{Spec}(\mathbb{Q}))$ and $\alpha \leq \beta$ be two integers. We have the following statements:*

1. *M lies in $\mathrm{DM}_{B,c}^{Ab}(\mathrm{Spec}(\mathbb{Q}))_{w=0}$ if and only if $H^n R_H(M)$ is pure of weight n, for all $n \in \mathbb{Z}$,*
2. *M lies in $\mathrm{DM}_{B,c}^{Ab}(\mathrm{Spec}(\mathbb{Q}))_{w\leq\alpha}$ if and only if $H^n R_H(M)$ is of weights $\leq n+\alpha$, for all $n \in \mathbb{Z}$,*
3. *M lies in $\mathrm{DM}_{B,c}^{Ab}(\mathrm{Spec}(\mathbb{Q}))_{w\geq\beta}$ if and only if $H^n R_H(M)$ is of weights $\geq n+\beta$, for all $n \in \mathbb{Z}$,*
4. *M is without weights $\alpha, \alpha+1, \ldots, \beta$ if and only if $H^n R_H(M)$ is without weights $n+\alpha, n+(\alpha+1), \ldots, n+\beta$, for all $n \in \mathbb{Z}$.*

6.4.2. *The proof.* The proof is parallel to the Hodge case, so it is also divided into three steps. Recall that we let $a\colon S^* \to \mathrm{Spec}(\mathbb{Q})$ be the structure morphism of S^*.

Step I. In the first step, the goal is to prove the exactness of the sequence in Proposition 6.28.

Lemma 6.25. *We have the following vanishing of motivic cohomology:*

$$\mathrm{Hom}_{\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))}(1_{\mathrm{Spec}(\mathbb{Q})}, C_{\geq(k+1)}[2]) = 0,$$

where $C_{\geq(k+1)} \in \mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))_{w\geq k+1}$ is the one in the exact triangle (6.2).

Proof. We first apply the Hodge realization functor R_H to the above Hom space and get

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHS}_{\mathbb{R}}^+)}(\mathbb{R}(0), N[2]),$$

where $N = R_H(C_{\geq(k+1)})$. By the fact that $\mathrm{MHS}_{\mathbb{R}}^+$ has cohomological dimension 1, we have the short exact sequence

$$0 \to \mathrm{Ext}^1_{\mathrm{MHS}_{\mathbb{R}}^+}(\mathbb{R}(0), \mathcal{H}^1(N)) \to \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHS}_{\mathbb{R}}^+)}(\mathbb{R}(0), N[2]) \to \mathrm{Hom}_{\mathrm{MHS}_{\mathbb{R}}^+}(\mathbb{R}(0), \mathcal{H}^2(N)) \to 0.$$

It follows from Theorem 6.24 that $\mathcal{H}^1(N)$ is of weights $\geq k+2$ and $\mathcal{H}^2(N)$ is of weights $\geq k+3$. Hence, we have $\mathrm{Hom}_{\mathrm{MHS}_{\mathbb{R}}^+}(\mathbb{R}(0), \mathcal{H}^2(N)) = 0$ for weight reasons and $\mathrm{Ext}^1_{\mathrm{MHS}_{\mathbb{R}}^+}(\mathbb{R}(0), \mathcal{H}^1(N)) = 0$ by [HW98, Theorem A.2.10]. Therefore, the following identity holds:

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHS}_{\mathbb{R}}^+)}(\mathbb{R}(0), N[2]) = 0.$$

Finally, the vanishing

$$\mathrm{Hom}_{\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))}(1_{\mathrm{Spec}(\mathbb{Q})}, C_{\geq(k+1)}[2]) = 0$$

follows from Corollary 6.23. □

Proposition 6.26. *We have the following exact sequence*

$$0 \longrightarrow \mathrm{H}^3_M(S^*, j_{!*}V(2)) \longrightarrow \mathrm{H}^3_M(S, V(2)) \xrightarrow{\partial_S^M} \mathrm{H}^3_M(\partial S, i^*j_*V(2)).$$

Proof. First, if we let $N = j_*V(2)$ in the exact triangle of Proposition 4.8 and apply the functor a_* to it, then we get the following exact triangle in $\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))$:

$$a_*j_!V(2) \to a_*j_*V(2) \to a_*i_*i^*j_*V(2) \to a_*j_!V(2)[1].$$

We then apply the functor $\mathrm{Hom}_{\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))}(1_{\mathrm{Spec}(\mathbb{Q})}, -[3])$ to it and get the long exact sequence

$$\cdots \to \mathrm{H}^3_M(S^*, j_!V(2)) \to \mathrm{H}^3_M(S, V(2)) \to \mathrm{H}^3_M(\partial S, i^*j_*V(2)) \to \cdots.$$

Second, if we apply the functor $\mathrm{Hom}_{\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))}(1_{\mathrm{Spec}(\mathbb{Q})}, -[3])$ to the commutative diagram (6.4), then we have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{H}^3_M(S^*, j_!V(2)) & \xrightarrow{\quad m_* \quad} & \mathrm{H}^3_M(S, V(2)) \\ & {\scriptstyle \iota_{2,*}} \searrow \quad \nearrow {\scriptstyle \iota_{1,*}} & \\ & \mathrm{H}^3_M(S^*, j_{!*}V(2)) & \end{array} , \tag{6.5}$$

where the map $\iota_{1,*}$ is induced from the map ι_1 in the commutative diagram (6.4).

Finally, after applying the functor

$$\mathrm{Hom}_{\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))}(1_{\mathrm{Spec}(\mathbb{Q})}, -[3])$$

to the exact triangle (6.3) we get the exact sequence

$$\mathrm{Hom}_{\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))}(1_{\mathrm{Spec}(\mathbb{Q})}, C_{\geq(k+1)}[2]) \longrightarrow \mathrm{H}^3_M(S^*, j_{!*}V(2)) \xrightarrow{\iota_{1,*}} \mathrm{H}^3_M(S, V(2)).$$

It follows from Lemma 6.25 that $\iota_{1,*}$ is injective. Therefore, we get the following exact sequence:

$$0 \to \mathrm{H}^3_M(S^*, j_{!*}V(2)) \to \mathrm{H}^3_M(S, V(2)) \to \mathrm{H}^3_M(\partial S, i^*j_*V(2)).$$

□

Definition 6.27. For the interior motive $\mathrm{Gr}_0M_{gm}(V(2)) \in \mathrm{DM}_{gm}(\mathbb{Q})$, we define its motivic cohomology $\mathrm{H}^3_M(\mathrm{Gr}_0M_{gm}(V(2)), \mathbb{Q}(0))$ as

$$\mathrm{H}^3_M(\mathrm{Gr}_0M_{gm}(V(2)), \mathbb{Q}(0)) := \mathrm{Hom}_{\mathrm{DM}_{gm}(\mathbb{Q})}(\mathrm{Gr}_0M_{gm}(V(2)), \mathbb{Q}(0)),$$

which is isomorphic to

$$\mathrm{H}^3_M(S^*, j_{!*}V(2)) = \mathrm{Hom}_{\mathrm{DM}_{B,c}(\mathrm{Spec}(\mathbb{Q}))}(1_{\mathrm{Spec}(\mathbb{Q})}, a_*j_{!*}V(2)[3]).$$

Hence, we get the following proposition.

Proposition 6.28. *We have the exact sequence*

$$0 \longrightarrow \mathrm{H}^3_M(\mathrm{Gr}_0M_{gm}(V(2)), \mathbb{Q}(0)) \longrightarrow \mathrm{H}^3_M(S, V(2)) \xrightarrow{\partial_S^M} \mathrm{H}^3_M(\partial S, i^*j_*V(2)).$$

Remark 6.29. The exact sequence in Proposition 6.28 is the "motivic lifting" of the exact sequence in Proposition 5.17.

Step II. In the second step, the goal is to prove the diagram in Proposition 6.30 commutes.

Proposition 6.30. *We have the following commutative diagram:*

$$\begin{array}{ccc} \mathcal{B}_n & & \\ \downarrow {\scriptstyle p^*\circ Eis^n_M} & & \\ \mathrm{H}^1_M(M, W(1)) & \xrightarrow{\partial_M^M} & \mathrm{H}^1_M(\partial M, i'^*j'_*W(1))) \\ \downarrow {\scriptstyle \iota_*^M} & & \downarrow {\scriptstyle \theta^M} \\ \mathrm{H}^3_M(S, V(2)) & \xrightarrow{\partial_S^M} & \mathrm{H}^3_M(\partial S, i^*j_*V(2)) \end{array},$$

where ι_*^M is the Gysin morphism defined in Proposition 4.16, ∂_M^M and ∂_S^M are the corresponding boundary maps and the map θ^M is induced from the map,

$$(\partial\iota_*^M\colon q_*i'^*j_*'W(1)[1] \to i^*j_*V(2)[3]) \in \mathrm{DM}_{B,c}(\partial S), \tag{6.6}$$

which is degenerated from ι_*^M.

Proof. The proof is parallel to the proof of Proposition 5.18. First, recall that we have the following diagram:

$$\begin{array}{ccccc} M & \xrightarrow{j'} & M^* & \xleftarrow{i'} & \partial M \\ \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ S & \xrightarrow{j} & S^* & \xleftarrow{i} & \partial S \end{array}.$$

It follows from Proposition 4.16 that the Gysin morphism $\iota_*: \mathrm{H}^1_M(M, W(1)) \to \mathrm{H}^3_M(S, V(2))$ is induced by

$$(\iota_*W(1)[1] \to V(2)[3]) \in \mathrm{DM}_{B,c}(S).$$

Second, after applying the functor j_* to the morphism and by the fact that $j_*\iota_* = p_*j_*'$, we have the morphism

$$(p_*j_*'W(1)[1] \to j_*V(2)[3]) \in \mathrm{DM}_{B,c}(S^*). \tag{6.7}$$

The morphism

$$(i_*i^*p_*j_*'W(1)[1] \to i_*i^*j_*V(2)[3]) \in \mathrm{DM}_{B,c}(S^*)$$

follows by applying the functor i_*i^* to the morphism in (6.7). It follows from the proper base change theorem for Beilinson motives [CD19, Theorem 2.2.14.(4)] and the fact that p is proper that $i^*p_* = q_*i'^*$. From it, we get

$$(i_*q_*i'^*j_*'W(1)[1] \to i_*i^*j_*V(2)[3]) \in \mathrm{DM}_{B,c}(S^*).$$

The natural transformation $\mathrm{id} \to i_*i^*$ from adjunction gives us the following commutative diagram in $\mathrm{DM}_{B,c}(S^*)$:

$$\begin{array}{ccc} p_*j_*'W(1)[1] & \longrightarrow & i_*q_*i'^*j_*'W(1)[1] \\ \downarrow & & \downarrow \\ j_*V(2)[3] & \longrightarrow & i_*i^*j_*V(2)[3]. \end{array}$$

Finally, we apply the functor $\mathrm{Hom}_{\mathrm{DM}_{B,c}(S^*)}(1_{S^*}, -)$ to the above commutative diagram and get the commutative diagram:

$$\begin{array}{ccc} \mathrm{H}^1_M(M, W(1)) & \longrightarrow & \mathrm{H}^1_M(\partial M, i'^*j_*'W(1)) \\ \downarrow & & \downarrow \\ \mathrm{H}^3_M(S, V(2)) & \longrightarrow & \mathrm{H}^3_M(\partial S, i^*j_*V(2)). \end{array}$$

□

Remark 6.31.
- From the construction, we can see the boundary map ∂_S^M in Proposition 6.30 is the same as the ∂_S^H in Proposition 6.28. Hence, we use the same notation for them.
- The commutative diagram in Proposition 6.30 is the "motivic lifting" of the commutative diagram in Proposition 5.18.

Step III. In the third step, the goal is to prove the map θ^M in Proposition 6.30 is zero, which is stated in Proposition 6.32.

Proposition 6.32. *The map θ^M in Proposition 6.30 is zero.*

Proof. First, it follows from Proposition 6.30 that in order to prove $\theta^M = 0$, it suffices to prove the following statement: for any

$$f \in \mathrm{H}^1_M(\partial M, i'^* j'_* W(1)) = \mathrm{Hom}_{\mathrm{DM}_{B,c}(S^*)}(1_{S^*}, i_* q_* i'^* j'_* W(1)[1]),$$

the element

$$i_*(\partial \iota_*^M) \circ f \in \mathrm{Hom}_{\mathrm{DM}_{B,c}(S^*)}(1_{S^*}, i_* i^* j_* V(2)[3]) = \mathrm{H}^3_M(\partial S, i^* j_* V(2))$$

is 0.

Second, it follows from the identity $R_H(i_*(\partial \iota_*^M) \circ f) = i_*(\partial \iota_*^H) \circ R_H(f)$ in

$$\mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S^*/\mathbb{R}))}(\mathbb{R}(0)_{S^*}, i_* i^* j_* V(2)[3]) = \mathrm{H}^1_H(\partial S, i^* j_* V(2))$$

where

$$R_H(f) \in \mathrm{H}^0_H(\partial M, i'^* j'_* W(1)) = \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S^*/\mathbb{R}))}(\mathbb{R}(0)_{S^*}, i_* q_* i'^* j'_* W(1)[2])$$

and Corollary 5.27 that $R_H(i_*(\partial \iota_*^M) \circ f) = 0$.

Finally, by Theorem 6.22, we have $i_*(\partial \iota_*^M) \circ f = 0$. □

Finally, by combining Proposition 6.28, Proposition 6.30 and Proposition 6.32, we finish the proof of Theorem 6.20.

Part 2. Connection to L-values

7. Constructing the differential forms and pairing with the motivic classes

Convention 7.1. In this section, we let $V = V^{a,b}\{r,s\} \in \mathrm{Rep}_{\mathbb{Q}}(V(2))$ and $n = a+b+r+s$, and Betti cohomology, compactly supported cohomology and interior cohomology all have $\mathbb{Q}$-coefficients, which is different from subsection 6.3.

7.1. **The use of Poincaré duality.** In this subsection, we will explain how the Poincaré duality pairing can be used to compute the Beilinson regulator. The idea is due to Beilinson [Bei86a] (see also [Kin98]).

Let us first recall a general result.

Lemma 7.2. [Lem17, Lemma 4.11] *Let E be a number field and M be an object of $\mathrm{MHS}^+_{\mathbb{R},E}$ with pure weight $w < 0$. Let M_{dR} be the $E\otimes_{\mathbb{Q}}\mathbb{R}$-submodule of $M_{\mathbb{C}}$ where the de Rham involution acts trivially and let M^- be the submodule of M where the infinitesimal Frobenius F_∞ acts by multiplication by -1. We write $M^-(-1) = \frac{1}{2\pi i}M^-$. Then there is an short exact sequence of $E\otimes_{\mathbb{Q}}\mathbb{R}$-modules*

$$0 \to F^0M_{dR} \to M^-(-1) \to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M) \to 0,$$

where the first map is the composite of the natural inclusions

$$F^0M_{dR} \to M_{dR} \to M_{\mathbb{C}}$$

and of the projections

$$M_{\mathbb{C}} \to M(-1) \to M^-(-1)$$

defined by $v \mapsto \frac{1}{2}(v-\bar{v})$ and $\frac{1}{2}(v - F_\infty(v))$, respectively.

We apply Lemma 7.2 to the situation that we are interested in to get the following lemma.

Lemma 7.3. *Let π_f be the non-archimedean part of an irreducible cuspidal automorphic representation π of G (see Definition 3.2) whose archimedean component π_∞ belongs to the discrete series L-packet $P(V_{\mathbb{C}}(2))$ and denote by π_f its rational model over $E(\pi_f)$ (see Theorem 3.37). Then there is a short exact sequence:*

$$0 \to F^0M_{dR}(\pi_f, V(2))_{\mathbb{R}} \to M_B(\pi_f, V(2))^-_{\mathbb{R}}(-1) \to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}}) \to 0, \tag{7.1}$$

where the second map is defined in Lemma 7.2.

Let $F^0M_{dR}(\pi_f, V(2))^*$ be the dual of $F^0M_{dR}(\pi_f, V(2))$. It follows from Lemma 7.3 that the one dimensional $E(\pi_f)$-vector space

$$\mathcal{B}(\pi_f, V(2)) = \det_{E(\pi_f)}F^0M_{dR}(\pi_f, V(2))^* \otimes_{E(\pi_f)} \det_{E(\pi_f)}M_B(\pi_f, V(2))^-(-1)$$

is an $E(\pi_f)$-structure of the $E(\pi_f)\otimes\mathbb{R}$-module

$$\det_{E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}} \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}}).$$

Here $\mathcal{B}(\pi_f, V(2))$ is the Beilinson $E(\pi_f)$-structure defined in [Nek94, §6.1]. We now define the Deligne $E(\pi_f)$-structure using the Beilinson $E(\pi_f)$-structure.

Definition 7.4 (Deligne $E(\pi_f)$-structure)**.** Let $\delta(\pi_f, V(2)) \in (E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C})^\times$ be the determinant of the isomorphism $I_\infty\colon M_B(\pi_f, V(2))_{\mathbb{C}} \to M_{dR}(\pi_f, V(2))_{\mathbb{C}}$ computed using the basis defined over $E(\pi_f)$ on both sides. Then the Deligne $E(\pi_f)$-structure of

$$\det_{E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}} \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}})$$

is

$$\mathcal{D}(\pi_f, V(2)) = (2\pi i)^{\dim_{E(\pi_f)} M_B(\pi_f, W)^-}\delta(\pi_f, W)^{-1}\mathcal{B}(\pi_f, V(2)).$$

Remark 7.5. This definition does not depend on the choice of the bases.

Lemma 7.6. *It can be seen that $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}})$ is a rank one $E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}$-module.*

Proof. It can be seen that

$$\mathrm{rank}_{E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}} F^0 M_{dR}(\pi_f, V(2))_{\mathbb{R}} = 2$$

from the Hodge decomposition of $M(\pi_f, V(2))$ in Corollary 3.42 and the condition of coefficients in equation (7.3). By the exact sequence (7.1) and

$$\mathrm{rank}_{E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}} M_B(\pi_f, V(2))^-_{\mathbb{R}}(-1) = 3,$$

we conclude that $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}})$ is a rank one $E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}$-module. □

Recall that by Theorem 5.15, the map $\mathcal{E}is^n_H \colon \mathcal{B}_{n,\mathbb{R}} \to \mathrm{H}^3_H(S, V(2))$ factors through the inclusion

$$\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbf{1}, \mathrm{H}^2_{B,!}(S, V(2))_{\mathbb{R}}) \hookrightarrow \mathrm{H}^3_H(S, V(2)).$$

Let $\mathcal{K}(V(2))$ be the sub $\mathbb{Q}[G(\mathbb{A}_f)]$-module of

$$\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^2_{B,!}(S, V(2))_{\mathbb{R}}) = \varinjlim_{L} \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^2_{B,!}(S(L), V(2))_{\mathbb{R}})$$

generated by the image of $\mathcal{E}is^n_H$ with domain $\mathcal{B}_n$ and let $\mathcal{K}(\pi_f, V(2))$ be defined by

$$\mathcal{K}(\pi_f, V(2)) := \mathrm{Hom}_{\mathbb{Q}[G(\mathbb{A}_f)]}(\mathrm{Res}_{E(\pi_f)/\mathbb{Q}}\, \pi_f, \mathcal{K}(V(2)). \tag{7.2}$$

This is an $E(\pi_f)$-submodule of $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}})$. Therefore, now we have two $E(\pi_f)$-submodules of $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}})$. The first one, which is called $\mathcal{D}(\pi_f, V(2))$, is defined using the comparison between Betti and De Rham cohomology, which is more elementary. The second one, called $\mathcal{K}(\pi_f, V(2))$, is defined using the Beilinson regulator r_H, which is more sophisticated. By definition, we know that $\mathcal{D}(\pi_f, V(2))$ is non-zero, so we hope to know whether $\mathcal{K}(\pi_f, V(2))$ is zero or not by a comparision between $\mathcal{K}(\pi_f, V(2))$ and $\mathcal{D}(\pi_f, V(2))$.

Definition 7.7. If μ and μ' are two elements of $E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C}$, we write $\mu \sim \mu'$ if there exists $\lambda \in E(\pi_f)^\times$ such that $\mu = \lambda\mu'$.

Lemma 7.8. *Let v_D be a non-zero vector in $\mathcal{D}(\pi_f, V(2))$ and v_K be a non-zero vector in $\mathcal{K}(\pi_f, V(2))$. Let $\tilde{v}_D$ and $\tilde{v}_K$ be any lifting of v_D and v_K in $M_B(\pi_f, V)^-_{\mathbb{R}}(-1)$ using the last arrow in the exact sequence (7.1). Then for any $E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}$-linear map $\psi : M_B(\pi_f, V(2))^-_{\mathbb{R}}(-1) \to E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C}$ which is trivial on $F^0 M_{dR}(\pi_f, V(2))_{\mathbb{R}}$, we have*

$$\mathcal{K}(\pi_f, V(2)) = \frac{\psi(\tilde{v}_K)}{\psi(\tilde{v}_D)}\mathcal{D}(\pi_f, V(2)).$$

Proof. It follows from the exact sequence (7.1). □

Our goal is to compute $\psi(\tilde{v}_K)$ and $\psi(\tilde{v}_D)$ seperately for a well-chosen linear functional ψ. One natural way to choose ψ is to use Poincaré duality. Hence, let us first recall properties of Poincaré duality for Picard modular surfaces. By previous computations, we assume that

$$\begin{cases} 0 \le -r \le a \\ 0 \le -s \le b \\ a > 0 \quad \text{and} \quad b > 0 \\ r \ne 0 \quad \text{or} \quad s \ne 0 \end{cases} \tag{7.3}$$

Recall for a representation $V = V^{a,b}\{r, s\}$, its contragredient representation is $D = V^* = V^{b,a}\{-a-b-r, -a-b-r\}$. In other words, we have a perfect pairing [16]

$$V \otimes D \to \mathbb{Q}(0).$$

The pairing induces a $G(\mathbb{A}_f)$-equivariant pairing

$$\langle \cdot, \cdot \rangle_B \colon \mathrm{H}^2_{B,!}(S, V(2)) \otimes \mathrm{H}^2_{B,!}(S, D) \to \mathrm{H}^4_{B,!}(S, \mathbb{Q}(2)) \to \mathbb{Q}(0),$$

[16] Both V and D are in $\mathrm{Rep}_{\mathbb{Q}}(G)$.

where the last map is defined by the trace. The pairing becomes perfect after restriction to the vectors which are invariant by a compact open subgroup of $G(\mathbb{A}_f)$.

Fact 7.9. *Here, $\mathbb{Q}(0)$ is given by an action of $G(\mathbb{A}_f)$ by $|\mu|^{-2}$.*

Proof. We have

$$\mathrm{H}^4_{B,!}(S,\mathbb{Q}(2)) \cong \mathrm{H}^4_{B,c}(S,\mathbb{Q}(2)) \cong \mathrm{H}^0_B(S,\mathbb{Q}(-2)) \cong \bigoplus(\chi|\mu|^{-2}), \tag{7.4}$$

where the sum is over finite Hecke characters $\chi\colon \mathbb{Q}^\times\backslash\mathbb{A}^\times_{f,\mathbb{Q}} \to \mathbb{C}^\times$. The first isomorphism in (7.4) comes from Corollary 4.27 (1) and the fact $\dim \partial S = 0$; the second isomorphism in (7.4) holds by Poincaré duality, and the last isomorphism in (7.4) can be deduced from [Mil17, Theorem 5.17]. The trace map is a projection onto the factor with χ trivial. □

Remark 7.10. This fact is similar to [Tay93, p295] in the Siegel 3-folds case.

Hence, this induces a $G(\mathbb{A}_f)$-equivariant morphism of Hodge structures [17]

$$\langle\cdot,\cdot\rangle_B\colon M_B(\pi_f,V(2))\otimes M_B(\tilde\pi_f|\mu|^{-2},D)\to E(\pi_f)(0). \tag{7.5}$$

Recall by Proposition 3.41 that we have the Hodge decomposition

$$\begin{aligned} &M_B(\tilde\pi_f|\mu|^{-2},D)_{\mathbb{C}}\\ \cong&M_B^{a+b+r+2,s}\oplus M_B^{a+r+1,b+s+1}\oplus M_B^{r,a+b+s+2}\\ \oplus&M_B^{s,a+b+r+2}\oplus M_B^{b+s+1,a+r+1}\oplus M_B^{a+b+s+2,r}. \end{aligned}$$

Lemma 7.11. *Let $\Omega\in M(\tilde\pi_f|\mu|^{-2},D)^+_{\mathbb{C}}$. Let*

$$M_B(\pi_f,V(2))^-_{\mathbb{R}}(-1)\xrightarrow{\langle\Omega,\cdot\rangle_B} E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C}$$

be the morphism defined by the composition of the inclusion $M_B(\pi_f,V(2))^-_{\mathbb{R}}(-1)\to M_B(\pi_f,V(2))_{\mathbb{C}}$ and the pairing with Ω. Assume Ω belongs to $M_B^{a+r+1,b+s+1}\oplus M_B^{b+s+1,a+r+1}$. Then

$$\mathcal{K}(\pi_f,V(2)) = \frac{\langle\Omega,\tilde v_K\rangle_B}{\langle\Omega,\tilde v_D\rangle_B}\mathcal{D}(\pi_f,V(2)) \tag{7.6}$$

Proof. This follows from a Hodge-type computation. □

Convention 7.12. From now on, we will use Sh_G (resp. Sh_H) to denote $\mathrm{Sh}^{an}_{G,\mathbb{C}}$ (resp., $\mathrm{Sh}^{an}_{H,\mathbb{C}}$).

Remark 7.13. Let $\mathbf{1}$ be the multiplicative identities of $E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C}$.

(1) By definition, we have the following isomorphisms

$$\begin{aligned} M_B(\pi_f,V(2))_{\mathbb{C}} &\cong ((\mathrm{H}^2_{B,!}(\mathrm{Sh}_G,V(2))_{\mathbb{C}}\oplus \mathrm{H}^2_{B,!}(\overline{\mathrm{Sh}_G},V(2))_{\mathbb{C}})[\pi_f])\otimes_{\mathbb{C}}(E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C}),\\ M_B(\tilde\pi_f|\mu|^{-2},D)_{\mathbb{C}} &\cong ((\mathrm{H}^2_{B,!}(\mathrm{Sh}_G,D)_{\mathbb{C}}\oplus \mathrm{H}^2_{B,!}(\overline{\mathrm{Sh}_G},D))_{\mathbb{C}})[\tilde\pi_f|\mu|^{-2}])\otimes_{\mathbb{C}}(E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C}), \end{aligned}$$

where the notation $[\cdot]$ means the corresponding isotypic subspaces.

(2) If we could construct a differential form

$$\omega_\Psi = \omega\otimes\Psi_f\in \mathrm{H}^2_{B,!}(\mathrm{Sh}_G,D)_{\mathbb{C}}[\pi_f] = \mathrm{H}^2(\mathfrak{g}_{\mathbb{C}},K_G;D_{\mathbb{C}}\otimes_{\mathbb{C}}\tilde\pi_\infty)\otimes\tilde\pi_f|\mu|^{-2}$$

associated to a $\Psi=\Psi_f\otimes\Psi_\infty$ that belongs to a cuspidal automorphic representation $\tilde\pi=\tilde\pi_f|\mu|^{-2}\otimes\tilde\pi_\infty$ of $G(\mathbb{A})$, then it is natural to let $\Omega\in M(\tilde\pi_f|\mu|^{-2},D)^+_{\mathbb{C}}$ be

$$\frac{1}{2}(\omega_\Psi+\overline{\omega_\Psi})\otimes\mathbf{1},$$

where $\overline{\omega_\Psi}\in \mathrm{H}^2_{B,!}(\overline{\mathrm{Sh}_G},D)_{\mathbb{C}}$ is the complex conjugate of ω_Ψ.

[17] It is a $\mathbb{Q}$-morphism.

(3) Since the Eisenstein symbol is defined over $\mathbb{Q}$ (see subsubsection 4.4.1), $\tilde{v}_K$ can be represented by

$$(\tilde{v}_K^1, \tilde{v}_K^1) \otimes \mathbf{1},$$

where the first $\tilde{v}_K^1$ belongs to $\mathrm{H}^2_{B,!}(\mathrm{Sh}_G, V(2))_{\mathbb{C}}[\pi_f]$ and the second $\tilde{v}_K^1$ belongs to

$$\mathrm{H}^2_{B,!}(\overline{\mathrm{Sh}_G}, V(2))_{\mathbb{C}}[\pi_f].$$

(4) The Poincaré duality pairing

$$\langle \cdot, \cdot \rangle_B \colon \mathrm{H}^2_{B,!}(S, V(2))_{\mathbb{C}} \otimes \mathrm{H}^2_{B,!}(S, D)_{\mathbb{C}} \to \mathbb{C}(0),$$

can be decomposed into

$$\langle \cdot, \cdot \rangle_B = \langle \cdot, \cdot \rangle_B^{\mathrm{Sh}_G} + \overline{\langle \cdot, \cdot \rangle_B^{\mathrm{Sh}_G}},$$

where $\langle \cdot, \cdot \rangle_B^{\mathrm{Sh}_G}$ is the Poincaré duality pairing on Sh_G:

$$\langle \cdot, \cdot \rangle_B^{\mathrm{Sh}_G} \colon \mathrm{H}^2_{B,!}(\mathrm{Sh}_G, V(2))_{\mathbb{C}} \otimes \mathrm{H}^2_{B,!}(\mathrm{Sh}_G, D)_{\mathbb{C}} \to \mathbb{C}(0),$$

and its complex conjugate $\overline{\langle \cdot, \cdot \rangle_B^{\mathrm{Sh}_G}}$ is t Poincaré duality pairing on $\overline{\mathrm{Sh}_G}$:

$$\overline{\langle \cdot, \cdot \rangle_B^{\mathrm{Sh}_G}} \colon \mathrm{H}^2_{B,!}(\overline{\mathrm{Sh}_G}, V(2))_{\mathbb{C}} \otimes \mathrm{H}^2_{B,!}(\overline{\mathrm{Sh}_G}, D)_{\mathbb{C}} \to \mathbb{C}(0).$$

(5) We can see that

$$\begin{aligned}
&\langle \Omega, \tilde{v}_K \rangle_B \\
=&\langle \frac{1}{2}(\omega_\Psi + \overline{\omega_\Psi}), (\tilde{v}_K^1, \tilde{v}_K^1) \rangle_B \otimes \mathbf{1} \\
=&\frac{1}{2}\langle \omega_\Psi, \tilde{v}_K^1 \rangle_B^{\mathrm{Sh}_G} \otimes \mathbf{1} + \frac{1}{2}\overline{\langle \overline{\omega_\Psi}, \tilde{v}_K^1 \rangle_B^{\mathrm{Sh}_G}} \otimes \mathbf{1} \\
=&\langle \omega_\Psi, \tilde{v}_K^1 \rangle_B^{\mathrm{Sh}_G} \otimes \mathbf{1}.
\end{aligned}$$

Therefore, in order to compute the pairing $\langle \Omega, \tilde{v}_K \rangle_B$, it suffices to construct a differential form ω_Ψ and compute $\langle \omega_\Psi, \tilde{v}_K^1 \rangle_B^{\mathrm{Sh}_G}$.

Definition 7.14. If μ and μ' are two elements of $E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{C}$ (resp., $\mathbb{C}$), we denote $\mu \sim \mu'$ if there exists $\lambda \in (E(\pi_f) \otimes_{\mathbb{Q}} \overline{\mathbb{Q}})^\times$ (resp., $\overline{\mathbb{Q}}^\times$) such that $\mu = \lambda\mu'$.

Convention 7.15. In this paper, because we can only compute $\langle \omega_\Psi, \tilde{v}_K^1 \rangle_B^{\mathrm{Sh}_G}$ up to $\overline{\mathbb{Q}}^\times$, we can freely choose $\tilde{v}_K$ up to $\overline{\mathbb{Q}}^\times$. In other words, we can use $\mathcal{B}_{n,\overline{\mathbb{Q}}}$ as the domain of $\mathcal{E}is_H$.

7.2. **Differential forms.** The next step is to explain how to construct a differential form ω_Ψ on S. It is based on some elementary representation-theoretic considerations.

By equation (3.11), Theorem 3.30 and Proposition 3.33, we have the decomposition

$$\mathrm{H}^2_{B,!}(S, D)_{\mathbb{C}} \cong \mathrm{H}^2_{dR,!}(S, D)_{\mathbb{C}} = \bigoplus_{\tilde{\pi} = \tilde{\pi}_\infty \otimes \tilde{\pi}_f} \mathrm{Hom}_{K_G}(\wedge^2 \mathfrak{p}_{G,\mathbb{C}}, D_{\mathbb{C}} \otimes_{\mathbb{C}} \tilde{\pi}_\infty) \otimes \tilde{\pi}_f.$$

Hence, in order to associate a differential form $\omega_\Psi = \omega \otimes \Psi_f$ to a cusp form $\Psi = \Psi_f \otimes \Psi_\infty$ in the cuspidal automorphic representation $\tilde{\pi} = \tilde{\pi}_f |\mu|^{-2} \otimes \tilde{\pi}_\infty$, it suffices to associate some

$$\omega \in \mathrm{Hom}_{K_G}(\wedge^2 \mathfrak{p}_{G,\mathbb{C}}, D_{\mathbb{C}} \otimes_{\mathbb{C}} \pi_\infty)$$

to $\Psi_\infty \in \tilde{\pi}_\infty$. Here Ψ_f is any vector in $\tilde{\pi}_f |\mu|^{-2}$, and the choice of $\Psi_\infty \in \tilde{\pi}_\infty$ is explained below.

Now we explain the way to contruct ω. Recall that the complexified Lie algebra of G can be decomposed as $\mathfrak{g}_{\mathbb{C}} = \mathfrak{k}_{G,\mathbb{C}} \oplus \mathfrak{p}^+_{G,\mathbb{C}} \oplus \mathfrak{p}^-_{G,\mathbb{C}}$. Let

$$X_{(1,-1,0)} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \ X_{(-1,1,0)} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
$$X_{(1,0,-1)} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \ X_{(-1,0,1)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$
$$X_{(0,1,-1)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \ X_{(0,-1,1)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

be some fixed root vectors in $\mathfrak{g}_{\mathbb{C}}$.

Let $\tilde{\pi}_\infty$ be the representation of $G(\mathbb{R})^+$ in the discrete series L-packet $P(D_{\mathbb{C}})$ with Blattner parameter $(1+a+r-s, -1-b+r-s, r-s)$ [18].

Proposition 7.16. *Let $v \in D_{\mathbb{C}}$ be a vector of weight $(-a+s-r, b+s-r, s-r; -b-2s+r)$ that is the lowest weight vector in the K_G-type $\tau_{(b+s-r,-a+s-r,s-r)}$ of the algebraic representation $D_{\mathbb{C}}$ and let $\Psi_\infty \in \tilde{\pi}_\infty$ be a vector of weight $(-1-b+r-s, 1+a+r-s, r-s; b+2s-r)$ that is the lowest weight vector of the minimal K_G-type $\tau_{(1+a+r-s,-1-b+r+s,r-s)}$ of $\tilde{\pi}_\infty$. Let*

$$X_{(1,0,-1)} \otimes X_{(0,-1,1)} \in \mathfrak{p}^+_{G,\mathbb{C}} \otimes \mathfrak{p}^-_{G,\mathbb{C}} = \tau_{(1,-1,0)} \oplus \tau_{(0,0,0)}$$

be a highest weight vector of the K_G-type $\tau_{(1,-1,0)}$. Then there exists a unique non-zero map

$$\omega \in \mathrm{Hom}_{K_G}(\mathfrak{p}^+_{G,\mathbb{C}} \otimes \mathfrak{p}^-_{G,\mathbb{C}}, D_{\mathbb{C}} \otimes \tilde{\pi}_\infty)$$

such that

$$\omega(X_{(1,0,-1)} \otimes X_{(0,-1,1)}) = \sum_{i=0}^{a+b} (-1)^i X^i_{(1,-1,0)} v \otimes X^{2+a+b-i}_{(1,-1,0)} \Psi_\infty. \tag{7.7}$$

Proof. First, the K_G-weight of $X^i_{(1,-1,0)} v \otimes X^{2+a+b-i}_{(1,-1,0)} \Psi_\infty$ is $(1,-1,0)$, which is the weight of $X_{(1,0,-1)} \otimes X_{(0,-1,1)}$. Hence, the map ω preserves K_G-weights.

Second, we have

$$\begin{aligned}
&X_{(1,-1,0)} \Big(\sum_{i=0}^{a+b} (-1)^i X^i_{(1,-1,0)} v \otimes X^{2+a+b-i}_{(1,-1,0)} \Psi_\infty \Big) \\
=& \sum_{i=0}^{a+b} (-1)^i X^{i+1}_{(1,-1,0)} v \otimes X^{2+a+b-i}_{(1,-1,0)} \Psi_\infty + \sum_{i=0}^{a+b} (-1)^i X^i_{(1,-1,0)} v \otimes X^{3+a+b-i}_{(1,-1,0)} \Psi_\infty \\
=& (-1) \sum_{i=1}^{a+b} (-1)^i X^i_{(1,-1,0)} v \otimes X^{3+a+b-i}_{(1,-1,0)} \Psi_\infty + \sum_{i=1}^{a+b} (-1)^i X^i_{(1,-1,0)} v \otimes X^{3+a+b-i}_{(1,-1,0)} \Psi_\infty \\
=& 0.
\end{aligned}$$

Therefore, the vector $\sum_{i=0}^{a+b} (-1)^i X^i_{(1,-1,0)} v \otimes X^{2+a+b-i}_{(1,-1,0)} \Psi_\infty$ is a highest weight vector of K_G-type $\tau(1,-1,0)$. So the map ω is non-zero and by

$$\dim \mathrm{Hom}_{K_G}(\mathfrak{p}^+_{G,\mathbb{C}} \otimes \mathfrak{p}^-_{G,\mathbb{C}}, D_{\mathbb{C}} \otimes \tilde{\pi}_\infty) = 1,$$

ω is uniquely determined. □

[18] This is the π_2 in the talble (3.12) in Proposition 3.34.

Remark 7.17.

- Since ω is uniquely determined by the equaiton (7.7), ω must satisfy
$$\omega(X_{(0,0,0)}) = 0,$$
for any $X_{(0,0,0)} \in \tau_{(0,0,0)}$.
- Since $v \in D_{\mathbb{C}}$ is a vector of weight $(-a+s-r, b+s-r, s-r; -b-2s+r)$ and the highest weight of $D_{\mathbb{C}}$ is $(b+s-r, s-r, -a+s-r, -a+s-r; -b-2s+r)$, we can see that v is lowest weight vector in $D_{\mathbb{C}}$ with respect to the positive root system
$$\{(1,-1,0),(0,-1,1),(-1,0,1)\},$$
which guarantees the existence of v. Hence, we have

$$X_{(1,0,-1)}^{a+b+1} v = 0. \tag{7.8}$$

7.3. **Restriction of differential forms.** Recall that we have the following inclusion of algebraic groups
$$\iota\colon H \hookrightarrow G, \quad (\begin{pmatrix} a & b \\ c & d \end{pmatrix}, z) \mapsto (\begin{pmatrix} a & & b \\ & z & \\ c & & d \end{pmatrix}, z\bar{z}),$$
which induces the closed immersion of Shimura varieties
$$\iota\colon \mathrm{Sh}_H \hookrightarrow \mathrm{Sh}_G.$$
Now, we compute the pullback of the differential form
$$\iota^*\omega_\Psi = \iota^*(\omega \otimes \Psi_f) = \iota^*(\omega) \otimes \Psi_f$$
along the map ι. Hence, it suffices to compute $\iota^*(\omega)$.

Recall the complexified Lie algebra of H is $\mathfrak{h}_{\mathbb{C}} = \mathfrak{k}_{H,\mathbb{C}} \oplus \mathfrak{p}_{H,\mathbb{C}}^+ \oplus \mathfrak{p}_{H,\mathbb{C}}^-$. Let

$$v^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{p}_{H,\mathbb{C}}^+, \ v^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \in \mathfrak{p}_{H,\mathbb{C}}^-. \tag{7.9}$$

be two root vectors in $\mathfrak{gl}_{2,\mathbb{C}}$. So under the tangent map ι_* of the map ι of Shimura varieties, we have
$$\iota_*(v^+) = X_{(1,0,-1)}, \ \iota_*(v^-) = X_{(-1,0,1)},$$
so
$$\iota_*(v^+ \otimes v^-) = \iota_* v^+ \otimes \iota_* v^- = X_{(1,0,-1)} \otimes X_{(-1,0,1)}.$$

Lemma 7.18. *If we set*
$$Y_{(1,-1,0)} := \mathrm{ad}_{X_{(-1,1,0)}}(X_{(1,0,-1)} \otimes X_{(0,-1,1)}).$$
and $Y_{(0,0,0)}$ is a vector in $\tau_{(0,0,0)}$, then we have
$$\iota_*(v^+ \otimes v^-) = -\frac{1}{2} Y_{(1,-1,0)} + \beta Y_{(0,0,0)}$$
for some $\beta \in \mathbb{C}$.

Proof. First, it follows from the fact that $X_{(1,0,-1)} \otimes X_{(0,-1,1)}$ is a highest weight vector of $\tau_{(1,-1,0)}$ as a K_G-representation that $Y_{(1,-1,0)}$ is vector in $\tau_{(1,-1,0)}$ with weight $(0,0,0)$. Hence, we could let
$$\iota_*(v^+ \otimes v^-) = \alpha Y_{(1,-1,0)} + \beta Y_{(0,0,0)}$$
for some $\alpha, \beta \in \mathbb{C}$, from which we have

$$\mathrm{ad}_{X_{(1,-1,0)}}(\iota_*(v^+ \otimes v^-)) = \alpha \mathrm{ad}_{X_{(1,-1,0)}}(\mathrm{ad}_{X_{(-1,1,0)}}(X_{(1,0,-1)} \otimes X_{(0,-1,1)})). \tag{7.10}$$

Second, we can compute the left hand side of the formula (7.10) aså

$$\begin{aligned}\mathrm{ad}_{\mathrm{X}_{(1,-1,0)}}(\iota_*(v^+\otimes v^-)) &= \mathrm{ad}_{\mathrm{X}_{(1,-1,0)}}(X_{(1,0,-1)}\otimes X_{(-1,0,1)})\\ &= X_{(1,0,-1)}\otimes \mathrm{ad}_{\mathrm{X}_{(1,-1,0)}}(X_{(-1,0,1)})\\ &= X_{(1,0,-1)}\otimes [X_{(1,-1,0)}, X_{(-1,0,1)}]\\ &= -X_{(1,0,-1)}\otimes X_{(0,-1,1)}.\end{aligned}$$

Third, if we let

$$H := [X_{(1,-1,0)}, X_{(-1,1,0)}] = \begin{pmatrix}1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & 0\end{pmatrix},$$

then we have

$$\begin{aligned}\alpha\mathrm{ad}_{\mathrm{X}_{(1,-1,0)}}(\mathrm{ad}_{X_{(-1,1,0)}}(X_{(1,0,-1)}\otimes X_{(0,-1,1)})) &= \alpha\mathrm{ad}_{\mathrm{X}_{(-1,1,0)}}(\mathrm{ad}_{X_{(1,-1,0)}}(X_{(1,0,-1)}\otimes X_{(0,-1,1)}))\\ &\quad + \alpha\mathrm{ad}_{[\mathrm{X}_{(1,-1,0)},\mathrm{X}_{(-1,1,0)}]}(X_{(1,0,-1)}\otimes X_{(0,-1,1)})\\ &= 0 + \alpha\mathrm{ad}_H(X_{(1,0,-1)}\otimes X_{(0,-1,1)})\\ &= \alpha X_{(1,0,-1)}\otimes X_{(0,-1,1)} + \alpha X_{(1,0,-1)}\otimes X_{(0,-1,1)}\\ &= 2\alpha X_{(1,0,-1)}\otimes X_{(0,-1,1)}.\end{aligned}$$

Finally, by equation (7.10), we have

$$-X_{(1,0,-1)}\otimes X_{(0,-1,1)} = 2\alpha X_{(1,0,-1)}\otimes X_{(0,-1,1)}.$$

Hence, we conclude that $\alpha = -\frac{1}{2}$.

□

Lemma 7.19. *For $0 \le i \le a+b$, we have the following identities:*

(1) $X_{(-1,1,0)}X_{(1,-1,0)}^{2+a+b-i}\Psi_\infty = (1+i)(2+a+b-i)X_{(1,-1,0)}^{1+a+b-i}\Psi_\infty$,

(2) $X_{(-1,1,0)}X_{(1,-1,0)}^{i}v = \begin{cases} i(a+b+1-i)X_{(1,-1,0)}^{i-1}v & i>0\\ 0 & i=0\end{cases}$.

Proof. Since Ψ_∞ is the lowest weight vector with weight $(-1-b+r-s, 1+a+r-s, r-s; b+2s-r)$ in the minimal K_G-type $\tau_{(1+a+r-s,-1-b+r-s,r-s)}$ of $\tilde{\pi}_\infty$, which is the vector ν_0 in the representation $\tau_{(1+a+r-s,-1-b+r-s,r-s)}$ in our setup of representations of K_G (see equation (3.5)), we have

$$\begin{aligned}X_{(-1,1,0)}X_{(1,-1,0)}^{2+a+b-i}\Psi_\infty &= X_{(-1,1,0)}X_{(1,-1,0)}^{2+a+b-i}\nu_0\\ &= (2+a+b-i)!X_{(-1,1,0)}\nu_{2+a+b-i}.\\ &= (1+i)(2+a+b-i)!\nu_{1+a+b-i}.\end{aligned}$$

We also have

$$X_{(1,-1,0)}^{1+a+b-i}\Psi_\infty = (1+a+b-i)!\nu_{1+a+b-i}.$$

Hence, we get

$$X_{(-1,1,0)}X_{(1,-1,0)}^{2+a+b-i}\Psi_\infty = (1+i)(2+a+b-i)X_{(1,-1,0)}^{1+a+b-i}\Psi_\infty.$$

The computation of $X_{(-1,1,0)}X_{(1,-1,0)}^{i}v$ is similar, which we finishes the proof of the lemma.

□

Proposition 7.20. *If we pullback the ω defined in Proposition 7.16 along the map ι, then we have the following formula:*

$$\iota^*\omega(v_+\otimes v_-) = -\sum_{i=0}^{a+b}(-1)^i(i+1)X_{(1,-1,0)}^{i}v\otimes X_{(1,-1,0)}^{1+a+b-i}\Psi_\infty.$$

Proof. The pullback functor ι^* preserves the type of the differential form and the type of a differential form on a Shimura variety is determined by the archimedean part, $\iota^*\omega$ is determined by its value on $v^+ \otimes v^-$. It follows from Lemma 7.18 that

$$\begin{aligned}
\iota^*\omega(v^+ \otimes v^-) &= \omega(\iota_*(v^+ \otimes v^-)) \\
&= \omega(-\frac{1}{2}\mathrm{ad}_{X_{(-1,1,0)}}(X_{(1,0,-1)} \otimes X_{(0,-1,1)})) \\
&= -\frac{1}{2}X_{(-1,1,0)}(\omega(X_{(1,0,-1)} \otimes X_{(0,-1,1)})) \\
&= -\frac{1}{2}X_{(-1,1,0)}(\sum_{i=0}^{a+b}(-1)^i X^i_{(1,-1,0)}v \otimes X^{2+a+b-i}_{(1,-1,0)}\Psi_\infty) \\
&= -\frac{1}{2}\sum_{i=0}^{a+b}(-1)^i X_{(-1,1,0)}X^i_{(1,-1,0)}v \otimes X^{2+a+b-i}_{(1,-1,0)}\Psi_\infty \\
&- \frac{1}{2}\sum_{i=0}^{a+b}(-1)^i X^i_{(1,-1,0)}v \otimes X_{(-1,1,0)}X^{2+a+b-i}_{(1,-1,0)}\Psi_\infty.
\end{aligned}$$

Plugging the formula in Lemma 7.19 into the formula for $\iota^*\omega(v^+ \otimes v^-)$ above, we get

$$\begin{aligned}
\iota^*\omega(v^+ \otimes v^-) &= -\frac{1}{2}\sum_{i=1}^{a+b}(-1)^i i(a+b+1-i)X^{i-1}_{(1,-1,0)}v \otimes X^{2+a+b-i}_{(1,-1,0)}\Psi_\infty \\
&- \frac{1}{2}\sum_{i=0}^{a+b}(-1)^i(i+1)(2+a+b-i)X^i_{(1,-1,0)}v \otimes X^{1+a+b-i}_{(1,-1,0)}\Psi_\infty \otimes \\
&= \frac{1}{2}\sum_{i=0}^{a+b-1}(-1)^i(i+1)(a+b-i)X^i_{(1,-1,0)}v \otimes X^{1+a+b-i}_{(1,-1,0)}\Psi_\infty \\
&- \frac{1}{2}\sum_{i=0}^{a+b}(-1)^i(i+1)(2+a+b-i)X^i_{(1,-1,0)}v \otimes X^{1+a+b-i}_{(1,-1,0)}\Psi_\infty \\
&= -\sum_{i=0}^{a+b}(-1)^i(i+1)X^i_{(1,-1,0)}v \otimes X^{1+a+b-i}_{(1,-1,0)}\Psi_\infty.
\end{aligned}$$

□

7.4. **Deligne–Beilinson cohomology and tempered currents.** In this subsection, we collect some facts about Deligne–Beilinson cohomology and tempered currents, mainly following [BGCLRJ24].

7.4.1. *Classical Definition of Deligne–Beilinson Cohomology.* We first recall the classical definition of Deligne–Beilinson Cohomology (DB-cohomology for short).

Let X be a smooth, quasi-projective complex analytic variety of pure dimension d. Let $\overline{X}$ be a smooth compactification of X such that $D = \overline{X} - X$ is a simple normal crossings divisor. We let $j\colon X \to \overline{X}$ be the open immersion. We will assume that X is defined as the analytification of the base change to $\mathbb{C}$ of a smooth, quasi-projective $\mathbb{R}$-scheme. For $p \in \mathbb{Z}$, let $\mathbb{R}(p)$ denote the subgroup $(2\pi i)^p\mathbb{R}$ of $\mathbb{C}$. We will also use the same notation to denote the constant sheaf with value $\mathbb{R}(p)$ on X.

Let Ω^*_X be the sheaf of holomorphic differential forms on X and let $\Omega^*_{\overline{X}}(\log D)$ be the sheaf of differential forms on X with logarithmic singularities along D. It is defined to be the sheaf of sub-$\mathcal{O}_{\overline{X}}$-module of $j_*\Omega_X$ locally generated by sections ξ_i, $1 \le i \le d$ (see [Del71, (3.1.2)]). Here,

we let

$$\xi_i = \begin{cases} \frac{dzi}{z_i} & 1 \le i \le m \\ dz_i & m < i \le d. \end{cases}$$

The Hodge filtration on $\Omega^*_{\overline{X}}(\log D)$ is defined as

$$F^p\Omega^*_{\overline{X}}(\log D) = \bigoplus_{p' \ge p} \Omega^{p'}_{\overline{X}}(\log D).$$

Fact 7.21. *There are natural quasi-isomorphisms:*

(1) $Rj_*\mathbb{C} \to Rj_*\Omega^*_X$,

(2) $\Omega^*_{\overline{X}}(\log D) \to Rj_*\Omega^*_X$ *[Del71, (3.1.8.2)].*

Definition 7.22. (1) For each $p \in \mathbb{Z}$, the *DB-cohomology group* $\mathrm{H}^n_D(X, \mathbb{R}(p))$ of X with coefficients in $\mathbb{R}(p)$ is defined as the n-th hypercohomology of the complex

$$\mathbb{R}(p)_D := \mathrm{Cone}(Rj_*\mathbb{R}(p) \oplus F^p\Omega^*_{\overline{X}}(\log D) \to Rj_*\Omega^*_X)[-1], \tag{7.11}$$

where the arrow is given by the difference of the natural maps.

(2) Let $\overline{F^*_\infty} = F^*_\infty \otimes c$ be the de Rham involution given by the complex conjugation F_∞ on X and c on the coefficients. The *real DB-cohomology groups* are defined as

$$\mathrm{H}^n_D(X/\mathbb{R}, \mathbb{R}(p)) := \mathrm{H}^n_D(X, \mathbb{R}(p))^{\overline{F^*_\infty}=1}. \tag{7.12}$$

7.4.2. *Tempered Currents.* We recall the definition of tempered currents in [BGCLRJ24] here.

Let $\Delta = \{z \in \mathbb{C} || z| < \frac{1}{2}\}$ be the open disc of radius $\frac{1}{2}$ and $\Delta^* = \{z \in \mathbb{C} | 0 < |z| < \frac{1}{2}\}$ be the open punctured disc of radius r.

For each point $x \in \overline{X}$, there is an open neighborhood U of x that is isomorphic to Δ^d with coordinates $(z_1, z_2, \ldots, z_d)$ for which $x = (0, 0, \ldots, 0)$, and such that there exist some integers m, n with $0 \le n, m \le d$ and $m + n = d$ such that

$$X \cap U = (\Delta^*)^m \times (\Delta)^n = \{(z_1, z_2, \cdots, z_d) \in \Delta^d | z_1 z_2 \cdots z_m \ne 0\}.$$

We call such a subset U an open coordinate neighborhood of x.

We denote by $\mathcal{A}^0_{\overline{X}}$ the sheaf of smooth functions on $\overline{X}$ and let $\mathcal{A}^*_{\overline{X}}$ be the complex of sheaves of smooth differential forms. The complex $\mathcal{A}^*_{\overline{X}}(\log D)$ of smooth differential forms on X with logarithmic growth at D is defined to be the $\mathcal{A}_{\overline{X}}$-algebra subsheaf of $j_*\mathcal{A}_X$ locally generated by sections $\log|z_i|\,(1 \le i \le m)$ and $\xi_i, \bar{\xi}_i\,(1 \le i \le d)$. [Bur94, §2]

We first give the definition of sheaves of differential forms with growth conditions.

Definition 7.23. We denote by $\mathcal{A}^0_{si}$ (resp., $\mathcal{A}^0_{rd}$) the sheaf on $\overline{X}$ whose sections on an open $U \subset \overline{X}$ is given by complex-valued functions on $U \cap X$, which are locally slowly increasing (resp. rapidly decreasing) [BGCLRJ24, Definition 2.3] at each point of U. The graded sheaf $\mathcal{A}^*_{si}$ of slowly increasing differential forms is defined to be the $\mathcal{A}^0_{si}$-subalgebra of $j_*\mathcal{A}^*_X$ locally generated by $\xi_i, \bar{\xi}_i$ for $1 \le i \le d$. Similarly, the graded sheaf $\mathcal{A}^*_{rd}$ of rapidly decreasing differential forms is defined to be the $\mathcal{A}^0_{rd}$-subalgebra of $j_*\mathcal{A}^*_X$ locally generated by $\xi_i, \bar{\xi}_i$ for $1 \le i \le d$. We denote by $\mathcal{A}_{si}(\overline{X})$ and $\mathcal{A}_{rd}(\overline{X})$ the corresponding complex of global sections.

Remark 7.24. (1) There are natural inclusions

$$\mathcal{A}^*_{rd} \subseteq \mathcal{A}^*_{\overline{X}}(\log D) \subseteq \mathcal{A}^*_{si} \subseteq j_*\mathcal{A}^*_X.$$

Moreover, all these sheaves are fine sheaves.

(2) The complex structure on $\overline{X}$ induces compatible bigradings

$$\mathcal{A}^n_{rd} = \bigoplus_{p+q=n} \mathcal{A}^{p,q}_{rd},\ \mathcal{A}^n_{si} = \bigoplus_{p+q=n} \mathcal{A}^{p,q}_{si},\ \mathcal{A}^n_{\overline{X}}(\log\ D) = \bigoplus_{p+q=n} \mathcal{A}^{p,q}_{\overline{X}}(\log D),$$

with correspoding Hodge filtrations

$$F^p\mathcal{A}^n_{rd} = \bigoplus_{p' \ge p} \mathcal{A}^{p',q}_{rd},\ F^p\mathcal{A}^n_{si} = \bigoplus_{p' \ge p} \mathcal{A}^{p',q}_{si},\ F^p\mathcal{A}^n_{\overline{X}}(\log\ D) = \bigoplus_{p' \ge p} \mathcal{A}^{p',q}_{\overline{X}}(\log\ D).$$

(3) We denote by

$$\mathcal{A}^*_{si,\mathbb{R}} \subseteq \mathcal{A}^*_{si}, \mathcal{A}^*_{rd,\mathbb{R}} \subseteq \mathcal{A}^*_{rd}, \mathcal{A}^*_{\overline{X},\mathbb{R}}(\log\ D) \subseteq \mathcal{A}^*_{\overline{X}}(\log\ D)$$

the subcomplexes of sheaves of $\mathbb{R}$-valued differential forms.

We now give the definition of sheaves of tempered currents.

Definition 7.25. (1) For any $0 \leq p, q \leq d$, the sheaf $\mathcal{D}^{p,q}_{si}$ of tempered currents is defined to be the sheaf on $\overline{X}$ assigning to any open coordinate neighborhood $U \subseteq \overline{X}$ the complex vector space $\mathcal{D}^{p,q}_{si}(U)$ of continuous linear forms on the compactly supported sections $\Gamma_c(U, \mathcal{A}^{d-p,d-q}_{rd})$. Similarly, We denote by $\mathcal{D}^{p,q}_{si,\mathbb{R}}$ the sheaf on $\overline{X}$ assigning to any open coordinate neighborhood $U \subseteq \overline{X}$ the real vector space $\mathcal{D}^{p,q}_{si,\mathbb{R}}(U)$ of continuous linear forms on the compactly supported sections $\Gamma_c(U, \mathcal{A}^{d-p,d-q}_{rd,\mathbb{R}})$.

(2) For $n \in \mathbb{Z}_{\geq 0}$, let

$$\mathcal{D}^n_{si} = \bigoplus_{p+q=n} \mathcal{D}^{p,q}_{si}, \mathcal{D}^n_{si,\mathbb{R}} = \bigoplus_{p+q=n} \mathcal{D}^{p,q}_{si,\mathbb{R}}.$$

The exterior differentials ∂ and $\overline{\partial}$ on $\mathcal{D}^*_{si}$ are defined as follows: for any open coordinate neighborhood $U \subseteq \overline{X}$, any $T \in \mathcal{D}^{p,q}_{si}(U)$ and any $\omega \in \Gamma_c(U, \mathcal{A}^{d-p,d-q}_{rd})$, we have

$$\partial T(w) := (-1)^{p+q+1} T(\partial w), \overline{\partial} T(w) := (-1)^{p+q+1} T(\overline{\partial} w).$$

The exterior differential $d\colon \mathcal{D}^n_{si} \to \mathcal{D}^{n+1}_{si}$ is defined as $d = \partial + \overline{\partial}$. This defines complexes of sheaves of tempered currents $\mathcal{D}^*_{si}, \mathcal{D}^{*,*}_{si}$ (with differential $d, \partial, \overline{\partial}$). We will denote by $\mathcal{D}^{*,*}_{si}(\overline{X})$ the corresponding complex of global sections.

(3) The complex D^*_{si} is equipped with a Hodge filtration given by

$$F^p D^*_{si} = \bigoplus_{p' \geq p} \mathcal{D}^{p',q}_{si}.$$

(4) For any open $U \subseteq \overline{X}$, there is a natural way to associate any form $\omega \in \mathcal{A}^{p,q}_{si}(U)$ a current $T_\omega \in \mathcal{D}^{p,q}_{si}(U)$ given by

$$T_\omega(\eta) = \frac{1}{(2\pi i)^d} \int_U \omega \wedge \eta,\ (\eta \in \Gamma_c(U, \mathcal{A}^{d-p,d-q}_{rd})).$$

Remark 7.26. Since currents are modules over $\mathcal{A}^0_{si}$, all these complexes are complexes of fine sheaves.

Proposition 7.27. [BGCLRJ24, Theorem 1.1] *The natural inclusions*

$$(\Omega^*_{\overline{X}}(\log\ D), F) \to (\mathcal{A}^*_{\overline{X}}(\log\ D), F) \to (\mathcal{A}^*_{si}, F) \to (\mathcal{D}^*_{si}, F)$$

are filtered quasi-isomorphisms. Futhermore, the last two quasi-isomorphisms are compatible for the corresponding real structures.

Remark 7.28. By Proposition 7.27, Fact 7.21 and, the fact that $\mathcal{D}^*_{si}$ is a fine sheaf, we can see that Betti cohomology classes of X can be represented by closed tempered currents.

7.4.3. *Deligne–Beilinson cohomology in terms of tempered currents.* In [BGCLRJ24], the authors give a new definition of Deligne–Beilinson cohomology in terms of smooth slowly increasing differential forms and tempered currents, which will be recalled here. This definition is convenient for computing Beilinson regulators.

Proposition 7.29. *There is a quasi-isomorphism*

$$\mathbb{R}(p)_D \simeq \mathrm{Cone}(F^p \mathcal{A}^*_{si} \to \mathcal{A}^*_{si,\mathbb{R}(p-1)})[-1],$$

where the arrow is induced by the projection $\pi_{p-1}\colon \mathbb{C} \to \mathbb{R}(p-1)$ *defined by* $\pi_{p-1}(z) = \frac{z+(-1)^{p-1}\overline{z}}{2}$. *In particular, we have the canonical isomorphism*

$$\mathrm{H}^n_D(X, \mathbb{R}(p)) \simeq \frac{\{(\phi, \phi') \in \mathcal{A}^{n-1}_{si,\mathbb{R}(p-1)}(\overline{X}) \oplus F^p\mathcal{A}^n_{si}(\overline{X}) | d\phi' = 0, d\phi = \pi_{p-1}(\phi')\}}{\{d(\tilde{\phi}, \tilde{\phi}')\}}, \tag{7.13}$$

where $d(\tilde{\phi}, \tilde{\phi}') = (d\tilde{\phi} - \pi_{p-1}(\tilde{\phi}'), d\tilde{\phi}')$.

Proof. See [BGCLRJ24, Proposition 2.23]. □

Proposition 7.30. *There is a quasi-isomorphism*

$$\mathbb{R}(p)_D \simeq \mathrm{Cone}(F^p\mathcal{D}^*_{si} \to \mathcal{D}^*_{si,\mathbb{R}(p-1)})[-1].$$

In particular, we have the canonical isomorphism

$$\mathrm{H}^n_D(X, \mathbb{R}(p)) \simeq \frac{\{(S, T) \in \mathcal{D}^{n-1}_{si,\mathbb{R}(p-1)}(\overline{X}) \oplus F^p\mathcal{D}^n_{si}(\overline{X}) | dT = 0, dS = \pi_{p-1}(T)\}}{\{d(\tilde{S}, \tilde{T})\}}, \tag{7.14}$$

where $d(\tilde{S}, \tilde{T}) = (d\tilde{S} - \pi_{p-1}(\tilde{T}), d\tilde{T})$.

Proof. See [BGCLRJ24, Theorem 2.25]. □

In what follows, we will use $[(S, T)] \in \mathrm{H}^n_D(X, \mathbb{R}(p))$ to denote the cohomology class of the pair (S, T).

Proposition 7.31. *Let* $x \in \mathrm{H}^n_D(X, \mathbb{R}(p))$ *be a DB-cohomology class which is represented by a pair* (ϕ, ϕ') *of smooth slowly increasing differential forms via Proposition 7.29. Then via the isomorphism of Proposition 7.30, the class* x *is represented by the pairs of tempered currents* $(T_\phi, T_{\phi'})$.

Proof. See [BGCLRJ24, Proposition 2.26]. □

The reinterpretation of DB-cohomology classes in terms of tempered currents allows us to describe explicitly the Gysin morphism as follows. Let $\iota\colon X' \hookrightarrow X$ be a closed immersion of pure codimension c. Let $\overline{X'}$ be a smooth compactification of X' such that $D' = \overline{X'} - X'$ is a simple normal crossings divisor. We further assume that ι extends to a morphism $\iota\colon \overline{X'} \to \overline{X}$ such that $\iota^{-1}(D) = D'$. Then we define the Gysin morphism

$$\iota_*\colon \mathrm{H}^n_D(X', \mathbb{R}(p)) \to \mathrm{H}^{n+2c}_D(X, \mathbb{R}(p+c)) \tag{7.15}$$

by $\iota_*[(S, T)] = [(\iota_* S, \iota_* T)]$. Here for a tempered current T, we denote by $\iota_* T$ the tempered current defined as: for any smooth rapidly decreasing differential form ω,

$$\iota_* T(\omega) = T(\iota^*\omega), \tag{7.16}$$

which makes sense because $\iota^*\omega$ is also rapidly decreasing.

Proposition 7.32. *Let* $n \in \mathbb{Z}_{\geq 0}$, $p \in \mathbb{Z}$ *and let* $\omega \in \mathcal{A}^{2d-n}_{rd}(\overline{X})$ *be a smooth closed rapidly decreasing differential form with Hodge type components inside* $\{(a, b) : a, b > d - p\}$. *Then the assignment* $(S, T) \mapsto S(\omega)$ *induces a linear map*

$$\langle -, \omega \rangle\colon \mathrm{H}^n_D(X', \mathbb{R}(p)) \to \mathbb{C}.$$

Proof. See [BGCLRJ24, Proposition 2.27] □

7.5. **DB-cohomology with coefficients.** In this subsection, we give the definition of DB-cohomology with coefficients using the "Liebermann's trick":

Let $p\colon A \to S$ be the universal abelian 3-fold over S.

Proposition 7.33. *We have a inclusion.*

$$\mathrm{H}^3_H(S, V(2)) \subset \mathrm{H}^{a+5b-3(r+s)+3}_H(A^{a+5b-3(r+s)}, \mathbb{R}(a+3b-r-s+2)). \tag{7.17}$$

Proof. Since $V = V^{a,b}\{a,b\} = \lambda(a+r-s, r-s, r-s-b; 2s-r+b)$ is in $\mathrm{Rep}(G)$, we have

$$V \subset (\mathrm{std})^{\otimes(a+5b-3(r+s))}(2s+2r-2b),$$

where std is the standard representation of G. We then get the inclusion

$$V \subset \mathcal{H}^{a+5b-3(r+s)} p_*\mathbb{R}(a+3b-r-s)_{A^{a+5b-3(r+s)}} \tag{7.18}$$

in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R}))$ by applying the functor μ_H to the inclusion (7.18). Recall that for the morphism $p\colon A^{a+5b-3(r+s)} \to S$, we have the following isomorphism in $\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R}))$ [Sai90b, (4.5.4)]:

$$p_*\mathbb{R}(a+3b-r-s) \cong \bigoplus_i \mathcal{H}^i p_*\mathbb{R}(a+3b-r-s)_{A^{a+5b-3(r+s)}}[-i].$$

Thus, we can see that

$$\mathcal{H}^{a+5b-3(r+s)} p_*\mathbb{R}(a+3b-r-s)_{A^{a+5b-3(r+s)}} \subset p_*\mathbb{R}(a+3b-r-s)[a+5b-3(r+s)].$$

Finally, if we let $\mathcal{A} = A^{a+5b-3(r+s)}$, then it can be seen from the definition of absolute Hodge cohomology that

$$\begin{aligned}
\mathrm{H}^3_H(S, V(2)) &= \mathrm{Hom}_{\mathrm{D}^b(\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R}))}(\mathbb{R}(0)_S, V(2)[3]) \\
&\subset \mathrm{Hom}_{\mathrm{MHM}_{\mathbb{R}}(S/\mathbb{R}))}(\mathbb{R}(0)_S, p_*\mathbb{R}(a+3b-r-s+2)[a+5b-3(r+s)+3]) \\
&= \mathrm{Hom}_{\mathrm{MHM}_{\mathbb{R}}(\mathcal{A}/\mathbb{R}))}(\mathbb{R}(0)_{\mathcal{A}}, \mathbb{R}(a+3b-r-s+2)[a+5b-3(r+s)+3]) \\
&= \mathrm{H}^{a+5b-3(r+s)+3}_H(\mathcal{A}, \mathbb{R}(a+3b-r-s+2)).
\end{aligned}$$

□

Definition 7.34. Let $i = a+5b-3(r+s)$ and $j = a+3b-r-s+2$.

(1) We define $\mathrm{H}^3_D(S, V(2))$ as

$$\mathrm{H}^3_D(S, V(2)) := r_{H\to D}(\mathrm{H}^3_H(S, V(2))),$$

where $r_{H\to D}$ is the natural map

$$r_{H\to D}\colon \mathrm{H}^{i+3}_H(A^i, \mathbb{R}(j)) \to \mathrm{H}^{i+3}_D(A^i, \mathbb{R}(j))$$

and A is the universal abelian 3-fold over S;

(2) We define $\mathrm{H}^1_D(M, \iota^*V(1))$ as

$$\mathrm{H}^1_D(M, \iota^*V(1)) := r_{H\to D}(\mathrm{H}^1_H(M, \iota^*V(1))),$$

where $r_{H\to D}$ is the natural map

$$r_{H\to D}\colon \mathrm{H}^{i+1}_H(A^i, \mathbb{R}(j-1)) \to \mathrm{H}^{i+1}_D(A^i, \mathbb{R}(j-1))$$

and A is the pullback of the universal abelian 3-fold over S to M through ι;

(3) We define $\mathrm{H}^1_D(M, W(1))$ as

$$\mathrm{H}^1_D(M, W(1)) := r_{H\to D}(\mathrm{H}_H(M, W(1))),$$

where $r_{H\to D}$ is the map

$$r_{H\to D}\colon \mathrm{H}^{n+1}_H(E^n, \mathbb{R}(n+1)) \to \mathrm{H}^{n+1}_D(E^n, \mathbb{R}(n+1))$$

and E is the universal elliptic curve over M;

(4) We define $\mathrm{H}^1_D(\mathrm{Sh}_{\mathrm{GL}_2}, W(1))$ as

$$\mathrm{H}^1_D(\mathrm{Sh}_{\mathrm{GL}_2}, W(1)) := r_{H\to D}(\mathrm{H}_H(\mathrm{Sh}_{\mathrm{GL}_2}, W(1))),$$

where $r_{H\to D}$ is the map

$$r_{H\to D}\colon \mathrm{H}^{n+1}_H(E^n, \mathbb{R}(n+1)) \to \mathrm{H}^{n+1}_D(E^n, \mathbb{R}(n+1))$$

and E is the universal elliptic curve over $\mathrm{Sh}_{\mathrm{GL}_2}$.

Remark 7.35.
- This definition is enough for our application to Beilinson's conjectures and can be easily generalized to all Shimura varieties of PEL type. However, we do not give a definition of "DB-cohomology with coefficients" for general coefficients over general analytic varieties.
- It follows from [Bei86b, Page 60] that the maps $r_{H\to D}$ in (3) and (4) are isomorphic, but the maps $r_{H\to D}$ in (1) and (2) may not be isomorphic.

Convention 7.36. Since we only work with real DB-cohomology in this paper, from now on, we will write $\mathrm{H}^n_D(X,\mathbb{R}(p))$ for $\mathrm{H}^n_D(X/\mathbb{R},\mathbb{R}(p))$.

7.6. **Deligne-Beilinson classes.** In this subsection, we compute the realization of the motivic classes constructed in Subsection 4.4 in Deligne–Beilinson cohomology.

7.6.1. *Eisenstein symbol in DB-cohomology.* We first compute the realization of Eisenstein symbols (see Subsubsection 4.4.1) in Deligne-Beilinson cohomology. We follow [Kin98, 6.3] closely but we change the setting to use the Hermitian form J_2'.

Convention 7.37. Recall that when we write $\mathrm{GL}_2(\mathbb{R})$, we always mean $\mathrm{GU}(J_2')'(\mathbb{R})$.

Notation 7.38.
- We let $\lambda'(n,c)$ be the character of the torus $T_c = Z_2(\mathbb{R})^+\,\mathrm{U}(1)(\mathbb{R})$ of $\mathrm{GL}_2(\mathbb{R})$
$$\begin{pmatrix} t_1 & \\ & \bar{t}_1 \end{pmatrix} \in T_c \longmapsto (x+iy)^n(x^2+y^2)^{\frac{c-n}{2}},$$
where Z_2 is the center of GL_2, $t_1 = x+iy$, n is the weight and c is the central character.
- Let $T_{spl} = Z_2(\mathbb{R})\{\begin{pmatrix}\cosh t & \sinh t\\ \sinh t & \cosh t\end{pmatrix}\}$, where $t\in\mathbb{R}$, $\cosh t = \frac{e^t+e^{-t}}{2}$ is the hyperbolic cosine function and $\sinh t = \frac{e^t-e^{-t}}{2}$ is the hyperbolic sine function. It can viewed as a split torus of the group $\mathrm{GL}_2(\mathbb{R})$ for the following reason. If we let $C = \begin{pmatrix} i & i\\ -1 & 1\end{pmatrix}$, for $\begin{pmatrix}\cosh t & \sinh t\\ \sinh t & \cosh t\end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})$, then we have
$$C\begin{pmatrix}\cosh t & \sinh t\\ \sinh t & \cosh t\end{pmatrix}C^{-1} = \begin{pmatrix} e^t & \\ & e^{-t}\end{pmatrix}.$$
- We denote by $\lambda(n,c)$ the character of T_{spl}
$$\lambda(n,c)\colon z\begin{pmatrix}\cosh t & \sinh t\\ \sinh t & \cosh t\end{pmatrix} \in T_{spl} \longmapsto (e^t)^n z^c,$$
where $z\in Z_2(\mathbb{R})$, n is the weight and c is the central character.

Notation 7.39.
- Let (X,Y) be a basis of the standard representation V_2 of GL_2 such that each $\begin{pmatrix} a & b\\ c & d\end{pmatrix}\in\mathrm{GL}_2(\mathbb{R})$ acts by
$$\begin{pmatrix} a & b\\ c & d\end{pmatrix}X = aX+bY,$$
$$\begin{pmatrix} a & b\\ c & d\end{pmatrix}Y = cX+dY.$$
- Let us view $\mathrm{Sym}^n V_{2,\mathbb{C}}$ as the n-dimensional vector space of homogeneous polynomials of degree n in the variables X and Y with coefficients in $\mathbb{C}$. For any integer $0\le j\le n$, let b_j^n be the vector $b_j^n = (-1)^j X^j Y^{n-j}$, so it has weight $\lambda'(2j-n,n)$. The family $(b_j^n)_{0\le j\le n}$ forms a basis of $\mathrm{Sym}^n V_{2,\mathbb{C}}$. We denote by $(a_j^n)_{0\le j\le n}$ the dual basis of $(b_j^n)_{0\le j\le n}$. Hence, the vector a_j^n has weight $\lambda'(n-2j,-n)$.

Definition 7.40. (1) We define the function z_2, z_2', w_2 and w_2' on $\mathrm{GL}_2(\mathbb{R})$ as:

$$z_2(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) := \frac{b+d}{-b+d}i,$$
$$z_2'(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) := \frac{a+c}{-a+c}i;$$
$$w_2(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) := -b+d,$$
$$w_2'(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) := a-c.$$

(2) For any integer r such that $r \equiv n (\mathrm{mod}\ 2)$, we define the function ϕ_r^n on $\mathrm{GL}_2(\mathbb{R})$ as

$$\phi_r^n = (z_2 - z_2')w_2^{-\frac{n-r}{2}} w_2'^{-\frac{n+r}{2}}.$$

Remark 7.41. Since we identify $\mathrm{GL}_2(\mathbb{R})$ with $\mathrm{GU}(J_2')'(\mathbb{R})$, $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})$ if and only if $a = \bar{d}$ and $c = \bar{b}$. Hence, it is impossible that $a = c$ or $b = d$, and the functions z_2 and z_2' are well-defined on $\mathrm{GL}_2(\mathbb{R})$.

Notation 7.42. Let B_2 be the standard Borel of GL_2. Since we identify $\mathrm{GL}_2(\mathbb{R})$ with $\mathrm{GU}(J_2')'(\mathbb{R})$, we have $B_2(\mathbb{R}) = T_{spl}N$ where $N = \left\{ \begin{pmatrix} 1-\frac{1}{2i}u & \frac{1}{2i}u \\ -\frac{1}{2i}u & 1+\frac{1}{2i}u \end{pmatrix} | u \in \mathbb{R} \right\}$.

Proposition 7.43. *We have*

$$\phi_r^n \in \mathrm{Ind}_{B_2(\mathbb{R})^+}^{\mathrm{GL}_2(\mathbb{R})^+} \lambda(n+2, -n).$$

Furthermore, the compact torus T_c acts on ϕ_r^n by right translation and ϕ_r^n has weight $\lambda'(-r, -n)$.

Proof. (1) We have the following identity:

$$\begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a\cosh t + c\sinh t & b\cosh t + d\sinh t \\ a\sinh t + c\cosh t & b\sinh t + d\cosh t \end{pmatrix},$$
$$\begin{pmatrix} 1-\frac{1}{2i}u & \frac{1}{2i}u \\ -\frac{1}{2i}u & 1+\frac{1}{2i}u \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a+(c-a)\frac{1}{2i}u & b+(d-b)\frac{1}{2i}u \\ c+(c-a)\frac{1}{2i}u & d+(d-b)\frac{1}{2i}u \end{pmatrix}.$$

Then by the identity

$$\cosh t \pm \sinh t = e^{\pm t}$$

and $\phi_r^n = (z_2 - z_2')w_2^{-\frac{n-r}{2}} w_2'^{-\frac{n+r}{2}}$, we can see that

$$\phi_r^n \in \mathrm{Ind}_{B_2(\mathbb{R})^+}^{\mathrm{GL}_2(\mathbb{R})^+} \lambda(n+2, -n).$$

(2) The weight of ϕ_r^n under T_c can be seen from the identity

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} (x+iy) & 0 \\ 0 & (x-iy) \end{pmatrix} = \begin{pmatrix} (x+iy)a & (x-iy)b \\ (x+iy)c & (x-iy)d \end{pmatrix}$$

and the definition of ϕ_r^n.

□

Recall that we have the decomposition

$$\mathfrak{gl}_{2,\mathbb{C}} = \mathfrak{k}_{\mathrm{GL}_2,\mathbb{C}} \oplus \mathfrak{p}^+_{\mathrm{GL}_2,\mathbb{C}} \oplus \mathfrak{p}^-_{\mathrm{GL}_2,\mathbb{C}},$$

and we let $v^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{p}^+_{\mathrm{GL}_2,\mathbb{C}}$ and $v^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \in \mathfrak{p}^-_{\mathrm{GL}_2,\mathbb{C}}$.

Let us define

$$w_n^+, w_n^- \in \mathrm{Hom}_{K_{\mathrm{GL}_2}}(\mathfrak{p}^+_{\mathrm{GL}_2,\mathbb{C}} \oplus \mathfrak{p}^-_{\mathrm{GL}_2,\mathbb{C}}, \mathrm{Sym}^n V_{2,\mathbb{C}} \otimes \mathrm{Ind}_{B_2(\mathbb{R})^+}^{\mathrm{GL}_2(\mathbb{R})^+} \lambda(n+2, -n))$$

by

$$\omega_n^+(v^+) = (2\pi i)^{n+1} b_0^n \otimes \phi^n_{-(n+2)},$$
$$\omega_n^+(v^-) = 0,$$
$$\omega_n^-(v^+) = 0,$$
$$\omega_n^-(v^-) = (2\pi i)^{n+1} b_n^n \otimes \phi^n_{n+2}.$$

Remark 7.44.
- The weight of b_0^n is $\lambda'(-n,n)$, the weight of $\phi^n_{-(n+2)}$ is $\lambda'(n+2,-n)$ and the weight v^+ is $\lambda'(2,0)$; the weight of b_n^n is $\lambda'(n,n)$, the weight of ϕ^n_{n+2} is $\lambda'(-(n+2),-n)$ and the weight v^- is $\lambda'(-2,0)$. So $\omega_n^\pm$ preserves K_{GL_2}-weight, so it is well-defined.
- The vector $\phi^n_{-(n+2)}$ is a minimal K_{GL_2}-type vector in the discrete series D_n^+.

Let us state some basic properties.

Lemma 7.45. *We have the following identities of the action of $v^\pm$:*

(1)

$$v^+(w_2) = -w_2',\ v^+(w_2') = 0,$$
$$v^+(z_2) = \frac{2(ad-bc)}{(-b+d)^2} i,\ v^+(z_2') = 0,$$
$$v^-(w_2) = 0,\ v^-(w_2') = -w_2,$$
$$v^-(z_2) = 0,\ v^-(z_2') = \frac{2(bc-ad)}{(-a+c)^2} i;$$

(2)

$$v^+(\phi_r^n) = \frac{n-r+2}{2}\phi^n_{r-2},\ v^-(\phi_r^n) = \frac{n+2+r}{2}\phi^n_{r+2},$$

(3)

$$v^+ b_r^n = -(n-r) b^n_{r+1},\ v^- b_r^n = -r b^n_{r-1}.$$

Proof. If f is a linear function on $\mathrm{GL}_2(\mathbb{R})$, then

$$(v^+f)(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) = \lim_{t\to 0} \frac{f(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}) - f(\begin{pmatrix} a & b \\ c & d \end{pmatrix})}{t}$$
$$= \lim_{t\to 0} \frac{f(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix})}{t} = f(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}).$$

Similarly, we have

$$(v^+f)(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) = f(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}).$$

(1) It follows from the fact that w_2 and w_2' are linear functions that

$$(v^+w_2)(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) = -a+c = w_2'(\begin{pmatrix} a & b \\ c & d \end{pmatrix}),$$
$$(v^+w_2')(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) = 0.$$

The proof for $v^-(w_2)$ and $v^-(w_2')$ is similar.
It can be seen from the quotient rule that

$$(v^+z_2) = \frac{(v^+(b+d))(-b+d) - (v^+(-b+d))(b+d)}{(-b+d)^2} i = \frac{2(ad-bc)}{(-b+d)^2} i,$$
$$(v^+z_2') = \frac{(v^+(a+c))(-a+c) - (v^+(-a+c))(a+c)}{(-a+c)^2} i = 0.$$

The proof for $v^-(z_2)$ and $v^-(z_2')$ is similar.

(2) It follows from explicit computation that

$$\begin{aligned} z_2 - z_2' &= \frac{2(bc-ad)}{(-b+d)(-a+c)}i \\ &= (v^+(z_2-z_2'))\frac{w_2}{w_2'}. \end{aligned}$$

From this, we get

$$v^+(z_2-z_2') = (z_2-z_2')\frac{w_2'}{w_2}.$$

Hence,

$$\begin{aligned} v^+\phi_r^n &= (z_2-z_2'){w_2}^{-(\frac{n-r}{2}+1)}{w_2'}^{-(\frac{n+r}{2}-1)} \\ &\quad + (\frac{n-r}{2})(z_2-z_2'){w_2}^{-(\frac{n-r}{2}+1)}{w_2'}^{-(\frac{n+r}{2}-1)} \\ &= \frac{n-r+2}{2}(z_2-z_2'){w_2}^{-(\frac{n-(r-2)}{2})}{w_2'}^{-(\frac{n+(r-2)}{2})} \\ &= \frac{n-r+2}{2}\phi_{r-2}^n. \end{aligned}$$

The computation of $v^-\phi_r^n$ is similar.

(3) Since X and Y are linear functions, we have $v^+(X)=0$ and $v^+(Y)=X$. Hence, by the product rule,

$$v^+(b_r^n) = v^+((-1)^r X^r Y^{n-r}) = (-1)^j(n-r)X^{r+1}Y^{n-r-1} = -(n-r)b_{r+1}^n.$$

Similar computations hold for $v^-(b_j^n)$.

□

Lemma 7.46. *We have the following identities with complex conjuagtion:*

$$\begin{aligned} &\overline{\phi_r^n} = (-1)\phi_{-r}^n,\ \overline{b_r^n} = b_{n-r}^b, \overline{v^+} = v^-, \\ &\overline{\omega^{\pm}} = (-1)^n\omega_n^{\mp}, \end{aligned}$$

where $\overline{(\cdot)}$ denotes the complex conjugation.

Proof. Recall that the Shimura data of $\mathrm{Sh}_{\mathrm{GL}_2}$ is defined on $\mathbb{R}$-points as

$$\begin{aligned} h\colon \mathbb{S}(\mathbb{R}) &\longrightarrow \mathrm{GL}_2(\mathbb{R}) \\ z = x+iy &\longmapsto \begin{pmatrix} z & 0 \\ 0 & \bar{z} \end{pmatrix}, \end{aligned}$$

and the reflex field of the Shimura variety $\mathrm{Sh}_{\mathrm{GL}_2}$ is $\mathbb{Q}$ and the complex conjugation on it is given by conjugation by $N = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$. From this we can see that

$$\overline{v^+} = \mathrm{Ad}_N(v^+) = v^-.$$

It follows from

$$\mathrm{Ad}_N(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) = \begin{pmatrix} d & c \\ b & a \end{pmatrix},$$

that $\overline{a} = d$ and $\overline{b} = c$. So we have

$$\begin{aligned} \overline{w_2} &= -\overline{b} + \overline{d} = a - c = w_2', \\ \overline{z_2} &= \frac{\overline{b}+\overline{d}}{-\overline{b}+\overline{d}}(-i) = \frac{a+c}{c-a}i = z_2'. \end{aligned}$$

We can see from this that

$$\overline{\phi_r^n} = (-1)(z_2 - z_2')w_2^{-(\frac{n-(-r)}{2})}w_2'^{-(\frac{n+(-r)}{2})} = (-1)\phi_{-r}^n.$$

The complex conjugate on X, Y is given by $\overline{(X,Y)} = (X,Y)N = (-Y,-X)$. Hence, we have

$$\overline{b_r^n} = \overline{(-1)^r X^r Y^{n-r}} = (-1)^{n+r} Y^r X^{n-r} = b_{n-r}^n.$$

Finally, we can deduce that

$$\begin{aligned}\overline{\omega_n^-}(v^-) &= \overline{(2\pi i)^{n+1}b_n^n} \otimes \overline{\phi_{n+2}^n} \\ &= (-1)^{n+1}(2\pi i)^{n+1} b_0^n \otimes (-1)\phi_{-(n+2)}^n \\ &= (-1)^n (2\pi i)^{n+1} b_0^n \otimes \phi_{-(n+2)}^n \\ &= (-1)^n \omega_n^+(v^+).\end{aligned}$$

Therefore, $\overline{\omega_n^\pm} = (-1)^n \omega_n^\mp$. □

Given $\phi_f \in \mathcal{B}_n$, we let

$$Eis_B^n(\phi_f) = \sum_{\gamma \in B_2(\mathbb{Q})\backslash \mathrm{GL}_2(\mathbb{Q})} \gamma^*(\omega_n^+ \otimes \phi_f),$$

which is absolutely convergent [Kin98, (6.3.1)] and defines an element of

$$\mathrm{Hom}_{K_{\mathrm{GL}_2}}(\mathfrak{p}^+_{\mathrm{GL}_2,\mathbb{C}} \oplus \mathfrak{p}^-_{\mathrm{GL}_2,\mathbb{C}}, \mathrm{Sym}^n V_{2,\mathbb{C}} \otimes C^\infty(\mathrm{GL}_2(\mathbb{Q})\backslash \mathrm{GL}_2(\mathbb{A})).$$

Consider

$$\theta_n = \frac{(2\pi i)^{n+1}}{2(n+1)} \sum_{j=0}^n b_{n-j}^n \otimes \phi_{n-2j}^n \in \mathrm{Sym}^n V_{2,\mathbb{C}} \otimes \mathrm{Ind}_{B_2(\mathbb{R})^+}^{\mathrm{GL}_2(\mathbb{R})^+} \lambda(n+2,-n),$$

Similar to before, for $\phi_f \in \mathcal{B}_n$ we define

$$Eis_H^n(\phi_f) = \sum_{\gamma \in B_2(\mathbb{Q})\backslash \mathrm{GL}_2(\mathbb{Q})} \gamma^*(\theta_n \otimes \phi_f). \tag{7.19}$$

This infinite series is absolutely convergent for $n \geq 1$. In this paper, we only treat the case $n \geq 1$.

Lemma 7.47. [Kin98, (6.3.5)] *We have the following identity:*

$$\mathrm{d}(Eis_H^n(\phi_f)) = \pi_n(Eis_B^n(\phi_f)).$$

In other words, for any $n \geq 1$ and $\phi_f \in \mathcal{B}_n$, the class $Eis_D^n(\phi_f)$ is represented by

$$(Eis_H^n(\phi_f), Eis_B^n(\phi_f)) \in \mathrm{H}^1_D(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)),$$

which is a direct factor of $\mathrm{H}_D^{n+1}(E^n, \mathbb{R}(n+1))$. Here $E \to \mathrm{Sh}_{\mathrm{GL}_2}$ is the universal elliptic curve over the Shimura variety $\mathrm{Sh}_{\mathrm{GL}_2}$.

Proof. We have

$$\begin{aligned}d\theta_n(v^+) = v^+\theta_n = \frac{(2\pi i)^{n+1}}{2(n+1)}(&b_n^n \otimes \phi_{n-2}^n \\ &+ 2b_{n-1}^n \otimes \phi_{n-4}^n - b_n^n \otimes \phi_{n-2}^n \\ &+ 3b_{n-2}^n \otimes \phi_{n-6}^n - 2b_{n-1}^n \otimes \phi_{n-4}^n \\ &+ \cdots \\ &+ (n+1)b_0^n \otimes \phi_{-(n+2)}^n - nb_1^n \otimes \phi_{-n}^n) \\ &= \frac{(2\pi i)^{n+1}}{2} b_0^n \otimes \phi_{-(n+2)}^n.\end{aligned}$$

By a similar computation, we have $d\theta_n(v^-) = \frac{(2\pi i)^{n+1}}{2} b_n^n \otimes \phi_{n+2}^n$. It follows from the definitions that

$$\pi_n(\omega_n^+) = \frac{\omega_n^+ + (-1)^n \overline{\omega_n^+}}{2} = \frac{\omega_n^+ + \omega_n^-}{2}.$$

Hence,

$$\pi_n(\omega_n^+)(v^+) = \frac{\omega_n^+ + \omega_n^-}{2}(v^+) = \frac{(2\pi i)^{n+1}}{2} b_0^n \otimes \phi_{-(n+2)}^n = d\theta_n(v^+),$$
$$\pi_n(\omega_n^+)(v^-) = \frac{\omega_n^+ + \omega_n^-}{2}(v^-) = \frac{(2\pi i)^{n+1}}{2} b_n^n \otimes \phi_{n+2}^n = d\theta_n(v^-).$$

In summary, we have shown $\pi_n(\omega_n^+) = d\theta_n$ so $\mathrm{d}(Eis_H^n(\phi_f)) = \pi_n(Eis_B^n(\phi_f))$. □

Remark 7.48. This proof is essentially the same as in [Kin98, §6.3], but we chose to identify $\mathrm{GL}_2(\mathbb{R})$ with $\mathrm{GU}(J_2')'$, so we reproduce the proof here.

7.6.2. *Motivic classes in DB-cohomology.* Now we give an expression of the realization of the motivic classes constructed in Construction 4.44 in Deligne–Beilinson cohomology using functoriality of motivic cohomology, DB-cohomology and the Beilinson regulator.

Notation 7.49. We use the following notations:

- $* := a + 5b - 3(r+s) + 1$, $** := a + 5b - 3(r+s) + 3$, $\square := a + 5b - 3(r+s)$, $\circ := a + 3b - r - s + 1$ and $\triangle := a + 3b - r - s + 2$;
- for $\diamond \in \{M, H, D\}$, $i_\diamond$ stands for the natural inclusion induced by $W \hookrightarrow \iota^* V$.

Proposition 7.50. *We have the following commutative diagram:*

$$\begin{array}{ccccccccc}
\mathcal{B}_n & \xrightarrow{Eis_M^n} & \mathrm{H}^1_M(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p_M^*} & \mathrm{H}^1_M(M, W(1)) & \xrightarrow{i_M} & \mathrm{H}^1_M(M, \iota^* V(1)) & \xrightarrow{\iota_{M,*}} & \mathrm{H}^3_M(S, V(2)) \\
\downarrow{\scriptstyle r_H} & & \downarrow{\scriptstyle r_H} & & \downarrow{\scriptstyle r_H} & & \downarrow{\scriptstyle r_H} & & \downarrow{\scriptstyle r_H} \\
\mathcal{B}_{n,\mathbb{R}} & \xrightarrow{Eis_H^n} & \mathrm{H}^1_H(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p_H^*} & \mathrm{H}^1_H(M, W(1)) & \xrightarrow{i_H} & \mathrm{H}^1_H(M, \iota^* V(1)) & \xrightarrow{\iota_{H,*}} & \mathrm{H}^3_H(S, V(2)) \\
\downarrow{\scriptstyle r_{H\to D}} & & \downarrow{\scriptstyle r_{H\to D}} & & \downarrow{\scriptstyle r_{H\to D}} & & \downarrow{\scriptstyle r_{H\to D}} & & \downarrow{\scriptstyle r_{H\to D}} \\
\mathcal{B}_{n,\mathbb{R}} & \xrightarrow{Eis_D^n} & \mathrm{H}^1_D(\mathrm{Sh}_{\mathrm{GL}_2}, \mathrm{Sym}^n V_2(1)) & \xrightarrow{p_D^*} & \mathrm{H}^1_D(M, W(1)) & \xrightarrow{i_D} & \mathrm{H}^1_D(M, \iota^* V(1)) & \xrightarrow{\iota_{D,*}} & \mathrm{H}^3_D(S, V(2)) \\
\downarrow{\scriptstyle \cong} & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
\mathcal{B}_{n,\mathbb{R}} & \xrightarrow{Eis_D^n} & \mathrm{H}^{n+1}_D(E^n, \mathbb{R}(n+1)) & \xrightarrow{p_D^*} & \mathrm{H}^{n+1}_D(E^n, \mathbb{R}(n+1)) & \xrightarrow{i_D} & \mathrm{H}^*_D(A^\square, \mathbb{R}(\circ)) & \xrightarrow{\iota_{D,*}} & \mathrm{H}^{**}_D(A^\square, \mathbb{R}(\triangle))
\end{array},$$

where in the last row, E^n in the first $\mathrm{H}^{n+1}_D(E^n, \mathbb{R}(n+1))$ is the n-th fiber product of the universal elliptic curve over $\mathrm{Sh}_{\mathrm{GL}_2}$, E^n in the second $\mathrm{H}^{n+1}_D(E^n, \mathbb{R}(n+1))$ is the n-th fiber product of the universal elliptic curve over M, $A^\square$ in $\mathrm{H}^*_D(A^\square, \mathbb{R}(\circ))$ is the $\square$-th fiber product of the pullback of the universal abelian 3-fold from S to M through ι, and $A^\square$ in $\mathrm{H}^{**}_D(A^\square, \mathbb{R}(\triangle))$ is the $\square$-th fiber product of the universal abelian 3-fold over S.

Proof. First, the commutativity of the diagram comes from the functoriality of r_H and $r_{H\to D}$.

Second, the decomposition of the Gysin morphism

$$\mathrm{H}^1_\diamond(M, W(1)) \to \mathrm{H}^3_\diamond(S, V(2))$$

into $\iota_{\diamond,*} \circ i_\diamond$ is natural from the construction of Gysin morphism (Proposition 4.16).

Finally, the embebding of the third row into the fourth row is from "Liebermann's trick" as explained in Proposition 7.33. □

Remark 7.51. From Construction 4.44, the Eisenstein classes $c \in \mathrm{H}^3_M(S, V(2))$ are

$$c = \iota_{M,*} \circ i_M \circ p_M^* \circ Eis_M^n(\phi_f),$$

for $\phi_f \in \mathcal{B}_n$. By commutativity of the above diagram, their image under the Deligne regulator $r_D := r_{H\to D} \circ r_H$ is

$$\begin{aligned} r_D(c) &= r_D(\iota_{M,*} \circ i_M \circ p_M^* \circ Eis_M^n(\phi_f)) \\ &= \iota_{D,*} \circ i_D \circ p_D^* \circ Eis_D^n(r_D(\phi_f)) \\ &= \iota_{D,*} \circ i_D \circ p_D^*(Eis_H^n(\phi_f), Eis_B^n(\phi_f)). \end{aligned}$$

By "Liebermann's trick", Proposition 7.30, Proposition 7.31 and the fact that $Eis_H^n(\phi_f)$ and $Eis_B^n(\phi_f)$ are slowly increasing as explained in [Kin98, page 120], we can view

$$i_D \circ p_D^* \circ (Eis_H^n(\phi_f), Eis_B^n(\phi_f)).$$

as a pair of tempered currents

$$(T_{p^*Eis_H^n(\phi_f)}, T_{p^*Eis_B^n(\phi_f)}) \in \mathrm{H}_D^1(M, \iota^*V(1)) \subset \mathrm{H}_D^{a+5b-3(r+s)+1}(A^{a+5b-3(r+s)}/M, \mathbb{R}(a+3b-r-s+1)),$$

where $A^{a+5b-3(r+s)}/M$ is the $a+5b-3(r+s)$-th fiber product of the pullback of the universal abelian 3-fold from S to M through ι. Hence, by the explicit description of the Gysin morphism for DB-cohomology (cf. equation (7.15)), we have

$$\begin{aligned} r_D(c) = (\iota_* T_{p^*Eis_H^n(\phi_f)}, \iota_* T_{p^*Eis_B^n(\phi_f)}) &\in \mathrm{H}_D^3(S, V(2)) \\ &\subset \mathrm{H}_D^{a+5b-3(r+s)+3}(A^{a+5b-3(r+s)}/S, \mathbb{R}(a+3b-r-s+2)), \end{aligned}$$

where $A^{a+5b-3(r+s)}/S$ is the $a+5b-3(r+s)$-th fiber product of the universal abelian 3-fold over S.

7.7. **The pairing.** In this subsection, we give an explicit formula for the Poincaré duality pairing with the tool of tempered currents.

7.7.1. *From Poincaré duality pairing to an integral of differential forms.* We first express the Poincaré duality pairing in terms of an integration of differential forms on Shimura varieties.

Lemma 7.52. *Let $\widetilde{A}^{a+5b-3(r+s)}$ be a smooth toroidal compactification of $A^{a+5b-3(r+s)}$ such that the complement $\widetilde{A}^{a+5b-3(r+s)} \backslash A^{a+5b-3(r+s)}$ is a simple normal crossings divisor. Then there exists a closed tempered current*

$$\rho \in \mathcal{D}^{a+5b-3(r+s)+2}_{si,\mathbb{R}(a+3b-r-s+1)}(\widetilde{A}^{a+5b-3(r+s)})$$

such that:

1. *the cohomology class $[\rho]$ of ρ belongs to $\mathrm{H}^2_{B,!}(S, V(2))^-(-1)$;*
2. *the class $[\rho]$ maps to $\mathcal{E}is_H^n(\phi_f) = r_H(\mathcal{E}is_M^n(\phi_f))$ through the third map in the exact sequence (7.1):*

$$\mathrm{H}^2_{B,!}(S, V(2))^-_{\mathbb{R}}(-1) \to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), \mathrm{H}^2_{B,!}(S, V(2))_{\mathbb{R}});$$

3. *the pair of tempered currents $(\iota_* T_{p^*Eis_H^n(\phi_f)}, \iota_* T_{p^*Eis_B^n(\phi_f)})$ and $(\rho, 0)$ represent the same cohomology class in $\mathrm{H}^3_D(S, V(2))$.*

Proof. This is a direct consequence of [Bei86b, §5.7.]. □

Corollary 7.53. *We have the following identity:*

$$\langle \Omega, \tilde{v}_K \rangle_B = \langle \omega_\Psi, [\rho] \rangle_B \otimes \mathbf{1},$$

where $\mathbf{1}$ be unit of $E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{C}$.

Proof. It is direct from Lemma 7.52 (2) and Remark 7.13 (5). □

We first take the dual of the map of representations

$$\mathrm{Sym}^n V_2 \hookrightarrow \iota^* V,$$

and get the map $\iota^* D \to \mathrm{Sym}^n V_2^\vee$. Composing the above map with the natural pairing $\mathrm{Sym}^n V_2^\vee \otimes \mathrm{Sym}^n V_2 \to \mathbb{Q}(0)$ defined by

$$a_j^n \otimes b_k^n \mapsto \begin{cases} 0 & \text{if } j \neq k, \\ \binom{n}{j} & \text{if } j = k, \end{cases}$$

we have a natural pairing

$$\langle \cdot, \cdot \rangle \colon \iota^* D \otimes \mathrm{Sym}^n V_2 \to \mathbb{Q}(0).$$

Convention 7.54. We normalize the embedding

$$\mathrm{H}^2_{B,!}(S, V(2)) \hookrightarrow \mathrm{H}^{a+5b-3(r+s)+2}_{B,!}(A^{a+5b-3(r+s)}, \mathbb{R}(a+3b-r-s+2))$$

and

$$\mathrm{H}^2_{B,!}(S, D) \hookrightarrow \mathrm{H}^{a+5b-3(r+s)+2}_{B,!}(A^{a+b-3(r+s)}, \mathbb{R}(2b-2r-2s))$$

such that we have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{H}^2_{B,!}(S, V(2)) \otimes \mathrm{H}^2_{B,!}(S, D) & \xrightarrow{\langle \cdot, \cdot \rangle_B} & \mathbb{Q}(0) \\ \downarrow & & \downarrow = \\ \mathrm{H}^*_{B,!}(A^\square, \mathbb{R}(a+3b-r-s+2)) \otimes \mathrm{H}^*_{B,!}(A^\square, \mathbb{R}(2b-2r-2s)) & \xrightarrow{\langle \cdot, \cdot \rangle_B} & \mathbb{Q}(0) \end{array},$$

where $* = a+5b-3(r+s)+2$ and $\square = a+5b-3(r+s)$.

Let $\Psi = \Psi_f \otimes \Psi_\infty$ be a factorizable cusp form in the irreducible cuspidal automorphic representation $\tilde{\pi} = \tilde{\pi}_f \otimes \tilde{\pi}_\infty$ of $G(\mathbb{A})$ with central character[19]

$$\begin{aligned} \omega_\pi \colon Z_G(\mathbb{Q}) \backslash Z_G(\mathbb{A}) &\longrightarrow \mathbb{C}^\times \\ z &\longmapsto |z|^n \nu(z)^{-1}, \end{aligned}$$

where ν is a finite order Hecke character of sign $(-1)^n$. Let

$$\omega_\Psi := \omega \otimes \Psi_f$$

be a differential form associated to Ψ, where ω is the Lie algebra cohomology class constructed in Proposition 7.16.

Recall that we let $n = a+b+r+s$ and we use $\mathrm{Sh}_G(L)$ (resp., $\mathrm{Sh}_H(K)$) to denote $\mathrm{Sh}_G(L)^{an}_{\mathbb{C}}$ (resp., $\mathrm{Sh}_H(K)^{an}_{\mathbb{C}}$).

Proposition 7.55. *Let $\phi_f \in I_n(\nu) \subset \mathcal{B}_{n,\overline{\mathbb{Q}}}$ defined in Definition 4.42 and let ρ be the closed tempered current associated to the class $\mathcal{E}is^n_D(\phi_f) = r_D(\mathcal{E}is^n_M(\phi_f))$ by Lemma 7.52. We have*

$$\langle \omega_\Psi, [\rho] \rangle_B = \frac{1}{(2\pi i)^{n+1}} \int_{\mathrm{Sh}_H(K)} p^* Eis_H(\phi_f) \iota^* \omega_\Psi,$$

where K is the level of ϕ_f.

Proof. For the pair of tempered currents $(\iota_* T_{p^* Eis^n_H(\phi_f)}, \iota_* T_{p^* Eis^n_B(\phi_f)})$ and $(\rho, 0)$ that represent the same cohomology class in $\mathrm{H}^3_D(\mathrm{Sh}_G, V(2))$, there exists

$$(S', T') \in \mathcal{D}^{a+5b-3(r+s)+1}_{si, \mathbb{R}(a+3b-r-s+1)}(\widetilde{A}^{a+5b-3(r+s)}) \oplus F^{a+3b-r-s+2} \mathcal{D}^{a+5b-3(r+s)+2}_{si}(\widetilde{A}^{a+5b-3(r+s)})$$

such that

$$\rho = \iota_* T_{p^* Eis^n_H(\phi_f)} + dS' + \pi_{a+3b-r-s+1}(T').$$

[19] Here $\tilde{\pi}_\infty$ is a $(\mathfrak{g}_{\mathbb{C}}, K_G)$-module, but $A_G \subset K_G$, so the word "central character of $\tilde{\pi}$ makes sense.

It follows from the fact that ω_Ψ is a diffferential form on Sh_G that

$$\langle \omega_\Psi, [\rho] \rangle_B = \langle \omega_\Psi, [\rho] \rangle_B^{\mathrm{Sh}_G}.$$

Then we can see that

$$\begin{aligned}
\langle \omega_\Psi, [\rho] \rangle_B &= \rho(\omega_\Psi) \\
&= \iota_* T_{p^* Eis_H^n(\phi_f)}(\omega_\Psi) + dS'(\omega_\Psi) + \pi_{a+2r-s+1}(T')(\omega_\Psi) \\
&= \iota_* T_{p^* Eis_H^n(\phi_f)}(\omega_\Psi) + \pi_{a+2r-s+1}(T')(\omega_\Psi) \\
&= \iota_* T_{p^* Eis_H^n(\phi_f)}(\omega_\Psi) \\
&= T_{p^* Eis_H^n(\phi_f)}(\iota^* \omega_\Psi) \\
&= \frac{1}{(2\pi i)^{n+1}} \int_{\mathrm{Sh}_H(K)} p^* Eis_H(\phi_f) \iota^* \omega_\Psi.
\end{aligned}$$

The first equality holds by the compatibility between Poincaré duality pairings and integrations (see [Har90, Prop.1.4.4.(c)]). The third equality follows from the fact that ω_Ψ is closed. The fourth equality follows from the fact that the Hodge type of ω_Ψ is $(a+r+1, b+s+1)$ and the fact that $T' \in F^{a+3b-r-s+2} \mathcal{D}_{si}^{a+5b-3(r+s)+2}(\widetilde{A}^{a+5b-3(r+s)})$. The fifth equality follows from the definition of the pushforward of tempered currents (see equation (7.16)). The last equality comes from the definition of tempered currents. This completes the proof of the proposition. □

7.7.2. *From the integral of differential forms to an adelic integral.* We now translate the above integral into an adelic integral. In order to do so, we need to make the following choices of measures.

Convention 7.56. (1) We normalize the Haar measure for a reductive or unipotent group G' over $\mathbb{Z}_p$ such that $\mathrm{Vol}(G'(\mathbb{Z}_p)) = 1$.

(2) At the archimedean place, we use the Iwasawa decomposition to determine the measure. More precisely, we define the measure on the unipotent radical of the Borels[20] of $G(\mathbb{R})$ and $H(\mathbb{R})$ so that the integral points have covolume 1. We choose the Haar measure on K_G (resp., K_H) such that the volume of $K_G/Z_G(\mathbb{R})$ (resp., $K_H/Z_H(\mathbb{R})$) is 1. On the maximal split torus $\mathbb{R}^{\times,n}$, we choose the multiplicative Haar measure $\frac{dx_1 dx_2 \cdots dx_n}{|x_1 x_2 \cdots x_n|}$.

(3) The measures on $G(\mathbb{A})$ and $H(\mathbb{A})$ are the product of the above nonarchimedean measures and archimedean measures.

(4) For quotients of groups, we use the unique measure compatible with these choices.

(5) Once we fix $\mathbf{1} = v^+ \otimes v^-$, we have fixed an equivalence of top diffferential forms on $\mathrm{Sh}_H(K)$ and invariant measures on $\mathrm{Sh}_H(K)$ (see [Har97, Page 83]).

Proposition 7.57. *For ϕ_f and ρ be as in Proposition 7.55. We have*

$$\langle \omega_\Psi, [\rho] \rangle_B = \frac{[H(\mathbb{Z}_p) : K]^{-1}}{(2\pi i)^{n+1}} \int_{H(\mathbb{Q}) Z(\mathbb{A}) \backslash H(\mathbb{A})} (p^* Eis_H(\phi_f) \iota^* \omega_\Psi(\mathbf{1}))(g) dg,$$

where $\mathbf{1}$ is the vector $v^+ \otimes v^-$ and the vectors v^+ and v^- are defined in equation (7.9).

Proof. By Proposition 7.55, we have

$$\langle \omega_\Psi, [\rho] \rangle_B = \frac{1}{(2\pi i)^{n+1}} \int_{\mathrm{Sh}_H(K)} p^* Eis_H(\phi_f) \iota^* \omega_\Psi.$$

Thanks to the equivalence of top differential forms on $\mathrm{Sh}_H(K)$ and invariant measures on $\mathrm{Sh}_H(K)$ explained above, we can write the integral as an adelic integral. More precisely, we

[20] The algebraic group G is quasi-split over $\mathbb{Q}$.

have

$$\begin{aligned}\langle \omega_\Psi, [\rho]\rangle_B &= \frac{1}{(2\pi i)^{n+1}} \int_{H(\mathbb{Q})\backslash H(\mathbb{A})/K_H K} (p^* Eis_H(\phi_f)\iota^*\omega_\Psi(\mathbf{1}))(g)dg \\ &= \frac{[H(\mathbb{Z}_p):K]^{-1}}{(2\pi i)^{n+1}} \int_{H(\mathbb{Q})Z_H(\mathbb{R})\backslash H(\mathbb{A})} (p^* Eis_H(\phi_f)\iota^*\omega_\Psi(\mathbf{1}))(g)dg \\ &= \frac{[H(\mathbb{Z}_p):K]^{-1}}{(2\pi i)^{n+1}} \int_{H(\mathbb{Q})Z(\mathbb{R})\backslash H(\mathbb{A})} (p^* Eis_H(\phi_f)\iota^*\omega_\Psi(\mathbf{1}))(g)dg \\ &= \frac{[H(\mathbb{Z}_p):K]^{-1}}{(2\pi i)^{n+1}} \int_{H(\mathbb{Q})Z(\mathbb{A})\backslash H(\mathbb{A})} (p^* Eis_H(\phi_f)\iota^*\omega_\Psi(\mathbf{1}))(g)dg.\end{aligned}$$

The second equality follows from $\mathrm{Vol}(K) = [H(\mathbb{Z}_p):K]^{-1}$ and $\mathrm{Vol}(K_H/Z_H(\mathbb{R})) = 1$. The third equality is from the fact that $\mathrm{Vol}(\mathbb{S}^1) = 1$ and the fourth comes from $\mathrm{Vol}(\mathbb{Z}_p^\times) = 1$. □

7.7.3. *An explicit form of the adelic integral.* Recall that we let $\Psi_\infty \in \tilde{\pi}_\infty$ be a vector of weight $(-1-b+r-s, 1+a+r-s, r-s; b+2s-r)$ and $v \in D$ be a vector of weight $(-a+s-r, b+s-r, s-r; -b-2s+r)$. We also have the differential form ω_Ψ associated to Ψ by Proposition 7.16. The following proposition gives a precise formula for $\langle \omega_\Psi, [\rho]\rangle_B$.

Notation 7.58.

- We let
$$C_j := [H(\mathbb{Z}_p):K]^{-1}\langle X_{(1,-1,0)}^{2j-b-r-s} v, b_{n-j}^{a+b+r+s}\rangle.$$
- We denote by $\Xi_{m,t}(\phi_f)$ the functions on $H(\mathbb{A})$ defined by
$$\Xi_{m,t}(\phi_f) := X_{(1,-1,0)}^m \Psi \sum_{\gamma \in B_2(\mathbb{Q})\backslash \mathrm{GU}(J_2')'(\mathbb{Q})} \gamma^*(\phi_t^{a+b+r+s} \otimes \phi_f).$$
- We let
$$A = \max\{0, \frac{b+r+s}{2}\},$$
$$B = \min\{\frac{a+2b+r+s}{2}, a+b+r+s\}.$$

Proposition 7.59. *The pairing $\langle \omega_\Psi, [\rho]\rangle_B$ is equal to*

(7.20)

$$-\frac{1}{2(a+b+r+s+1)} \sum_{j=A}^{B} (-1)^{b+r+s}(2j-b-r-s+1)C_j \int \Xi_{1+a+2b-2j+r+s, a+b+r+s-2j}(\phi_f),$$

where the integrals are over $H(\mathbb{Q})Z(\mathbb{A})\backslash H(\mathbb{A})$.

Proof. By Proposition 7.57, we have

$$\langle \omega_\Psi, [\rho]\rangle_B = \frac{[H(\mathbb{Z}_p):K]^{-1}}{(2\pi i)^{n+1}} \int_{H(\mathbb{Q})Z(\mathbb{A})\backslash H(\mathbb{A})} (p^* Eis_H(\phi_f)\iota^*\omega_\Psi(\mathbf{1}))(g)dg,$$

where $\mathbf{1}$ is the vector $v^+ \otimes v^-$.

It follows from the formula for $Eis_H(\phi_f)$ in equation (7.19) and the formula for $\iota^*\omega_\Psi$ in Proposition 7.20 that for $g \in H(\mathbb{A})$,

$$\begin{aligned}&(p^* Eis_H(\phi_f)\iota^*\omega_\Psi)(\mathbf{1})(g) \\ =& -\frac{(2\pi i)^{n+1}}{2(n+1)} (\sum_{i=0}^{a+b} (-1)^i (i+1) X_{(1,-1,0)}^{1+a+b-i} \Psi_\infty \otimes \Psi_f)(g) \\ \times& (\sum_{\gamma \in B_2(\mathbb{Q})\backslash \mathrm{GL}_2(\mathbb{Q})} \sum_{j=0}^{n} \gamma^*(\phi_{n-2j}^n \otimes \phi_f))(g)\langle X_{(1,-1,0)}^i v, b_{n-j}^n\rangle.\end{aligned}$$

The weight of $X^i_{(1,-1,0)}v$ is $\lambda'(i-a,-a-b-r-s)$ when restricting to GL_2 since its weight is $(i-a+s-r,-i+b+s-r,s-r;-b-2s+r)$. It can be seen from the definition of the pairing and the fact that the weight of b^n_{n-j} is $\lambda'(a+b+r+s-2j,a+b+r+s)$ that if $a+b+r+s-2j\neq -(i-a)$, then

$$\langle X^i_{(1,-1,0)}v, b^n_{n-j}\rangle = 0.$$

Thus only the term satisfying $i=2j-b-r-s$ remains. By the inequality $0\leq i\leq a+b$, we have

$$\frac{b+r+s}{2}\leq j\leq \frac{a+2b+r+s}{2}.$$

By $0\leq j\leq a+b+r+s$ and the definition of A and B, we have $A\leq j\leq B$. Hence, we get

$$\begin{aligned}
&(p^*Eis_H(\phi_f)\iota^*\omega_\Psi)(\mathbf{1})(g)\\
=&-\frac{(2\pi i)^{n+1}}{2(n+1)}\sum_{j=A}^{B}(-1)^{b+r+s}(2j-b-r-s+1)[H(\mathbb{Z}_p):K]C_j\\
&\times(X^{1+a+2b-2j+r+s}_{(1,-1,0)}\Psi_\infty\otimes\Psi_f)\times(\sum_{\gamma\in B_2(\mathbb{Q})\backslash \mathrm{GL}_2(\mathbb{Q})}\gamma^*(\phi^{a+b+r+s}_{a+b+r+s-2j}\otimes\phi_f)).
\end{aligned}$$

Plugging this into the integral, we get

$$\langle\omega_\Psi,[\rho]\rangle_B=-\frac{1}{2(n+1)}\sum_{j=A}^{B}(-1)^{b+r+s}(2j-b-r-s+1)C_j\int\Xi_{1+a+2b-2j+r+s,a+b+r+s-2j}(\phi_f).$$

□

Remark 7.60. (1) By the inequality $A\leq j\leq B$, we can see that $2j-b-r-s+1>0$, which is important for later applications.

(2) The K_{GL_2}-weight of $\Xi_{1+a+2b-2j+r+s,a+b+r+s-2j}(\phi_f)$ is 0, so the integral

$$\int\Xi_{1+a+2b-2j+r+s,a+b+r+s-2j}(\phi_f)$$

could be non-zero.

Lemma 7.61. *The constant C_{b+s} is non-zero.*

Proof. First, recall that

$$C_{b+s}=[H(\mathbb{Z}_p):K]^{-1}\langle X^{b+s-r}_{(1,-1,0)}v,b^{a+b+r+s}_{a+r}\rangle.$$

The weight of $X^{b+s-r}_{(1,-1,0)}v$ is $(b-a+2(s-r),0,s-r;-b-2s+r)$, so the weight of it is $\lambda'(b+s-(a+r),-(a+b+r+s))$ when restricting to GL_2. By the fact that the weight of the representation $\mathrm{Sym}^n V_2^\vee$ of GL_2 is multiplicity -free, we have

$$X^{b+s-r}_{(1,-1,0)}v|_{\mathrm{GL}_2}=\lambda^{b+s}(v)a^{a+b+r+s}_{a+r},$$

where $a^{a+b+r+s}_{a+r}$ is $(b^{a+b+r+s}_{a+r})^\vee$ and $\lambda^{b+s}(v)\in\mathbb{Z}_{\geq 0}$. By the definition of the pairing $\langle\cdot,\cdot\rangle$, in order to show that $C_{b+s}\neq 0$, it suffices to show that

$$\rho(X^{b+s-r}_{(1,-1,0)}v)=X^{b+s-r}_{(1,-1,0)}v|_{\mathrm{GL}_2}\neq 0.$$

Second, if we untwist by the character $\mu^{-b-2s+r}$, the weight of $X^{a+r}_{(1,0,-1)}X^{b+s-r}_{(1,-1,0)}v$ is $(b+2s-r,0,-a+s-2r)$. After the untwist, the highest weight of

$$\mathrm{Sym}^n V_2^\vee\hookrightarrow\iota^*D$$

is also $(b+2s-r, 0, -a+s-2r)$, which is the same as the weight of $X_{(1,0,-1)}^{a+r} X_{(1,-1,0)}^{b+s-r} v$. Thirdly, we have the following two formulas: for a representation V of the Lie algebra $\mathfrak{g}_{\mathbb{C}}$, for any vector $v \in V$, we have

$$\begin{aligned} X_{(1,0,-1)} X_{(1,-1,0)} v &= [X_{(1,0,-1)}, X_{(1,-1,0)}] v + X_{(1,-1,0)} X_{(1,0,-1)} v \\ &= X_{(1,-1,0)} X_{(1,0,-1)} v. \end{aligned}$$

Finally, by the above formula, we have

$$X_{(1,0,-1)} (X_{(1,0,-1)}^{a+r} X_{(1,-1,0)}^{b+s-r} v) = X_{(1,-1,0)}^{b+s-r} (X_{(1,0,-1)}^{a+r+1} v) = 0,$$

where the last equality follows from equation (7.8). Hence, $X_{(1,0,-1)}^{a+r} X_{(1,-1,0)}^{b+s-r} v$ is a highest weight vector of

$$\mathrm{Sym}^n V_2^\vee \hookrightarrow \iota^* D.$$

By the branching law of $\mathrm{GL}_2 \subset \mathrm{GL}_3$, the highest weight vector is unique up to a non-zero constant, so $X_{(1,0,-1)}^{a+r} X_{(1,-1,0)}^{b+s-r} v \neq 0$. Thus

$$\rho(X_{(1,0,-1)}^{a+r} X_{(1,-1,0)}^{b+s-r}) v = X_{(1,0,-1)}^{a+r} \rho(X_{(1,-1,0)}^{b+s-r} v) \neq 0.$$

Hence, we have

$$\rho(X_{(1,-1,0)}^{b+s-r} v) \neq 0.$$

□

Remark 7.62.
(1) From the computation of an archimedean zeta integral in the next section, we can see that the only non-zero summand in the summation of Lemma 7.59 is associated to $j = b+s$, so we only need to verify that C_{b+s} is non-zero.
(2) We can replace v by $C_{b+s}^{-1} v$ so that we replace ω_Ψ by $C_{b+s}^{-1} \omega_\Psi$. This will not affect the result because our goal is compute the quotient

$$\frac{\langle \omega_\Psi, \tilde{\nu}_K \rangle_B}{\langle \omega_\Psi, \tilde{\nu}_D \rangle_B}.$$

8. Computation of the integral

Convention 8.1.
- In this section, we do not identify $\mathrm{GL}_2(\mathbb{R})$ with $\mathrm{GU}(J_2)'(\mathbb{R})$ automatically.
- In this section, we choose measures as in Convention 7.56.
- Let $n = a+b+r+s$.

8.1. The zeta integral of Gelbart and Piatetski-Shapiro.

In this subsection, we recall the zeta integral constructed in [GPS84] and modified in [PS18].

Convention 8.2.
- Let dt_∞ be the Lebesgue measure on the additive group $\mathbb{R}$. If v is a nonarchimedean place of $\mathbb{Q}$, let dt_v be the Haar measure on $\mathbb{Q}_v$ such that $\mathbb{Z}_v$ has volume one.
- Let $d^\times t_v$ be the measure on $\mathbb{Q}_v^\times$ by

$$d^\times t_v = \begin{cases} \frac{dt_v}{|t_v|} & \text{if } v \text{ is archimedean,} \\ \frac{p}{p-1} \frac{dt_v}{|t_v|} & \text{if } v \text{ is nonarchimedean.} \end{cases}$$

- Let dt (resp., $d^\times t$) denote the restricted product measure $\prod_v dt_v$ on $\mathbb{A}$ (resp., the restricted product measure $\prod_v d^\times t_v$ on $\mathbb{A}^\times$).

Definition 8.3. Let V_2 be the standard representation of GL_2 on row vectors and let $B_2 = T_2 U_2$ be the standard Borel of GL_2. Let Φ be a Schwartz-Bruhat function on $V_2(\mathbb{A})$ and let $\nu = (\nu_1, \nu_2) \colon T(\mathbb{A})/T(\mathbb{Q}) \to \mathbb{C}^\times$ be a character of the diagonal torus of GL_2. We now define

$$f(g, \Phi, \nu, s) = \nu_1(\det(g)) |\det(g)|^s \int_{\mathrm{GL}_1(\mathbb{A})} \Phi((0,t)g) \nu_1 \nu_2^{-1}(t) |t|^{2s} dt$$

and set the Eisenstein series on $\mathrm{GL}_2(\mathbb{A})$ to be

$$E(g,\Phi,\nu,s)=\sum_{\gamma\in B(\mathbb{Q})\backslash GL_2(\mathbb{Q})} f(\gamma g,\Phi,\nu,s).$$

We write $f(g_1,\Phi,\nu,s)$ and $E(g_1,\Phi,\nu,s)$ as functions on $H(\mathbb{A})$ via the projection $H(\mathbb{A})\twoheadrightarrow \mathrm{GL}_2(\mathbb{A})$.

Remark 8.4.

- The general theorem [Lan76, Lemma 4.1] implies that the Eisenstein series

$$E(g,\Phi,\nu,s)=\sum_{\gamma\in B_2(\mathbb{Q})\backslash \mathrm{GL}_2(\mathbb{Q})} f(\gamma g,\Phi,\nu,s)$$

  is absolutely convergent for $\mathrm{Re}(s)$ big enough and satisfies a functional equation.
- By the explicit formula for $f(g,\Phi,\nu,s)$, we can see that the central character is $\nu_1\nu_2$.

Definition 8.5 (Global zeta integral)**.** For an irreducible cuspidal automorphic representation π of $G(\mathbb{A})$ and a cusp form φ in the space of π, we define the global zeta integral as follows:

$$I(\varphi,\Phi,\nu,s)=\int_{H(\mathbb{Q})Z(\mathbb{A})\backslash H(\mathbb{A})}\varphi(g)E(g_1,\Phi,\nu,s)dg.$$

8.2. **Comparison of the integral and the zeta integral.** In this subsection, we express the integral

$$\int_{H(\mathbb{Q})Z(\mathbb{A})\backslash H(\mathbb{A})}\Xi_{m,t}(\phi_f)(g)dg$$

in Proposition 7.59 in terms of the zeta integral in Definition 8.5.

Proposition 8.6. *Let $\nu_1,\nu_2\colon \mathbb{Q}^\times\backslash\mathbb{A}^\times\to\mathbb{C}^\times$ be continuous characters and let $s\in\mathbb{C}$. Let $\chi_{\nu_1,\nu_2,s}$ be the character of $B_2(\mathbb{A})$ defined by*

$$\chi_{\nu_1,\nu_2,s}(\begin{pmatrix}a & b\\ & d\end{pmatrix})=\nu_1(a)\nu_2(d)\left|\frac{a}{d}\right|^s.$$

Then for any Schwartz-Bruhat function Φ on $\mathbb{A}^2$, the function $f(g,\Phi,\nu,s)$ on $\mathrm{GL}_2(\mathbb{A})$ defined by

$$f(g,\Phi,\nu,s)=\nu_1(\det(g))|\det(g)|^s\int_{\mathrm{GL}_1(\mathbb{A})}\Phi((0,t)g)\nu_1\nu_2^{-1}(t)|t|^{2s}dt \tag{8.1}$$

belongs to $\mathrm{Ind}_{B_2(\mathbb{A})}^{\mathrm{GL}_2(\mathbb{A})}\chi_{\nu_1,\nu_2,s}$, where Ind *means unnormalized induction.*

Proof. The proposition follows from the following computations:

$$f(\begin{pmatrix}1 & b\\ & 1\end{pmatrix}g,\Phi,\nu,s)=f(g,\Phi,\nu,s),$$

$$\begin{aligned}f(\begin{pmatrix}a_1 & \\ & d_1\end{pmatrix}g,\Phi,\nu,s)&=\nu_1(a_1d_1)|a_1d_1|^s\nu_1\nu_2^{-1}(d_1^{-1})|d_1|^{-2s}f(g,\Phi,\nu,s)\\&=\nu_1(a_1)\nu_2(d_1)\left|\frac{a_1}{d_1}\right|^s f(g,\Phi,\nu,s).\end{aligned}$$

□

Proposition 8.7. *Let ν_1 be the Hecke character $||^{-n}\nu$ and ν_2 be the trivial Hecke character, where ν is a finite-order Hecke character of sign $(-1)^n$. Then the following statements hold.*

- *The nonarchimedean part χ_f of $\chi_{\nu_1,\nu_2,1+n}$ satisfies*

$$\chi_f(\begin{pmatrix}a\alpha & b\\ & d\delta\end{pmatrix})=a^{-1}d^{n+1}\nu(\alpha),$$

  for any $a,d\in\mathbb{Q}_+^\times$ and $\alpha,\delta\in\widehat{\mathbb{Z}}^\times$.

- *The restriction of the archimedean part χ_∞ of $\chi_{\nu_1,\nu_2,1+n}$ to the identity component of the diagonal maximal torus of $\mathrm{GL}_2(\mathbb{R})^+$ is identified with $\lambda(n+2,-n)$ (see notation 7.38) through the isomorphism $\mathrm{GL}_2(\mathbb{R})^+ \cong \mathrm{GU}(J_2)'(\mathbb{R})^+$.*

Proof. For the nonarchimedean part, we have

$$\chi_f(\begin{pmatrix} a\alpha & b \\ & d\delta \end{pmatrix}) = |a|_f^{-n}\nu(\alpha)|\frac{a}{d}|_f^{1+n} = a^{-1}d^{n+1}\nu(\alpha).$$

For the archimedean part, for $\begin{pmatrix} a & \\ & a^{-1}v \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})^+$, we have

$$\chi_\infty(\begin{pmatrix} a & \\ & a^{-1}v \end{pmatrix}) = |a|_\infty^{-n}|a^2v^{-1}|_\infty^{1+n} = a^{n+2}v^{\frac{-n-(n+2)}{2}}.$$

Hence, the action of χ_∞ is $\lambda(n+2,-n)$ via the isomorphism $\mathrm{GL}_2(\mathbb{R})^+ \cong \mathrm{GU}(J_2)'(\mathbb{R})^+$. □

Proposition 8.8. *Let ν_1 be the Hecke character $||^{-n}\nu$ and ν_2 be the trivial Hecke character, where ν is a finite-order Hecke character of sign $(-1)^n$. Let Φ be a factorizable Schwartz-Bruhat function $\Phi = \otimes_v'\Phi_v$ on $V_2(\mathbb{A})$ such that*

(1) The function $\Phi_f = \otimes_{v<\infty}'\Phi_v$ is $\overline{\mathbb{Q}}$-valued,
(2) The function $\Phi_\infty(x,y)$ is defined by

$$\Phi_\infty(x,y) := (ix-y)^{\frac{n-k}{2}}(ix+y)^{\frac{n+k}{2}}e^{-\pi(x^2+y^2)}.$$

Then there exists a $\phi_f \in \mathcal{B}_{n,\overline{\mathbb{Q}}}$ such that for any $g \in \mathrm{GL}_2(\mathbb{A})$, we have

$$(\phi_k^n \otimes \phi_f)(g) = (-1)^{\frac{a+b+r+s-k}{2}}2i\pi^{1+a+b+r+s}\Gamma(1+a+b+r+s)^{-1}f(g,\Phi,\nu,1+a+b+r+s),$$

where the ϕ_k^n on the left-hand side the pullback of the function ϕ_k^n defined in Definition 7.40 through the isomorphism $\mathrm{GL}_2(\mathbb{R}) \cong \mathrm{GU}(J_2)'(\mathbb{R})$ and f is defined in equation (8.1).

Proof. First, we can factor $f(g,\Phi,\nu,s)$ into an Euler product of local Tate integrals

$$f(g,\Phi,\nu,s) = \otimes_v' f(g_v,\Phi_v,\nu_v,s),$$

where

$$f(g_v,\Phi_v,\nu_v,s) = \nu_{1,v}(\det(g_v))|\det(g_v)|^s\int_{\mathrm{GL}_1(\mathbb{Q}_v)}\Phi((0,t_v)g_v)(\nu_{1,v}\nu_{2,v})^{-1}(t_v)|t_v|_v^{2s}dt_v.$$

Second, it follows from Proposition 8.6, Proposition 8.7 and Defintion 4.42 that

$$f^\infty(g,\Phi,\nu,1+n) = \otimes_{v<\infty}'f(g_v,\Phi_v,\nu_v,1+n)$$

belongs to $I_n(\nu) \subset \mathcal{B}_{n,\overline{\mathbb{Q}}}$ for $g = (g_v)_{v<\infty} \in G(\mathbb{A}_f)$. Hence, there exists a $\phi_f \in \mathcal{B}_{n,\overline{\mathbb{Q}}}$ such that

$$\phi_f = f^\infty(g,\Phi,\nu,1+n)$$

for $g \in G(\mathbb{A}_f)$.

Third, it can seen from the fact that ϕ_k^n (see Proposition 7.43) and Φ_∞ (see Lemma 8.25) has $\mathrm{U}(1)(\mathbb{R})$-weight $-k$ and $\mathrm{Ind}_{B_2(\mathbb{R})^+}^{\mathrm{GL}_2(\mathbb{R})^+}\lambda(2+n,-n)$ is $\mathrm{U}(1)(\mathbb{R})$-multiplicity-free that there is a $c \in \mathbb{C}$ such that for any $g \in G(\mathbb{R})$,

$$\phi_k^n(g) = c\cdot f(g,\Phi_\infty,\nu_\infty,1+n),$$

where $\nu_\infty = (\nu_{1,\infty} = ||^{-1}\mathrm{sgn}^n, \nu_{2,\infty} = 1)$ is the archimedean component of $\nu = (\nu_1,\nu_2)$. Since $\phi_k^n(1) = 2i$ (see Proposition 7.40) and

$$f(1,\Phi_\infty,\nu_\infty,1+n) = (-1)^{\frac{a+b+r+s-k}{2}}\pi^{-1-a-b-r-s}\Gamma(1+a+b+s)$$

(see Lemma 8.26), we get

$$c = (-1)^{\frac{a+b+r+s-k}{2}}2i\pi^{1+a+b+r+s}\Gamma(1+a+b+r+s)^{-1}.$$

□

Corollary 8.9. *There exists a $\phi_f \in \mathcal{B}_{n,\overline{\mathbb{Q}}}$ such that the integral*

$$\int_{H(\mathbb{Q})Z(\mathbb{A})\backslash H(\mathbb{A})} \Xi_{m,k}(\phi_f)(g)dg$$

defined in notation 7.58 is equal to

$$(-1)^{\frac{a+b+r+s-k}{2}} 2i\pi^{1+a+b+r+s}\Gamma(1+a+b+r+s)^{-1} I(\varphi, \Phi, \nu, 1+a+b+r+s),$$

where $\varphi = X^m_{(1,-1,0)}\Psi$ and the choice of Φ and ν is explained in Proposition 8.8.

Proof. This is direct from Proposition 8.8. □

8.3. **The unfolding.** In this subsection, we unfold the zeta integral defined in Subsection 8.1 and factor it into the product of the standard automorphic L-function and a local integral of Whittaker functions.

Definition 8.10. (1) (Global Whittaker functionals) We define a non-zero global Whittaker functional $\Lambda\colon \pi \to \mathbb{C}$ on an irreducible cuspidal automorphic representation π by the following integral: for $\varphi \in \pi$,

$$\Lambda(\varphi) := \int_{U_B(\mathbb{Q})\backslash U_B(\mathbb{A})} \chi^{-1}(u)\varphi(u)du.$$

Here, U_B is the unipotent radical of the upper-triangular Borel subgroup of G and define the character $\chi\colon U_B(\mathbb{Q})\backslash U_B(\mathbb{A}) \to \mathbb{C}^\times$ by $\chi(u) = \psi(\mathrm{Tr}_{E/\mathbb{Q}}(\delta^{-1}u_{23}))$, where ψ is the standard additive character $\psi\colon \mathbb{Q}\backslash\mathbb{A} \to \mathbb{C}^\times$ and u_{23} is the entry of u in the position of the second row and the third column.

(2) (Global Whittaker functions) For a cusp form φ in the space of π, denote by

$$W_\varphi(g) = \Lambda(\pi(g)\varphi) = \int_{U_B(\mathbb{Q})\backslash U_B(\mathbb{A})} \chi^{-1}(u)\varphi(ug)du$$

the global Whittaker function associated to φ.

(3) (Local Whittaker functions) We assume that $\varphi = \otimes'_v \varphi_v$ is a pure tensor in the decomposition $\pi = \otimes'_v \pi_v$ and such that $\Lambda(\varphi) = W_\varphi(1) \neq 0$. Then for each place v of $\mathbb{Q}$, we define a local Whittaker functional $W_{\varphi,v}\colon G(\mathbb{Q}_v) \to \mathbb{C}$ by setting $W_{\varphi,v}(g_v) = \Lambda(\pi(g_v)\varphi)/\Lambda(\varphi)$ for $g_v \in G(\mathbb{Q}_v)$ and $\varphi \in \pi$. We then have

$$W_\varphi(g) = W_\varphi(1)\prod_v W_{\varphi,v}(g_v),$$

where the product runs over all the places v of $\mathbb{Q}$.

Convention 8.11.
- Let $\omega_\pi\colon Z_G(\mathbb{Q})\backslash Z_G(\mathbb{A}) \to \mathbb{C}^\times$ be the central character of π. For later sections, we assume that $\omega_\pi|_{Z(\mathbb{A})} = (\nu_1\nu_2)^{-1}$.
- When φ is clear, we use W_v to denote $W_{\varphi,v}$.

Definition 8.12 (Local zeta integral)**.** Let U_2 be the upper-triangular unipotent subgroup $\{\begin{pmatrix} 1 & * \\ & 1 \end{pmatrix}\}$ of GL_2. For a Whittaker function W_v on $G(\mathbb{Q}_v)$ and a Schwartz-Bruhat function Φ_v on $V_2(\mathbb{Q}_v)$, we define the local zeta integral as

$$I_v(W_v, \Phi_v, \nu_v, s) := \int_{Z(\mathbb{Q}_v)U_2(\mathbb{Q}_v)\backslash H(\mathbb{Q}_v)} f(g_{1,v}, \Phi_v, \nu_v, s)W_v(g_v)dg_v, \tag{8.2}$$

$$= \int_{U_2(\mathbb{Q}_v)\backslash H(\mathbb{Q}_v)} \nu_1(\det(g_{1,v}))\Phi_v((0,1)g_{1,v})W_v(g_v)|\det(g_{1,v})|_v^s dg_v, \tag{8.3}$$

where $g_{1,v}$ is projection of g_v onto $\mathrm{GL}_2(\mathbb{Q}_v)$ and

$$f(g_{1,v}, \Phi_v, \nu_v, s) = \nu_1(\det(g_{1,v}))|\det(g_{1,v})|^s \int_{\mathrm{GL}_1(\mathbb{Q}_v)} \Phi((0,t)g_{1,v})(\nu_1\nu_2)^{-1}(t)|t|_v^{2s}dt.$$

We now recall the definition of the standard automorphic L-function attached to π.

Definition 8.13 (Standard automorphic L-function)**.**

- The Langlands dual group of G is $\widehat{G} = \mathrm{G_m}(\mathbb{C}) \times \mathrm{GL}_3(\mathbb{C})$, with an action of the non-trivial element $\sigma \in \mathrm{Gal}(E/\mathbb{Q})$ given by
$$\sigma\colon (x, g) \in \mathrm{G_m}(\mathbb{C}) \times \mathrm{GL}_3(\mathbb{C}) \mapsto (x \det g, \Phi_3^{-1\,t} g^{-1} \Phi_3) \in \mathrm{G_m}(\mathbb{C}) \times \mathrm{GL}_3(\mathbb{C}),$$
where
$$\Phi_3 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$
The L-group of G is ${}^L G = \widehat{G} \rtimes \mathrm{Gal}(E/\mathbb{Q})$.
- Define $G' = \mathrm{Res}_{E/\mathbb{Q}}(\mathrm{G_m} \times \mathrm{GL}_3)$. The Langlands dual group of G' is $\widehat{G'} = \widehat{G} \times \widehat{G}$ with an action of the non-trivial element $\sigma \in \mathrm{Gal}(E/\mathbb{Q})$ given by
$$\sigma\colon (g, h) \in \widehat{G'} \mapsto (\sigma(h), \sigma(g)) \in \widehat{G'},$$
where $\sigma(g)$ and $\sigma(h)$ is the action of σ on $\widehat{G}$ defined before. Define the L-group of G' as ${}^L G' = \widehat{G'} \rtimes \mathrm{Gal}(E/\mathbb{Q})$.
- It can be seen from the definition that we have the diagonal embedding of Langlands dual groups $\widehat{G} \to \widehat{G'}$, and we can extend the map to the L-groups ${}^L G \to {}^L G'$. We define the standard representation $\mathrm{std}\colon {}^L G' \to \mathrm{GL}_6(\mathbb{C})$ as
$$((x_1, g_1), (x_2, g_2)) \rtimes 1 \mapsto \begin{pmatrix} x_1 g_1 & \\ & x_2 \det(g_2) \Phi_3 {}^t g_2^{-1} \Phi_3^{-1} \end{pmatrix},$$
$$1 \rtimes \sigma \mapsto \begin{pmatrix} & 1_3 \\ 1_3 & \end{pmatrix}.$$
- For unramified π_p, using the Satake transform of Cartier [Car79] or Haines–Rostami [HR10], the well-defined local standard L-function $L_p(s, \pi_p, \mathrm{std})$ is a meromorphic function of $s \in \mathbb{C}$.
- For a Hecke character $\nu\colon \mathbb{Q}^\times \backslash \mathbb{A}^\times \to \mathbb{C}^\times$, we define the twisted standard L-function $L_p(s, \pi_p \times \nu_p, \mathrm{std})$ of $\pi \times \nu$ as the standard L-function of $(\nu \circ \mu)\pi$, where $(\nu \circ \mu)\pi$ is the product of the representation π by the automorphic character $\nu \circ \mu$ of G, where μ is the similitude character of G.

We now state a result regarding twisted base change that can be used to define L-factors at all places.

Proposition 8.14 ([Rog90], §13)**.** *Let π be a cuspidal automorphic representation of G over $\mathbb{Q}$ such that π_∞ is a discrete series. Then there exists a unique automorphic representation τ on $\mathrm{Res}_{E/\mathbb{Q}}(\mathrm{G_m} \times \mathrm{GL}_3)$ that is compatible with π at every place v of $\mathbb{Q}$ such that*

- *v splits in E, or*
- *v is inert in E and π_v is unramified, or*
- *v is ramified in E and π_v has a vector fixed by $G(\mathbb{Z}_p)$.*

Remark 8.15.

- The results in [Rog90] are only for the ordinary unitary group. Results for the similitude setting are also constructed in [Mor10] and [Shi14].
- Since τ is uniquely determined by π, one can use τ to make a well-defined L-factor at *all* places by using the L-group representation in Definition 8.13 and the known local L-parameters of representations of $\mathrm{Res}_{E/\mathbb{Q}}(\mathrm{G_m} \times \mathrm{GL}_3)$.

Proposition 8.16. [GPS84, Bar97] [PS18, Proposition 3.11.] *With the factorizable data and notation as above, we have*
$$I(\varphi, \Phi, \nu, s) = W_\varphi(1) \prod_v I_v(W_v, \Phi_v, \nu_v, s).$$

For all finite places p of $\mathbb{Q}$, the local integral is absolutely convergent for $\mathrm{Re}(s) \gg 0$ and has meromorphic continuation to a rational function of p^{-s}. If the finite place p is not 2, p is unramified in E, π_p and ν_p are unramified, φ is fixed by the maximal compact subgroup $G(\mathbb{Z}_p)$ and Φ_p is the characteristic function of $V_2(\mathbb{Z}_p)$, then

$$I_p(W_p, \Phi_p, \nu_p, s) = L_p(s, \pi_p \times \nu_{1,p}, \mathrm{std}).$$

If we let S be the completement set of the above finite places, then we have

$$\begin{aligned} I(\varphi, \Phi, \nu, s) &= W_\varphi(1) \prod_{p \in S} I_p(W_p, \Phi_p, \nu_p, s) \times I_\infty(W_\infty, \Phi_\infty, \nu_\infty, s) \times \prod_{p \notin S, p < \infty} L_p(s, \pi_p \times \nu_{1,p}, \mathrm{std}) \\ &= W_\varphi(1) \prod_{p \in S} \frac{I_p(W_p, \Phi_p, \nu_p, s)}{L_p(s, \pi_p \times \nu_{1,p}, \mathrm{std})} \times I_\infty(W_\infty, \Phi_\infty, \nu_\infty, s) \times L(s, \pi \times \nu_1, \mathrm{std}). \end{aligned}$$

8.4. **The nonarchimedean local zeta integral.** In this subsection, we recall results about algebraicity of the nonarchimedean local zeta-integral $I_p(W_p, \Phi_p, \nu_p, s)$ and the nonarchimedean local L-factor $L_p(s, \pi_p \times \nu_{1,p}, \mathrm{std})$.

Proposition 8.17 ([PS18], Proposition 7.1)**.** *Suppose that the Schwartz-Bruhat function Φ_p on $\mathbb{Q}_p^2$ and the Whittaker function W_p are both $\overline{\mathbb{Q}}$-valued. Then the local integral $I_p(W_p, \Phi_p, \nu_p, s)$ has a meromorphic continuation to a rational function $\frac{P(X)}{Q(X)}$ of $X = p^{-s}$, where P and Q have algebraic coefficients.*
In particular, the meromorphic continuation of $I_p(W_p, \Phi_p, \nu_p, s)$ to any $s \in \mathbb{Q}$ is $\overline{\mathbb{Q}}$-valued if it is finite.

We now address the hypothesis that W_p is $\overline{\mathbb{Q}}$-valued in the previous proposition. In order to use it in the global setting later, we work in the general setting.

Definition 8.18. Assume that π is a representation of a group G on a $\mathbb{C}$-vector space V. We say that π is defined over $L \subset \mathbb{C}$ if there exists a representation π_0 on a L-vector space V_0 such that $V_0 \otimes_L \mathbb{C}$ is isomorphic to π. We call V_0 a model of V over L.

Proposition 8.19 ([GS18], Proposition 3.3.2)**.** *Suppose that a representation π of a group G is defined over L and let V and V_0 be as in Definition 8.18. Assume that $\Lambda \colon V \to \mathbb{C}$ is a non-zero functional on V that has a left transformation property with respect to a subgroup H and a character $\psi \colon H \to L^\times$ that characterizes it uniquely up an element of $\mathbb{C}^\times$. (For instance, Λ can be a Whittaker functional.) Then there exists a functional $\Lambda_0 \colon V \to \mathbb{C}$ with $\mathrm{Im}(\Lambda_0|_{V_0}) \subseteq L$.*

Notation 8.20. Let $\Lambda : \pi_p \to \mathbb{C}$ be a fixed Whittaker functional of π_p. If π_p is defined over a number field, we can assume that Λ is $\overline{\mathbb{Q}}$-valued by Proposition 8.19. For any $\varphi \in \pi_p$, we define the Whittaker function W_φ by

$$W_\varphi(g) = \Lambda(g\varphi),$$

for any $g \in G(\mathbb{Q}_p)$. (Note that this is slightly different from Definition 8.10, which we use since we will vary our choice of φ.)

Proposition 8.21. *Let π_p be a representation of $G(\mathbb{Q}_p)$ defined over a number field L. For any $\overline{\mathbb{Q}}$-valued Schwartz-Bruhat function Φ_p on $V_2(\mathbb{Q}_p)$ and φ_p in a model of π_p over L such that $\Lambda(\varphi_p) \neq 0$, there exists a function η in the Hecke algebra $C_c^\infty(G(\mathbb{Q}_p), \mathbb{Q})$ such that*

$$I_p(W_{\pi_p(\eta)\varphi_p}, \Phi_p, \nu_p, s) \in \overline{\mathbb{Q}}^\times,$$

for any $s \in \mathbb{Q}$.

Proof. The algebraicity follows from Proposition 8.17 and the nonvanishing is from [PS18, Lemma 7.4]. □

Definition 8.22. For any finite place p, we could also define the local L-factor $L_p(s, \pi_p \times \nu_{1,p}, \mathrm{std})$ to be the greatest common divisor of the zeta integrals $I_p(W_p, \Phi_p, \nu_p, s)$ as Φ_p and W_p vary, whose existence has been proved in [Bar97].

The following corollary is a direct consequence of Proposition 8.21.

Corollary 8.23. *The local L-factor* $L_p(s, \pi_p \times \nu_{1,p}, \mathrm{std})$ *is equal to*

$$L_p(s, \pi_p \times \nu_{1,p}, \mathrm{std}) = \frac{1}{Q(p^{-s})},$$

where $Q(X)$ *is a polynomial with* $\overline{\mathbb{Q}}$*-coefficients and constant* 1*. Hence, for any* $s \in \mathbb{Q}$*, we have*

$$L_p(s, \pi_p \times \nu_{1,p}, \mathrm{std}) \in \overline{\mathbb{Q}}^\times.$$

8.5. **The archimedean local zeta integral.** In this subsection, we compute the archimedean zeta integral for some "nice" test vectors.

Convention 8.24.

- Let $\nu_{1,\infty}, \nu_{2,\infty} \colon \mathbb{R}^\times \to \mathbb{C}^\times$ be the characters defined by
$$\nu_{1,\infty}(x) = |x|^{-n} \operatorname{sgn}(x)^n,$$
$$\nu_{2,\infty}(x) = 1,$$
and let $\nu_\infty = (\nu_{1,\infty}, \nu_{2,\infty})$.
- For $(x, y) \in V_2(\mathbb{R})$, we let

$$\Phi_\infty(x, y) := (ix - y)^{\frac{n-k}{2}} (ix + y)^{\frac{n+k}{2}} e^{-\pi(x^2+y^2)}. \tag{8.4}$$

- Let π_∞ be a discrete series representation of $G(\mathbb{R})^+$ with Blattner parameter
$$(1 + s - r + b, -1 - a + s - r, s - r).$$
- Let $\tilde{\pi}_\infty$ be the contragredient of π_∞, which is a discrete series representation with Blattner parameter $\mu = (1 + a + r - s, -1 - b + r - s, r - s)$.
- Let $\Psi_\infty \in \tilde{\pi}_\infty$ be a vector of weight $(-1 - b + r - s, 1 + a + r - s, r - s; b + 2s - r - 2)$, which is the lowest weight vector of the minimal K_G-type $\tau_{(1+a+r-s,-1-b+r+s,r-s)}$ of $\tilde{\pi}_\infty$.

Lemma 8.25. *The function* $\Phi_\infty(x, y)$ *has* $\mathrm{U}(1)(\mathbb{R})$*-weight* $-k$*.*

Proof. For any $g = \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \in \mathrm{U}(1)(\mathbb{R}) \hookrightarrow \mathrm{GL}_2(\mathbb{R})$, we have

$$(x, y)g = (ax - by, bx + ay).$$

We can see from this that:

$$\Phi_\infty((x, y) \begin{pmatrix} a & b \\ -b & a \end{pmatrix}) = (a + ib)^{-k} \Phi_\infty(x, y).$$

Hence, the function $\Phi_\infty(x, y)$ has $\mathrm{U}(1)(\mathbb{R})$-weight $-k$. □

Lemma 8.26. *For any* $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})$*, the section* $f(g, \Phi_\infty, \nu_\infty, z)$ *is equal to*

$$\operatorname{sgn}(\det(g))^n |\det(g)|^{z-n} (ic - d)^{\frac{n-k}{2}} (ic + d)^{\frac{n+k}{2}} \pi^{-z} (c^2 + d^2)^{-z} \Gamma(z)$$

for $z \in \mathbb{C}$*.*

Proof. It follows from the identity $(0, t) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = (ct, dt)$ that

$$\begin{aligned} \Phi_\infty(ct, dt) &= (ict - dt)^{\frac{n-k}{2}} (ict + dt)^{\frac{n+k}{2}} e^{-\pi(c^2+d^2)t^2} \\ &= (ic - d)^{\frac{n-k}{2}} (ic + d)^{\frac{n+k}{2}} t^n e^{-\pi(c^2+d^2)t^2}, \end{aligned}$$

so we have

$$\begin{aligned} f(g, \Phi_\infty, \nu_\infty, z) &= \operatorname{sgn}(\det(g))^n |\det(g)|^{z-n} (ic - d)^{\frac{n-k}{2}} (ic + d)^{\frac{n+k}{2}} \int_{\mathbb{R}^\times} e^{-\pi(c^2+d^2)t^2} t^n |t|^{2z-n} \operatorname{sgn}(t)^n \frac{dt}{t} \\ &= 2 \operatorname{sgn}(\det(g))^n |\det(g)|^{z-n} (ic - d)^{\frac{n-k}{2}} (ic + d)^{\frac{n+k}{2}} \int_0^\infty e^{-\pi(c^2+d^2)t^2} t^{2z} \frac{dt}{t}. \end{aligned}$$

It follows from a change of variables that

$$2\int_0^\infty e^{-\pi(c^2+d^2)t^2}t^{2z}\frac{dt}{t}=\left(\frac{1}{\pi(c^2+d^2)}\right)^z\Gamma(z),$$

where $\Gamma(z)$ is the Gamma function. So we get

$$f(g,\Phi_\infty,\nu_\infty,z)=\mathrm{sgn}(\det(g))^n|\det(g)|^{z-n}(ic-d)^{\frac{n-k}{2}}(ic+d)^{\frac{n+k}{2}}\pi^{-z}(c^2+d^2)^{-z}\Gamma(z).$$

□

Notation 8.27.

- If we use the convention in [KO95, page 965], we could write the Blattner parameter of $\tilde{\pi}_\infty$ as

$$\mu=(1+a,-1-b).$$

- Let $d=2+a+b$.
- It follows from the Iwasawa decomposition $G(\mathbb{R})=U_B(\mathbb{R})T_B(\mathbb{R})K_G$ that the Whittaker function with a fixed K_G-type is determined by its value on

$$T_B(\mathbb{R})=\begin{pmatrix}t&0&0\\0&1&0\\0&0&t^{-1}\end{pmatrix},$$

  for all $t\in\mathbb{R}^\times$.
- Recall that we have uniqueness of Whittaker models for $G(\mathbb{R})$ (see [KO95, §3.2]). Let $C_m^{(1+a,-1-b)}$ be the Whittaker function defined on $\mathbb{R}^\times$ in [KO95, P974] for $0\leq m\leq d$, which is associated to the vector ν_{d-m} in the K_G-type τ_μ (see subsubsection 3.2.2).

Definition 8.28. For an integer l, we define the classical Whittaker function $W_{0,l}$ on $\mathbb{R}^\times$ by

$$W_{0,l}(x)=\frac{e^{-\frac{1}{2}x}}{\Gamma(\frac{1}{2}+|l|)}\int_0^\infty t^{-\frac{1}{2}+|l|}(1+\frac{t}{x})^{-\frac{1}{2}+|l|}e^{-t}dt. \tag{8.5}$$

Remark 8.29. We have the following identity:

$$W_{0,l}(x)=\pi^{-\frac{1}{2}}x^{1/2}K_l(\frac{x}{2}),$$

where $K_l(\cdot)$ is the modified K-Bessel function.

Lemma 8.30. *The function $C_m^{(1+a,-1-b)}$ is related to a classical Whittaker function by*

$$C_m^{(1+a,-1-b)}(t)=(-i)^{2+a+b-m}t^{a+b+\frac{7}{2}}W_{0,m-(1+a)}(8\sqrt{2}\pi D^{-\frac{3}{4}}t),t\in\mathbb{R}^\times$$

Proof. In this section, we use the Hermitian form J to define the unitary group, but in [KO95] the Hermitian form J' is used. The corresponding transform is

$$g\in\mathrm{GU}(J')(\mathbb{R})\mapsto CgC^{-1}\in\mathrm{GU}(J)(\mathbb{R}),$$

where $C=\begin{pmatrix}\frac{D^{\frac{1}{4}}}{\sqrt{2}}&0&\frac{D^{\frac{1}{4}}}{\sqrt{2}}\\0&1&0\\-\frac{D^{\frac{1}{4}}}{\sqrt{-2}}&0&\frac{D^{\frac{1}{4}}}{\sqrt{-2}}\end{pmatrix}$.

Recall that in the setup of [KO95], $E_{2,+} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$ and $E_{2,-} = \begin{pmatrix} 0 & -i & 0 \\ -i & 0 & i \\ 0 & -i & 0 \end{pmatrix}$. So in our setting, we have

$$E_{2,+} = C \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} C^{-1} = \begin{pmatrix} 0 & -\sqrt{2}D^{\frac{1}{4}} & 0 \\ 0 & 0 & -\sqrt{-2}D^{-\frac{1}{4}} \\ 0 & 0 & 0 \end{pmatrix},$$

$$E_{2,-} = C \begin{pmatrix} 0 & -i & 0 \\ -i & 0 & i \\ 0 & -i & 0 \end{pmatrix} C^{-1} = \begin{pmatrix} 0 & -\sqrt{2}D^{\frac{1}{4}} & 0 \\ 0 & 0 & -\sqrt{2}D^{-\frac{1}{4}} \\ 0 & -\sqrt{2}D^{\frac{1}{4}} & 0 \end{pmatrix}.$$

It follows from Definition 8.10 that

$$\chi(E_{2,+}) = \psi(\mathrm{Tr}_{E/\mathbb{Q}}(\delta^{-1}(-\sqrt{-2}D^{-\frac{1}{4}}))) = -4\pi i\sqrt{2}D^{-\frac{3}{4}},$$
$$\chi(E_{2,+}) = \psi(\mathrm{Tr}_{E/\mathbb{Q}}(\delta^{-1}(-\sqrt{2}D^{-\frac{1}{4}}))) = 0.$$

Plugging these into the formulas in [KO95, Theorem (4.5), Page 977], we have

$$\eta_{\pm} = \chi(E_{2,+}) \pm \sqrt{-1}\chi(E_{2,-}) = -4\sqrt{2}\pi i D^{-\frac{3}{4}},$$
$$b = -\eta_+\eta_- = 32\pi^2 D^{-\frac{3}{2}}, \gamma_m = (-i)^{2+a+b-m},$$

and

$$C_m^{(1+a,-1-b)}(t) = (-i)^{2+a+b-m} t^{a+b+\frac{7}{2}} W_{0,m-(1+a)}(8\sqrt{2}\pi D^{-\frac{3}{4}}t).$$

□

Lemma 8.31. *The Mellin transform of the Whittaker function $C_m^{(1+a,-1-b)}$ at $l \in \mathbb{C}$ is*

$$\int_0^\infty C_m^{(1+a,-1-b)}(t)t^l d^\times t = (-i)^{2+a+b-m}\frac{1}{2}\left(\frac{1}{2\sqrt{2}D^{-\frac{3}{4}}}\right)^{l+a+b+\frac{7}{2}}\pi^{-(l+a+b+4)}\Gamma\left(\frac{1}{2}(l+m+b+3)\right)$$
$$\times\, \Gamma\left(\frac{1}{2}(l+b-m+5)+a\right).$$

Proof. It follows from Lemma 8.30 that

$$\int_0^\infty C_m^{(1+a,-1-b)}(t)t^l d^\times t = (-i)^{2+a+b-m}\int_0^\infty t^{l+a+b+\frac{7}{2}} W_{0,m-(1+a)}(8\sqrt{2}\pi D^{-\frac{3}{4}}t) d^\times t.$$

If we change variables by $t \mapsto \frac{t}{8\sqrt{2}\pi D^{-\frac{3}{4}}}$, then this integral is

$$(-i)^{2+a+b-m}\left(\frac{1}{8\sqrt{2}\pi D^{-\frac{3}{4}}}\right)^{l+a+b+\frac{7}{2}}\int_0^\infty t^{l+a+b+\frac{7}{2}} W_{0,m-(1+a)}(t) d^\times t.$$

Using the formula in [KO95, §5.2, P979], we find it is equal to

$$(-i)^{2+a+b-m}\frac{1}{2}\left(\frac{1}{2\sqrt{2}D^{-\frac{3}{4}}}\right)^{l+a+b+\frac{7}{2}}\pi^{-(l+a+b+4)}\Gamma\left(\frac{1}{2}(l+m+b+3)\right)\Gamma\left(\frac{1}{2}(l+b-m+5)+a\right).$$

□

Now we compute the archimedean zeta integral with specific chosen test vectors.

Convention 8.32.

- Let $\Psi_\infty \in \tilde{\pi}_\infty$ be a vector of weight $(-1-b+r-s, 1+a+r-s, r-s; b+2s-r)$, which is the lowest weight vector of the minimal K_G-type τ_μ.
- We write the basis $\nu_0, \nu_1, \ldots, \nu_d$ in the K_G-type μ defined in subsubsection 3.2.2 as $\nu_0^\mu, \nu_1^\mu, \ldots, \nu_d^\mu$.
- We identify Ψ_∞ with the vector ν_0^μ, so $X_{(1,-1,0)}^m \Psi_\infty$ is $m!\nu_m^\mu$.
- By the uniqueness of Whittaker models (see [KO95, §3.2]), we choose W_∞ to be the Whittaker function corresponding to $X_{(1,-1,0)}^m \Psi_\infty$ such that $W_\infty(1) = 1$.

- Because $W_\infty(1) = 1$ and $C_{d-m}^{(1+a,-1-b)}(1) = (-i)^m W_{0,1+b-m}(8\sqrt{2}\pi D^{-\frac{3}{4}})$ (see Lemma 8.30), it can be seen from uniqueness of Whittaker models that if we let

$$C = (-i)^m W_{0,1+b-m}(8\sqrt{2}\pi D^{-\frac{3}{4}}),$$

then we have

$$W_\infty(t) = C^{-1} C_{d-m}^{(1+a,-1-b)}(t), \tag{8.6}$$

for all $t \in \mathbb{R}^\times$[21].
- Let $K_\infty = \mathrm{U}(1)(\mathbb{R})$.

Lemma 8.33. *If $m = 1 + a + r - s$, then the weight of W_∞ is*

$$(a - b + 2(r - s), 0, r - s).$$

In this case, the integral

$$I_1 = \int_{\mathrm{GL}_1(\mathbb{R})} t^{2z-n-2} W_\infty\left(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}\right) dt$$

is equal to

$$C^{-1}(-i)^{1+a-r-s}\left(\frac{1}{2\sqrt{2}D^{-\frac{3}{4}}}\right)^{(2z-r-s+\frac{3}{2})}\pi^{-(2z-r-s+2)}\Gamma(z-r+1+\frac{1}{2}(b-a))\Gamma(z-s+1-\frac{1}{2}(a+b)).$$

Proof. First, we have

$$\begin{aligned} I_1 &= \int_{\mathrm{GL}_1(\mathbb{R})} t^{2z-n-2} W_\infty\left(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}\right) dt \\ &= \int_{\mathbb{R}_{>0}^\times} t^{2z-n-2} W_\infty\left(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}\right) dt + \int_{\mathbb{R}_{<0}^\times} t^{2z-n-2} W_\infty\left(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}\right) dt. \end{aligned}$$

By a change of variables and the fact that W_∞ has weight $(a - b + 2(r - s), 0, r - s)$, we get

$$\int_{\mathbb{R}_{>0}^\times} t^{2z-n-2} W_\infty\left(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}\right) dt = \int_{\mathbb{R}_{<0}^\times} t^{2z-n-2} W_\infty\left(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}\right) dt.$$

Hence, we get

$$I_1 = 2\int_{\mathbb{R}_{>0}^\times} t^{2z-n-2} W_\infty\left(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}\right) dt,$$

which is a Mellin transform of a Whittaker function.

Second, it follows from equation (8.6) that

$$I_1 = 2C^{-1}\int_0^\infty t^{2z-n-2} C_{1+b-r+s}^{(1+a,-1-b)}(t) d^\times t.$$

Finally, using the formula in Lemma 8.31, we have

$$I_1 = C^{-1}(-i)^{1+a-r-s}\left(\frac{1}{2\sqrt{2}D^{-\frac{3}{4}}}\right)^{(2z-r-s+\frac{3}{2})}\pi^{-(2z-r-s+2)}\Gamma(z-r+1+\frac{1}{2}(b-a))\Gamma(z-s+1-\frac{1}{2}(a+b)).$$

□

Proposition 8.34. *(1) If $m \neq 1 + a + r - s$, then the archimedean local zeta integral*

$$I_\infty(W_\infty, \Phi_\infty, \nu_\infty, z) = \int_{Z(\mathbb{R})U_2(\mathbb{R})\backslash H(\mathbb{R})} f(g_{1,\infty}, \Phi_\infty, \nu_\infty, z) W_\infty(g_\infty) dg_\infty$$

is equal to 0.

[21] Here we view W_∞ as a function on $\mathbb{R}^\times$.

(2) *If* $m = 1+a+r-s$, *then the zeta integral* $I_\infty(W_\infty, \Phi_\infty, \nu_\infty, 1+a+b+r+s)$ *is only non-zero when the* K_∞*-type of* Φ_∞ *is* $-(a+r-b-s)$. *In this case,* $I_\infty(W_\infty, \Phi_\infty, \nu_\infty, 1+a+b+r+s)$ *is equal to*

$$(-1)^a \pi^{-(3a+3b+2r+2s+5)} W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1} \left(\frac{1}{2\sqrt{2}D^{-\frac{3}{4}}}\right)^{2a+2b+r+s+\frac{7}{2}}$$
$$\times \Gamma(a+b+r+s+1)\Gamma\left(\frac{1}{2}a+\frac{3}{2}b+s+2\right)\Gamma\left(\frac{1}{2}a+\frac{1}{2}b+r+2\right),$$

where $W_{0,b+s-a-r}$ *is the classical Whittaker function on* $\mathbb{R}$ *defined in Definition 8.28 and* Γ *is the Gamma function.*

Proof. Recall that $H = \mathrm{GL}_2 \boxtimes E^\times$ and $Z(\mathbb{R}) = \left\{ \left(\begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}, z \right) | z \in \mathbb{R} \right\}$, so it follows from the Iwasawa decomposition that

$$Z(\mathbb{R})U_2(\mathbb{R})\backslash H(\mathbb{R})/K_\infty \cong \mathrm{GL}_1(\mathbb{R}) \times \mathbb{S}^1 = \left\{ \left(\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}, c \right) | t \in \mathrm{GL}_1(\mathbb{R}), c \in \mathbb{S}^1 \right\},$$

where $\mathbb{S}^1$ is the unit circle $\{z \in \mathbb{C} | z\bar{z} = 1\}$. There is an embedding $\iota\colon \mathrm{GL}_1(\mathbb{R}) \times \mathbb{S}^1 \hookrightarrow G(\mathbb{R})$ that maps $\left(\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}, c \right)$ to $\begin{pmatrix} t & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}$.

Since the K_∞-weight of Φ_∞ is $-k$ and the K_∞-weight of W_∞ is $m-1-b$, we have

$$I_\infty(W_\infty, \Phi_\infty, \nu_\infty, z) = 0$$

unless $k = m-1-b$. In this case, the action of K_∞ on $f(g_{1,\infty}, \Phi_\infty, \nu_\infty, z)W_\infty(g_\infty)$ is trivial, so

$$I_\infty(W_\infty, \Phi_\infty, \nu_\infty, z) \tag{8.7}$$
$$= \int_{Z(\mathbb{R})U_2(\mathbb{R})\backslash H(\mathbb{R})} f(g_{1,\infty}, \Phi_\infty, \nu_\infty, z) W_\infty(g_\infty) dg_\infty \tag{8.8}$$
$$= \pi^{-z}\Gamma(z) \int_{\mathrm{GL}_1(\mathbb{R})\times\mathbb{S}^1} (-t^{-1})^{\frac{n-k}{2}} (t^{-1})^{\frac{n+k}{2}} t^{2z} W_\infty(\iota(\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}, c)) |t|^{-2} dt dc \tag{8.9}$$
$$= \pi^{-z}\Gamma(z)(-1)^{\frac{n-k}{2}} \int_{\mathrm{GL}_1(\mathbb{R})\times\mathbb{S}^1} t^{2z-n-2} W_\infty(\begin{pmatrix} t & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}) dt dc. \tag{8.10}$$
$$= \pi^{-z}\Gamma(z)(-1)^{\frac{n-k}{2}} \int_{\mathrm{GL}_1(\mathbb{R})} t^{2z-n-2} W_\infty(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}) dt, \times \int_{\mathbb{S}^1} c^{1+a-m+r-s} dc \tag{8.11}$$

where the last identity holds since

$$W_\infty(\begin{pmatrix} t & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}) = W_\infty(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{pmatrix}) = c^{1+a-m+r-s} W_\infty(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}).$$

Now we set $I_1 = \int_{\mathrm{GL}_1(\mathbb{R})} t^{2z-n-2} W_\infty(\begin{pmatrix} t & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}) dt$ and $I_2 = \int_{\mathbb{S}^1} c^{1+a-m+r-s} dc$, and will compute the integral I_1 and I_2, respectively.

The integral I_2 is computed as follows. We have

$$\begin{aligned} I_2 &= \int_{\mathbb{S}^1} c^{1+a-m+r-s} dc \\ &= \int_0^1 e^{2\pi i(1+a-m+r-s)x} dx \\ &= \begin{cases} 1 & \text{If } m = 1+a+r-s, \\ 0 & \text{If } m \neq 1+a+r-s. \end{cases} \end{aligned}$$

When $m = 1+a+r-s$, the integral I_1 is computed in Lemma 8.33.

Finally, plugging the formulas for I_1 and I_2 into equation (8.11), we have:

- if $m \neq 1+a+r-s$, then
$$I_\infty(W_\infty, \Phi_\infty, \nu_\infty, z) = 0;$$
- if $m = 1+a+r-s$, then
$$\begin{aligned} &I_\infty(W_\infty, \Phi_\infty, \nu_\infty, z) \\ =&(-1)^{a+r} C^{-1} (-i)^{1+a-r-s} \left(\frac{1}{2\sqrt{2} D^{-\frac{3}{4}}}\right)^{(2z-r-s+\frac{3}{2})} \pi^{-(3z-r-s+2)} \\ &\times \Gamma(z-r+1+\frac{1}{2}(b-a))\Gamma(z-s+1-\frac{1}{2}(a+b))\Gamma(z). \end{aligned}$$
Plugging $z = 1+a+b+r+s$ and $C = (-i)^{1+a+r-s} W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})$ into the formula, we have
$$\begin{aligned} &I_\infty(W_\infty, \Phi_\infty, \nu_\infty, 1+a+b+r+s) \\ =& (-1)^a \pi^{-(3a+3b+2r+2s+5)} W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1} \left(\frac{1}{2\sqrt{2} D^{-\frac{3}{4}}}\right)^{2a+2b+r+s+\frac{7}{2}} \\ &\times \Gamma(a+b+r+s+1)\Gamma(\frac{1}{2}a+\frac{3}{2}b+s+2)\Gamma(\frac{1}{2}a+\frac{1}{2}b+r+2). \end{aligned}$$

□

Remark 8.35. (1) If we let $m = 1+a+r-s = 1+a+2b-2j+r+s$ as in Proposition 7.59, then $j = b+s$. So we can see from Corollary 8.9 only one term in the summation of equation (7.20) is left. So the pairing $\langle \omega_\Psi, [\rho] \rangle_B$ is Eulerian by our previous discussion of zeta integral.

(2) The integer $j = b+s$ satisfies the condition $A \leq j \leq B$ in Proposition 7.59.

9. The periods

9.1. **The motivic periods.** In this subsection, we compute the pairing $\langle \Omega, \tilde{v}_D \rangle_B$ defined in Lemma 7.11.

Recall that $V = V^{a,b}\{r,s\}$ and we have the following exact sequence:

$$0 \to F^0 M_{dR}(\pi_f, V(2))_{\mathbb{R}} \to M_B(\pi_f, V(2))^-_{\mathbb{R}}(-1) \to \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}}) \to 0. \tag{9.1}$$

The Beilinson $E(\pi_f)$-structure of $\det_{E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}} \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}})$ is defined as

$$\mathcal{B}(\pi_f, V(2)) = \det_{E(\pi_f)} F^0 M_{dR}(\pi_f, V(2))^* \otimes_{E(\pi_f)} \det_{E(\pi_f)} M_B(\pi_f, V(2))^-_{\mathbb{R}}(-1).$$

Let $\delta(\pi_f, V(2)) \in (E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{C})^\times$ be the determinant of the isomorphism

$$I_\infty \colon M_B(\pi_f, V(2))_{\mathbb{C}} \to M_{dR}(\pi_f, V(2))_{\mathbb{C}}$$

computed using the basis defined over $E(\pi_f)$ on both sides. The Deligne $E(\pi_f)$-structure of $\det_{E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}} \mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(\mathbb{R}(0), M_B(\pi_f, V(2))_{\mathbb{R}})$ is

$$\mathcal{D}(\pi_f, V(2)) := (2\pi i)^3 \delta(\pi_f, V(2))^{-1} \mathcal{B}(\pi_f, V(2)).$$

Let v_D be a generator of $\mathcal{D}(\pi_f, V(2))$, $\tilde{v}_D$ be a lifting of v_D via the last arrow in the exact sequence (9.1) and $\Omega = \frac{1}{2}(\omega_\Psi + \overline{\omega_\Psi}) \cdot \mathbf{1}$, where ω_Ψ is constructed in Subsection 7.2 and $\mathbf{1}$ the multiplicative identities of $E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{C}$.

It can be seen from Remark 3.9 and Proposition 3.41 that we have the following Hodge decompositions:

$$\begin{aligned} & M_B(\pi_f, V(2))_{\mathbb{C}} \\ \cong & M_B^{-r,-2-a-b-s} \oplus M_B^{-1-a-r,-1-b-s} \oplus M_B^{-2-a-b-r,-s} \\ \oplus & M_B^{-2-a-b-s,-r} \oplus M_B^{-1-b-s,-1-a-r} \oplus M_B^{-s,-2-a-b-r}, \end{aligned}$$

and

$$\begin{aligned} & M_B(\tilde{\pi}_f|\mu|^{-2}, D)_{\mathbb{C}} \\ \cong & M_B^{a+b+r+2,s} \oplus M_B^{a+r+1,b+s+1} \oplus M_B^{r,a+b+s+2} \\ \oplus & M_B^{s,a+b+r+2} \oplus M_B^{b+s+1,a+r+1} \oplus M_B^{a+b+s+2,r}, \end{aligned}$$

from which we can see that $M_B(\pi_f, V(2))^-_{\mathbb{R}}(-1)$ is 3-dimensional. By the condition of the parameters stated in (7.3), we get that $F^0 M_{dR}(\pi_f, V(2))_{\mathbb{R}}$ is 2-dimensional.

Notation 9.1. We use $M_{dR}^{s,t}$ to denote $I_\infty^{-1}(M_B^{s,t})$ for $s, t \in \mathbb{Z}$.

We now recall the definition of Deligne periods as in [Del79, §1.7].

Definition 9.2 (Deligne period)**.** Assume that $a + r \neq b + s$[22] and let

$$\begin{aligned} & F^+(\tilde{\pi}_f|\mu|^{-2}, D) \\ = & F^-(\tilde{\pi}_f|\mu|^{-2}, D) \\ = & M_{dR}^{a+b+r+2,s} \oplus M_{dR}^{a+b+s+2,r} \oplus M_{dR}^{A,B}, \end{aligned}$$

where $A = \max\{a+r+1, b+s+1\}$, $B = \min\{a+r+1, b+s+1\}$. It is shown in [Del79, §1.7] that the map

$$I_\infty^{\pm} \colon M_B(\tilde{\pi}_f|\mu|^{-2}, D)^{\pm}_{\mathbb{C}} \longrightarrow M_{dR}(\tilde{\pi}_f|\mu|^{-2}, D)_{\mathbb{C}} / F^{\pm}(\tilde{\pi}_f|\mu|^{-2}, D)$$

is an isomorphism.

The Deligne period $c^{\pm}(\tilde{\pi}_f|\mu|^{-2}, D)$ is defined as $\det(I_\infty^{\pm}) \in (E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{C})^\times$.

Remark 9.3. We can define the Deligne period $c^{\pm}(\pi_f, V(2))$ similarly. This is left to the reader.

We have the following commutative diagram:

$$\begin{array}{ccc} M_B(\pi_f, V(2))_{\mathbb{C}} \otimes M_B(\tilde{\pi}_f|\mu|^{-2}, D)_{\mathbb{C}} & \xrightarrow{\langle\cdot,\cdot\rangle_B} & E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{C} \\ \downarrow I_\infty & & \downarrow I_\infty \\ M_{dR}(\pi_f, V(2))_{\mathbb{C}} \otimes M_{dR}(\tilde{\pi}_f|\mu|^{-2}, D)_{\mathbb{C}} & \xrightarrow{\langle\cdot,\cdot\rangle_{dR}} & E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{C} \end{array} \tag{9.2}$$

where the vertical arrows are isomorphisms.

Construction 9.4. We contruct a "nice" basis of the spaces in the commutative diagram (9.2) as follows.

- Let v_1, v_2, v_3 be an $E(\pi_f)$-basis of $M_B(\pi_f, V(2))^-_{\mathbb{R}}(-1)$ such that $v_1 \in M_B^{-r,-2-a-b-s}$ and $v_2 \in M_B^{-s,-2-a-b-r}$. Then $2\pi i v_1, 2\pi i v_2, 2\pi i v_3$ is an $E(\pi_f)$-basis of $M_B(\pi_f, V(2))^+$.

[22] This assumption implies that condition that the infinite Frobenius F_∞ acts on the component with Hodge type (n, n) for all $n \in \mathbb{Z}$ by a scalar 1 or -1(see [Del79, §1.7]) is satisfied.

- Let w_1, w_2, w_3 be an $E(\pi_f)$-basis of $M_B(\tilde{\pi}_f|\mu|^{-2}, D)^+_{\mathbb{C}}$ such that
$$\langle 2\pi i v_i, w_j\rangle_B = \delta_{ij},$$
for $i, j \in \{1, 2, 3\}$.
- Let θ_1, θ_2 be the $E(\pi_f)$-basis of $F^0 M_{dR}(\pi_f, V(2))$ such that the image of θ_i for $i = 1, 2$ under the map
$$F^0 M_{dR}(\pi_f, V(2))_{\mathbb{R}} \to M_B(\pi_f, V(2))^-_{\mathbb{R}}(-1)$$
is v_i.
- Let $\theta_1', \theta_2' \in M_{dR}(\tilde{\pi}_f|\mu|^{-2}, D)_{\mathbb{C}}$ be the dual vectors of θ_1, θ_2, which means that
$$\langle \theta_1, \theta_1'\rangle_{dR} = \langle \theta_2, \theta_2'\rangle_{dR} = 1_{dR},$$
$$\langle \theta_1, \theta_2'\rangle_{dR} = \langle \theta_2, \theta_1'\rangle_{dR} = 0.$$
It follows that $\theta_1' \in M_{dR}^{r,a+b+s+2}$ and $\theta_2' \in M_{dR}^{s,a+b+r+2}$. If we let $\Omega' = I_\infty(\Omega)$, then it follows from the fact that
$$\Omega \in M_B^{a+r+1,b+s+1} \oplus M_B^{b+s+1,a+r+1}.$$
that
$$\Omega' \in M_{dR}^{a+r+1,b+s+1} \oplus M_{dR}^{b+s+1,a+r+1}.$$
Hence, $\Omega', \theta_1', \theta_2'$ is a basis of $M_{dR}(\tilde{\pi}_f|\mu|^{-2}, D)_{\mathbb{C}}/F^{\pm}(\tilde{\pi}_f|\mu|^{-2}, D)$.
- It follows from $\theta_1 \in M_{dR}^{-r,-2-a-b-s}$, $\theta_2 \in M_{dR}^{-s,-2-a-b-r}$ and
$$\Omega' \in M_{dR}^{a+r+1,b+s+1} \oplus M_{dR}^{b+s+1,a+r+1},$$
that
$$\langle \Omega', \theta_1\rangle_{dR} = \langle \Omega', \theta_2\rangle_{dR} = 0.$$

Lemma 9.5. *Using the basis w_1, w_2, w_3 of $M_B(\tilde{\pi}_f|\mu|^{-2}, D)^+_{\mathbb{C}}$ and basis $\Omega', \theta_1', \theta_2'$ of*
$$M_{dR}(\tilde{\pi}_f|\mu|^{-2}, D)_{\mathbb{C}}/F^{\pm}(\tilde{\pi}_f|\mu|^{-2}, D),$$
the inverse matrix of
$$I_\infty^+ \colon M_B(\tilde{\pi}_f|\mu|^{-2}, D)^+_{\mathbb{C}} \longrightarrow M_{dR}(\tilde{\pi}_f|\mu|^{-2}, D)_{\mathbb{C}}/F^+(\tilde{\pi}_f|\mu|^{-2}, D)$$
can be written as
$$c^+(\tilde{\pi}_f|\mu|^{-2}, D)^{-1} \begin{pmatrix} C & * & * \\ B & * & * \\ (2\pi i)^{-2} & * & * \end{pmatrix}.$$

Proof. Using the basis w_1, w_2, w_3 and $\Omega', \theta_1', \theta_2'$, if we write the map
$$I_\infty^+ \colon M_B(\tilde{\pi}_f|\mu|^{-2}, D)^+_{\mathbb{C}} \longrightarrow M_{dR}(\tilde{\pi}_f|\mu|^{-2}, D)_{\mathbb{C}}/F^+(\tilde{\pi}_f|\mu|^{-2}, D)$$
explicitly as
$$I_\infty^+ \colon (w_1, w_2, w_3) \mapsto (\Omega', \theta_1', \theta_2') \begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \\ \gamma_1 & \gamma_2 & \gamma_3 \end{pmatrix},$$
then $(I_\infty^+)^{-1}$ can be written as
$$(I_\infty^+)^{-1} \colon (\Omega', \theta_1', \theta_2') \mapsto (w_1, w_2, w_3)(\det(I_\infty^+))^{-1} \begin{pmatrix} C & * & * \\ B & * & * \\ A & * & * \end{pmatrix},$$
where $A = \det \begin{pmatrix} \beta_1 & \beta_2 \\ \gamma_1 & \gamma_2 \end{pmatrix} = \beta_1\gamma_2 - \beta_2\gamma_1$ and $\det(I_\infty^+) = c^+(\tilde{\pi}_f|\mu|^{-2}, D)$.

Now we compute A. We have
$$I_\infty^+(w_1) = \alpha_1\Omega' + \beta_1\theta_1' + \gamma_1\theta_2',$$
$$I_\infty^+(w_2) = \alpha_2\Omega' + \beta_2\theta_1' + \gamma_2\theta_2'.$$

By the definition of w_i and commutativity of the diagram (9.2), we have the following identities:

$$\begin{aligned}\beta_1 &= \langle \theta_1, I_\infty^+(w_1)\rangle_{dR} = \langle v_1, w_1\rangle_B = (2\pi i)^{-1},\\ \gamma_1 &= \langle \theta_2, I_\infty^+(w_1)\rangle_{dR} = \langle v_2, w_1\rangle_B = 0,\\ \beta_2 &= \langle \theta_1, I_\infty^+(w_2)\rangle_{dR} = \langle v_1, w_2\rangle_B = 0,\\ \gamma_2 &= \langle \theta_2, I_\infty^+(w_2)\rangle_{dR} = \langle v_2, w_2\rangle_B = (2\pi i)^{-1}.\end{aligned}$$

So we have $\begin{pmatrix}\beta_1 & \beta_2\\ \gamma_1 & \gamma_2\end{pmatrix} = \begin{pmatrix}(2\pi i)^{-1} & 0\\ 0 & (2\pi i)^{-1}\end{pmatrix}$ and $A = (2\pi i)^{-2}$. □

Notation 9.6. Recall that we had introduced the following notation: if μ and μ' are two elements of $E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{C}$, we write $\mu \sim \mu'$ if there exists $\lambda \in E(\pi_f)^\times$ such that $\mu = \lambda\mu'$.

Proposition 9.7. *Assuming that $a + r \neq b + s$, we have the following equality:*

$$\langle \Omega, \tilde{v}_D\rangle_B \sim c^-(\pi_f, V(2))^{-1}.$$

Proof. First, it can be seen from the construction that v_3 is a generator of the Beilinson $E(\pi_f)$-structure $\mathcal{B}(\pi_f, V(2))$. Therefore, we have

$$\langle \Omega, \tilde{v}_D\rangle_B \sim (2\pi i)^3\delta(\pi_f, V(2))^{-1}\langle \Omega, v_3\rangle_B.$$

Second, we compute $\langle \Omega, v_3\rangle_B$. It can seen from Lemma 9.5 that

$$\begin{aligned}\langle \Omega, v_3\rangle_B &= c^+(\tilde{\pi}_f|\mu|^{-2}, D)^{-1}\langle Cw_1 + Bw_2 + (2\pi i)^{-2}w_3, v_3\rangle_B\\ &= c^+(\tilde{\pi}_f|\mu|^{-2}, D)^{-1}(2\pi i)^{-2}\langle w_3, v_3\rangle_B\\ &= c^+(\tilde{\pi}_f|\mu|^{-2}, D)^{-1}(2\pi i)^{-3}.\end{aligned}$$

Finally, we can see that

$$\begin{aligned}\langle \Omega, \tilde{v}_D\rangle_B &\sim (2\pi i)^3\delta(\pi_f, V(2))^{-1}\langle \Omega, v_3\rangle_B\\ &= c^+(\tilde{\pi}_f|\mu|^{-2}, D)^{-1}\delta(\pi_f, V(2))^{-1}.\end{aligned}$$

It follows from [Del79, (5.1.7)] that

$$c^+(\tilde{\pi}_f|\mu|^{-2}, D)^{-1}\delta(\pi_f, V(2))^{-1} \sim c^-(\pi_f, V(2))^{-1}.$$

So we conclude that

$$\langle \Omega, \tilde{v}_d\rangle_B \sim c^-(\pi_f, V(2))^{-1}.$$

□

Remark 9.8. The condition $a+r \neq b+s$ guarantees that the weight of $M_B(\pi_f, V(2))$ is ≤ -3.

9.2. **The Whittaker period.** In this subsection, we define a special automorphic period called the Whittaker period.

For any cuspidal automorphic representation $\pi = \pi_f \otimes \pi_\infty$ of $G(\mathbb{A})$ such that π_∞ is a discrete series representation of $G(\mathbb{R})^+$, π_f is defined over a number field $E(\pi_f) \subseteq \overline{\mathbb{Q}}$ by Theorem 3.37. Hence, for the vector space V_{π_f} under π_f, there is a model $V_{\overline{\mathbb{Q}}}$ of V_{π_f} such that

$$V_{\pi_f} = V_{\overline{\mathbb{Q}}}\otimes_{\overline{\mathbb{Q}}}\mathbb{C}.$$

Therefore, by Proposition 8.19, for the Whittaker functional Λ associated to π_f by Definition 8.10, there exists a $\Lambda_f\colon V_{\pi_f} \to \mathbb{C}$ sending $V_{\overline{\mathbb{Q}}}$ to $\overline{\mathbb{Q}}$, which is unique up to multiplication by an element of $\overline{\mathbb{Q}}^\times$.

Definition 9.9 (Whittaker period)**.** By uniqueness of Whittaker models, there exists a $c \in \mathbb{C}^\times$ such that

$$\Lambda = c\cdot\Lambda_f.$$

The *Whittaker period* $W(\pi) \in \mathbb{C}^\times/\mathbb{Q}^\times$ is defined to be the class of c in $\mathbb{C}^\times/\mathbb{Q}^\times$.

10. The main results on connection to L-values

Notation 10.1. We use the same notations as Section 8.

10.1. **Connection to automorphic L-functions.** In this subsection, we state the main result in terms of automorphic L-functions.

Recall that we let $n = a + b + r + s$. Let $\pi = \pi_f \otimes \pi_\infty$ be a *generic* cuspidal automorphic representation of $G(\mathbb{A})$ whose central character is

$$\begin{aligned}\omega_\pi \colon Z_G(\mathbb{Q})\backslash Z_G(\mathbb{A}) &\to \mathbb{C}^\times \\ z &\longmapsto |z|^{-n}\nu(z),\end{aligned}$$

where ν is a finite-order Hecke character of sign $(-1)^n$. Moreover, let the archimedean component π_∞ of π be a discrete series with Blattner parameter $(1+s-r+b, -1-a+s-r, s-r)$. The cuspidal automorphic representation π can be viewed as a direct summand of

$$\mathrm{H}^2_{B,!}(\mathrm{Sh}_G, V(2))_{\mathbb{C}} := \mathrm{Im}(\mathrm{H}^2_{\mathrm{dR},c}(\mathrm{Sh}, V(2))_{\mathbb{C}} \to \mathrm{H}^2_{\mathrm{dR}}(\mathrm{Sh}_G, V(2))_{\mathbb{C}},$$

which is the interior cohomology of the Shimura variety of infinite level with coefficients in the local system associated to the representation $V(2)$. Let $E(\pi_f)$ be the rational field of π, which is a number field.

Let $M_B(\pi_f, V(2))$ (see Definition 3.39) be the π_f-isotypic subspace of $\mathrm{H}^2_{B,!}(\mathrm{Sh}_G, V(2))$. Similarly, let $M_{\mathrm{dR}}(\pi_f, V(2))$ (see Definition 3.39) be the π_f-isotypic subspace of

$$\mathrm{H}^2_{\mathrm{dR},!}(\mathrm{Sh}_G, V(2)) := \mathrm{Im}(\mathrm{H}^2_{\mathrm{dR},c}(\mathrm{Sh}, V(2)) \to \mathrm{H}^2_{\mathrm{dR}}(\mathrm{Sh}_G, V(2)).$$

Let $\mathcal{K}(V(2))$ (see equation (7.2)) be the $\mathbb{Q}[G(\mathbb{A}_f)]$-submodule of $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(1, \mathrm{H}^2_{B,!}(\mathrm{Sh}_G, V(2))_{\mathbb{R}})$ generated by the image of $\mathcal{E}is^n_H$ with domain $\mathcal{B}_{n,\mathbb{Q}}$. Let $\mathcal{K}(\pi_f, V(2))$ be the π_f-isotypic subspace of $\mathcal{K}(V(2))$ which is a 1-dimensional $E(\pi_f)$-subspace of the rank one $E(\pi_f)\otimes_{\mathbb{Q}}\mathbb{R}$-module $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(1, M_B(\pi_f, V(2))_{\mathbb{R}})$. In $\mathrm{Ext}^1_{\mathrm{MHS}^+_{\mathbb{R}}}(1, M_B(\pi_f, V(2))_{\mathbb{R}})$, there is another $E(\pi_f)$-subspace $\mathcal{D}(\pi_f, V(2))$ called the Deligne $E(\pi_f)$-structure (see Definition 7.4) coming from the comparison between Betti and de Rham realizations of the motive. The main result in this section gives a measure of the difference between $\mathcal{K}(\pi_f, V(2))$ and $\mathcal{D}(\pi_f, V(2))$ in terms of a non-critical automorphic L-value.

Notation 10.2. We use the notation $|\cdot|_f$ to denote the product over all finite absolute values:

$$\prod_{v<\infty} |\cdot|_v.$$

Lemma 10.3. *Let S be a finite set of places of $\mathbb{Q}$ (excluding $p=2$ and $p=\infty$) such that π_p is unramified for $p \notin S$ and let*

$$L^S(s, \pi|\mu|_f^{-2}, \mathrm{std}) = \prod_{p\notin S} L_p(s, \pi_p|\mu|_p^{-2}, \mathrm{std})$$

be the partial standard L-function of $\pi|\mu|_f^{-1}$. Then we have for $s \in \mathbb{C}$

$$L^S(s, (\tilde{\pi}\times\nu_1)|\mu|_f^{-2}, \mathrm{std}) = L^S(s, \pi, \mathrm{std}).$$

Proof. First, let η be the endomorphism on E_p^3 that maps $(x_1, x_2, x_3) \in E_p^3$ to $(\overline{x_1}, \overline{x_2}, -\overline{x_3})$. It induces an endomorphism on $G(\mathbb{Q}_p)$ and $H(\mathbb{Q}_p) = \mathrm{GL}_2(\mathbb{Q}_p)\boxtimes E_p^\times$ by

$$\eta\colon \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33}\end{pmatrix} \in G(\mathbb{Q}_p) \mapsto \begin{pmatrix} \overline{a_{11}} & \overline{a_{12}} & -\overline{a_{13}} \\ \overline{a_{21}} & \overline{a_{22}} & -\overline{a_{23}} \\ -\overline{a_{31}} & -\overline{a_{32}} & \overline{a_{33}}\end{pmatrix},$$

$$\eta\colon \left(\begin{pmatrix} a & b \\ c & d\end{pmatrix}, z\right) \in \mathrm{GL}_2(\mathbb{Q}_p)\boxtimes E_p^\times \mapsto \left(\begin{pmatrix} a & -b \\ -c & d\end{pmatrix}, \overline{z}\right).$$

Hence, the action of η on $\mathrm{GL}_2(\mathbb{Q}_p)$ is given by conjugating by $t = \begin{pmatrix} 1 & 0 \\ 0 & -1\end{pmatrix}$.

Second, it follows from [PS18, Proposition 7.9] that if we define $\pi_p^\eta(g) = \pi_p(\eta g \eta^{-1})$ and let ω_{π_p} be the central character of π_p, then we have

$$\tilde{\pi}_p \cong \pi_p^\eta \otimes (\omega_{\pi_p}^{-1}|_{Z(\mathbb{Q}_p)} \circ \mu).$$

This shows that

$$\begin{aligned}(\tilde{\pi}_p \times \nu_1)|\mu|_f^{-2} &\cong \pi_p^\eta \otimes (\nu_1 \circ \mu) \otimes (\omega_{\pi_p}^{-1}|_{Z(\mathbb{Q}_p)} \circ \mu)|\mu|_f^{-2} \\ &\cong \pi_p^\eta,\end{aligned}$$

where the last isomorphism holds for $\nu_1 \circ \mu = (\omega_{\pi_p}|_{Z(\mathbb{Q}_p)} \circ \mu)|\mu|^2$.

Third, by the unramified computation carried out in [PS18, §3.4 and §3.6], we have that for unramified W_p, Φ_p and ν_p,

$$\begin{aligned}L_p(s, \pi_p, \mathrm{std}) &= I_p(W_p, \Phi_p, \nu_p, s) \\ &= \int_{U_2(\mathbb{Q}_p)\backslash H(\mathbb{Q}_p)} \Phi_p((0,1)g_{1,p}) W_p(g_p) |\det(g_{1,p})|_p^s dg_p.\end{aligned}$$

Hence, for W_p associated to π_p, we have

$$\begin{aligned}L_p(s, (\tilde{\pi}_p \times \nu_1)|\mu|_f^{-2}, \mathrm{std}) &= L_p(s, \pi_p^\eta, \mathrm{std}) \\ &= \int_{U_2(\mathbb{Q}_p)\backslash H(\mathbb{Q}_p)} \Phi_p((0,1)g_{1,p}) W_p(t g_p t^{-1}) |\det(g_{1,p})|_p^s dg_p \\ &= \int_{U_2(\mathbb{Q}_p)\backslash H(\mathbb{Q}_p)} \Phi_p((0,1)t g_{1,p} t^{-1}) W_p(g_p) |\det(g_{1,p})|_p^s dg_p \\ &= \int_{U_2(\mathbb{Q}_p)\backslash H(\mathbb{Q}_p)} \Phi_p((0,1)g_{1,p}) W_p(g_p) |\det(g_{1,p})|_p^s dg_p \\ &= L_p(s, \pi_p, \mathrm{std}),\end{aligned}$$

where the fourth identity holds for unramified Φ_p. □

Theorem 10.4. *Let $n = a + b + r + s$. Under the conditions*

(1) $0 \le -r \le a$ and $0 \le -s \le b$,
(2) $a + r \ne b + s$,
(3) $a > 0$ and $b > 0$,
(4) $r \ne 0$ or $s \ne 0$,

the relation between $\mathcal{K}(\pi_f, V(2))$ and $\mathcal{D}(\pi_f, V(2))$ is as follows:

- *if $a \equiv b + 1 \pmod 2$, then*

$$\mathcal{K}(\pi_f, V(2)) = W(\pi) c^-(\pi_f, V(2)) \pi^{-(2a+2b+r+s+3)} W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1} L(n+1, \pi, \mathrm{std}) \mathcal{D}(\pi_f, V(2));$$

- *if $a \equiv b \pmod 2$, then*

$$\mathcal{K}(\pi_f, V(2)) = W(\pi) c^-(\pi_f, V(2)) \pi^{-(2a+2b+r+s+4)} W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1} L(n+1, \pi, \mathrm{std}) \mathcal{D}(\pi_f, V(2)).$$

The notations in the above formulas are as follows:

- *$W(\pi) \in \mathbb{C}^\times/\overline{\mathbb{Q}}^\times$ is a non-zero constant up to non-zero algebraic numbers depending on the normalization of the Whittaker–Fourier coeffcients of automorphic forms belongs to π,*
- *$W_{0,b+s-a-r}$ is the classical Whittaker function on $\mathbb{R}$,*
- *$c^-(\pi_f, V(2))$ is the Deligne period [Del79, §1.7] of the motive $M(\pi_f, V(2))$, which is non-zero,*
- *$L(s, \pi, \mathrm{std})$ is the standard L-function of the automorphic representation π.*

Proof. First, it follows from Corollary 7.53 that

$$\langle \Omega, \tilde{v}_K \rangle_B = \langle \omega_\Psi, [\rho] \rangle_B \otimes \mathbf{1}.$$

By equation (7.20), $\langle \omega_\Psi, [\rho] \rangle_B$ is equal to

(10.1)
$$-\frac{1}{2(a+b+r+s+1)} \sum_{j=A}^{B} (-1)^{b+r+s}(2j-b-r-s+1) C_j \int \Xi_{1+a+2b-2j+r+s, a+b+r+s-2j}(\phi_f).$$

Corollary 8.9 implies that

$$\int \Xi_{1+a+2b-2j+r+s, a+b+r+s-2j}(\phi_f) = c \cdot I(\varphi, \Phi, \nu, 1+a+b+r+s),$$

for $\Phi = \Phi_f \otimes \Phi_\infty, \varphi = \Psi_f \otimes X_{(1,-1,0)}^{1+a+2b-2j+r+s} \Psi_\infty, \nu = (\nu_1 = |\cdot|^{-n}\nu, \nu_2 = 1)$ and

$$c = (-1)^j 2i\pi^{1+a+b+r+s} \Gamma(1+a+b+r+s)^{-1}, \tag{10.2}$$

in which

$$\Phi_\infty(x,y) := (ix-y)^j (ix+y)^{a+b+r+s+j} e^{-\pi(x^2+y^2)}.$$

By Remark 8.35, only the term $j = b+s$ is left in the summation of formula (10.1). Hence, the pairing $\langle \omega_\Psi, [\rho] \rangle_B$ is equal to

$$\frac{(-1)^{b+s+r+1}(b+s-r+1)}{2(a+b+r+s+1)} C_{b+s} \cdot c \cdot I(\varphi, \Phi, \nu, 1+a+b+r+s),$$

for $\Phi = \Phi_f \otimes \Phi_\infty, \varphi = \Psi_f \otimes X_{(1,-1,0)}^{1+a+r-s} \Psi_\infty, \nu = (\nu_1 = |\cdot|^{-n}\nu, \nu_2 = 1)$ in which

$$\Phi_\infty(x,y) := (ix-y)^{b+s} (ix+y)^{a+r+2b+2s} e^{-\pi(x^2+y^2)}.$$

Second, it follows from Proposition 8.16 that the pairing $\langle \omega, [\rho] \rangle_B$ equals

$$\frac{(-1)^{b+r+s+1}(b+s-r+1)}{2(a+b+r+s+1)} c \cdot C_{b+s} W_\varphi(1) \prod_{p \in S} I_p(W_{\varphi,p}, \Phi_p, \nu_p, 1+a+b+r+s)$$
$$\times I_\infty(W_{\varphi,\infty}, \Phi_\infty, \nu_\infty, 1+a+b+r+s) L^S(1+a+b+r+s, (\tilde{\pi} \times \nu_1)|\mu|_f^{-2}, \mathrm{std}).$$

By the definition of the Whittaker period (see Definition 9.9), we can see that

$$W_\varphi(1) \sim W(\pi).$$

It follows from Lemma 7.61 and Lemma 10.3 that

$$\langle \omega_\Psi, [\rho] \rangle_B \sim c \cdot W(\pi) \prod_{p \in S} I_p(W_{\varphi,p}, \Phi_p, \nu_p, 1+a+b+r+s) I_\infty(W_{\varphi,\infty}, \Phi_\infty, \nu_\infty, 1+a+b+r+s)$$
$$\times L^S(a+b+r+s+1, \pi, \mathrm{std}).$$

Then by Proposition 8.21 and Corollary 8.23, if we make a careful choice of $W_{\varphi,p}$ and Φ_p, according to the proposition, we have

$$\langle \omega_\Psi, [\rho] \rangle_B \sim c \cdot W(\pi) I_\infty(W_{\varphi,\infty}, \Phi_\infty, \nu_\infty, 1+a+b+r+s) L(a+b+r+s+1, \pi, \mathrm{std}).$$

Third, we can see from the archimedean computation in Proposition 8.34 that

$$\langle \omega_\Psi, [\rho] \rangle_B \sim c \cdot W(\pi) \pi^{-(3a+3b+2r+2s+5)} W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1} \Gamma(a+b+r+s+1)$$
$$\times \Gamma(\frac{1}{2}a + \frac{3}{2}b + s + 2) \Gamma(\frac{1}{2}a + \frac{1}{2}b + r + 2) L(a+b+r+s+1, \pi, \mathrm{std}).$$

By the formula 10.2 for c, we have

$$\langle \omega_\Psi, [\rho] \rangle_B \sim W(\pi) \pi^{-(2a+2b+r+s+4)} W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1}$$
$$\times \Gamma(\frac{1}{2}a + \frac{3}{2}b + s + 2) \Gamma(\frac{1}{2}a + \frac{1}{2}b + r + 2) L(a+b+r+s+1, \pi, \mathrm{std}).$$

Recall that we have the following two formulas for Γ-values: for $n \in \mathbb{Z}_{\geq 0}$,

$$\Gamma(\frac{1}{2}+n) = \frac{(2n)!}{4^n n!}\sqrt{\pi},$$
$$\Gamma(n) = n!.$$

So it can be seen that if $a \equiv b+1 \,(\mathrm{mod}\, 2)$,

$$\Gamma\left(\frac{1}{2}a+\frac{3}{2}b+s+2\right) \sim \pi^{\frac{1}{2}},$$
$$\Gamma\left(\frac{1}{2}a+\frac{1}{2}b+r+2\right) \sim \pi^{\frac{1}{2}};$$

and if $a \equiv b \,(\mathrm{mod}\, 2)$,

$$\Gamma\left(\frac{1}{2}a+\frac{3}{2}b+s+2\right) \in \mathbb{Z},$$
$$\Gamma\left(\frac{1}{2}a+\frac{1}{2}b+r+2\right) \in \mathbb{Z}.$$

Hence, we can get that if $a \equiv b+1 \,(\mathrm{mod}\, 2)$

$$\langle \omega_\Psi, [\rho] \rangle_B \sim W(\pi)\pi^{-(2a+2b+r+s+3)} W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1} L(a+b+r+s+1, \pi, \mathrm{std});$$

and if $a \equiv b \,(\mathrm{mod}\, 2)$

$$\langle \omega_\Psi, [\rho] \rangle_B \sim W(\pi)\pi^{-(2a+2b+r+s+4)} W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1} L(a+b+r+s+1, \pi, \mathrm{std}).$$

Finally, combine it with the formula

$$\mathcal{K}(\pi_f, V(2)) = \frac{\langle \Omega, \tilde{v}_K \rangle_B}{\langle \Omega, \tilde{v}_D \rangle_B} \mathcal{D}(\pi_f, V(2)).$$

and Proposition 9.7, we complete the proof. □

Remark 10.5.
- The condition (1) guarantees that we have the branching from G to H. The condition (2) makes sure that we are in the weight ≤ -3 situation of Beilinson's conjectures (see [Nek94, Conjecture (6.1)]). The condition (3) implies the local system associated to V be regular so that we have the vanishing theorem. Condition (4) guarantees that we have the vanishing on the boundary theorem.
- Under the condition of the theroem, it follows from Proposition 8.14 and [GJ72, Page 151] that the point $s = 1+n = 1+a+b+r+s$ lies in the region of absolute convergent of $L(s, \pi, \mathrm{std})$.

10.2. **Connection to motivic L-functions.** In this subsection, we relate the automorphic L-functions to motivic L-functions and compare Theorem 10.4 with Beilinson's conjectures.

Notation 10.6.
- Recall that for any $V \in \mathrm{Rep}_{\mathbb{Q}}(G)$, there is a canonical l-adic local system $\mu_l(V)$ on S (with infinite level structure). We use V to denote the l-adic local system.
- We can define l-adic cohomology $\mathrm{H}^2_{\text{ét}}(S_{\overline{\mathbb{Q}}}, V(2))$, compactly supported l-adic cohomology $\mathrm{H}^2_{\text{ét},c}(S_{\overline{\mathbb{Q}}}, V(2))$ and interior l-adic cohomology

$$\mathrm{H}^2_{\text{ét},!}(S_{\overline{\mathbb{Q}}}, V(2)) := \mathrm{Im}(\mathrm{H}^2_{\text{ét},c}(S_{\overline{\mathbb{Q}}}, V(2)) \to \mathrm{H}^2_{\text{ét}}(S_{\overline{\mathbb{Q}}}, V(2))).$$

Remark 10.7. The interior cohomology $\mathrm{H}^2_{!,\text{ét}}(S_{\overline{\mathbb{Q}}}, V(2))$ is isomorphic to the intersection cohomology $\mathrm{IH}^2_{\text{ét}}(S^*_{\overline{\mathbb{Q}}}, V(2))$, where S^* is the Baily-Borel compactification of S. (see [Cav19, Corollary 2.3.3.4].)

Definition 10.8. The comparison isomorphism

$$\mathrm{H}^2_{\text{ét},!}(S_{\overline{\mathbb{Q}}}, V(2)) \cong \mathrm{H}^2_{B,!}(S(\mathbb{C}), V(2)) \otimes_{\mathbb{Q}} \mathbb{Q}_l,$$

suggests we define the following $E(\pi_f) \otimes_{\mathbb{Q}} \mathbb{Q}_l$-module as in Definition 3.39:

$$M_{\text{ét}}(\pi_f, V(2)) := \mathrm{Hom}_{\mathbb{Q}[G(\mathbb{A}_f)]}(\mathrm{Res}_{E(\pi_f)/\mathbb{Q}}\, \pi_f, \mathrm{H}^2_{\text{ét},!}(S_{\overline{\mathbb{Q}}}, V(2))).$$

This has an action of the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ induced by the action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on $S_{\overline{\mathbb{Q}}}$.

Remark 10.9. The $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-module $M_{\text{ét}}(\pi_f, V(2))$ is the l-adic étale realization of the Grothendieck motive associated to π constructed in [Wil15, Theorem 5.6].

Definition 10.10 (Motivic L-function)**.** Define $L(M_{\text{ét}}(\pi_f, V(2)), s)$, the motivic L-function for $s \in \mathbb{C}$, as the following partial Euler product (without archimedean factor):

$$L(M_{\text{ét}}(\pi_f, V(2)), s) := \prod_{v<\infty} L_v(M_{\text{ét}}(\pi_f, V(2)), s),$$

where

- if $v = p \neq l$, we define the local L-factor as
$$L_v(M_{\text{ét}}(\pi_f, V(2)), s) := \det(1 - Fr_p p^{-s} | M_{\text{ét}}(\pi_f, V(2))^{I_p})^{-1},$$
and I_p is the inertia group at the prime p and Fr_p is geometric Frobenius at p;
- if $v = l$, we choose a prime $l' \neq l$ and use l' to define $M_{\text{ét}}(\pi_f, V(2))$ and the motivic L-factor $L_v(M_{\text{ét}}(\pi_f, V(2)), s)$. It makes sense by the *independence of l* in Proposition 10.12.

Convention 10.11. As explained in Remark 7.13, we can view the motivic L-function as

$$L_v(M_{\text{ét}}(\pi_f, V(2)), s) \cdot \mathbf{1}$$

where $\mathbf{1}$ is unit of $E(\pi_f) \otimes_{\mathbb{Q}} \overline{\mathbb{Q}_l}$. Hence, we now view $L_v(M_{\text{ét}}(\pi_f, V(2)), s)$ as scalar-valued.

Proposition 10.12. *For any finite place v, the L-factor $L_v(M_{ét}(\pi_f, V(2)), s)$ is a polynomial in $\frac{1}{(Nv)^s}$ with coefficients in $\overline{\mathbb{Q}}$. In other words, the definition of the L-factor is independent of the choice of l.*

Proof. This follows from a theorem of Gabber (see [Fuj02, Theorem 1]). □

Now if we fix an embedding $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$, then the L-factor $L_v(M_{\text{ét}}(\pi_f, V(2)), s)$ makes sense as a $\mathbb{C}$-valued function.

Proposition 10.13. *The infinite product in the definition of $L(M_{ét}(\pi_f, V(2)), s)$ is absolutely convergent when $\mathrm{Res}(s) > \frac{-a-b-r-s}{2}$.*

Proof. It follows from the fact the weight of $M_{\text{ét}}(\pi_f, V(2))$ is $-2-a-b-r-s$ (see Corollary 3.42) and [Nek94, (1.5)]. □

Proposition 10.14. *If we let $L^S(M_{ét}(\pi_f, V(2)), s)$ be the partial motivic L-function away from finitely many bad primes in S, then we have*

$$L^S(M_{ét}(\pi_f, V(2)), s) = L^S(s + n + 1, \pi, std),$$

for $s \in \mathbb{C}$.

Proof. In our setting, we use the "homological normalization" of $V(2)$, but in [LR92], "cohomological normalization" is used. Hence, the motivic L-function in [LR92] is the

$$L^S(M_{\text{ét}}(\pi_f, V(2))(-2-n), s)$$

in our setting. It is proved in [LR92, Theorem A, P291] that

$$L^S(M_{\text{ét}}(\pi_f, V(2))(-2-n), s) = L^S(s-1, \pi, \mathrm{std}).$$

Hence, we have

$$L^S(M_{\text{ét}}(\pi_f, V(2)), s-n-2) = L^S(M_{\text{ét}}(\pi_f, V(2))(-2-n), s) = L^S(s-1, \pi, \text{std}).$$

This implies

$$L^S(M_{\text{ét}}(\pi_f, V(2)), s) = L^S(s+n+1, \pi, \text{std}).$$

□

Remark 10.15. Let us check the compatibility of motivic normalization with automorphic normalization.

First, we check from the motivic side. It follows from the weight of $M_B(\pi_f, V(2))$ is $w = -2-n$ that[23]

$$L^S(M_{\text{ét}}(\pi_f, V(2)), s) \leftrightarrow L^S(M_{\text{ét}}(\pi_f, V(2)), -1-n-s).$$

Then by Proposition 10.14, we can see that

$$L^S(M_{\text{ét}}(\pi_f, V(2)), s) = L^S(s+n+1, \pi, \text{std}),$$
$$L^S(M_{\text{ét}}(\pi_f, V(2)), -1-n-s) = L^S(-s, \pi, \text{std}).$$

So by the argument in Lemma 10.3, we get

$$L^S(s+n+1, \pi, \text{std}) = L^S(1+s, \tilde{\pi}, \text{std}) \leftrightarrow L^S(-s, \pi, \text{std}). \tag{10.3}$$

Hence, the center of $L(s, \pi, \text{std})$ is at $s = \frac{1}{2}$.

Second, we check from the automorphic side. The central character of π is

$$z \in Z(\mathbb{A}_{\mathbb{Q}}) \mapsto |z|^{-n}\nu,$$

where ν is a finite order Hecke character of sign $(-1)^n$. Hence, the automorphic representation $\pi|\mu|^{\frac{n}{2}}$ is unitary. Therefore, by the argument in Lemma 10.3, we have

$$L^S(s, \pi, \text{std}) = L^S(s, (\pi|\mu|^{\frac{n}{2}})|\mu|^{-\frac{n}{2}}, \text{std}) = L^S\left(s-\frac{n}{2}, \pi|\mu|^{\frac{n}{2}}, \text{std}\right).$$

The automorphic representation $\pi|\mu|^{\frac{n}{2}}$ is unitary, so by [GPS84, (5.1), P163], we have

$$L^S\left(s-\frac{n}{2}, \pi|\mu|^{\frac{n}{2}}, \text{std}\right) \leftrightarrow L^S\left(1+\frac{n}{2}-s, \tilde{\pi}|\mu|^{-\frac{n}{2}}, \text{std}\right).$$

It can be seen from the argument in Lemma 10.3 that

$$L^S\left(1+\frac{n}{2}-s, \tilde{\pi}|\mu|^{-\frac{n}{2}}, \text{std}\right) = L^S(1-s, \pi|\mu|^{\frac{n}{2}}, \text{std}).$$

Thus

$$L^S(s, \pi, \text{std}) \leftrightarrow L^S(1-s, \pi, \text{std}), \tag{10.4}$$

which means the center of $L(s, \pi, \text{std})$ is at $s = \frac{1}{2}$. Hence, the automorphic normalization is compatible with the motivic normalization.

Using the previous proposition, we can state our main theorem in terms of motivic L-functions.

Theorem 10.16. *Let $n = a+b+r+s$. Under the conditions*

(1) $0 \le -r \le a$ and $0 \le -s \le b$,
(2) $a+r \ne b+s$,
(3) $a > 0$ and $b > 0$,
(4) $r \ne 0$ or $s \ne 0$,

the relation between $\mathcal{K}(\pi_f, V(2))$ and $\mathcal{D}(\pi_f, V(2))$ is as follows:

- *if $a \equiv b+1 \pmod 2$, then*

$$\mathcal{K}(\pi_f, V(2)) = W(\pi)c^-(\pi_f, V(2))\pi^{-(2a+2b+r+s+3)}W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1}L(M_{\acute{e}t}(\pi_f, V(2)), 0)\mathcal{D}(\pi_f, V(2));$$

[23] The notation "$\leftrightarrow$" means that equality holds after multiply by the remaining Euler factors.

- *if* $a \equiv b \pmod 2$, *then*

$$\mathcal{K}(\pi_f, V(2)) = W(\pi)c^-(\pi_f, V(2))\pi^{-(2a+2b+r+s+4)}W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1}L(M_{\acute{e}t}(\pi_f, V(2)), 0)\mathcal{D}(\pi_f, V(2)).$$

The notations in the above formulas are as follows:

- $W(\pi) \in \mathbb{C}^\times/\overline{\mathbb{Q}}^\times$ *is a non-zero constant up to nonzero algebraic numbers depending on the normalization of the Whittaker–Fourier coefficients of automorphic forms belonging to* π,
- $W_{0,b+s-a-r}$ *is the classical Whittaker function on* $\mathbb{R}$,
- $\mathrm{c}^-(\pi_f, V(2))$ *is the Deligne period [Del79, §1.7] of the motive* $M(\pi_f, V(2))$ *which is non-zero,*
- $L(M_{\acute{e}t}(\pi_f, V(2)), s)$ *is the motivic L-function of the automorphic representation* π.

Proof. Since for a finite place v of $\mathbb{Q}$, the value of local L-factor $L_v(M_{\text{ét}}(\pi_f, V(2), s)$ at $s = 0$ and the value of $L_v(s, \pi, \mathrm{std})$ at $s = 1 + a + b + r + s$ belongs to $\overline{\mathbb{Q}}^\times$, we only need to consider the partial L-functions. By Proposition 10.14, we have

$$L^S(M_{\text{ét}}(\pi_f, V(2)), 0) = L^S(1 + n, \pi, std).$$

□

Remark 10.17. The point $s = 0$ of the motivic L-function $L(M(\pi_f, V(2)), s)$ is exactly what Beilinson conjectured in his conjectures [Nek94, Conjecture (6.1)], and the shape of the theorem is similar to the rank 1 case of the weak Beilinson's conjectures [Nek94, (6.6)]. It only remains to prove

- if $a \equiv b + 1 \pmod 2$, then

$$W(\pi)c^-(\pi_f, V(2))\pi^{-(2a+2b+r+s+3)}W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1} \in \overline{\mathbb{Q}}^\times;$$

- if $a \equiv b \pmod 2$, then

$$W(\pi)c^-(\pi_f, V(2))\pi^{-(2a+2b+r+s+4)}W_{0,b+s-a-r}(8\sqrt{2}\pi D^{-\frac{3}{4}})^{-1} \in \overline{\mathbb{Q}}^\times.$$

We will leave this to future work.

Corollary 10.18. *The* $E(\pi_f)$*-rational structure* $\mathcal{K}(\pi_f, V(2))$ *is non-trivial. In other words, the morphism in Construction 4.44*

$$\mathcal{E}is^n_M \colon \mathcal{B}_n \to \mathrm{H}^3_M(S, V(2))$$

is non-trivial.

Proof. It follows from Proposition 10.13 (or Remark 10.5) that

$$L(M(\pi_f, V(2)), 0) \neq 0.$$

It follows from the fact that $W(\pi) \neq 0$, $W_{0,r-s}(8\sqrt{2}D^{-\frac{3}{4}})^{-1} \neq 0$, $\mathrm{c}^-(\pi_f, V(2)) \neq 0$ and $\mathcal{D}(\pi_f, V(2))$ is non-trivial and Theorem 10.16 that $\mathcal{K}(\pi_f, V(2))$ is non-trivial. □

Remark 10.19.

- Theorem 10.16 is a $w \leq -3$ counterpart of what is proved in [PS18, Theorem 8.18], where they proved a case that the motivic sheaf V is trivial and the weight of the motive is -2.
- In [LSZ22], the authors constructed an Euler system for $\mathrm{GU}(2,1)$ based on the motivic class $\mathcal{E}is^n_M(\phi_f)$ for $\phi_f \in \mathcal{B}_n$ being non-trivial, so our theorem of non-triviality of $\mathcal{K}(\pi_f, V(2))$ answers a question raised in their paper, which gives a necessary condition for their Euler system to work.

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